

THE LANCET

Global Health

Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed.
We post it as supplied by the authors.

Supplement to: Smith ER, Shankar AH, Wu LS-F, et al. Modifiers of the effect of maternal multiple micronutrient supplementation on stillbirth, birth outcomes, and infant mortality: a meta-analysis of individual patient data from 17 randomised trials in low-income and middle-income countries. *Lancet Glob Health* 2017; **5**: e1090–100.

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Appendix A
Supplemental Tables

Supplemental Table 1. Nutrient composition of control and intervention group by study

	Control Group		MMS Intervention Group																
	Iron (mg)	Folic Acid (mcg)	Iron (mg)	Folic Acid (mcg)	A (mcg)	D (IU)	E (mg)	C (mg)	B1 (mg)	B2 (mg)	Niacin (mg)	B6 (mg)	B12 (mcg)	Zinc (mg)	Copper (mg)	Selenium (mcg)	Iodine (mcg)	K (mcg)	Others
Fawzi 1998	120	5 mg folate	120	5 mg folate	A: 5000 IU B: 0		30	500	1.4	20	100	25	50			-	-	-	-
Christian 2003	60	400	60	400	1000	10 mcg	10	100	1.6	1.8	20	2.2	2.6	-	2	-	-	65	^a
Ramakrishnan 2003	60	215	60	215	2150 IU	309	5.73 IU	66.5	0.93	1.87	15.5	1.94	2.04	12.9		-	-	-	^b
Friis 2004					3000	10 mcg	10	80	1.5	1.6	17	2.2	4	15	1.2	65	-	-	^c
Fawzi 2007	60	250	60	250	-	25 mg	30	500	-	-	100	-	50			-	-	-	
Gupta 2007	60	500	60	500	2500 IU	1 mg	7.5	50	1.5	1	-	-	1	15	2	-	-	-	^d
Kaestel 2005	60	400	30	A:400 B:800	A:800 B:1600	A:200 B:400	A:10 B:20	A:70 B:140	A:1.4 B:2.8	A:1.4 B:2.8	A:18 B:36	A:1.9 B:3.8	A:2.6 B:5.2	A:15 B:30	A:2 B:4	A:65 B:130	A:150 B:300	-	-
Osrin 2005	60	400	30	400	800	5 mcg	10	70	1.4	1.4	18	1.9	2.6	15	2	65	150	-	-
Zagre 2007	60	400	30	400	800	200	10	70	1.4	1.4	18	1.9	2.6	15	2	65	150	-	-
Roberfroid 2008	60	400	30	400	800	200	10	70	1.4	1.4	18	1.9	2.6	15	2	65	150	-	-
Shankar 2008	30	400	30	400	800	200	10	70	1.4	1.4	18	1.9	2.6	15	2	65	150	-	-
Zeng 2008	60	400	30	400	800	200	10	70	1.4	1.4	18	1.9	2.6	15	2	65	150	-	-
Bhutta 2009	60	400	30	400	800	200	10	70	1.4	1.4	18	1.9	2.6	15	2	65	150	-	-
Persson 2012	A: 60 B: 30	400	30	400	800	200	10	70	1.4	1.4	18	1.9	2.6	15	2	65	150	-	-
West 2014	27	600	27	600	800	200	10	70	1.4	1.4	18	1.9	2.6	15	2	65	150	-	-
Ashorn 2015	60	400	20	400	800	400	20	100	2.8	2.8	36	3.8	5.2	30	4	130	250	45	^e
Adu-Afarwuah2015	60	400	20	400	800	400	20	100	2.8	2.8	36	3.8	5.2	30	4	130	250	45	^e

^a magnesium 100 mg

^b magnesium 252 mg

^c beta carotene 3.5 mg

^d vitamin D 200IU, vitamin E 7.5mg, calcium patothenate 5mg, ferrous fumarate 10mg, folic acid 0.15 mg, nicotinamide 20mg, biotin 30 mcg, potassium iodide 0.15mg, manganese sulfate 2.5 mg, calcium 162 mg, phosphorus 125 mg, potassium 40 mg, chloride 36.3mg, chromium 25mcg, molybdenum, 25 mcg, sodium selenate 30 mcg, nickel 5 µg, silicon dioxide 2 mg, vanadium 10 mcg, boron 150mcg

^e pantothenic acid 7mg, manganese 2.6mg

Supplemental Table 2. Study Quality*

Study	Randomization Procedures	If cluster RCT, >12 clusters?	Allocation concealment	Assessment blinded?	Loss to follow up?	Overall risk of bias?
Fawzi 1998	Block randomization at individual level		Yes	Yes: Participants, field staff, researchers blinded	Moderate limitation ~10% LTFU for birth outcomes	Moderate
Christian 2003	Block randomization for village clusters	Yes (169 clusters)	Unclear	Yes: Participants, field staff, researchers blinded	<5% LTFU for birth outcomes	Low
Ramakrishnan 2003	Individual randomization using color coding & pre-generated list		Yes	Yes: Participants, field staff, researchers blinded	Serious limitation due to ~25% LTFU for birth outcomes	Moderate
Friis 2004	Block randomization at individual level		Yes	Yes: Participants, field staff, researchers blinded	Serious limitation due to ~34% LTFU for birth outcomes	Moderate
Kaestel 2005	Block randomization at individual level		No (due to coloring)	Yes: Participants, field staff, researchers blinded	Serious limitation due to 20.4% LTFU for birth outcomes	Moderate
Osrin 2005	Block randomization at individual level		Yes	Yes: Participants, field staff, researchers blinded	5% LTFU for birth outcomes	Low
Gupta 2007	Block randomization at individual level		Yes	Yes; Participants, field staff & most researchers blinded	Serious limitation >27% LTFU for birth outcomes	Moderate
Zagre 2007	Unclear method of randomization at village level	Yes (78 clusters)	Unclear	Yes; Participants & field staff blinded	Serious limitation ~18% LTFU and additional 7% missing birthweights	Moderate
Fawzi 2007	Block randomization at individual level		Yes	Yes: Participants, field staff, researchers blinded	~5% LTFU for birth outcomes	Low
SUMMIT 2008	Block randomization of midwives	Yes (262 clusters)	Yes	Yes: Participants, field staff, researchers blinded	~5% LTFU for birth outcomes	Low
Zeng 2008	Block randomization at cluster level	Yes (531 clusters)	Yes	Yes: Participants, field staff, researchers blinded	<5% LTFU for birth and mortality outcomes	Low
Roberfroid 2008	Block randomization at individual level		Yes	Yes: Participants & field staff	~7.5% LTFU for birth outcomes and additional ~15% missing birth weight	Moderate
Bhutta 2009	Randomization of CHW cluster	Yes (28 clusters)	Yes	Yes: Participants & field staff	~10% LTFU for birth outcomes and additional ~5% missing birth weight	Moderate
Persson 2012	Block randomization at individual level		Unclear	Yes: Participants & field staff	~19% LTFU for birth outcomes	Moderate
West 2014	Randomization at sector level	Yes (596 clusters)	Yes	Yes: Participants, field staff, researchers blinded	<5% LTFU for birth outcomes	Low
Ashorn 2015	Randomization at individual level		Yes	Yes: Participants, field staff, researchers blinded	~6% LTFU with additional 18% missing birth weight	Moderate
Adu-Afarwuah 2015	Randomization at individual level		Yes	Yes: Participants, field staff, researchers blinded	~20% LTFU for birth outcomes	Moderate

*All studies conducted ITT analysis per meta-analysis protocol

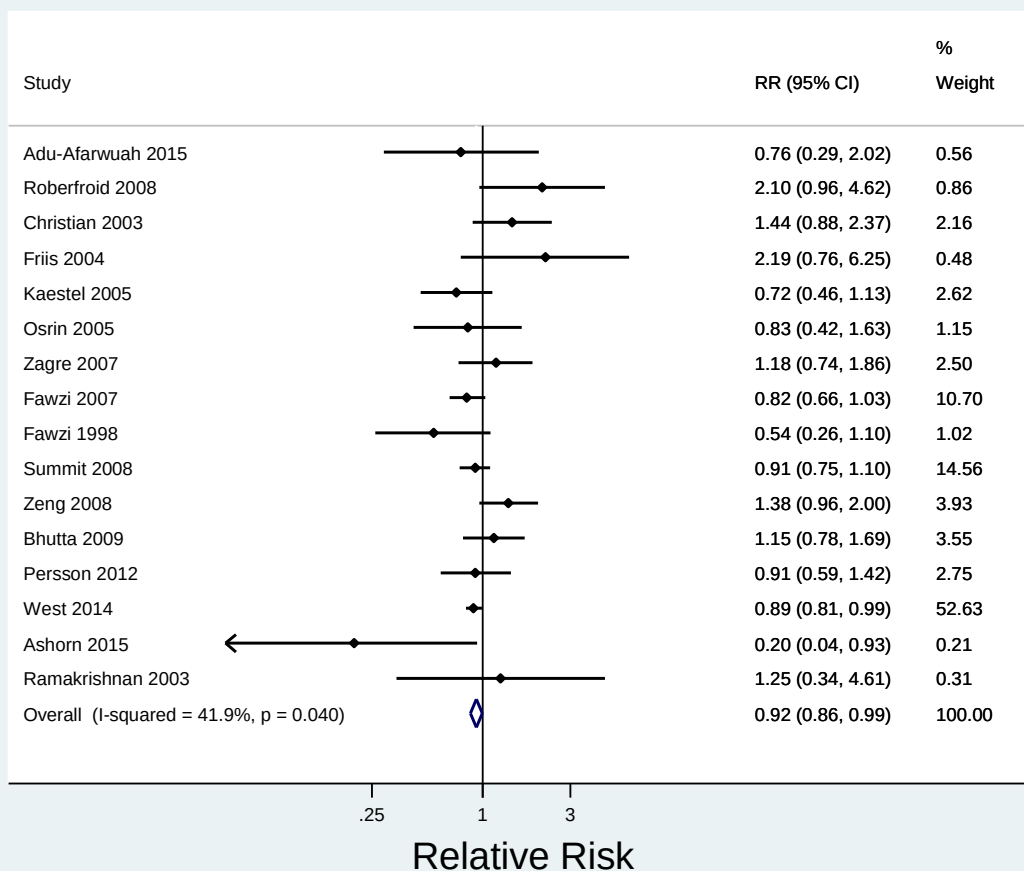
Supplemental Table 3. Prevalence of potential effect modifiers by study site

Study	1st Trimester at enrollment	2nd Trimester at enrollment	3rd Trimester at enrollment	<20 weeks gestation at enrollment	Maternal age <18 years	Maternal age <20 years	Maternal BMI <18.5 kg/m ²	Maternal stature <150 cm	Firstborn child	No formal maternal education	Skilled birth attendant at delivery	Maternal hemoglobin <110 g/L	Adherence >95%
Fawzi 1998	1%	99%	0%	35%	4%	14%	3%	9%	34%	8%	-	58%	-
Christian 2003	73%	26%	1%	93%	15%	29%	36%	49%	26%	82%	47%	34%	32%
Ramakrishnan 2003	89%	11%	0%	0%	10%	16%	18%	4%	60%	5%	-	4%	50%
Friis 2004	0%	37%	63%	0%	8%	23%	0%	2%	41%	-	-	29%	-
Kaestel 2005	8%	68%	24%	31%	7%	21%	5%	2%	32%	30%	56%	60%	14%
Osrin 2005	10%	90%	0%	93%	8%	30%	28%	44%	45%	46%	-	35%	66%
Gupta 2007	0%	39%	61%	100%	0%	4%	77%	4%	39%	23%	79%	86%	90%
Zagre 2007	91%	9%	0%	100%	11%	21%	19%	5%	20%	91%	-	-	70%
Fawzi 2007	1%	99%	0%	33%	0%	16%	2%	16%	45%	7%	2%	68%	51%
Summit 2008	25%	47%	28%	49%	4%	14%	6%	45%	35%	12%	57%	29%	22%
Zeng 2008	50%	50%	1%	86%	8%	7%	12%	4%	64%	2%	82%	-	64%
Roberfroid 2008	34%	55%	11%	60%	11%	26%	11%	1%	21%	83%	-	50%	-
Bhutta 2009	79%	21%	0%	98%	3%	9%	24%	27%	20%	79%	32%	49%	1%
Persson 2012	93%	7%	0%	99%	4%	12%	3%	48%	34%	32%	61%	29%	-
West 2014	82%	17%	1%	95%	13%	28%	40%	52%	30%	31%	59%	22%	50%
Ashom 2015	3%	97%	0%	97%	10%	25%	5%	14%	22%	24%	82%	44%	24%
Adu-Afarwuah 2015	15%	85%	0%	86%	7%	7%	3%	5%	24%	9%	89%	42%	4%

Supplemental Table 4. Prevalence of outcomes stratified by study site

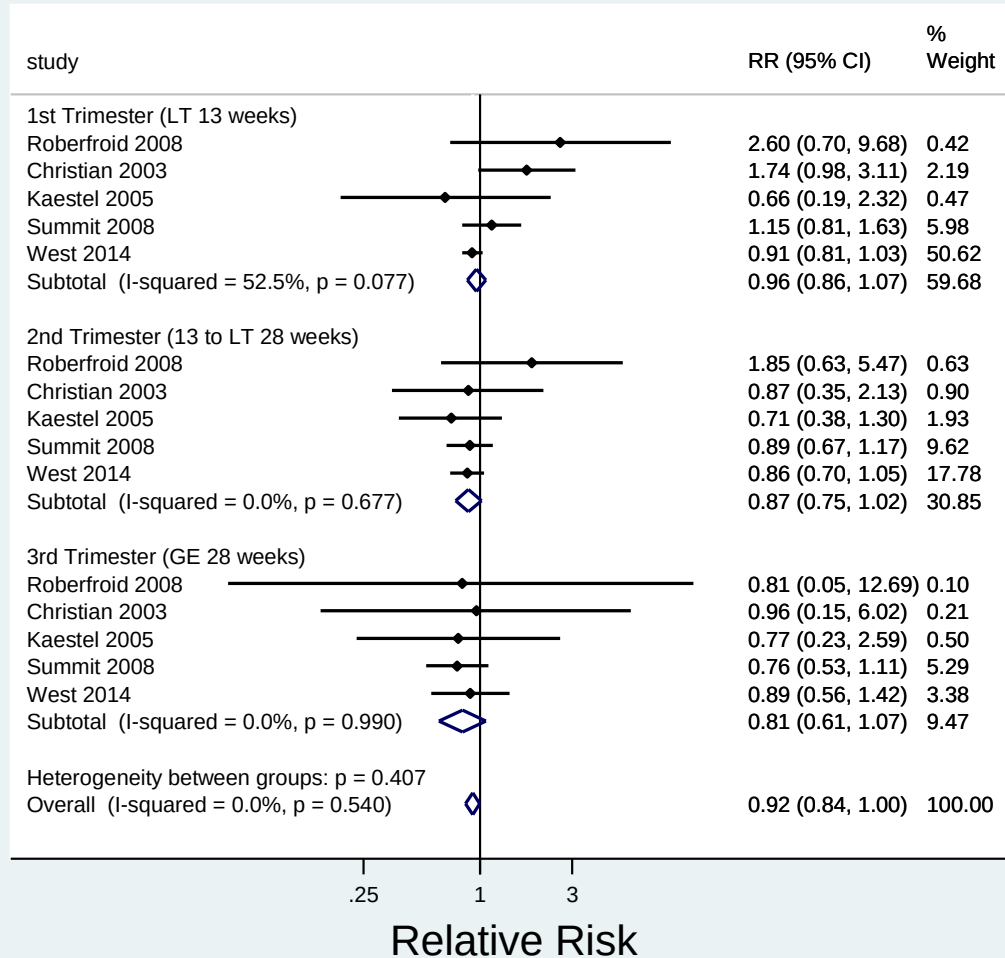
<i>Study</i>	Stillbirth	Death <7 days	Death <28 days	Death <180 days	Death <365 days	Very preterm (<32 weeks)	Preterm (<37 weeks)	SGA (Oken <10th %)	SGA (Intergrowth <10th %)	LGA (Oken >90th %)	LGA (Intergrowth >90th %)	Low birthweight (<2500g)	Very low birthweight (<2000g)
Fawzi 1998	6%	-	-	-	-	5%	24%	28%	17%	4%	14%	11%	0%
Christian 2003	4%	3%	5%	6%	6%	3%	22%	56%	43%	2%	5%	34%	6%
Ramakrishnan 2003	1%	1%	-	-	-	1%	7%	36%	28%	1%	4%	10%	1%
Friis 2004	1%	-	-	-	-	2%	16%	25%	16%	4%	11%	10%	1%
Kaestel 2005	4%	6%	6%	-	-	4%	24%	24%	17%	5%	17%	11%	1%
Osrin 2005	3%	2%	3%	3%	3%	0%	5%	61%	45%	0%	1%	22%	3%
Gupta 2007	0%	5%	-	-	-	0%	1%	80%	68%	0%	0%	30%	3%
Zagre 2007	4%	-	-	-	-	-	-	-	-	-	-	8%	1%
Fawzi 2007	4%	2%	3%	-	-	2%	16%	22%	15%	5%	15%	7%	2%
Summit 2008	2%	2%	2%	4%	-	3%	26%	17%	12%	9%	23%	10%	1%
Zeng 2008	4%	1%	1%	1%	2%	0%	5%	21%	13%	3%	9%	4%	0%
Roberfroid 2008	2%	1%	2%	4%	6%	3%	14%	38%	23%	2%	4%	15%	3%
Bhutta 2009	6%	3%	5%	-	-	-	-	-	-	-	-	18%	3%
Persson 2012	2%	3%	3%	3%	4%	1%	12%	60%	43%	0%	2%	31%	4%
West 2014	5%	4%	4%	5%	6%	3%	20%	66%	52%	2%	2%	43%	8%
Ashorn 2015	1%	3%	4%	5%	6%	2%	9%	34%	19%	1%	4%	13%	2%
Adu-Afarwuah 2015	2%	1%	1%	2%	2%	1%	7%	30%	18%	2%	4%	11%	2%

Relative risk of stillbirth - OVERALL



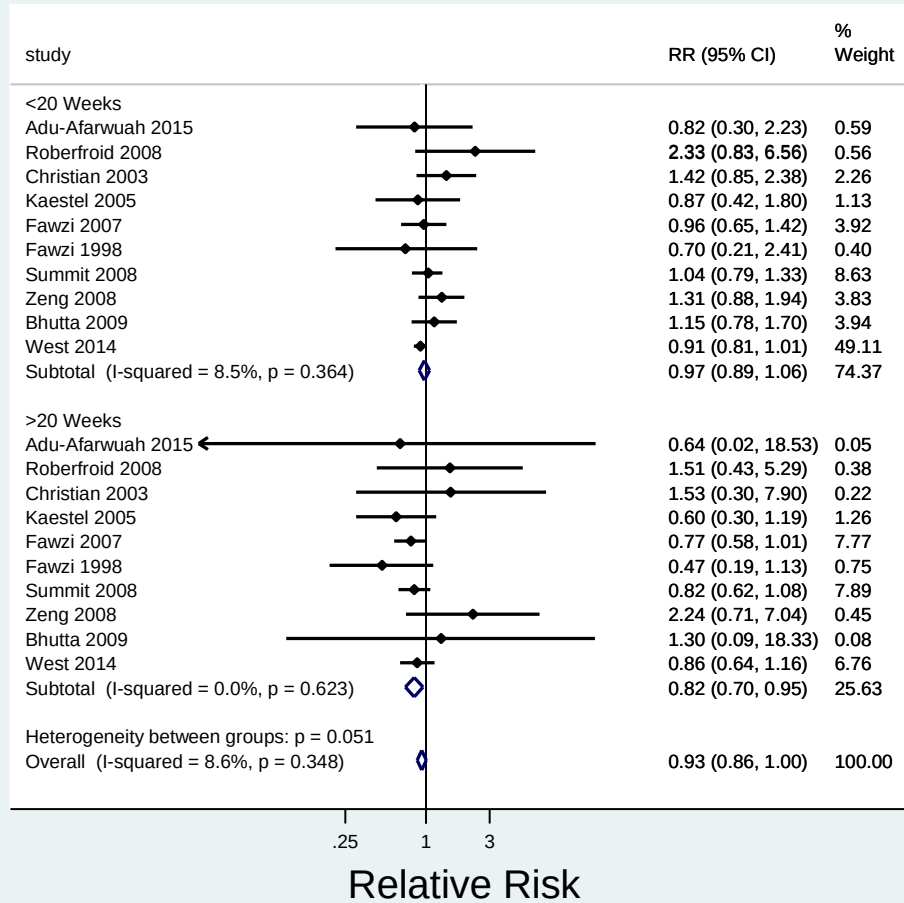
Outcome = Stillbirth

Relative risk of stillbirth by trimester at enrollment



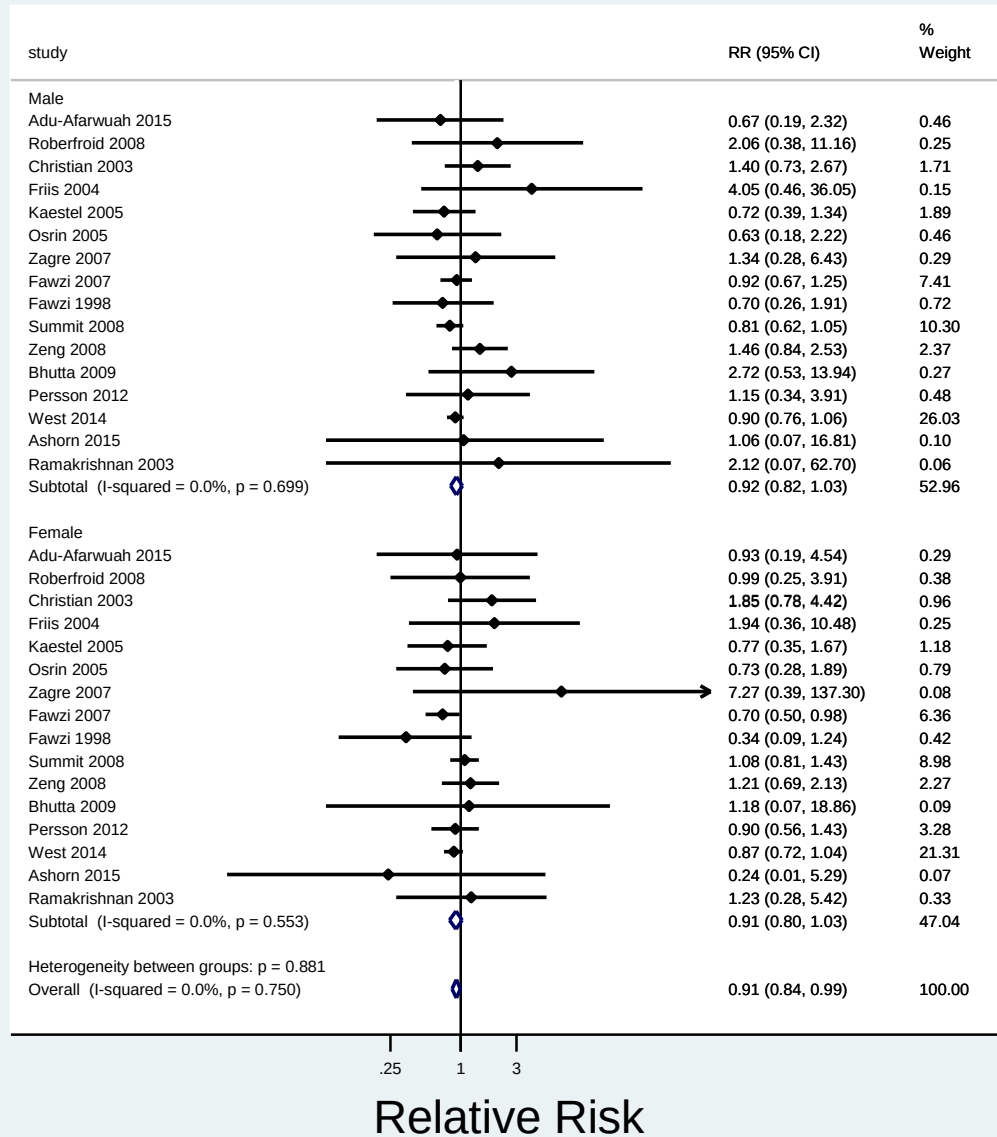
Outcome = Stillbirth

Relative risk of stillbirth by gestational age at enrollment



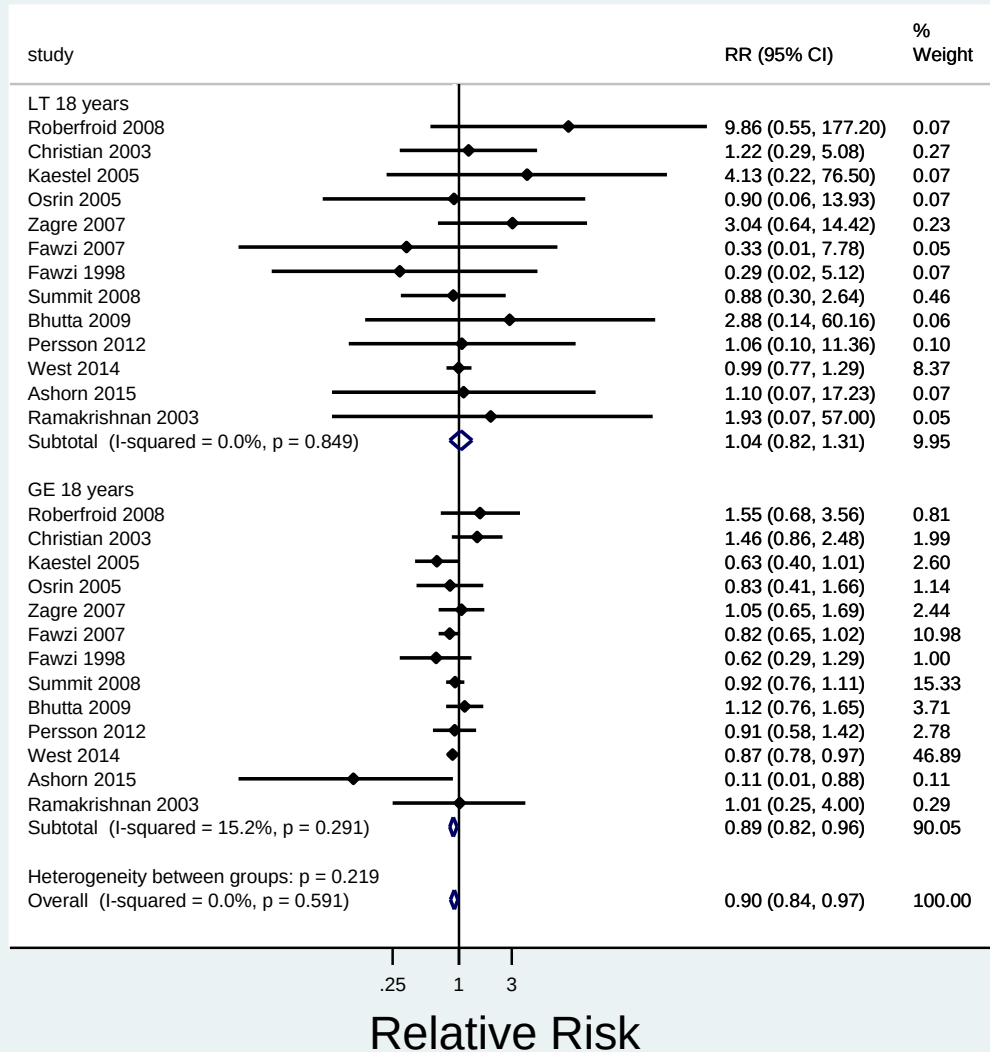
Outcome = Stillbirth

Relative risk of stillbirth by sex



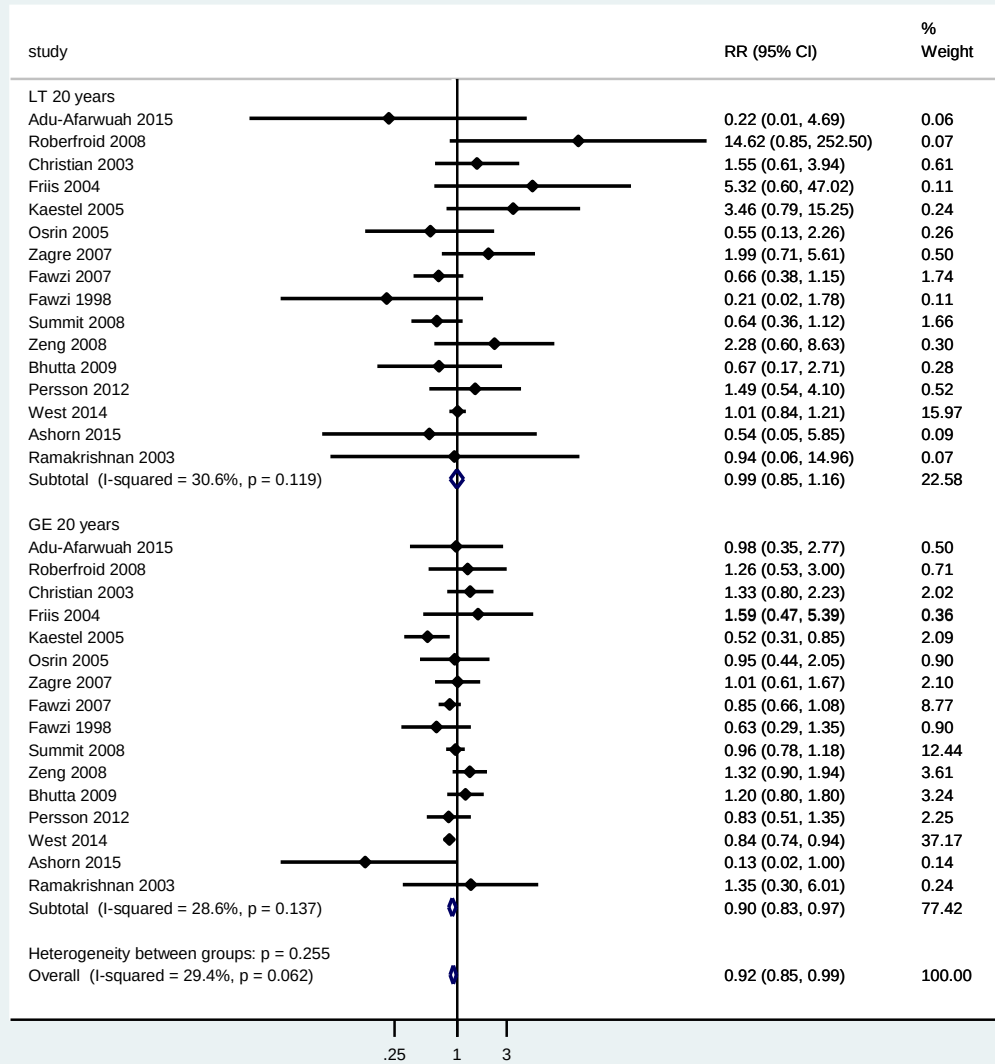
Outcome = Stillbirth

Relative risk of stillbirth by maternal age <18



Outcome = Stillbirth

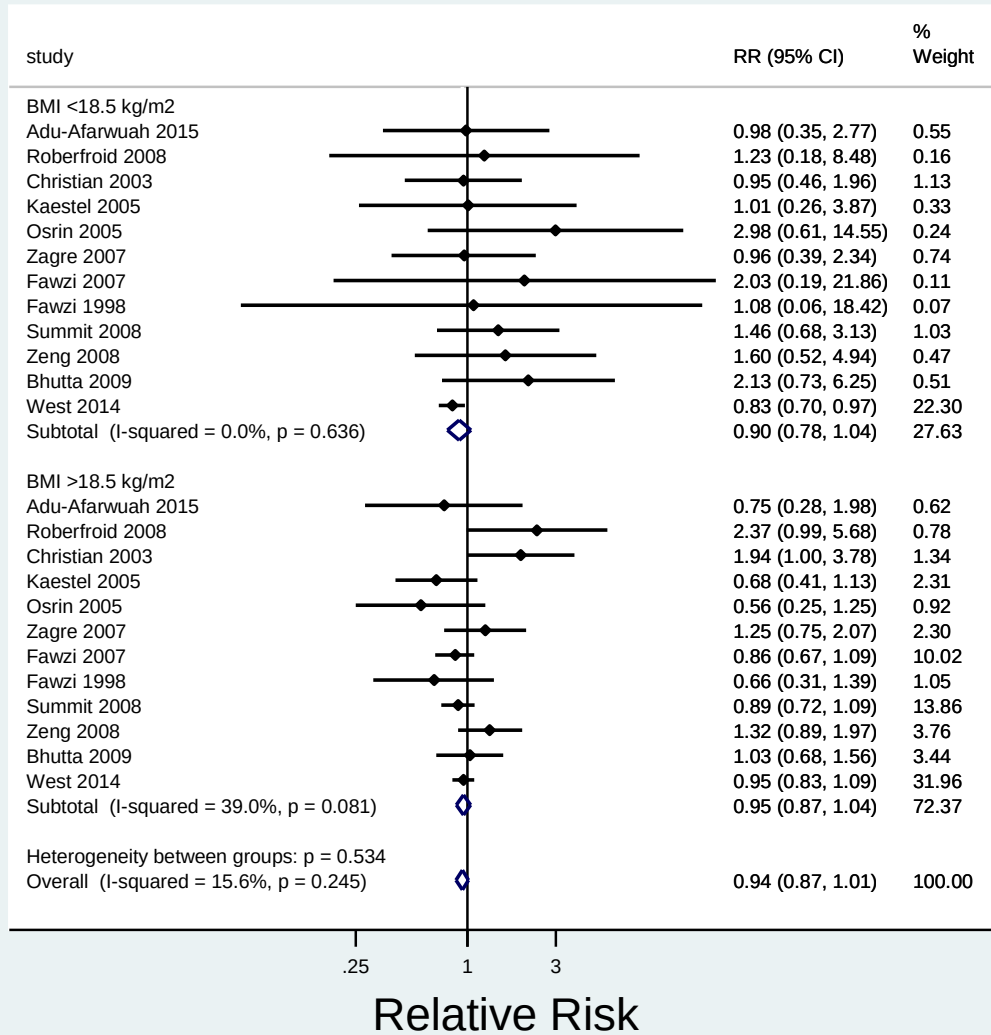
Relative risk of stillbirth by maternal age <20



Relative Risk

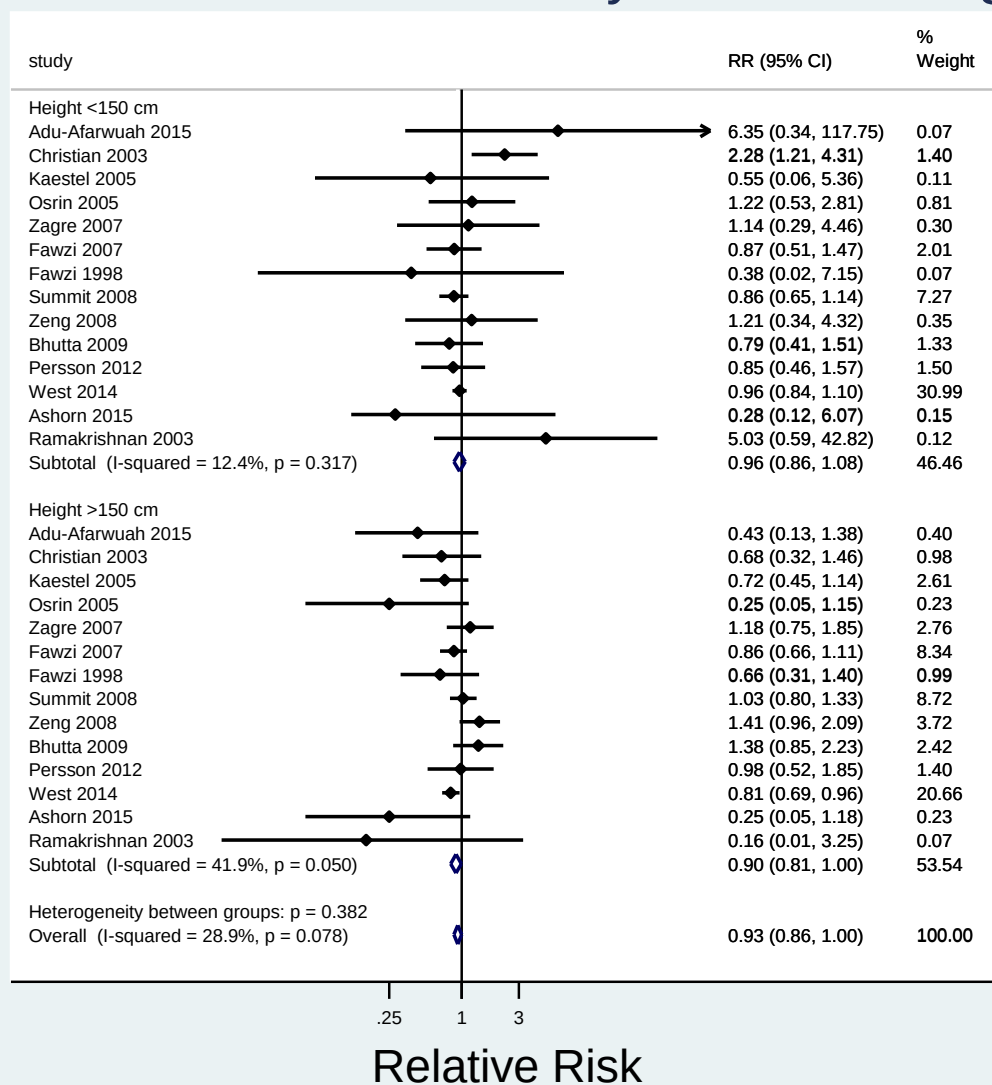
Outcome = Stillbirth

Relative risk of stillbirth by mat weight



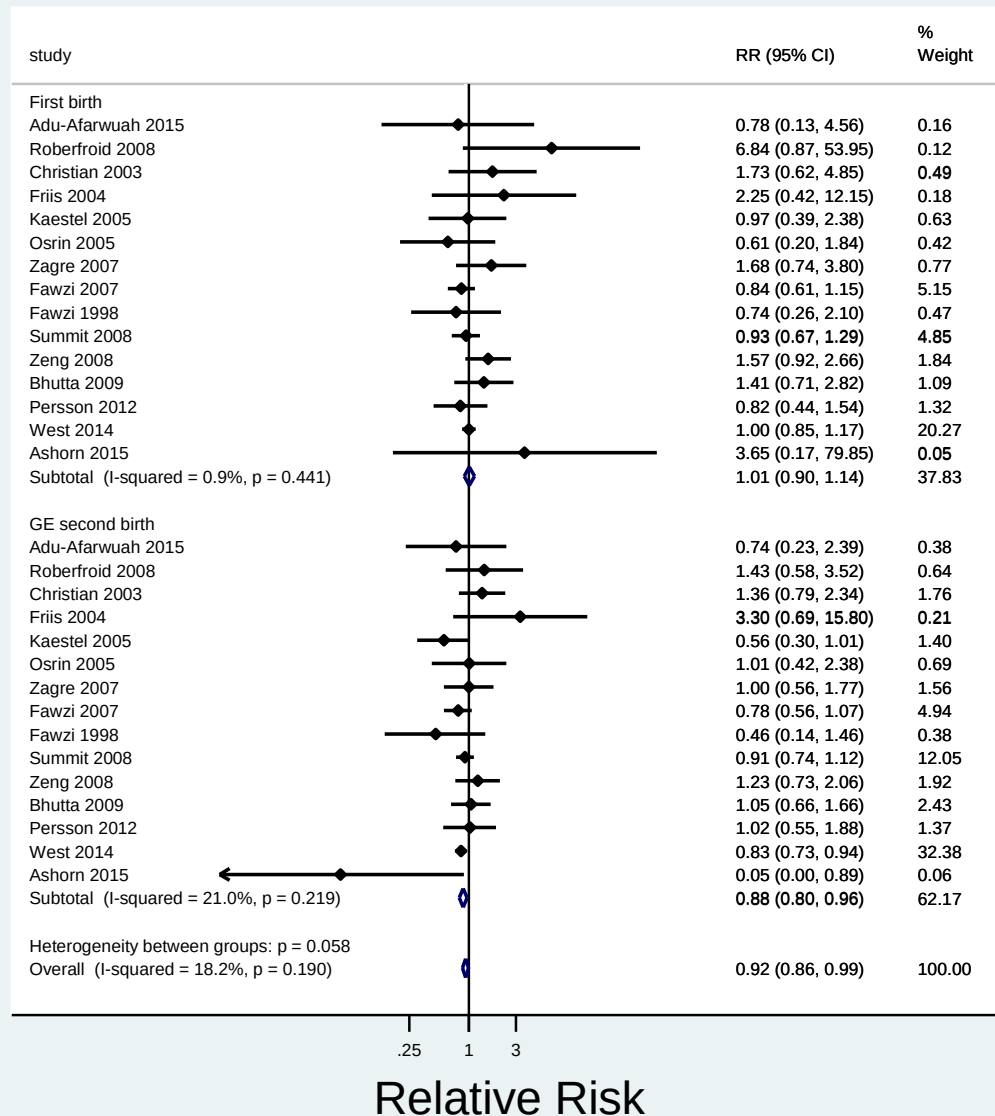
Outcome = Stillbirth

Relative risk of stillbirth by maternal height



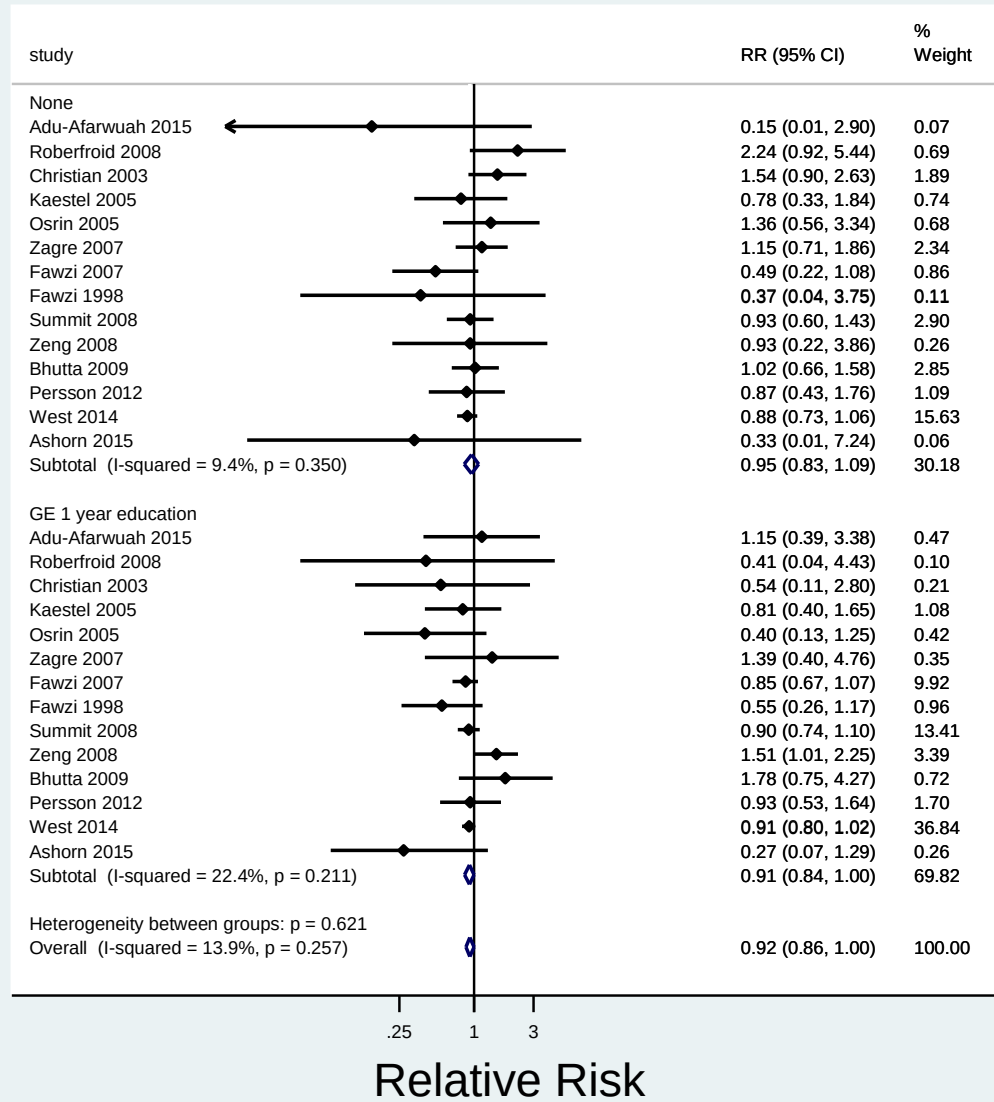
Outcome = Stillbirth

Relative risk of stillbirth by parity



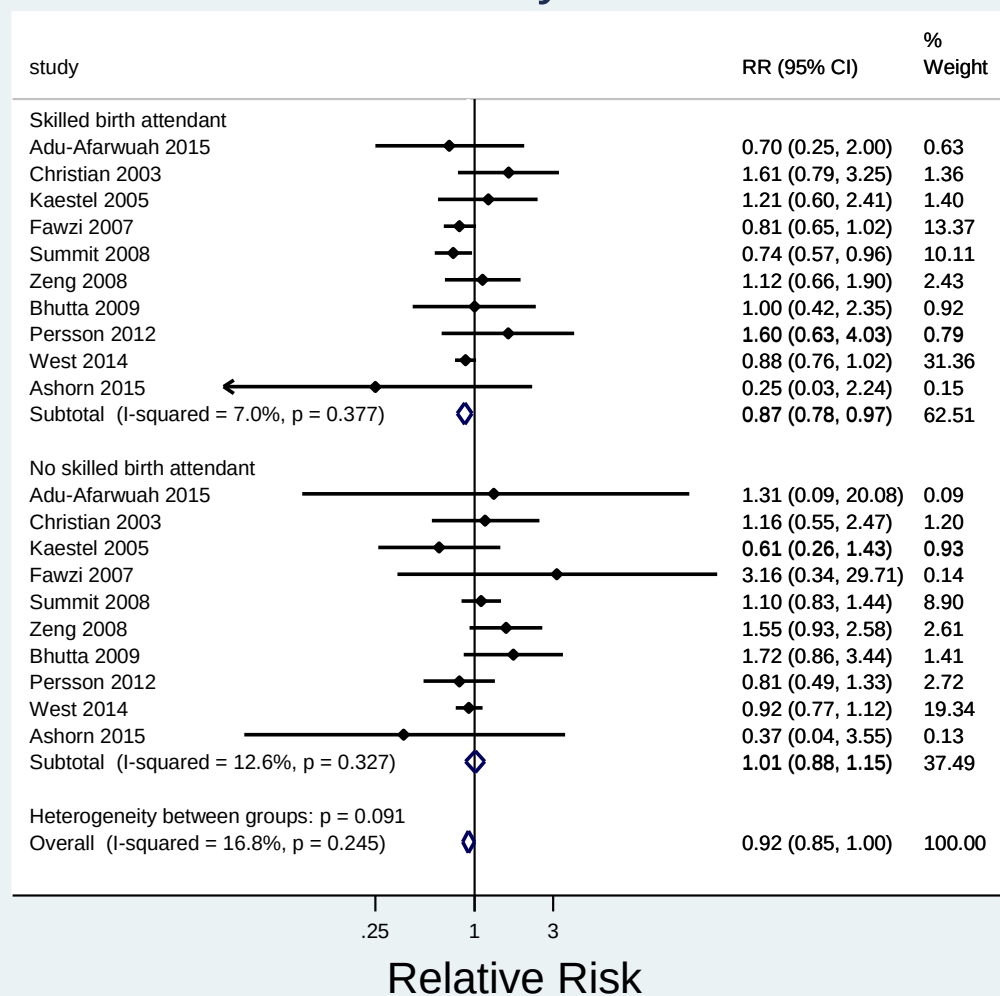
Outcome = Stillbirth

Relative risk of stillbirth by maternal education



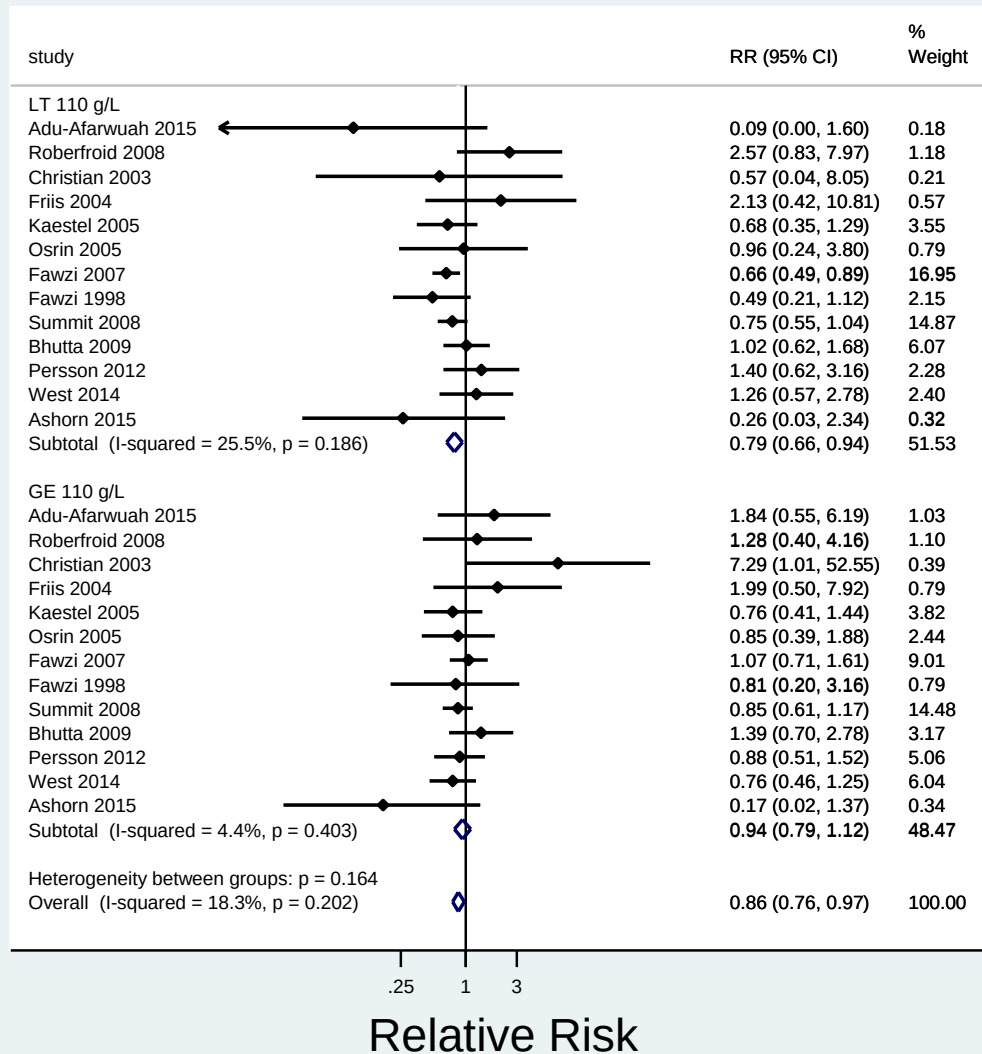
Outcome = Stillbirth

Relative risk of stillbirth by skilled birth attendant



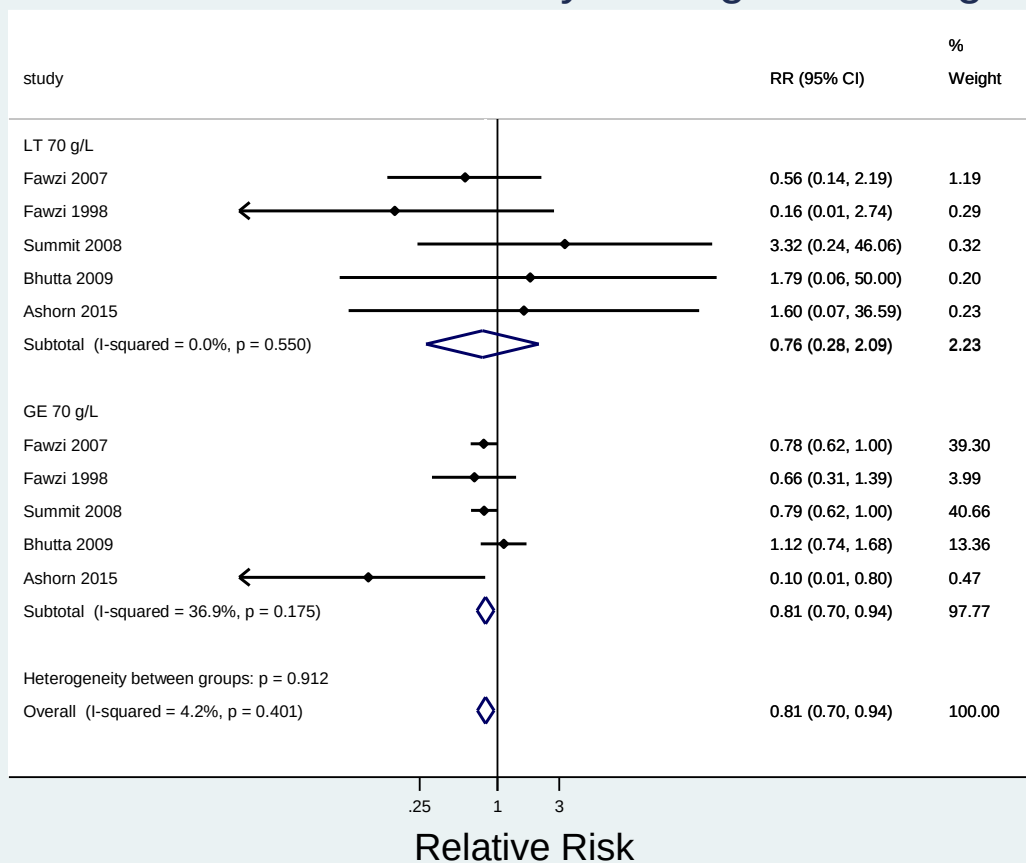
Outcome = Stillbirth

Relative risk of stillbirth by hemoglobin <110 g/L



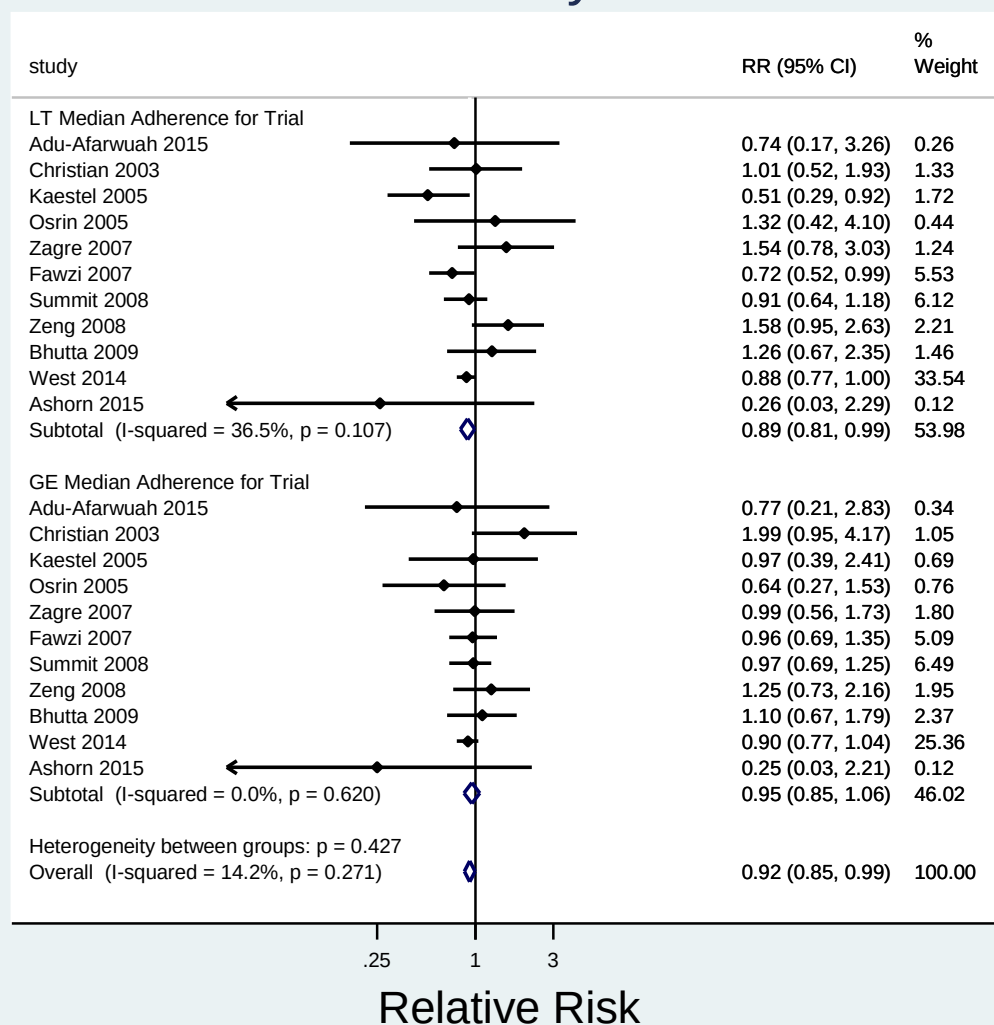
Outcome = Stillbirth

Relative risk of stillbirth by hemoglobin <70 g/L



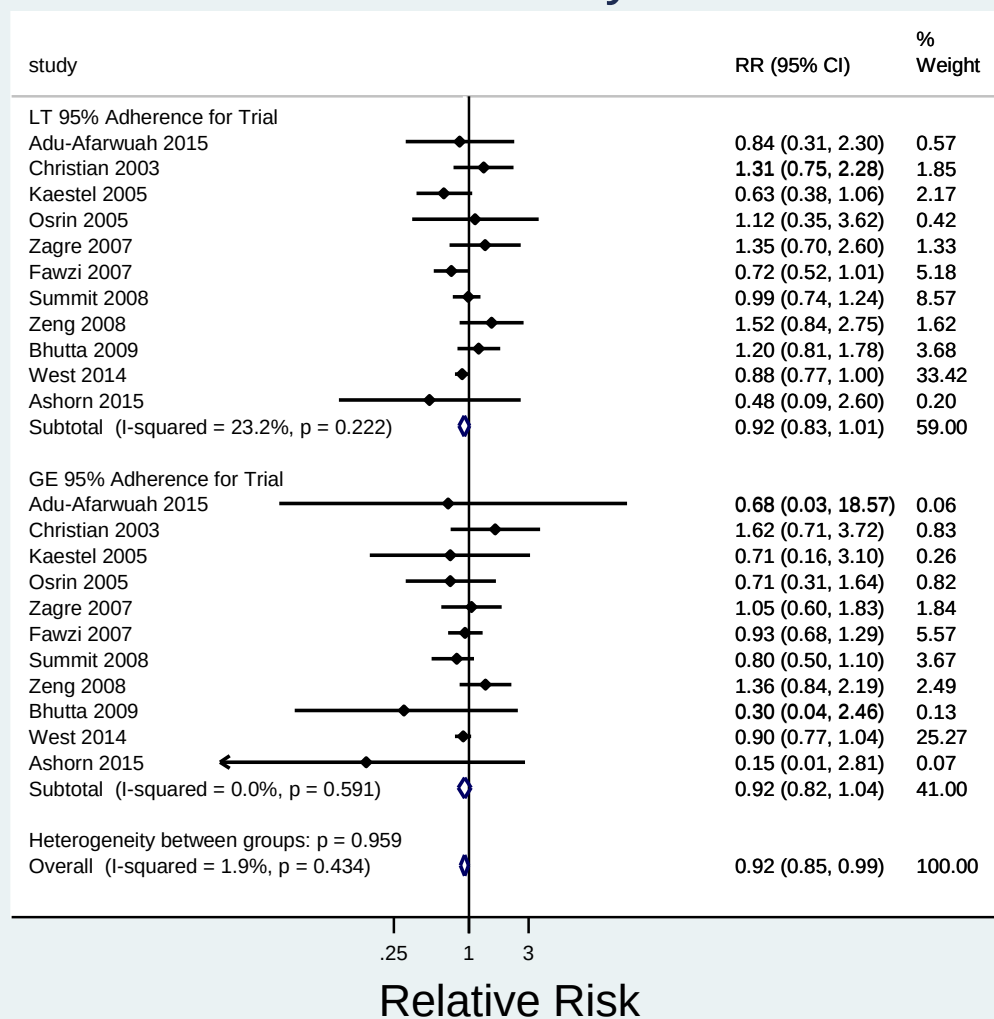
Outcome = Stillbirth

Relative risk of stillbirth by median adherence



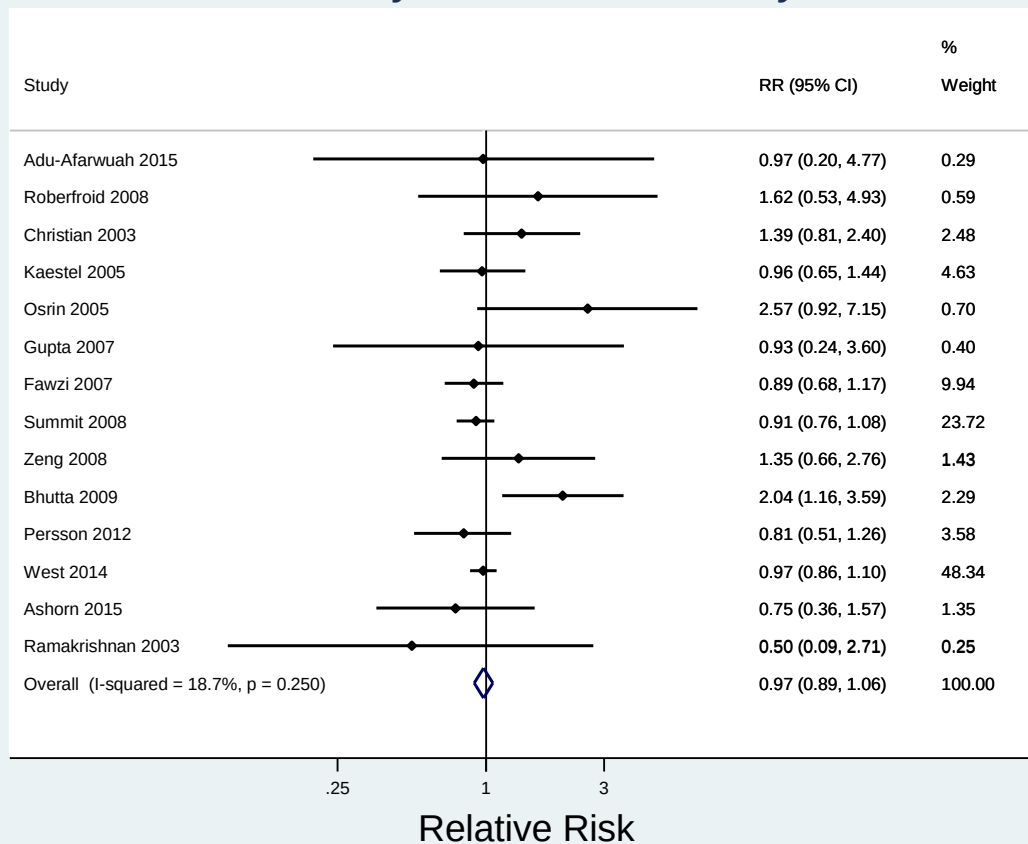
Outcome = Stillbirth

Relative risk of stillbirth by 95% adherence



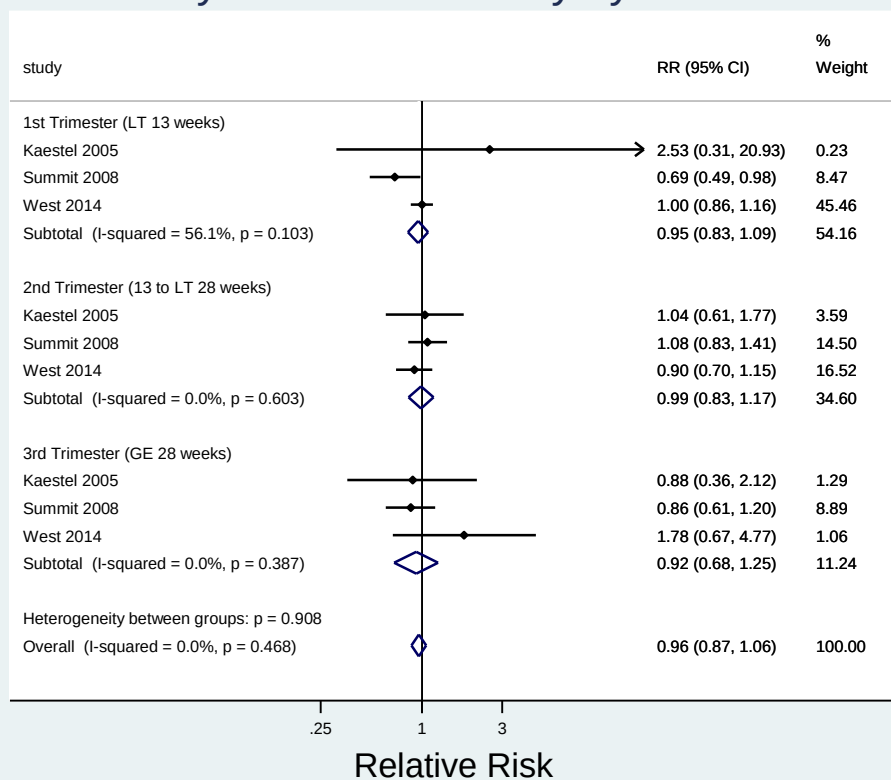
Outcome = Stillbirth

Relative risk of early neonatal mortality - OVERALL



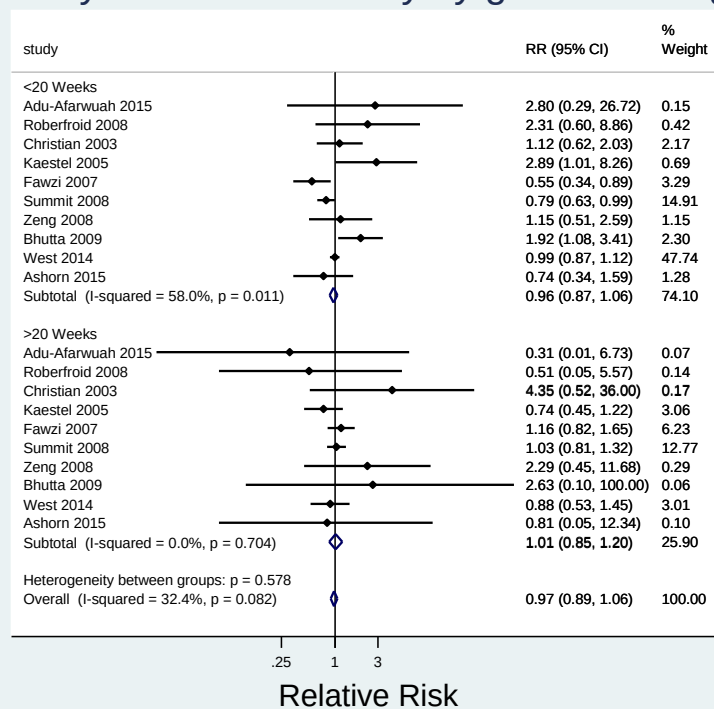
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by trimester at enrollment



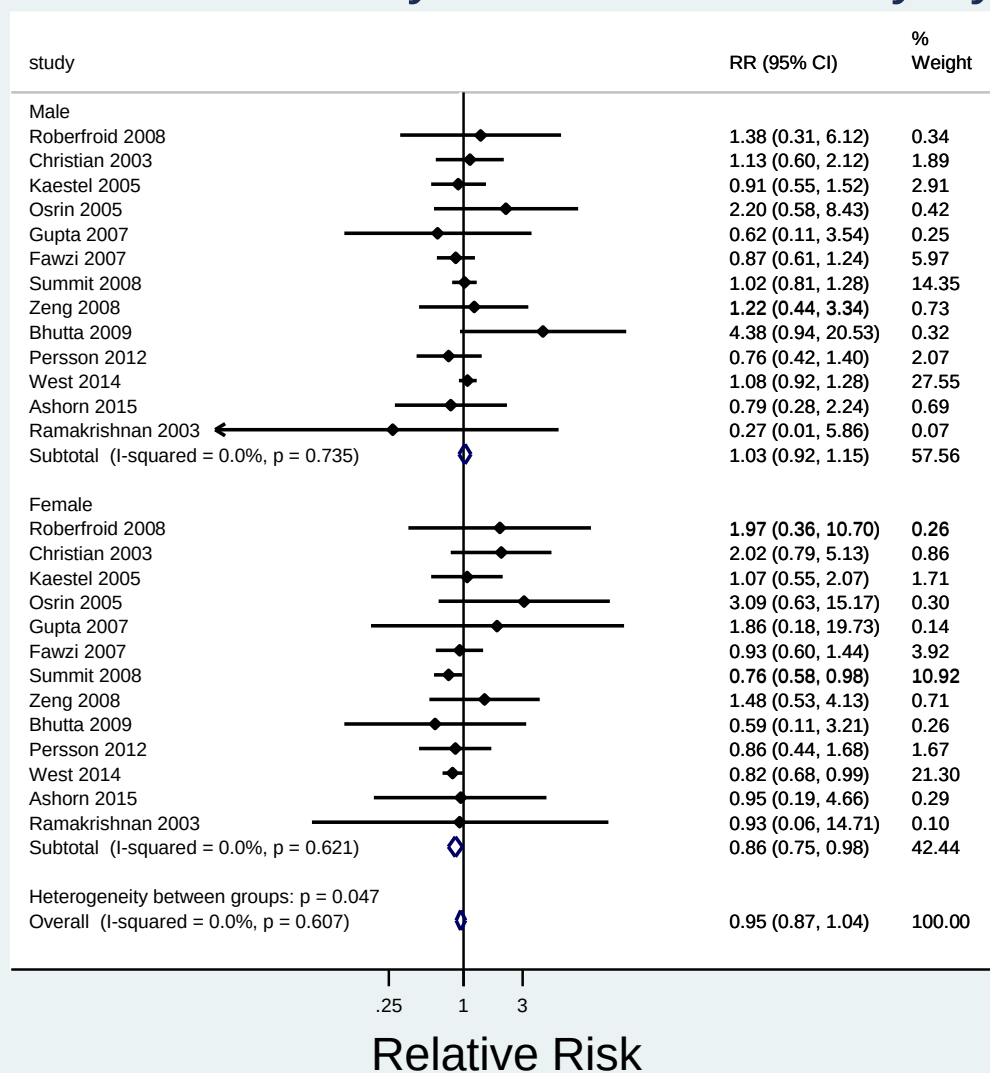
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by gestational age at enrollment



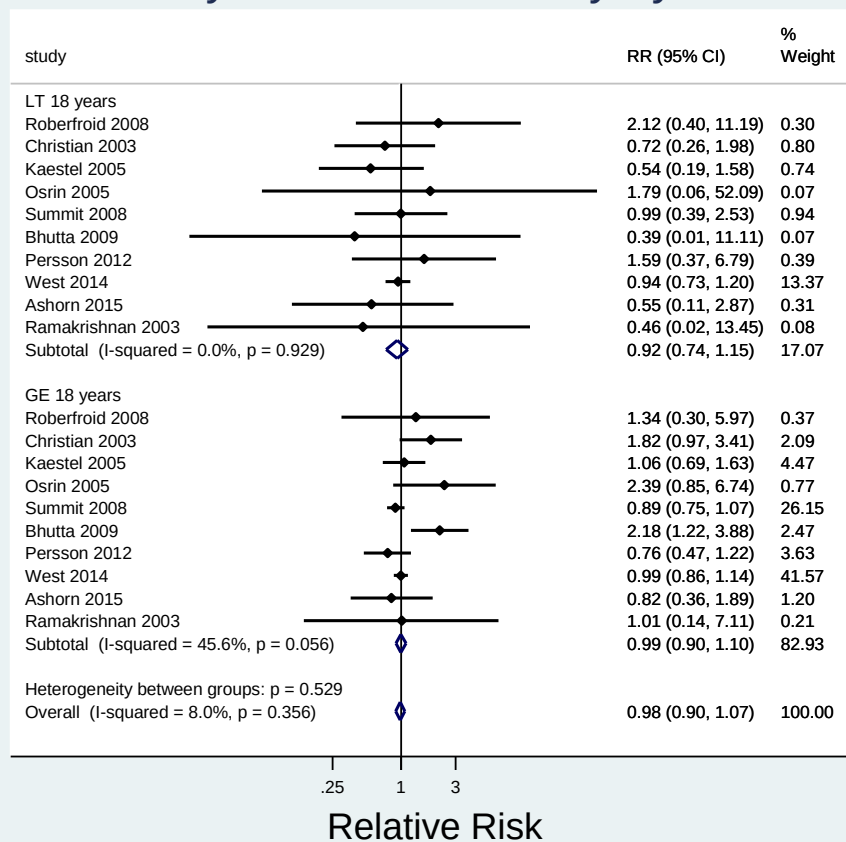
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by sex



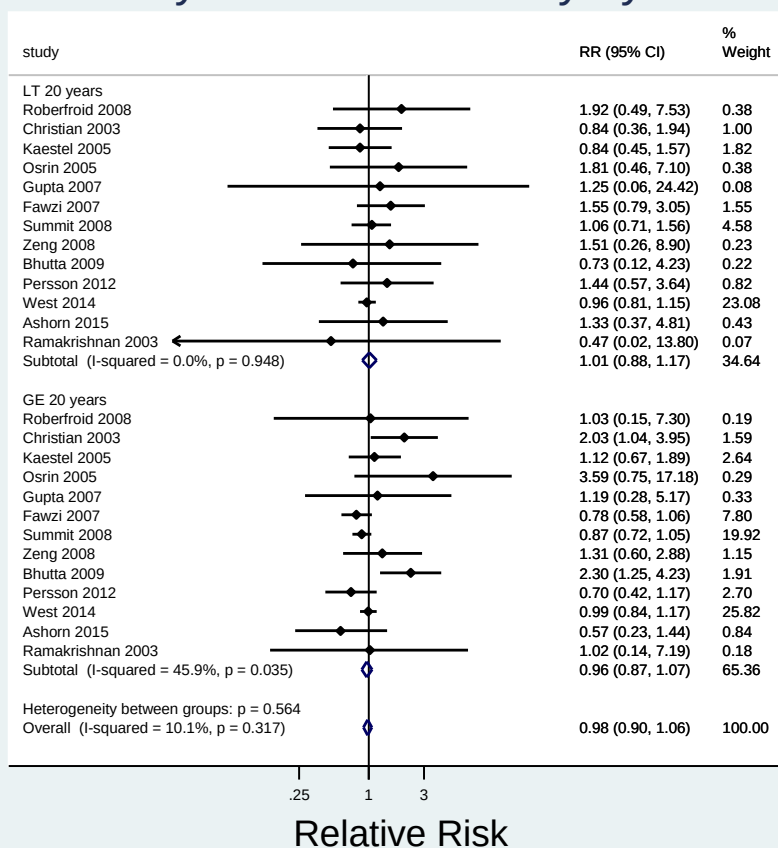
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by maternal age <18



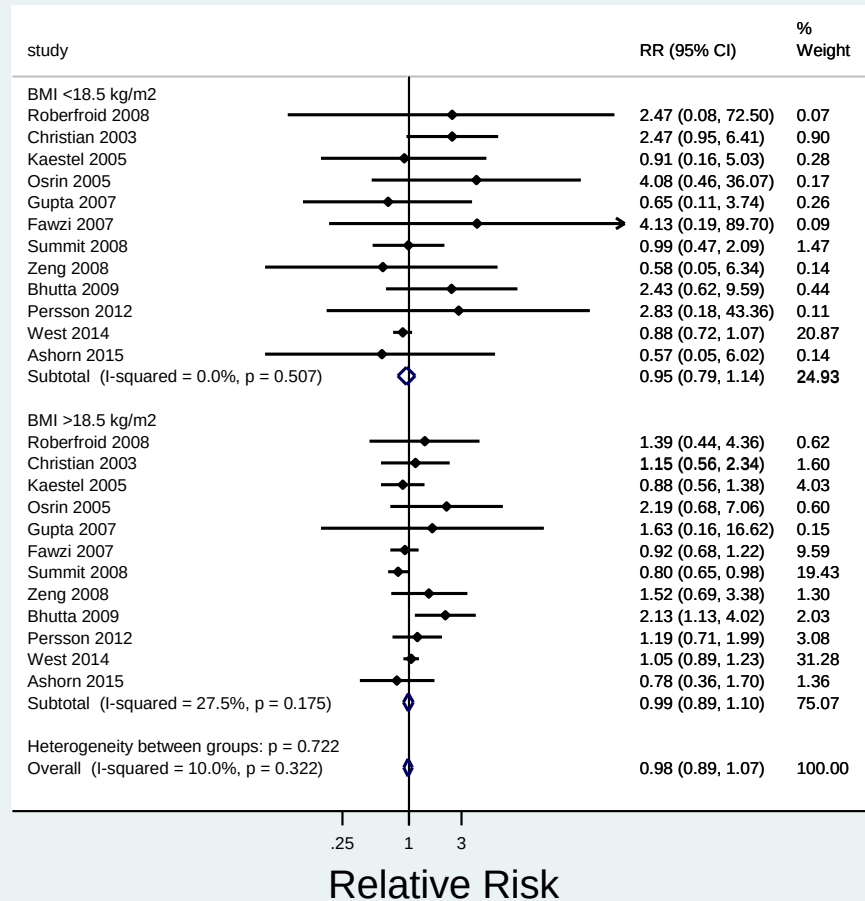
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by maternal age <20



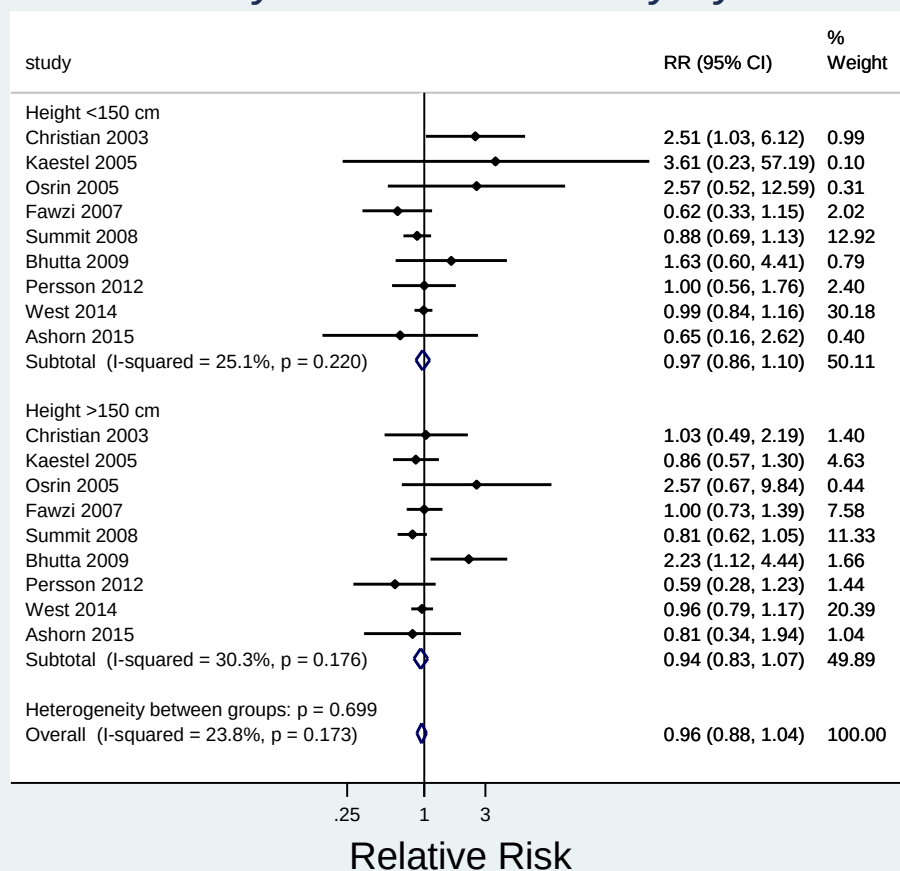
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by maternal BMI



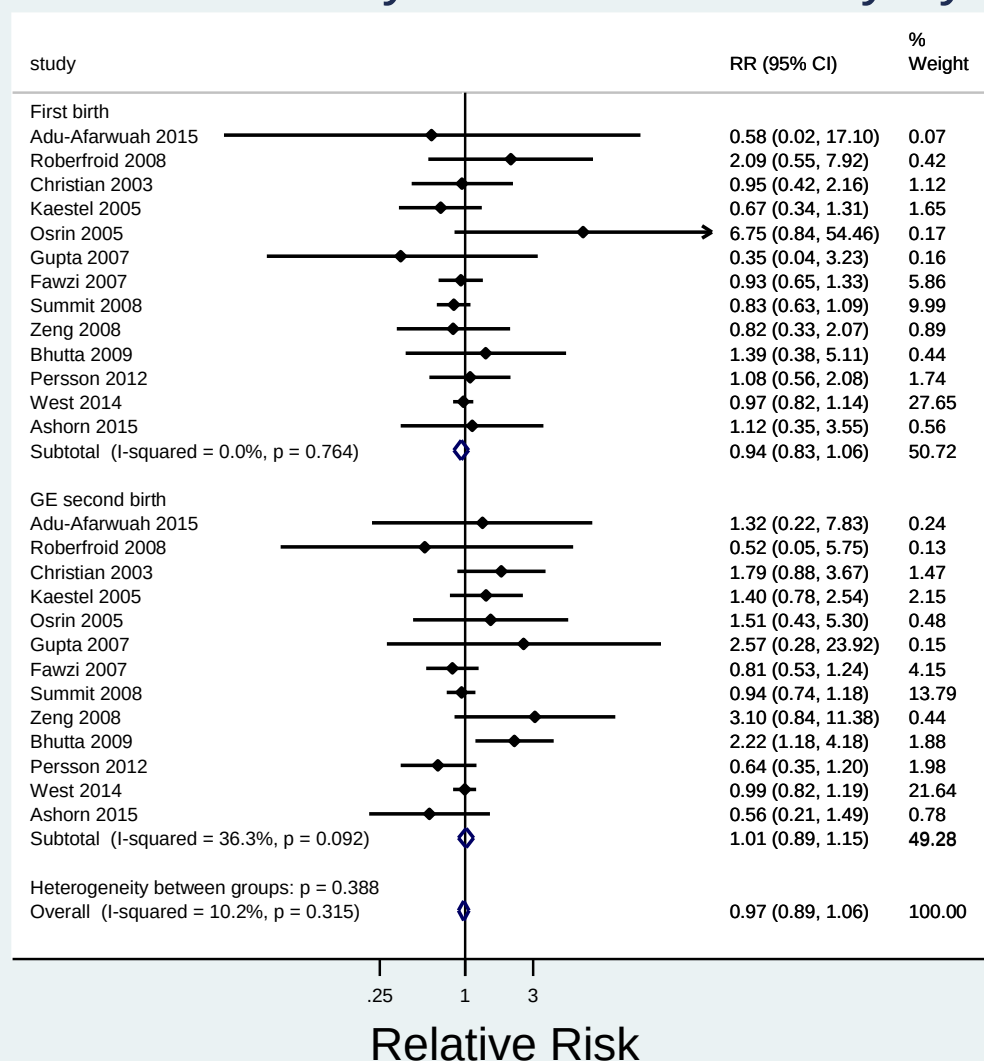
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by maternal height



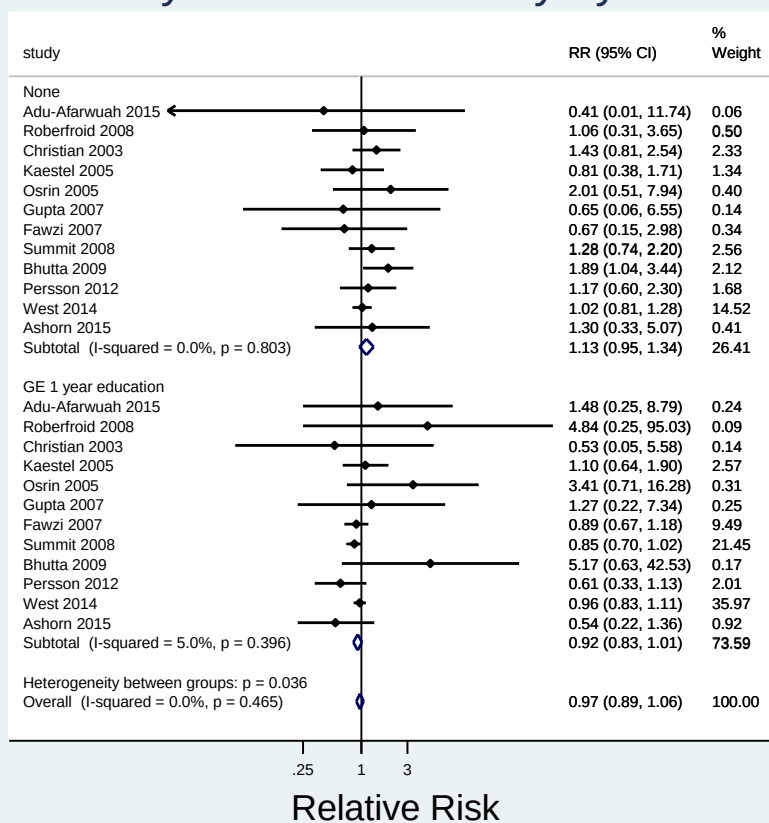
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by parity



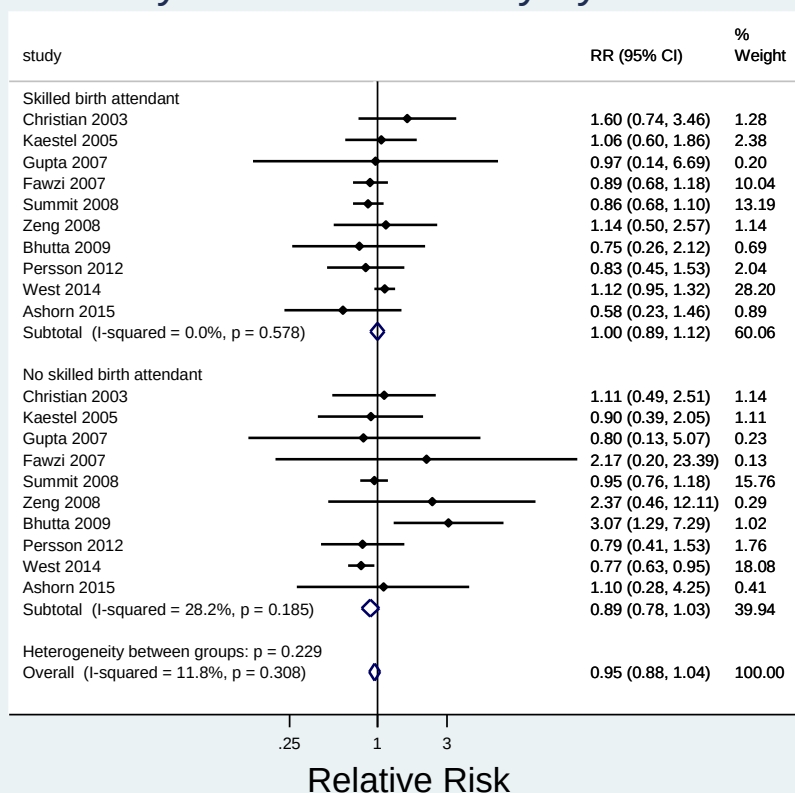
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by maternal education



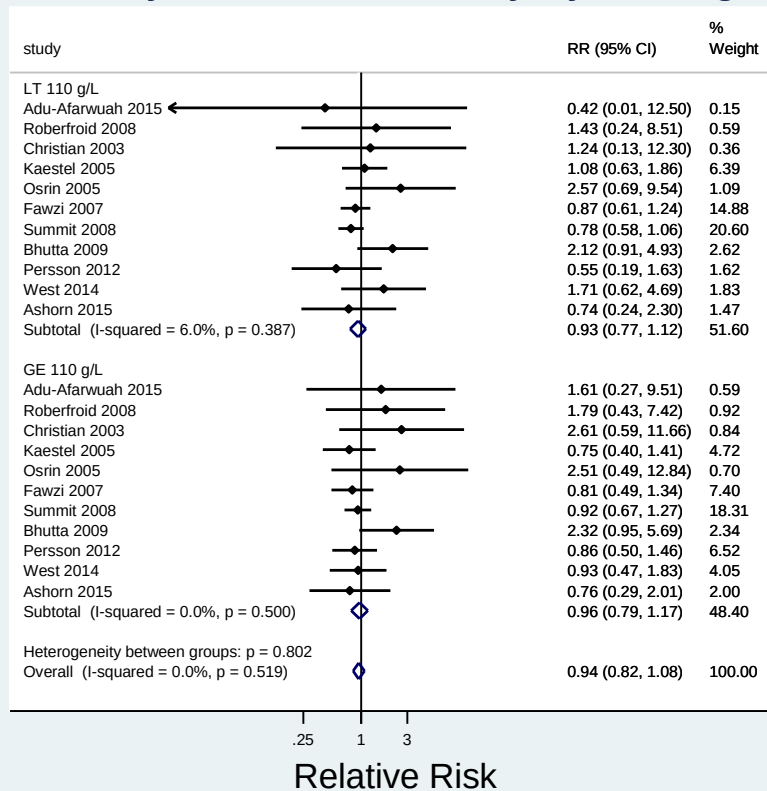
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by skilled birth attendant



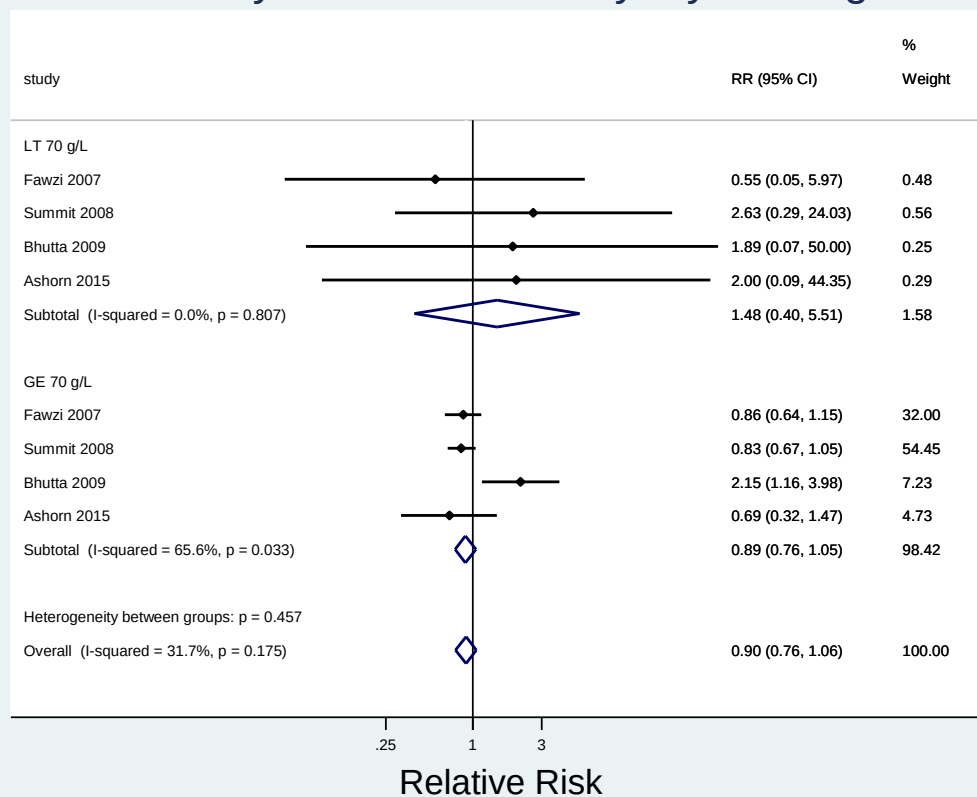
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by hemoglobin <110 g/L



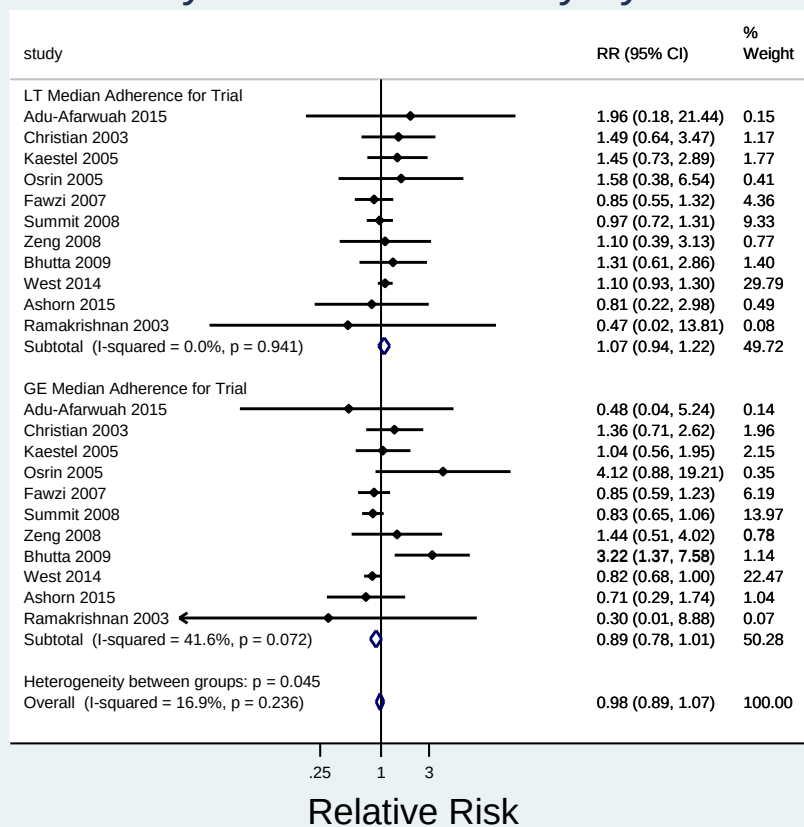
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by hemoglobin <70 g/L



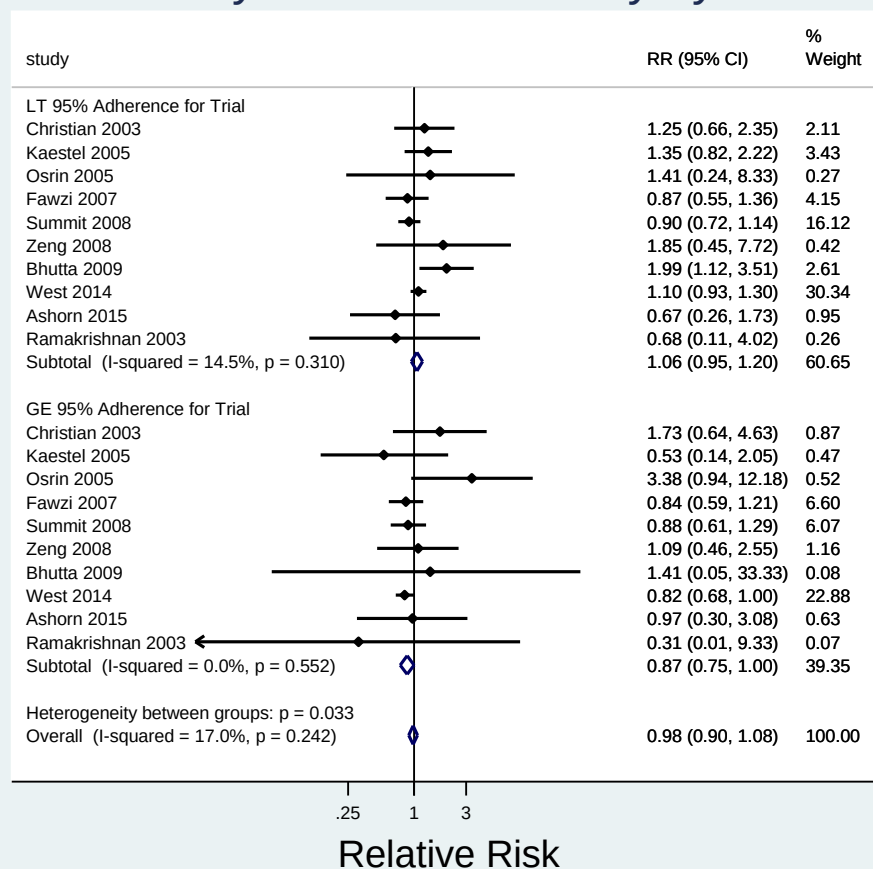
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by median adherence



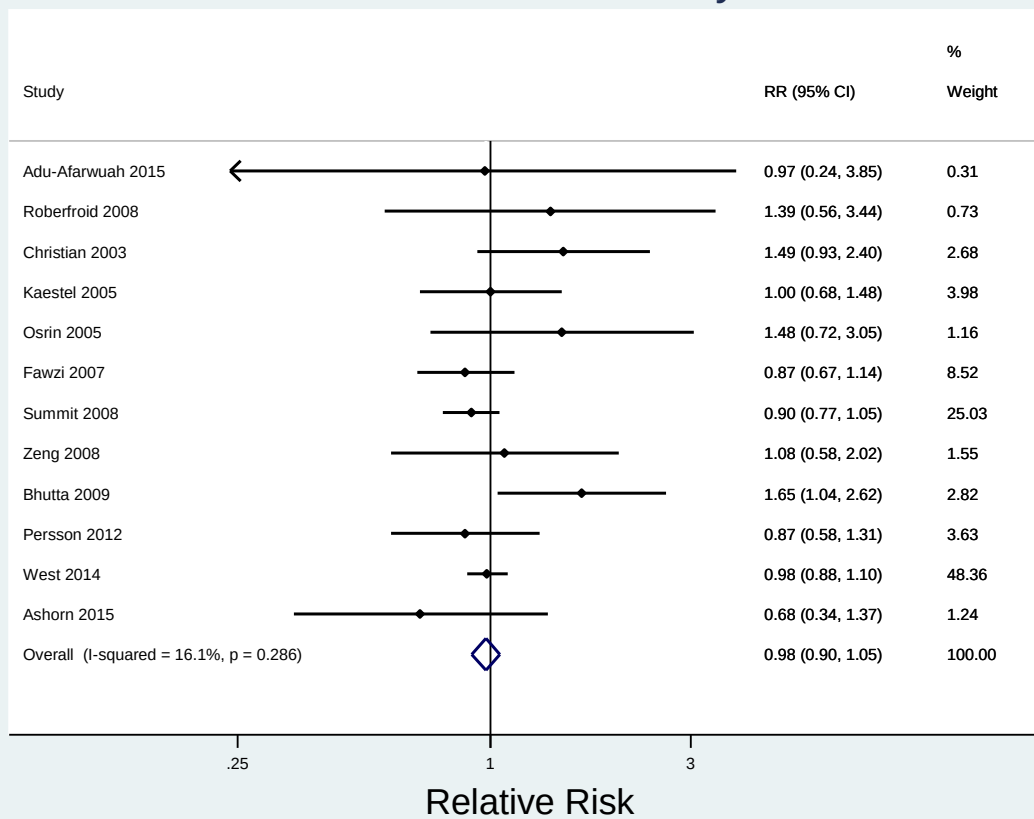
Outcome = Early Neonatal Mortality

Relative risk of early neonatal mortality by 95% adherence



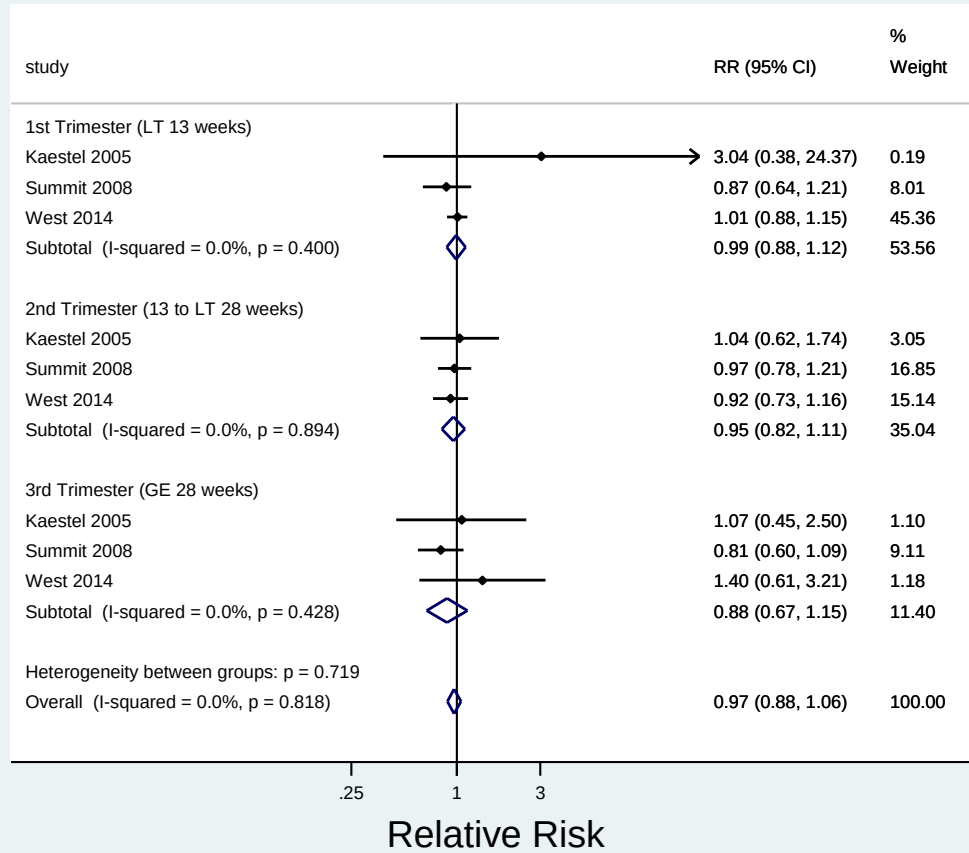
Outcome = Early Neonatal Mortality

Relative risk of neonatal mortality - OVERALL



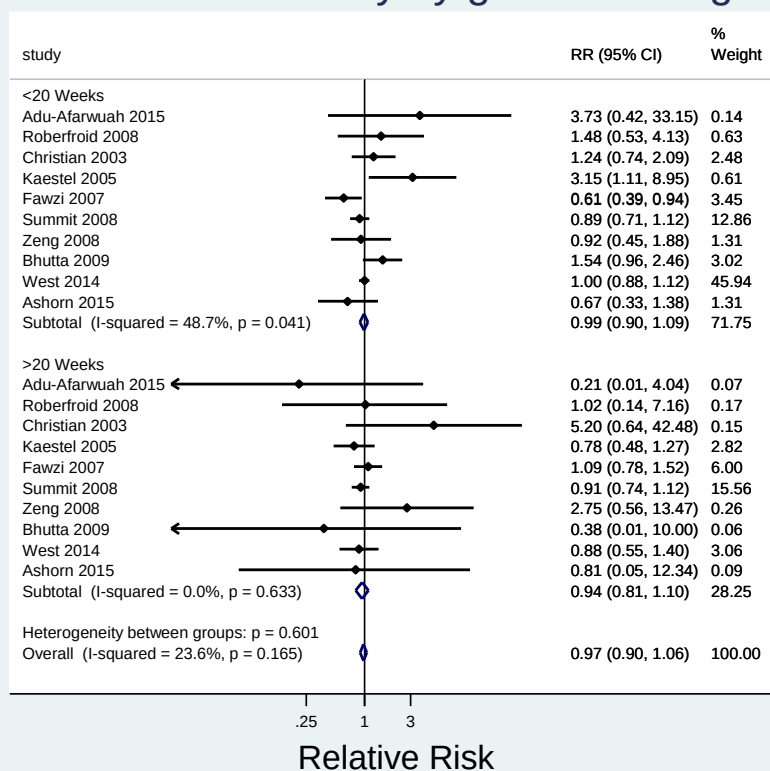
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by trimester at enrollment



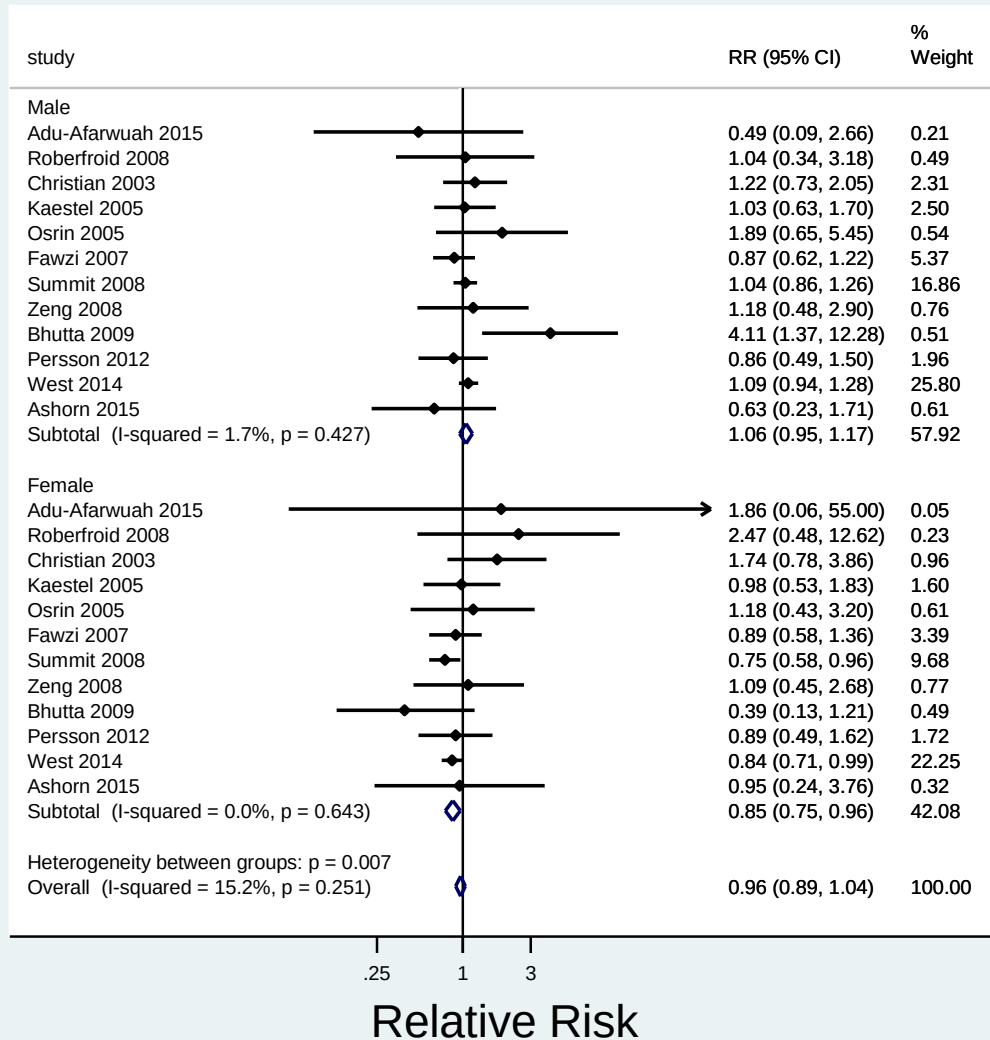
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Relative risk of neonatal mortality by gestational age at enrollment



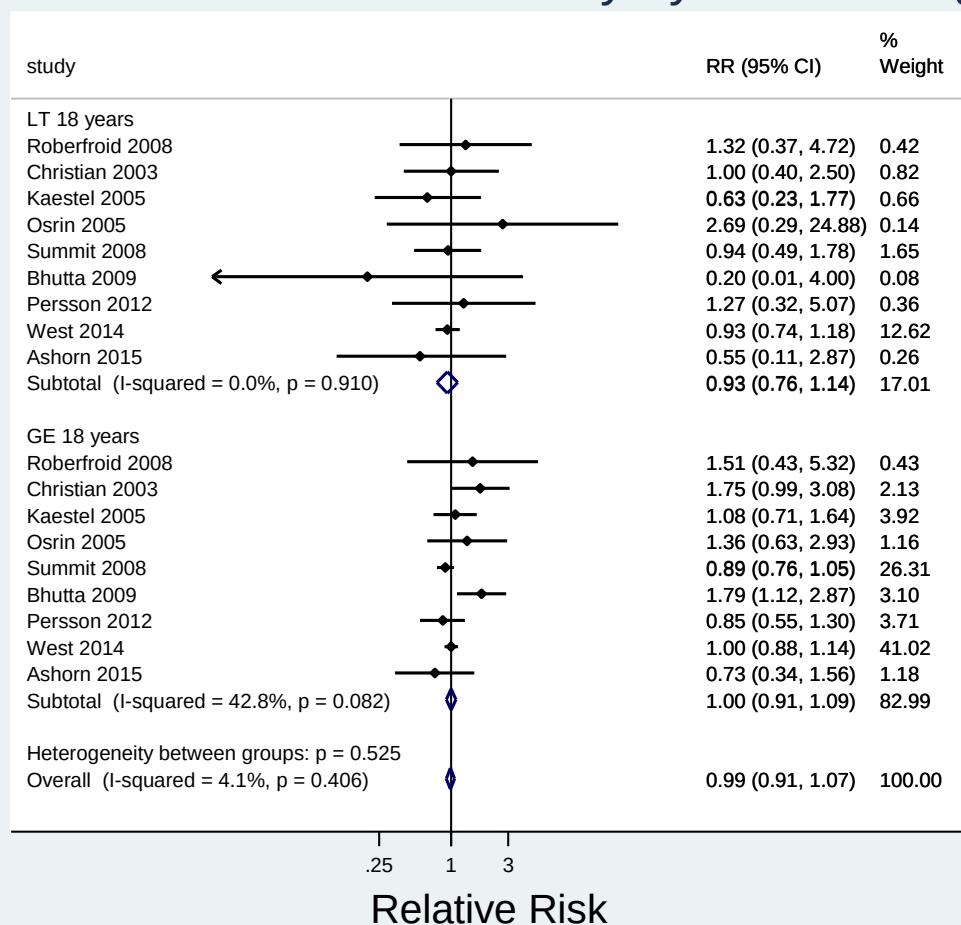
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by sex



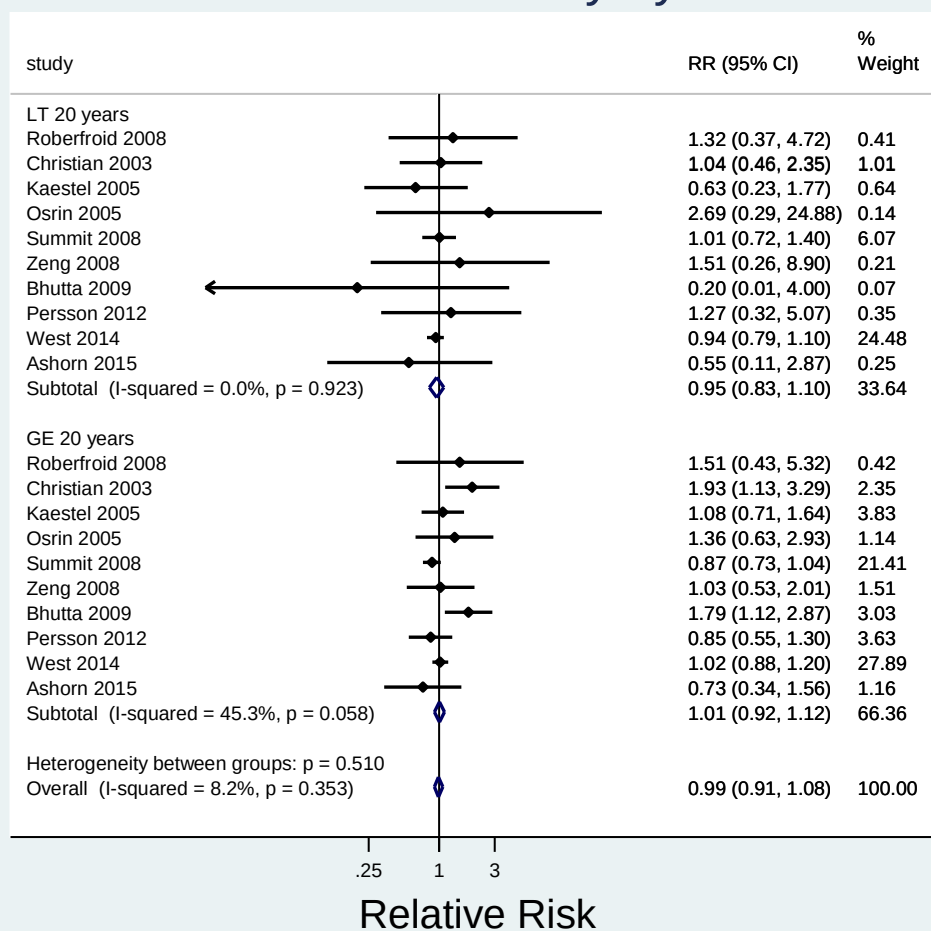
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by maternal age <18



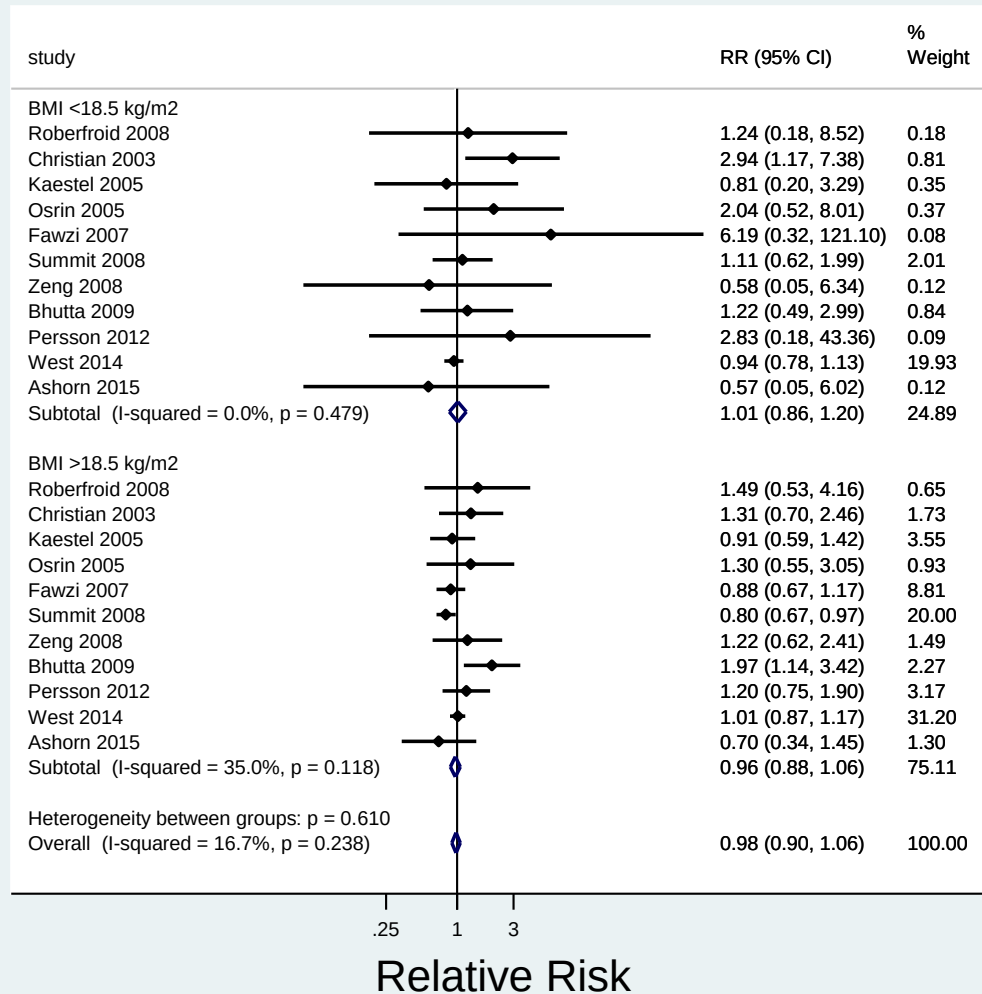
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by maternal age <20



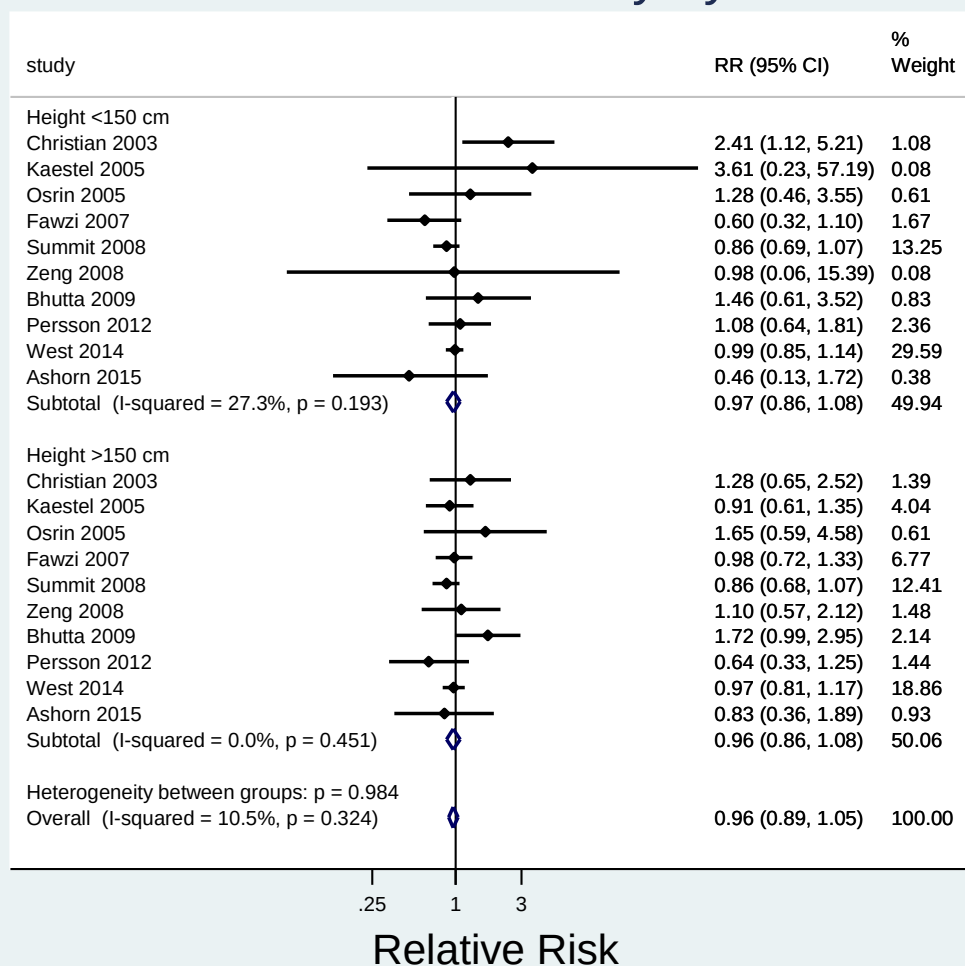
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by maternal BMI



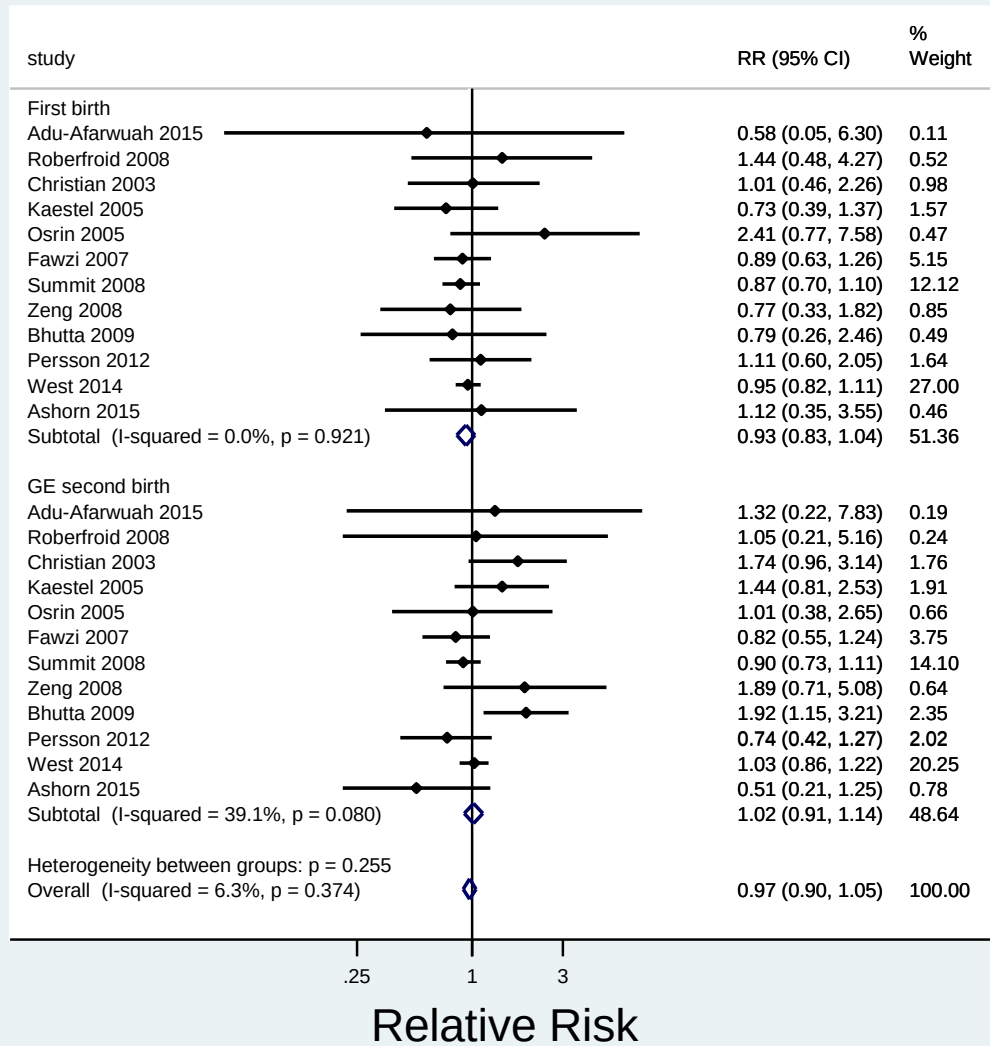
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by maternal height



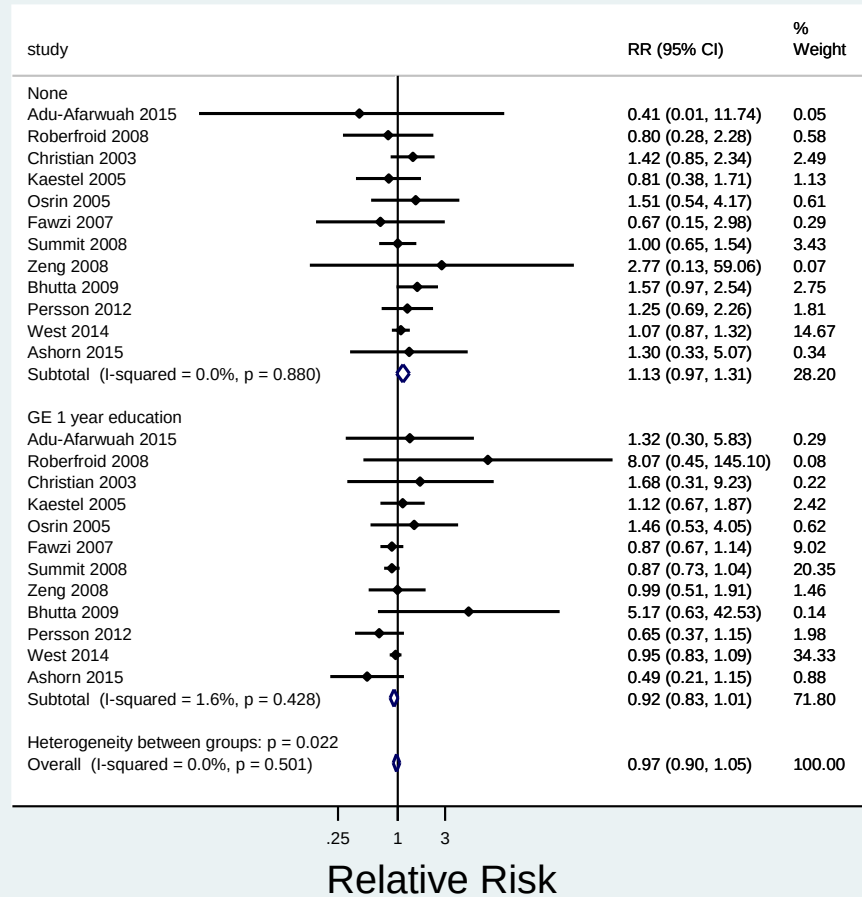
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by parity



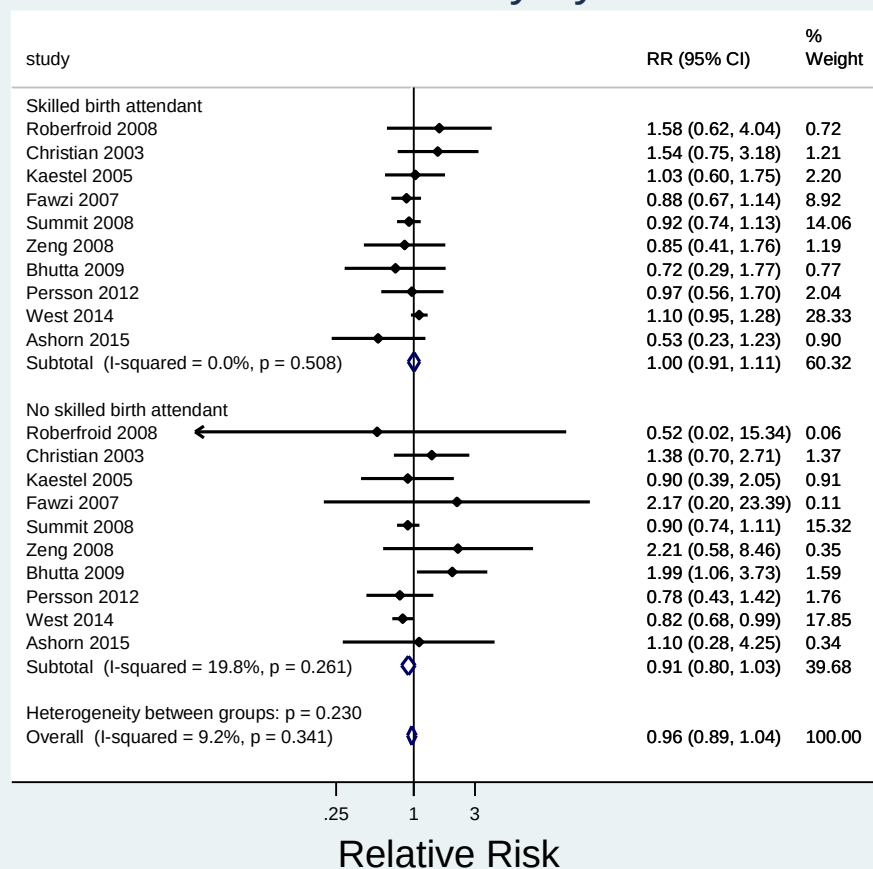
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by maternal education



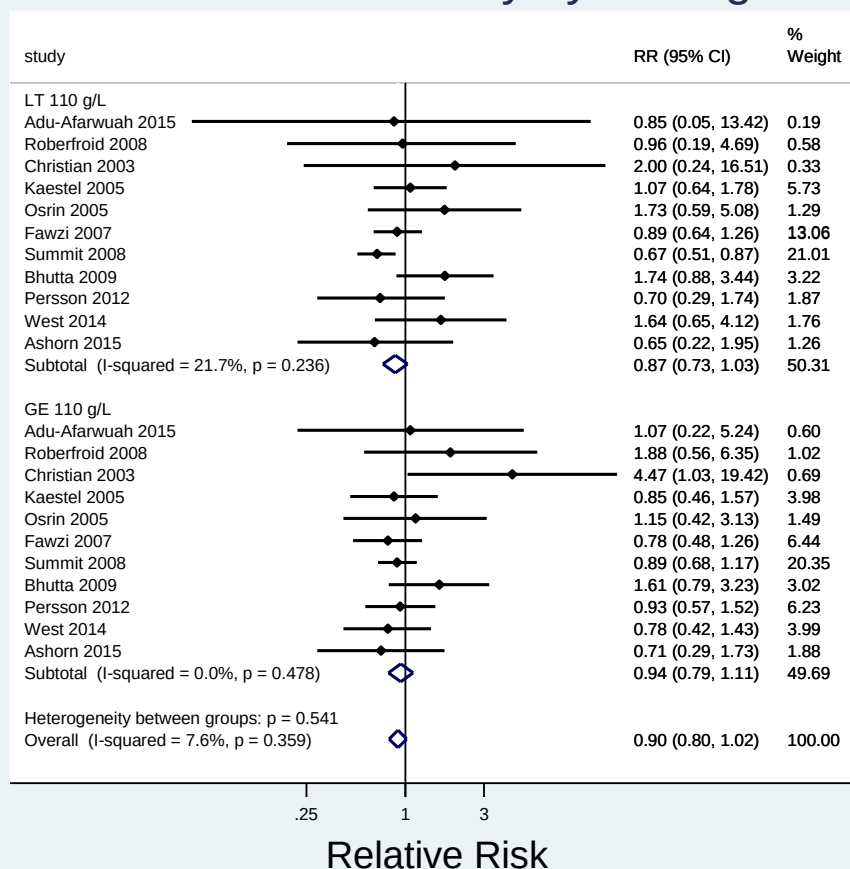
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by skilled birth attendant



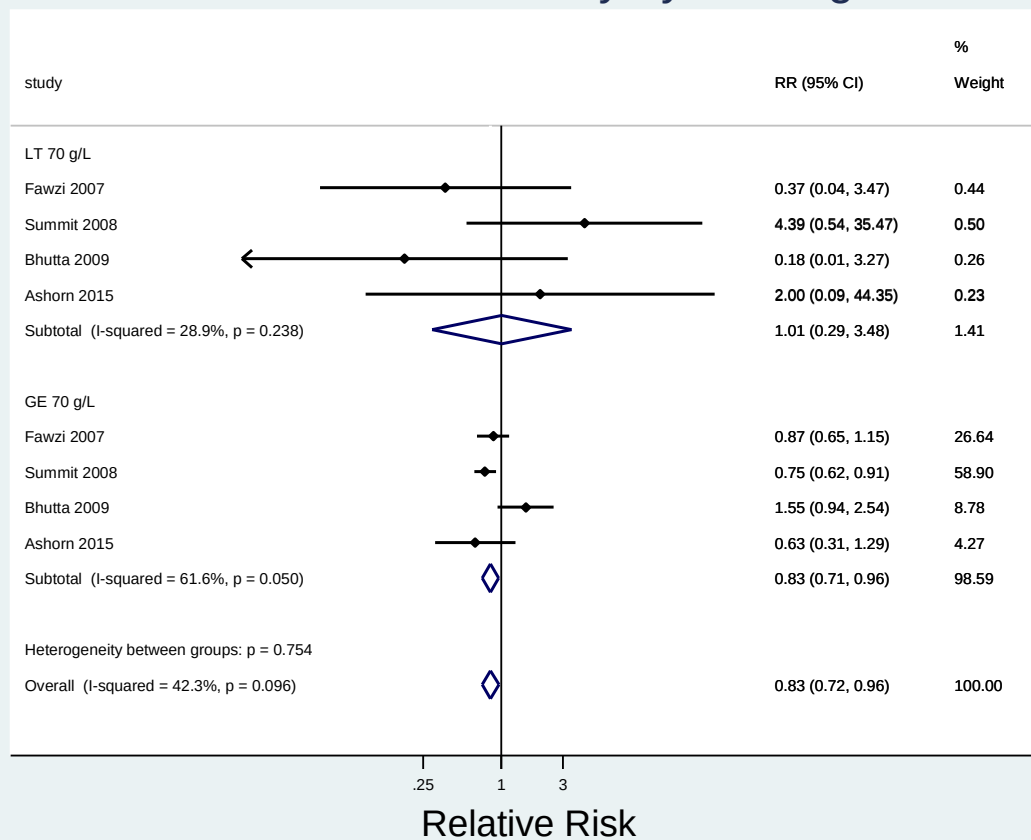
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by hemoglobin <110 g/L



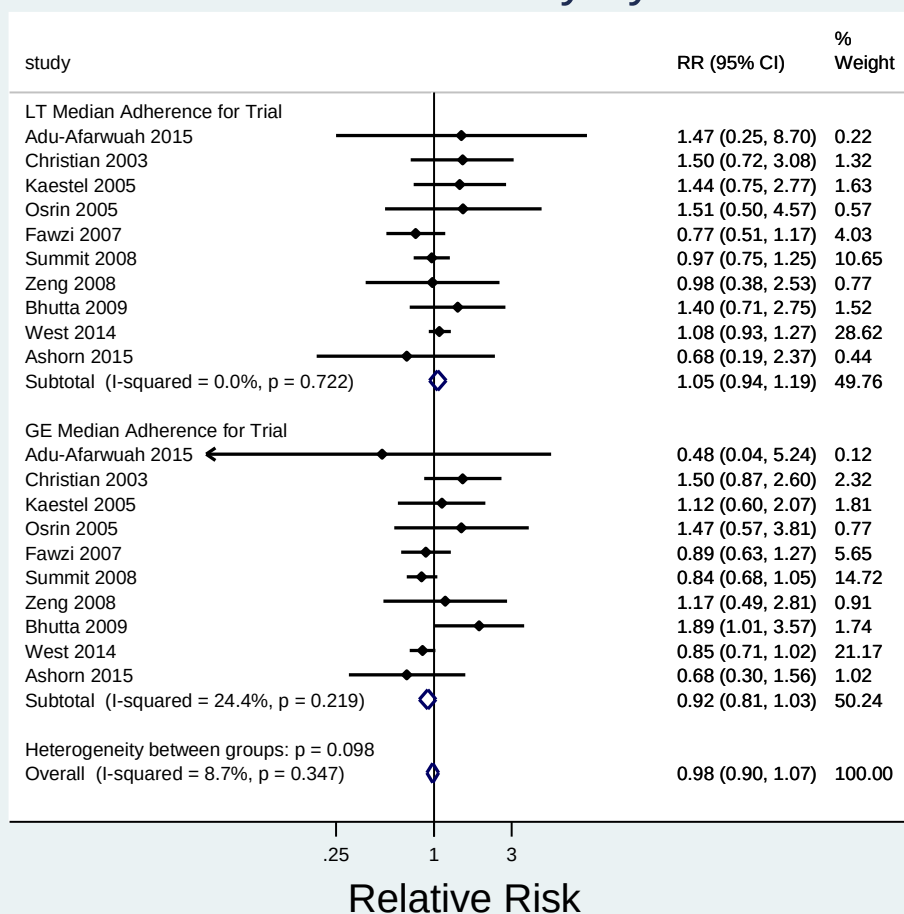
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by hemoglobin <70 g/L



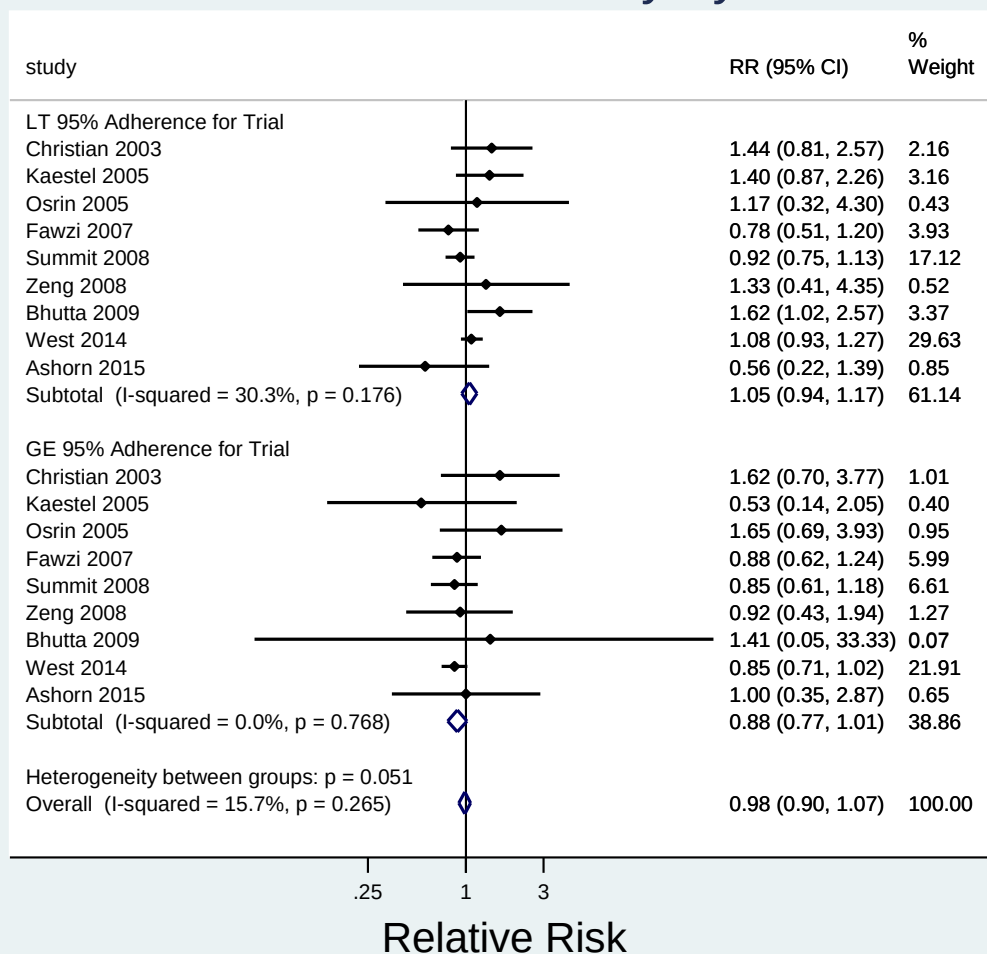
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by median adherence



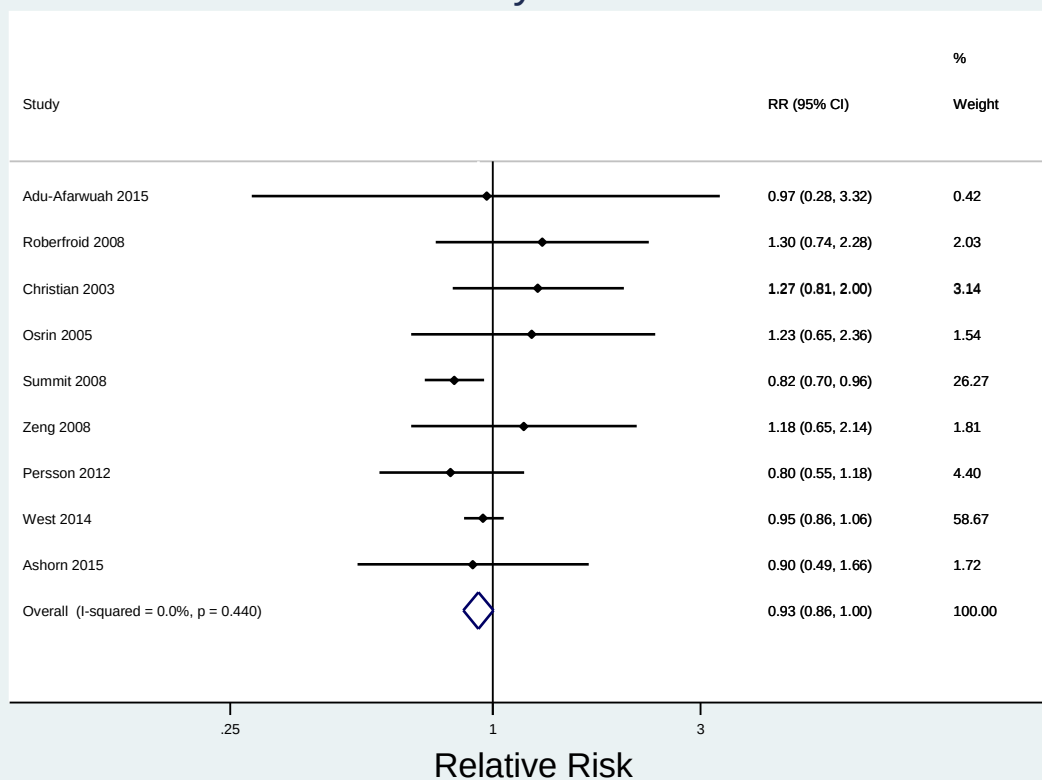
Outcome = Neonatal Mortality

Relative risk of neonatal mortality by 95% adherence



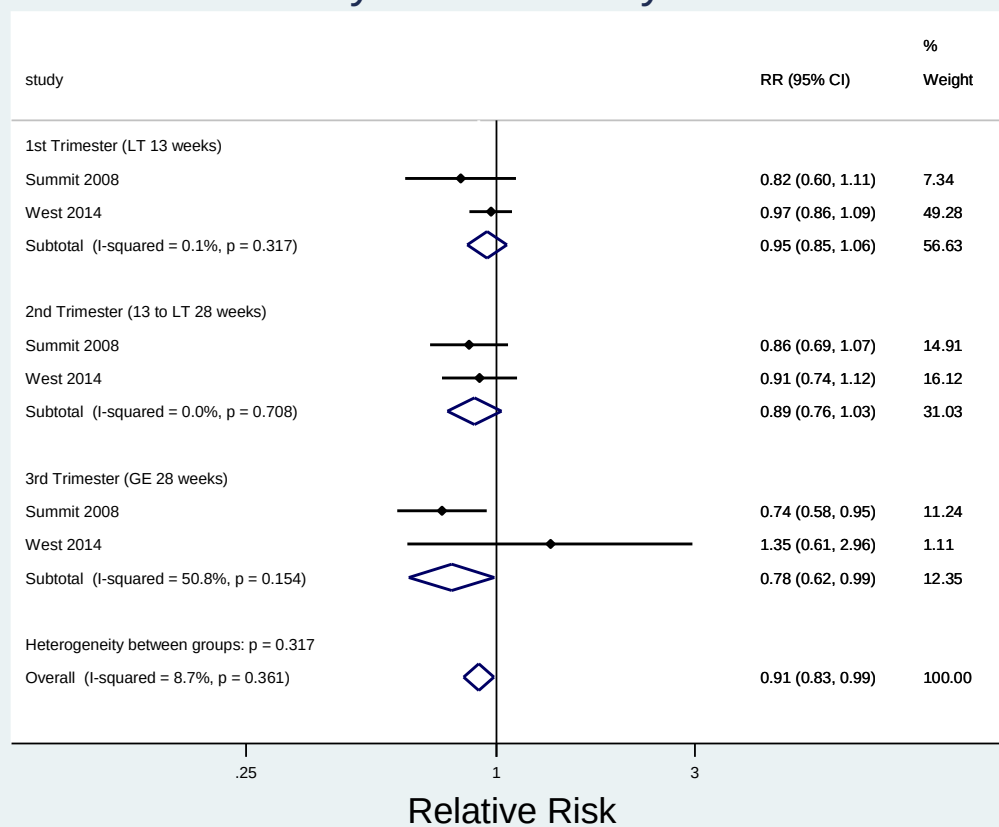
Outcome = Neonatal Mortality

Relative risk of mortality to 6 month - OVERALL



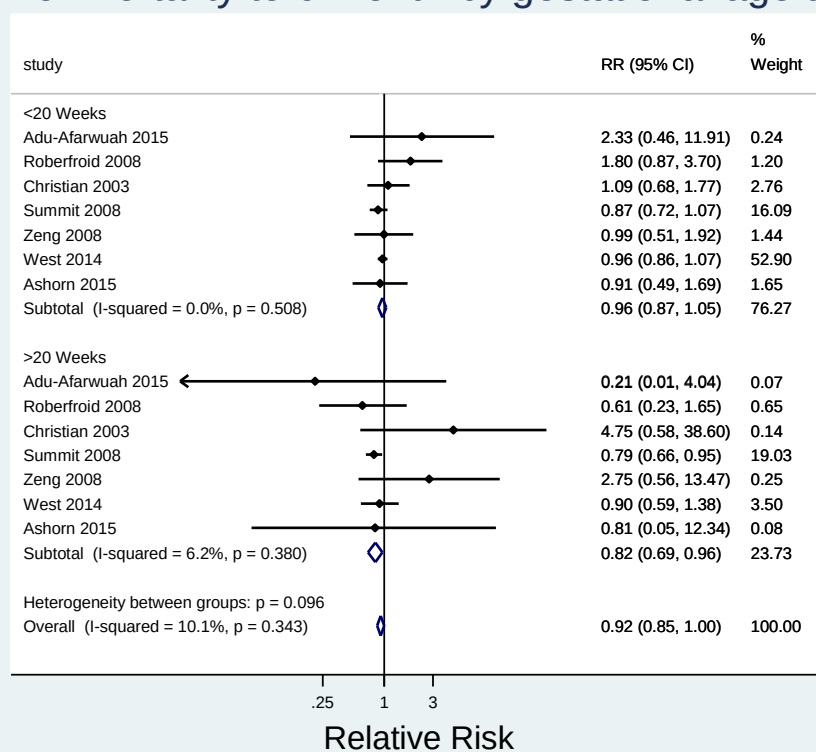
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by trimester at enrollment



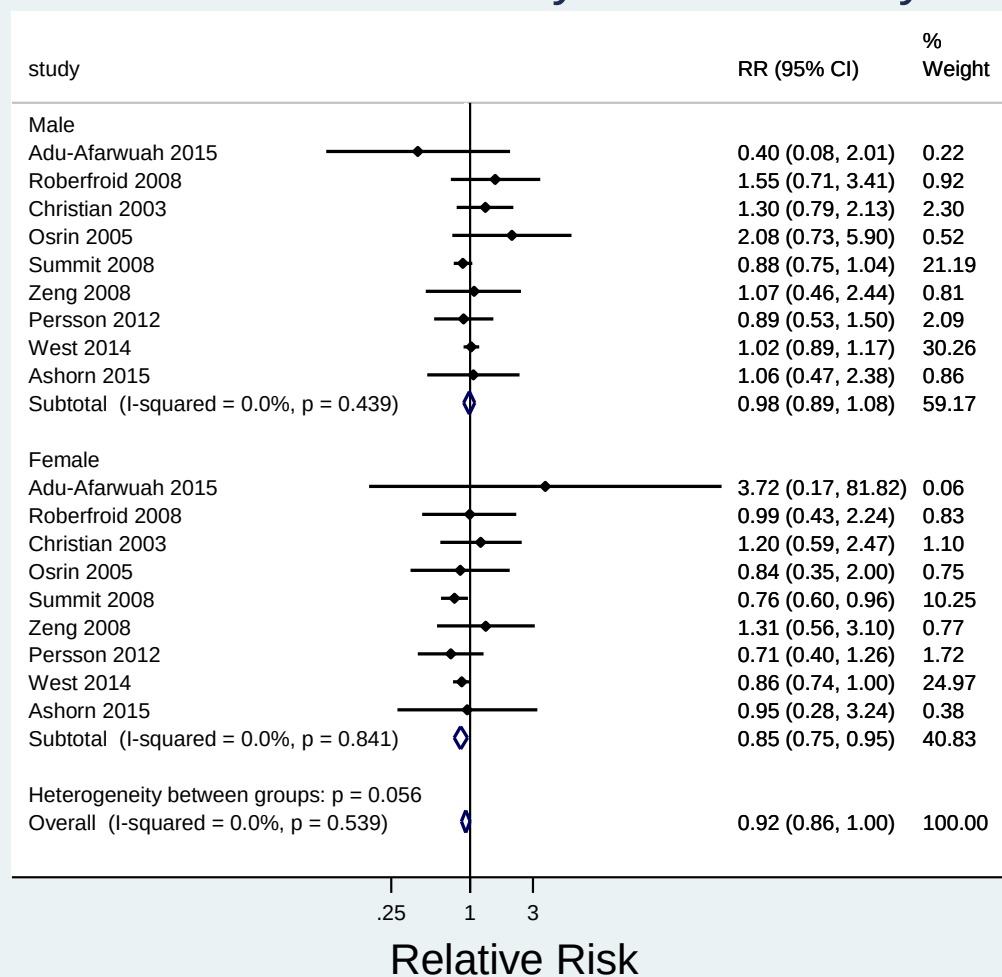
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by gestational age at enrollment



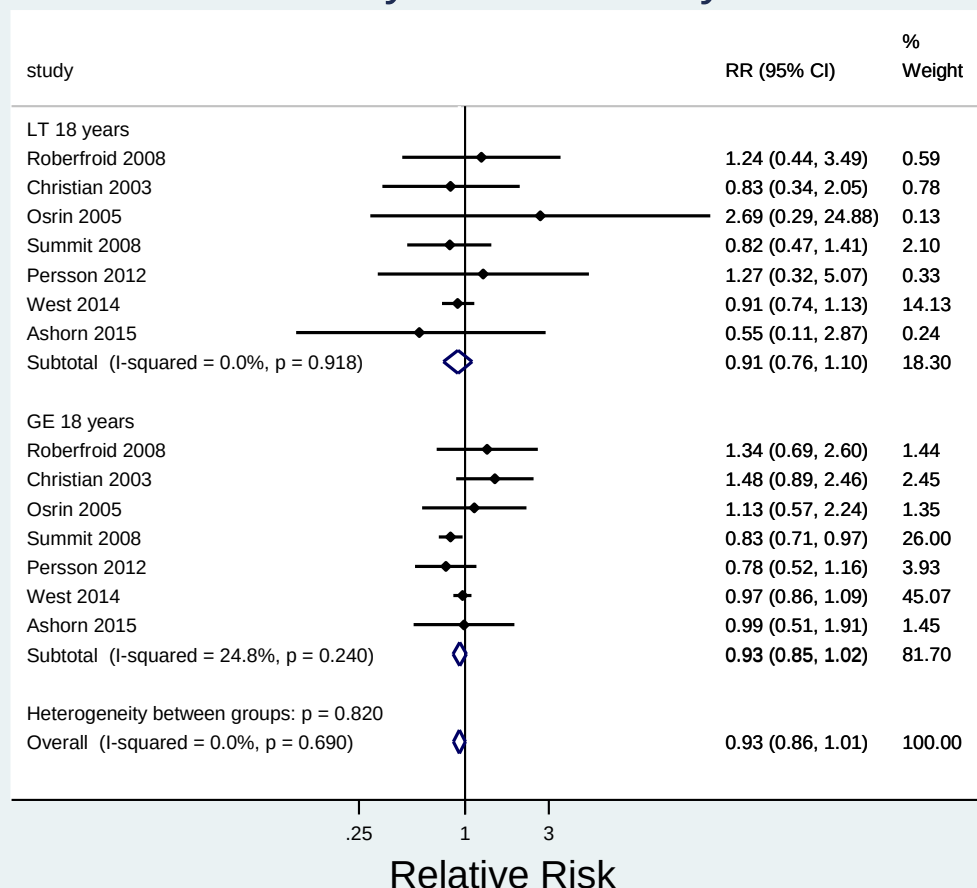
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by sex



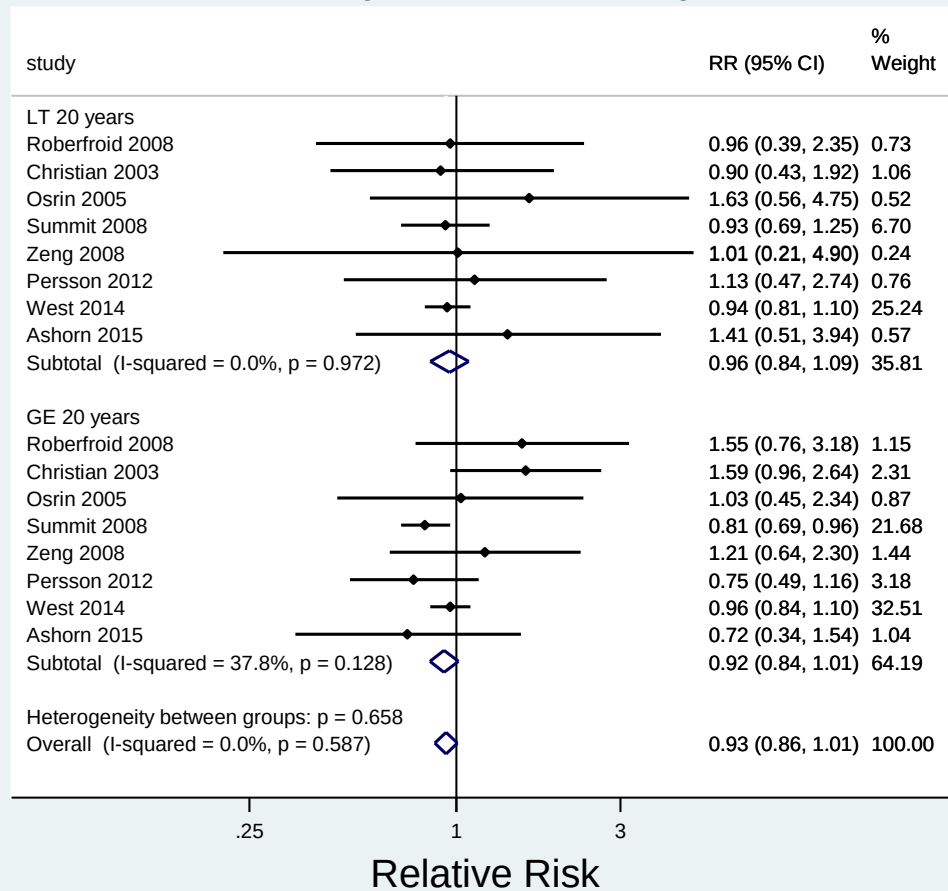
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by maternal age <18



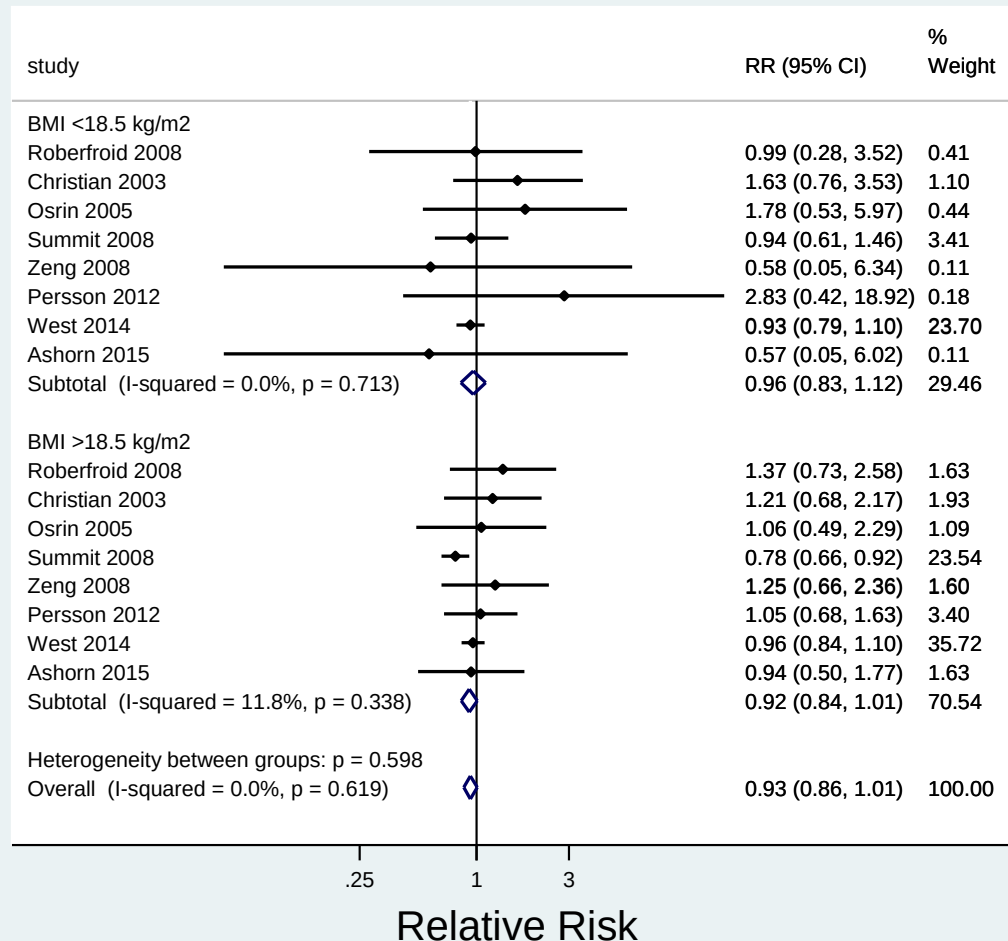
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by maternal age <20



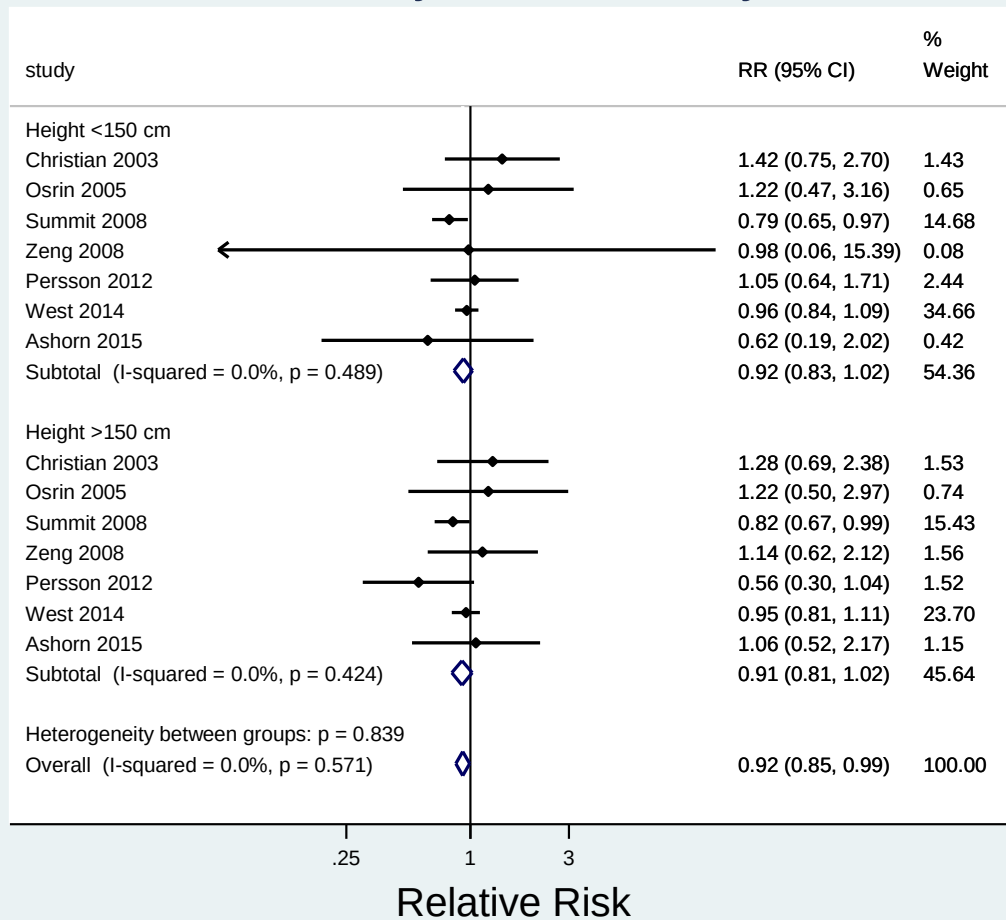
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by maternal BMI



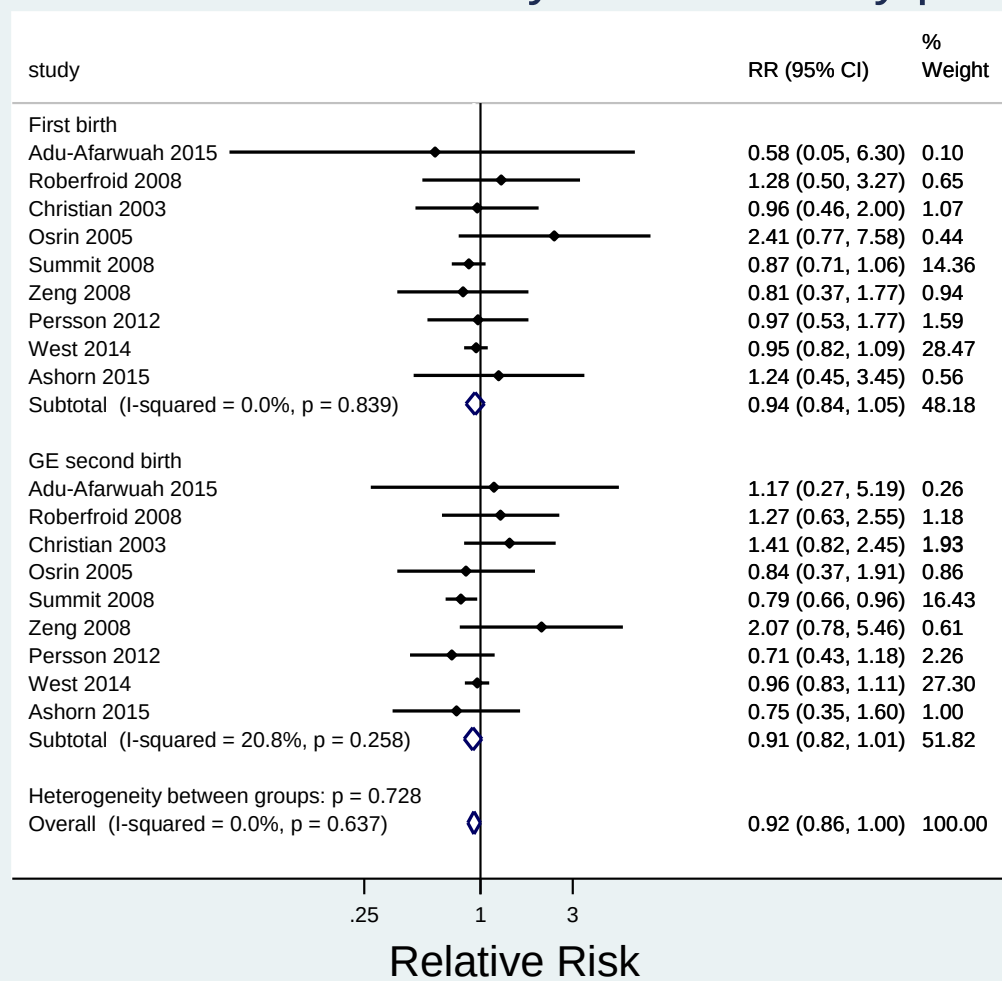
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by maternal height



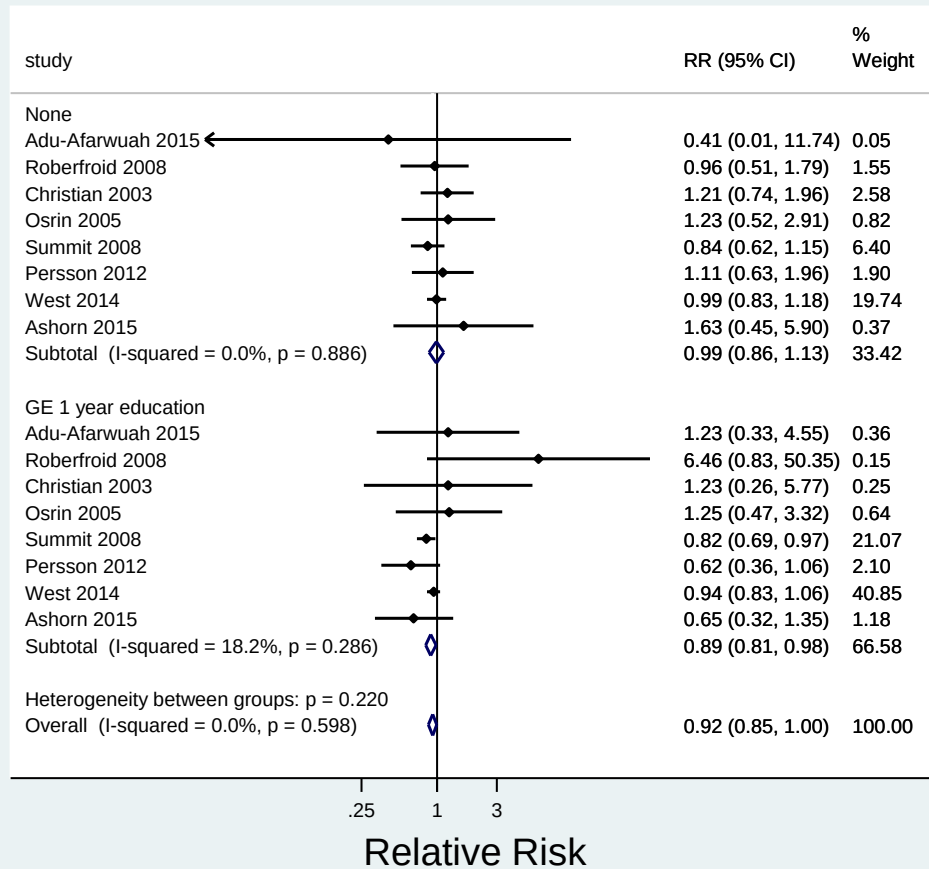
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by parity



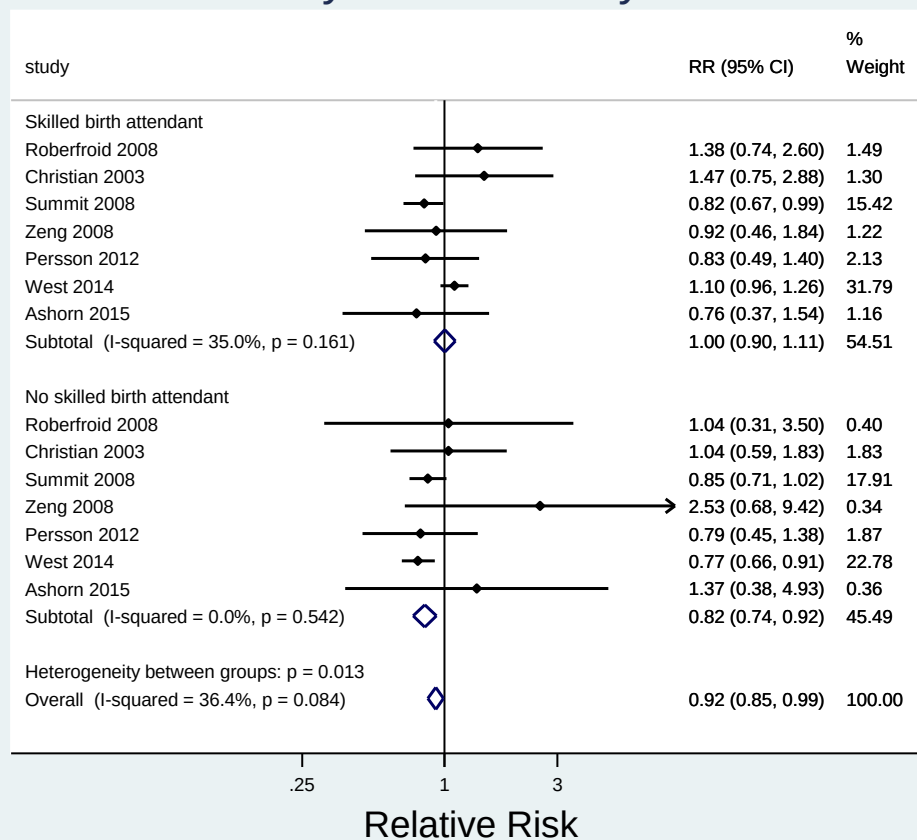
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by maternal education



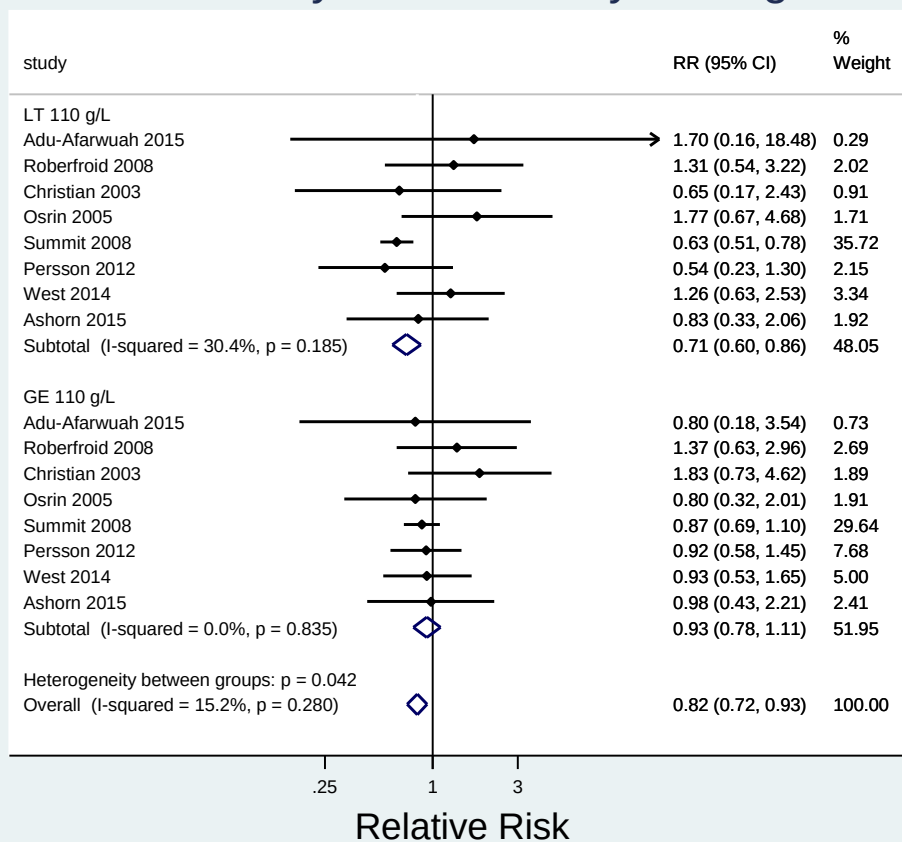
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by skilled birth attendant



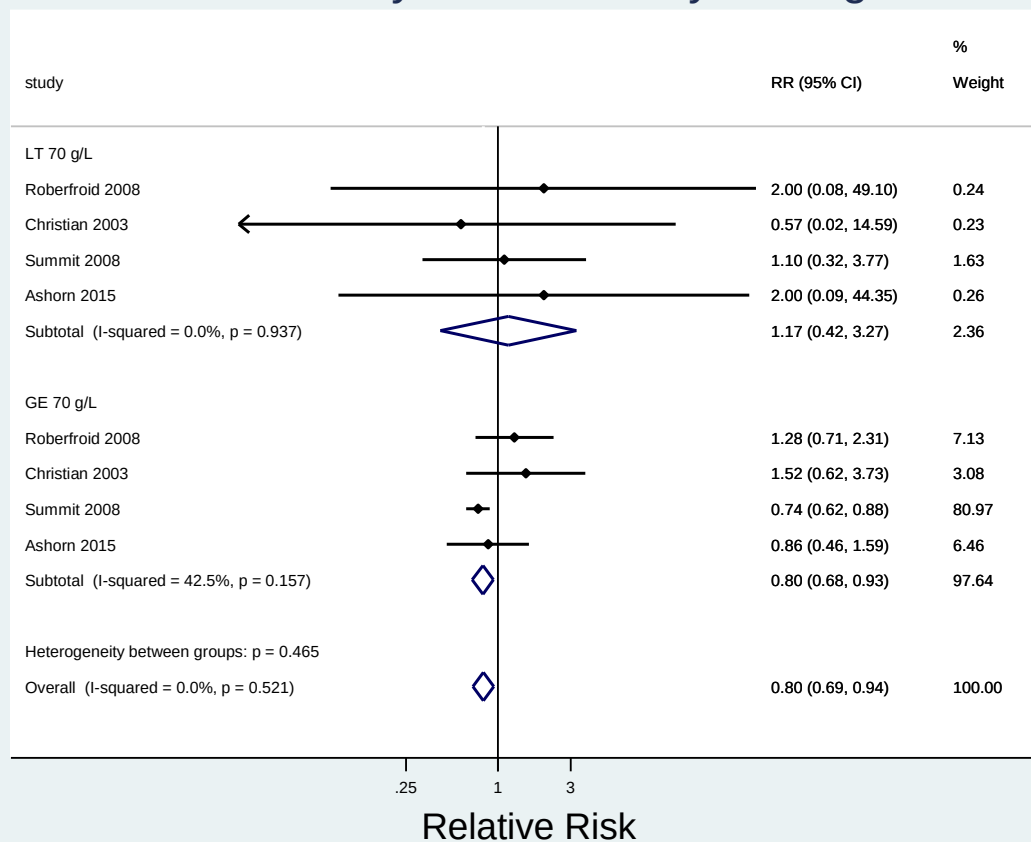
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by hemoglobin <110 g/L



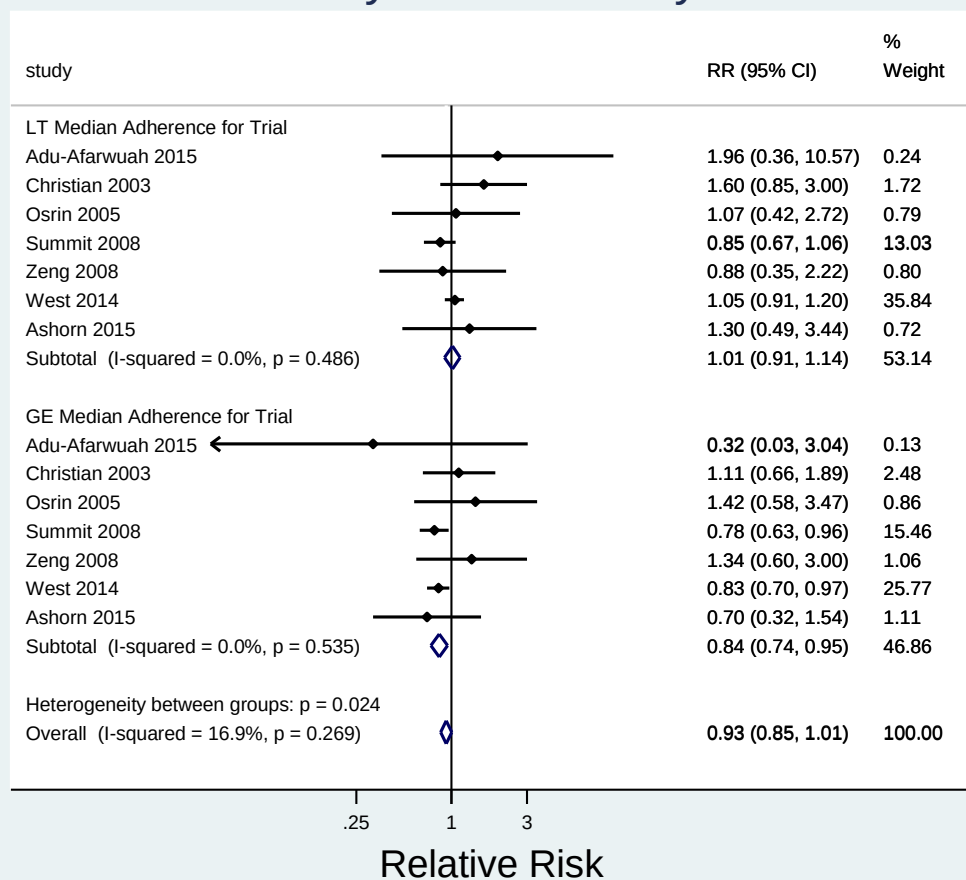
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by hemoglobin <70 g/L



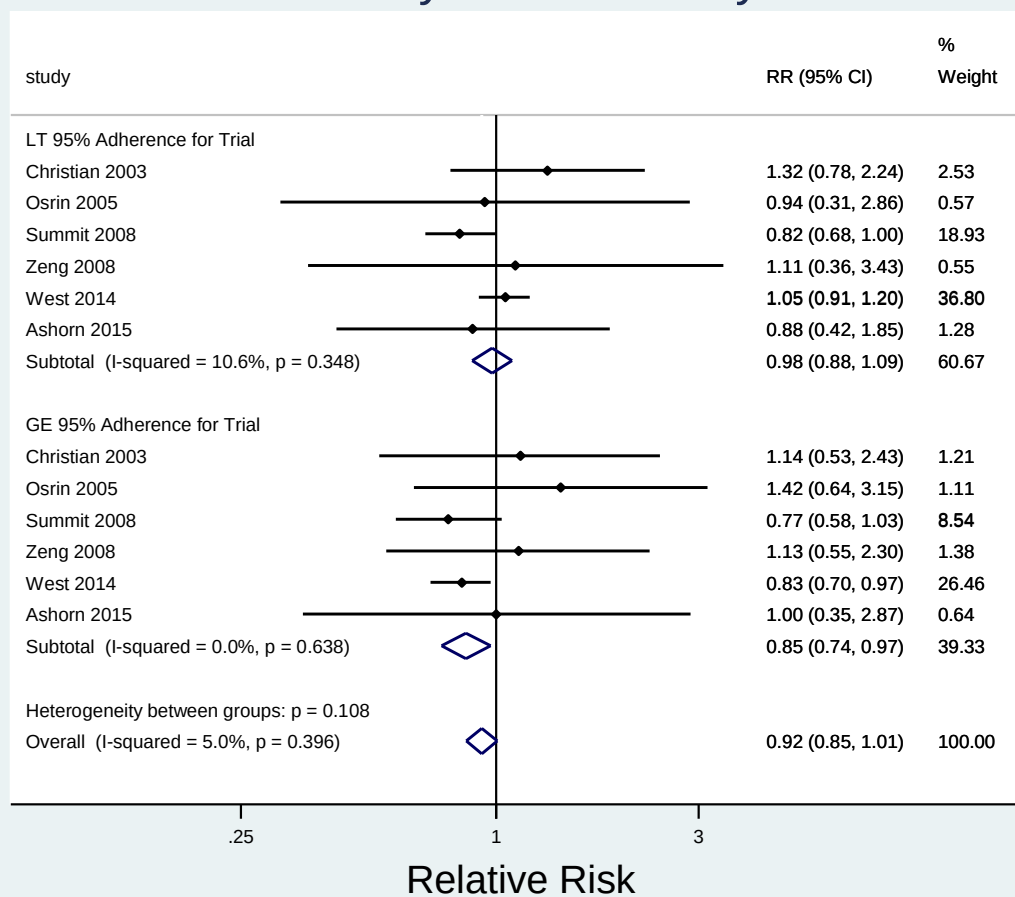
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by median adherence



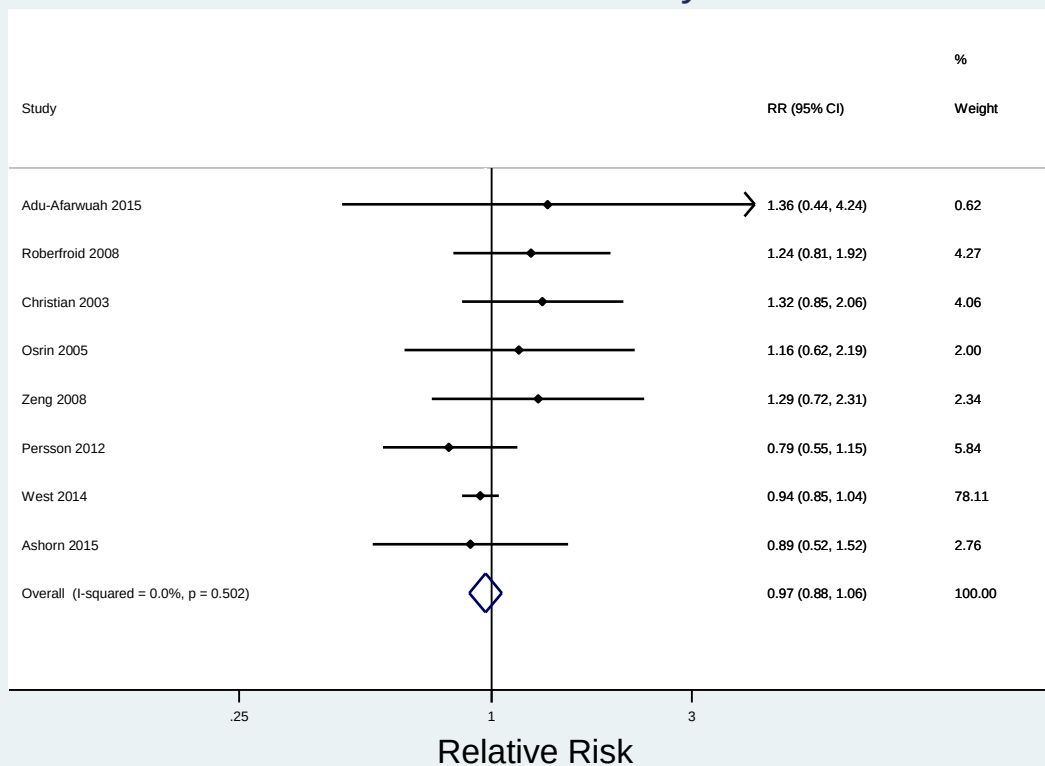
Outcome = Mortality to 6 Months

Relative risk of mortality to 6 month by 95% adherence



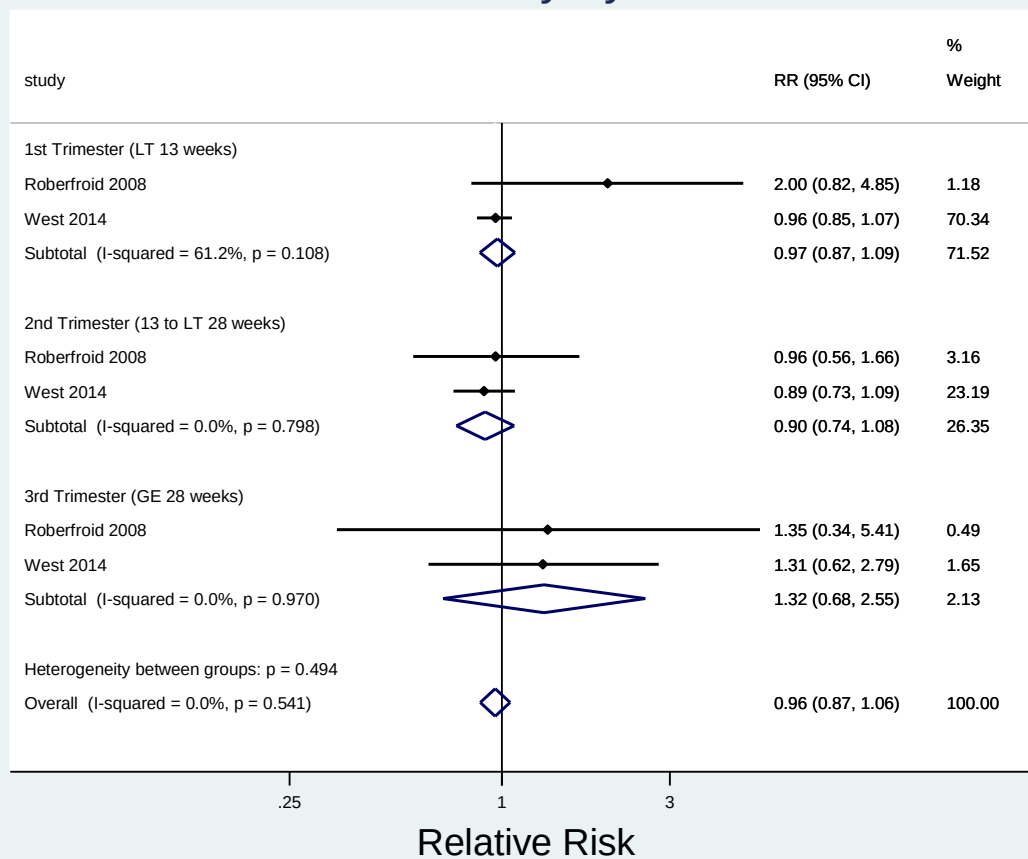
Outcome = Mortality to 6 Months

Relative risk of infant mortality - OVERALL



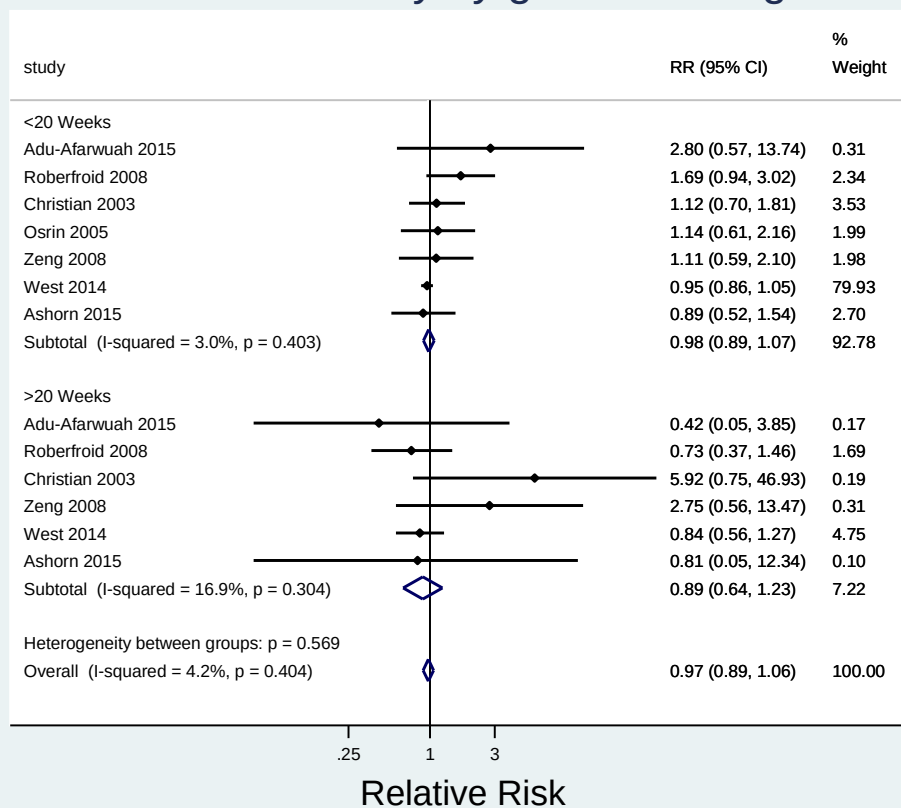
Outcome = Infant Mortality

Relative risk of infant mortality by trimester at enrollment



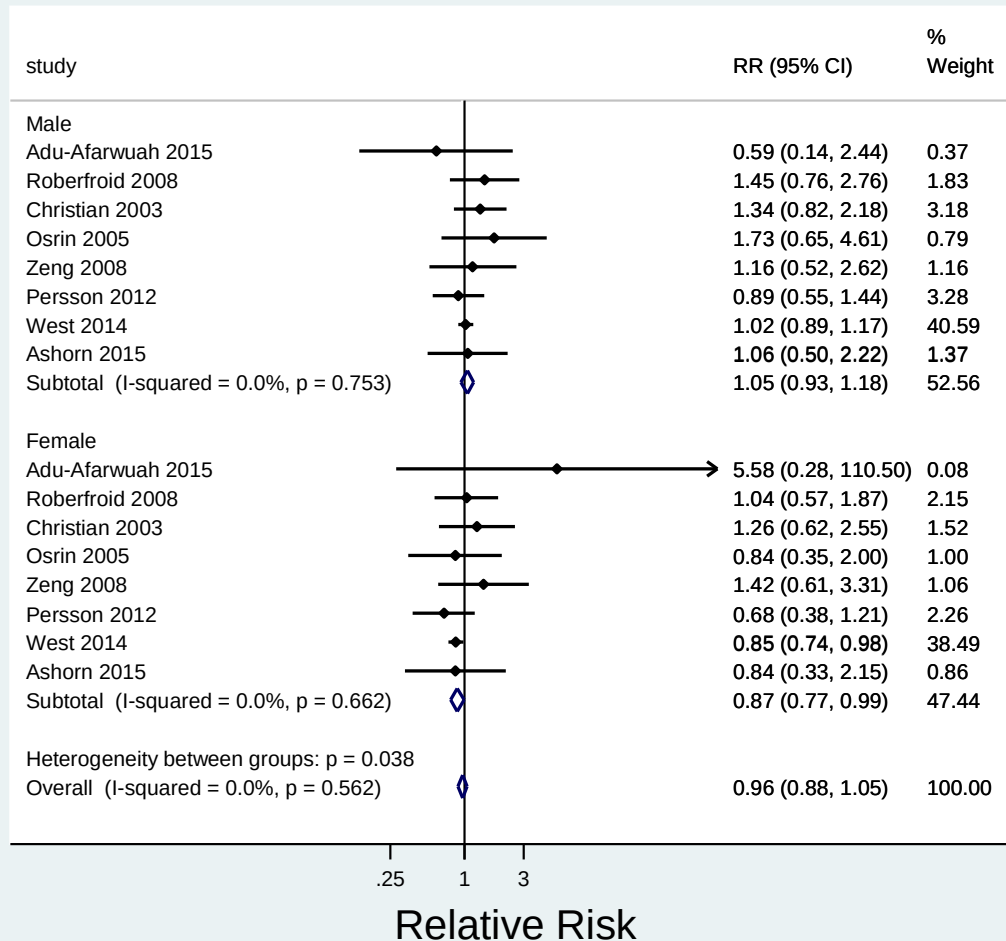
Outcome = Infant Mortality

Relative risk of infant mortality by gestational age at enrollment



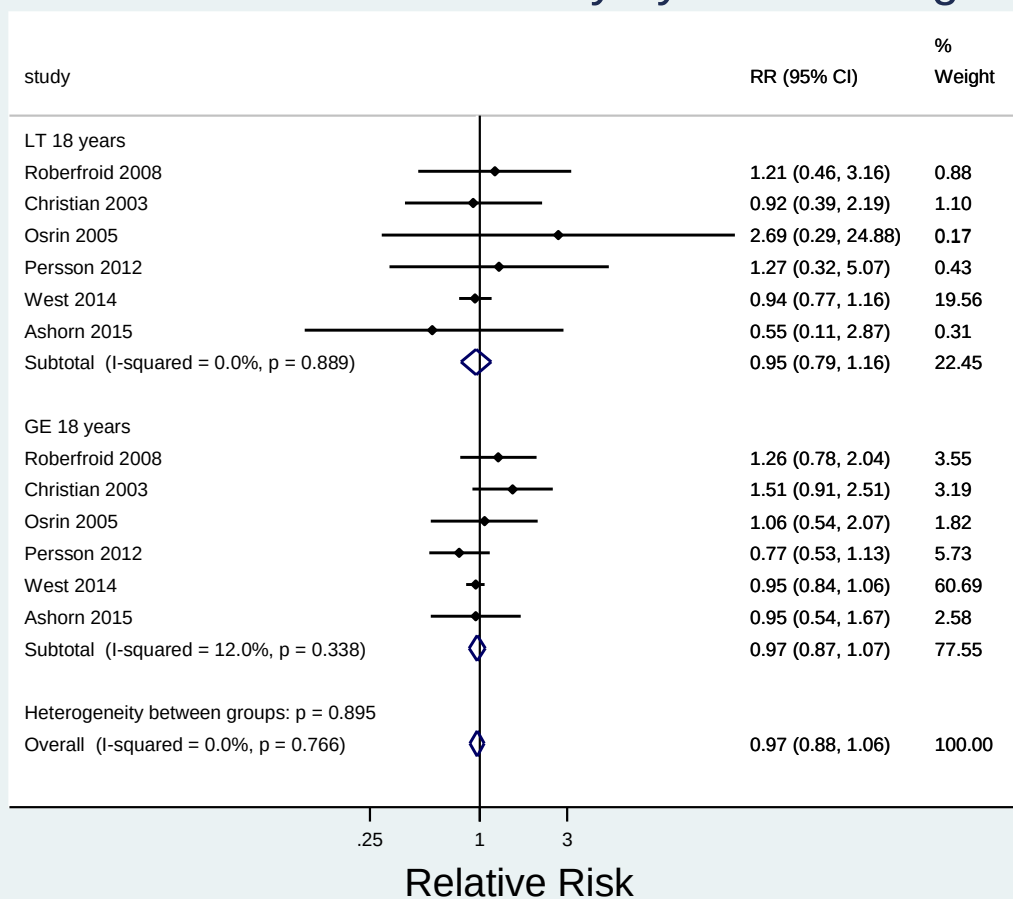
Outcome = Infant Mortality

Relative risk of infant mortality by sex



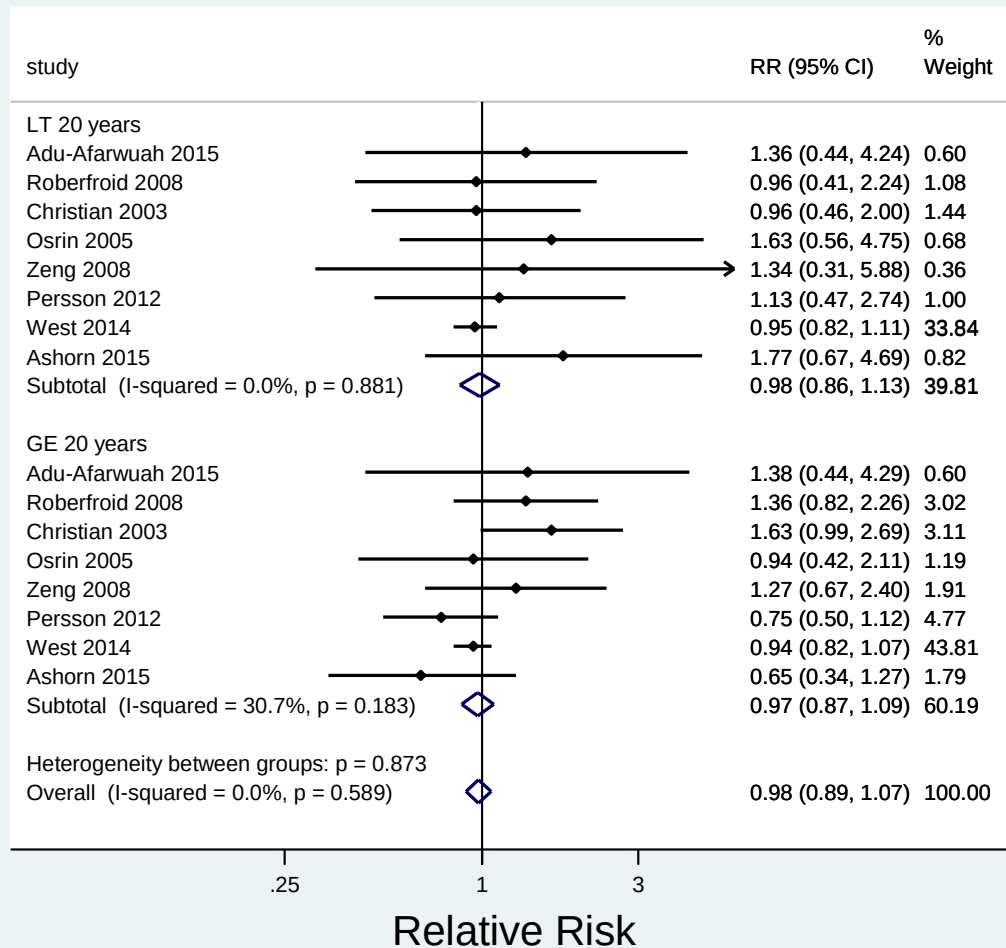
Outcome = Infant Mortality

Relative risk of infant mortality by maternal age <18



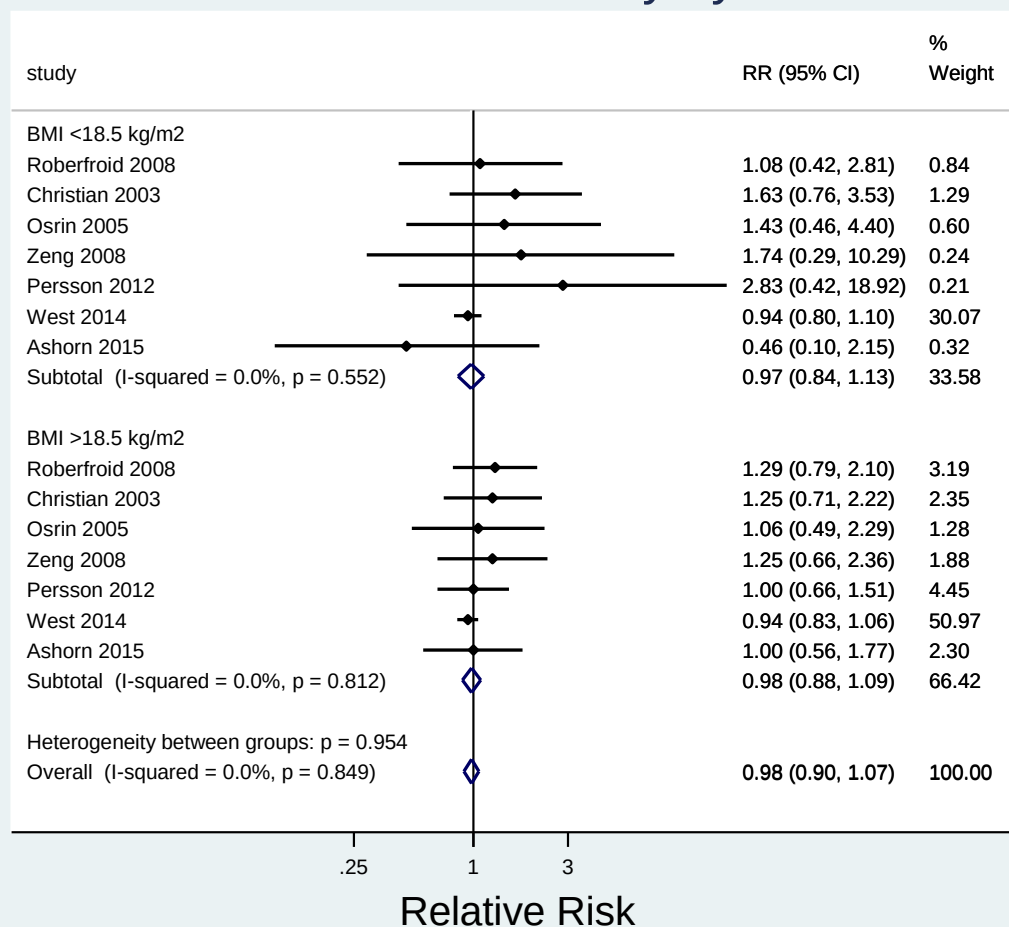
Outcome = Infant Mortality

Relative risk of infant mortality by maternal age <20



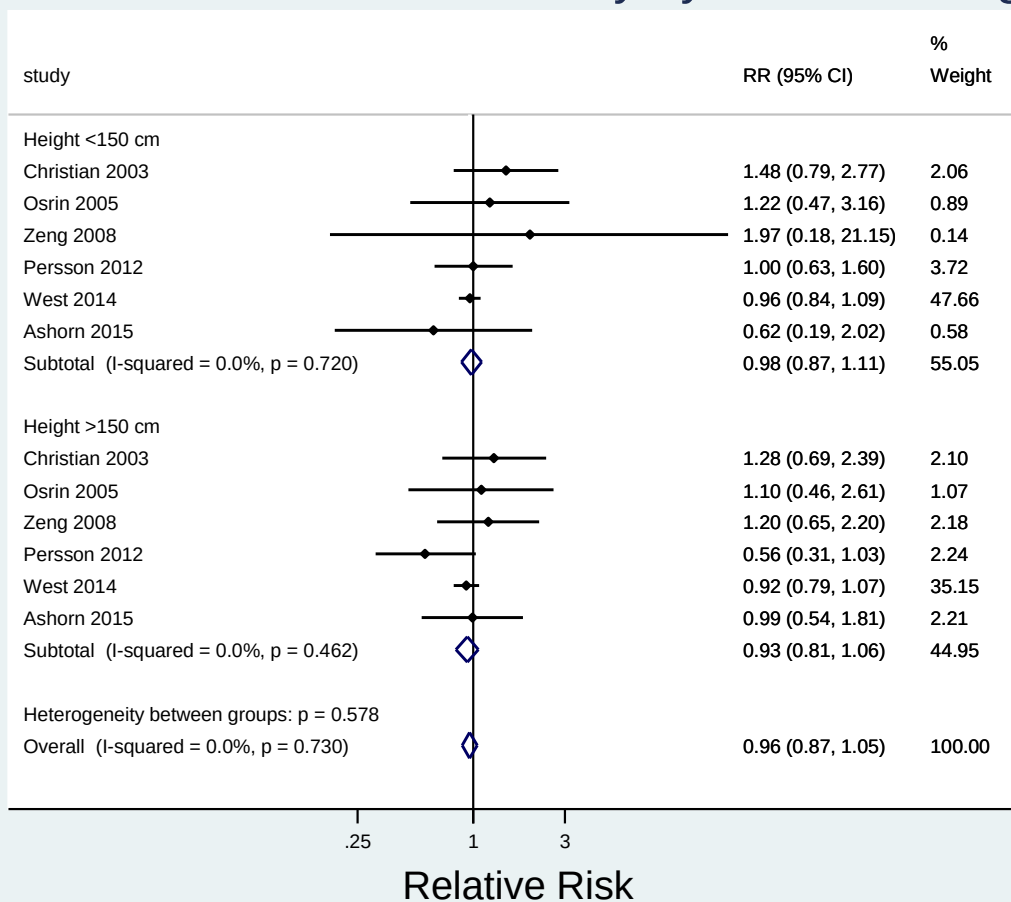
Outcome = Infant Mortality

Relative risk of infant mortality by maternal BMI



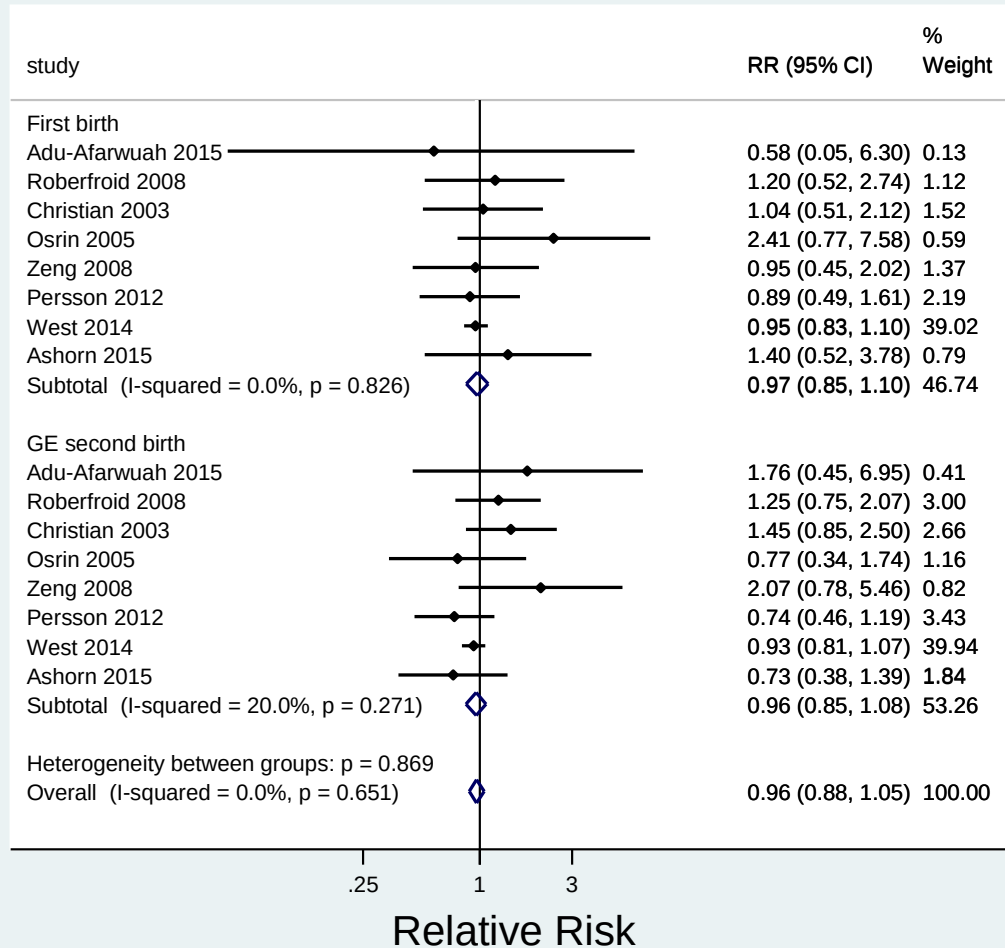
Outcome = Infant Mortality

Relative risk of infant mortality by maternal height



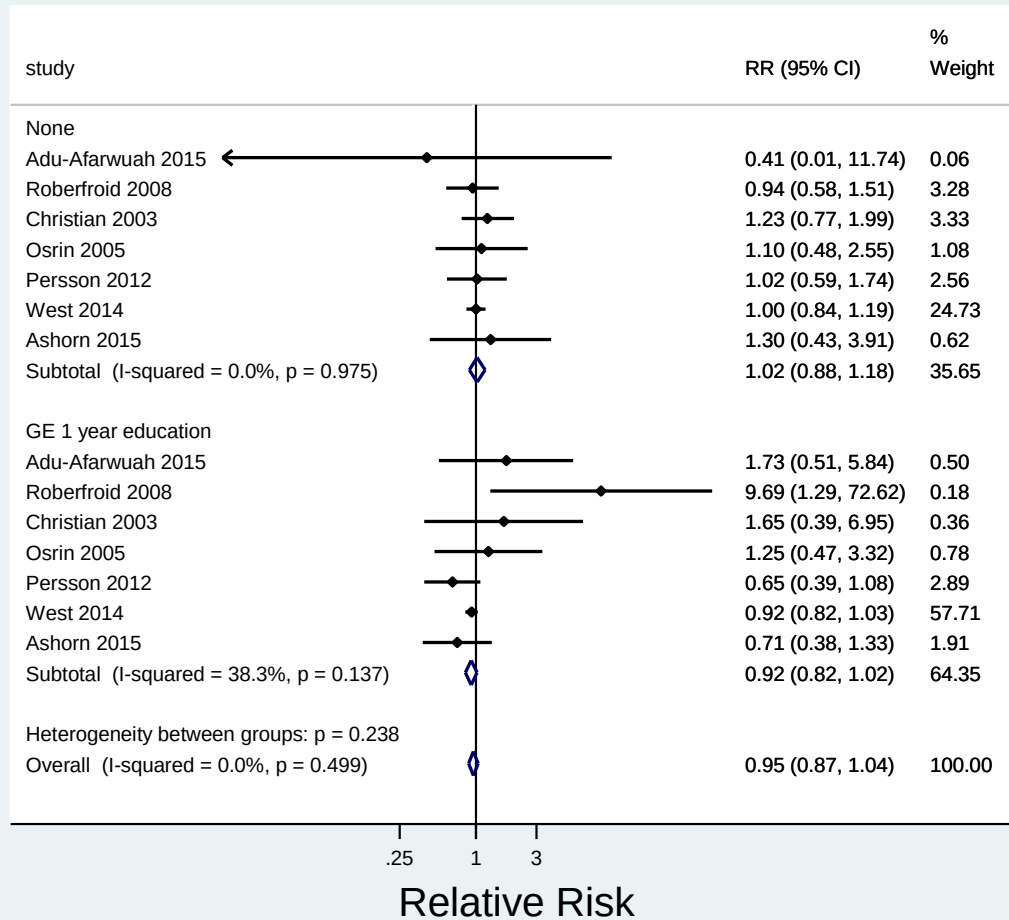
Outcome = Infant Mortality

Relative risk of infant mortality by parity



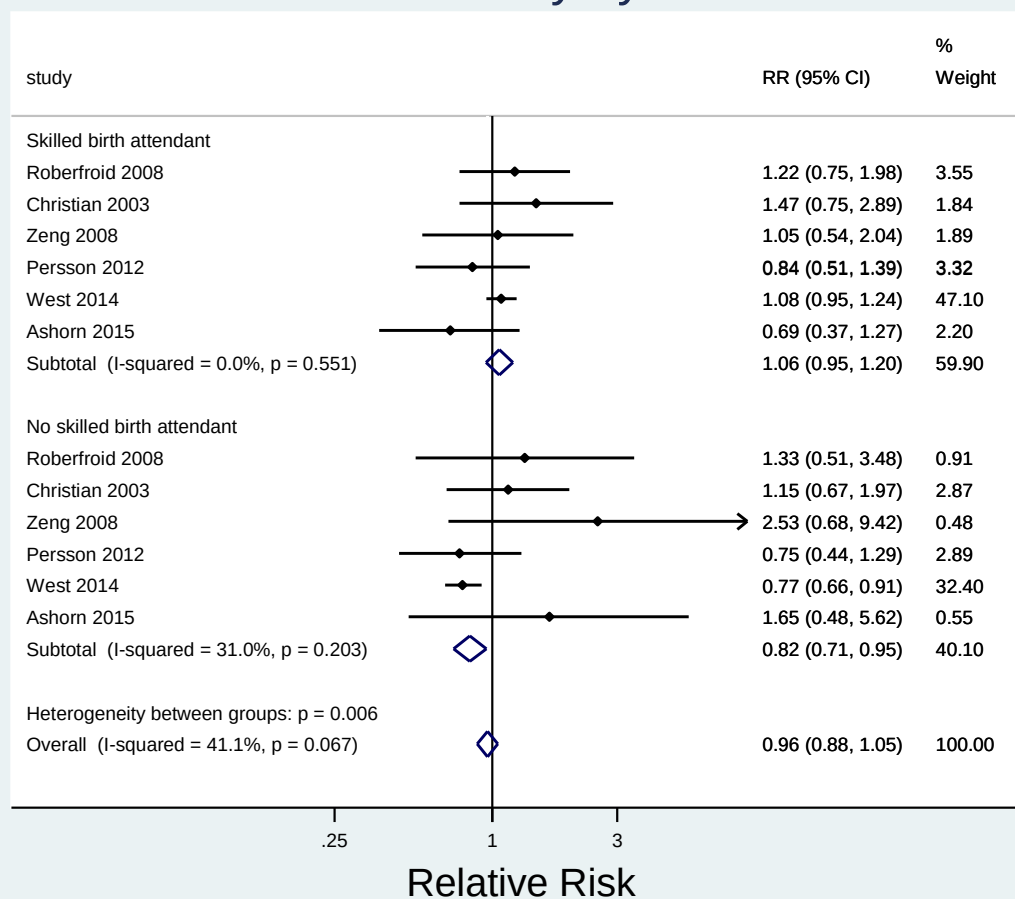
Outcome = Infant Mortality

Relative risk of infant mortality by maternal education



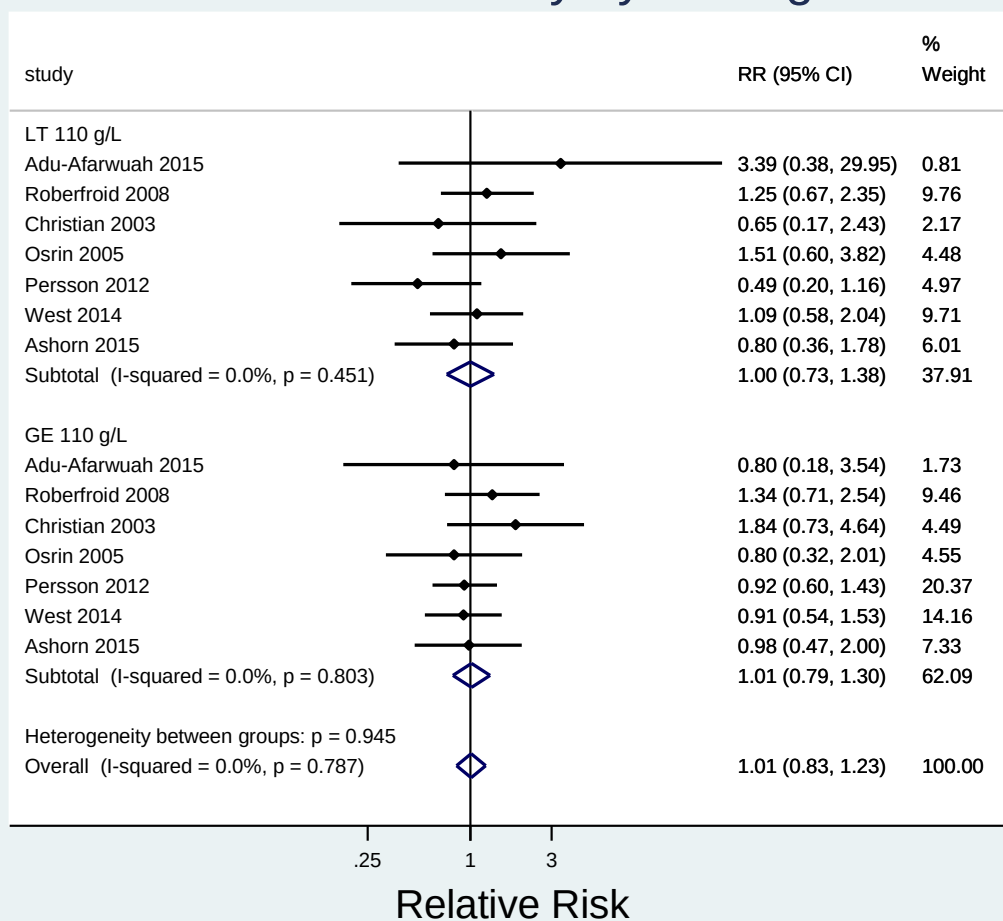
Outcome = Infant Mortality

Relative risk of infant mortality by skilled birth attendant



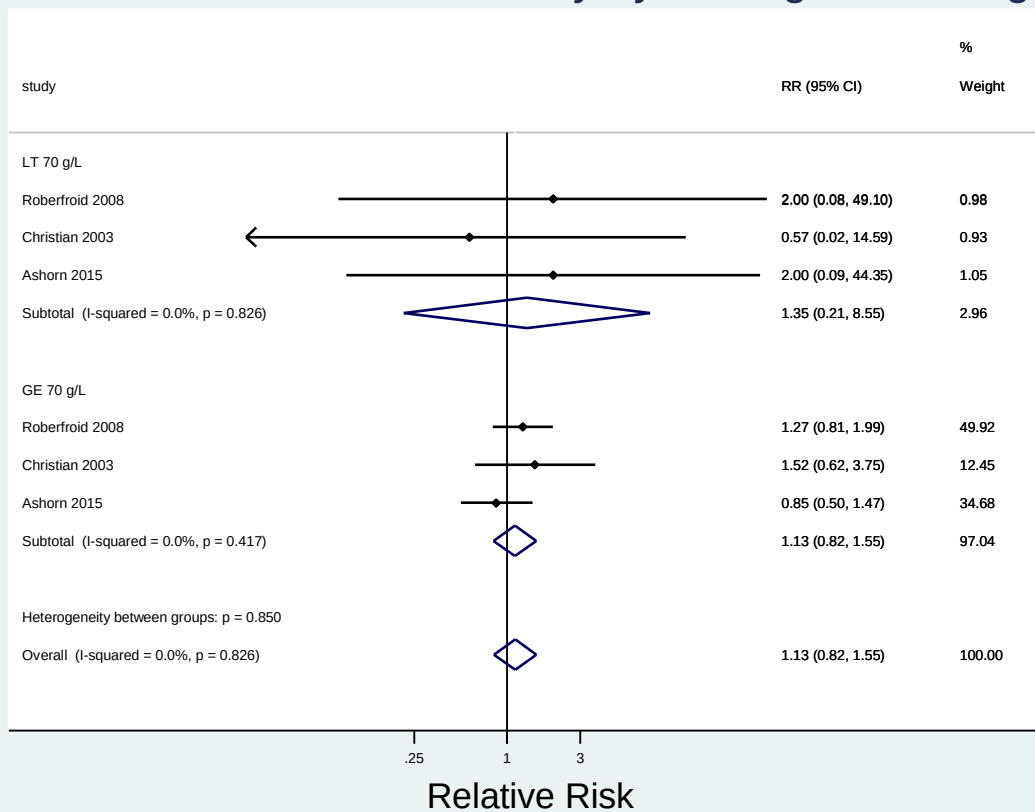
Outcome = Infant Mortality

Relative risk of infant mortality by hemoglobin <110 g/L



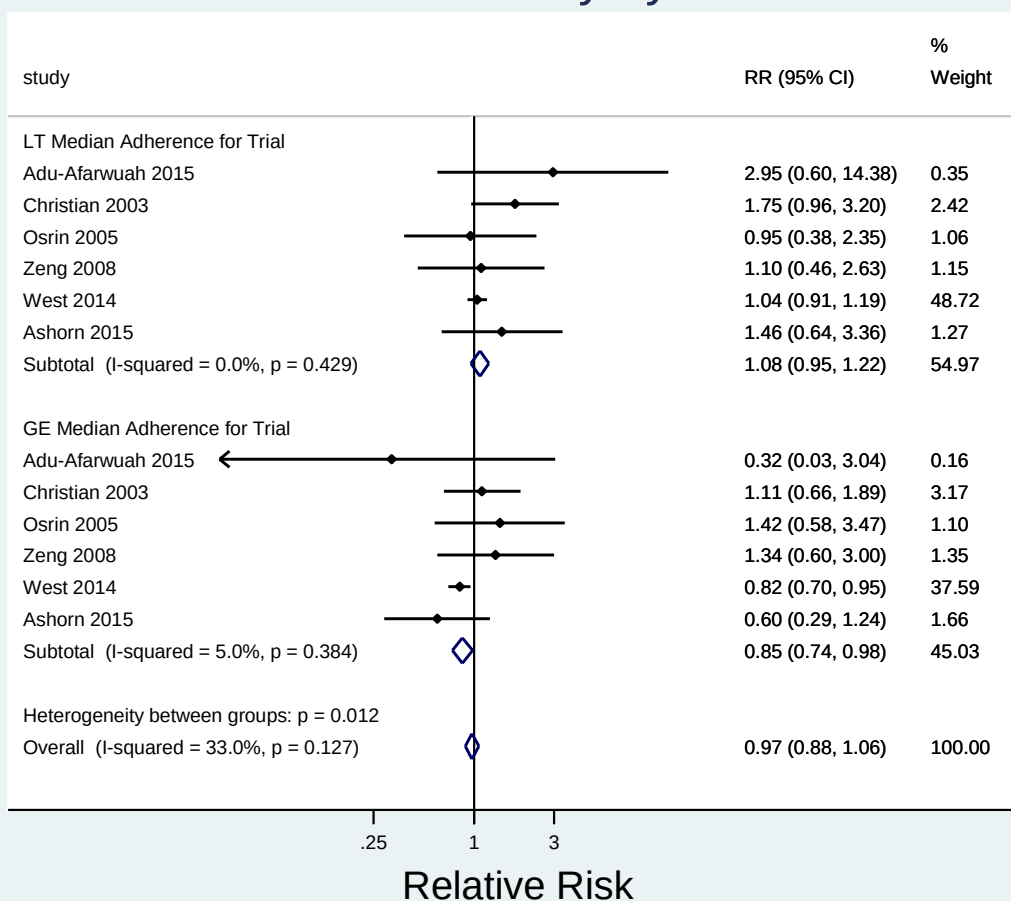
Outcome = Infant Mortality

Relative risk of infant mortality by hemoglobin <70 g/L



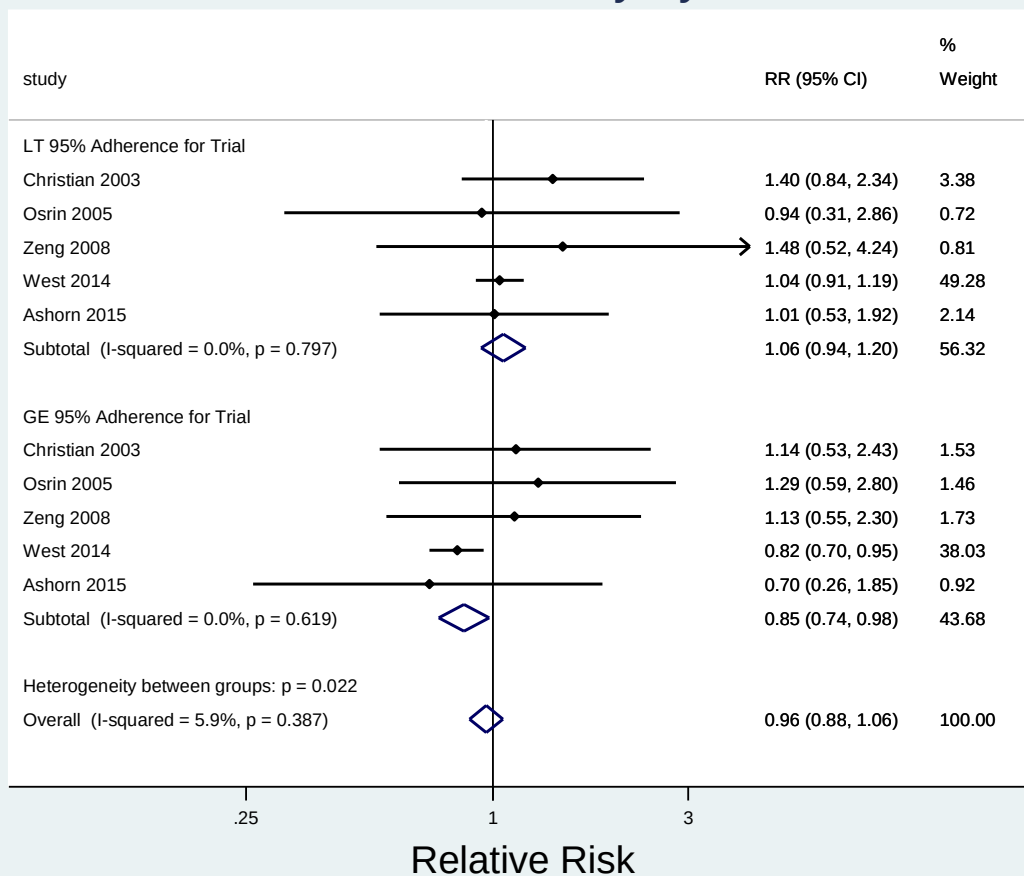
Outcome = Infant Mortality

Relative risk of infant mortality by median adherence



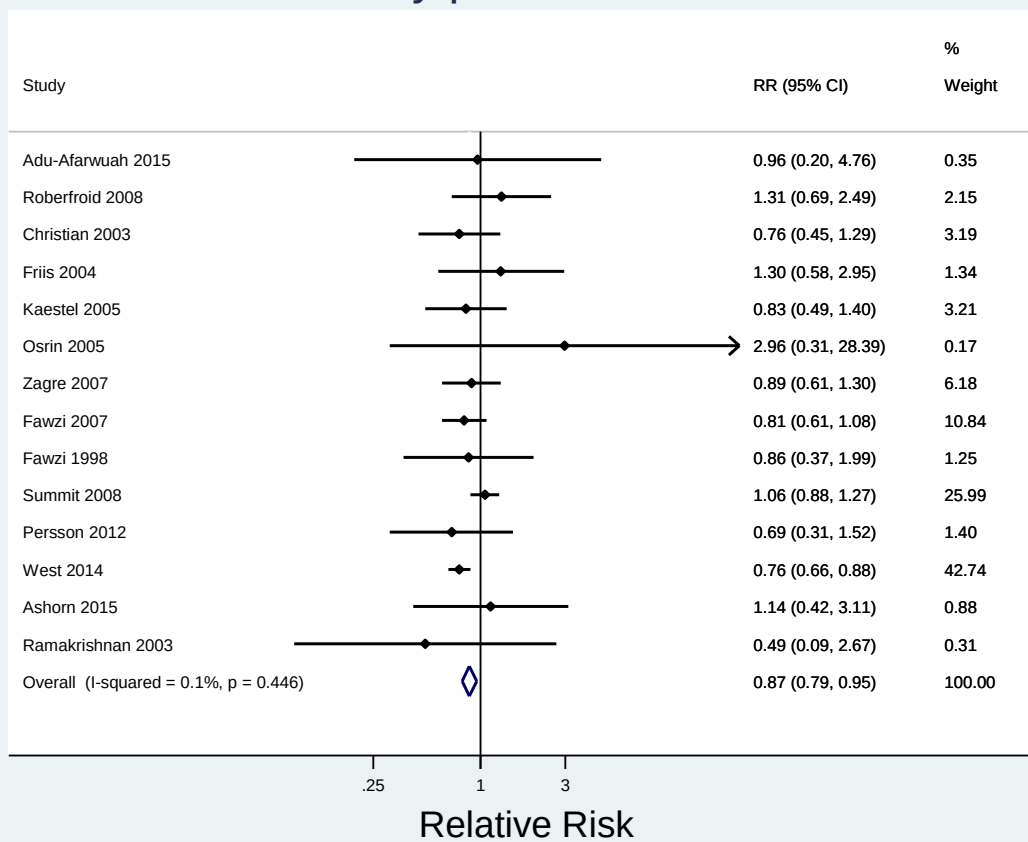
Outcome = Infant Mortality

Relative risk of infant mortality by 95% adherence



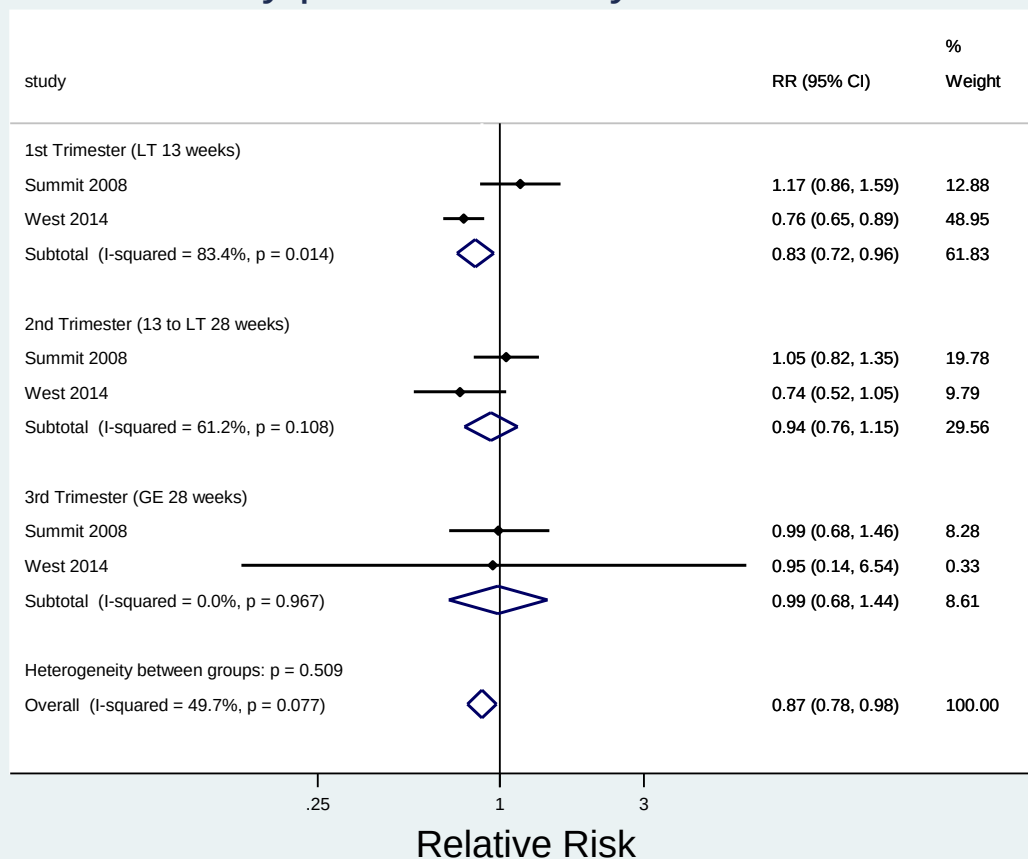
Outcome = Infant Mortality

Relative risk very preterm birth - OVERALL



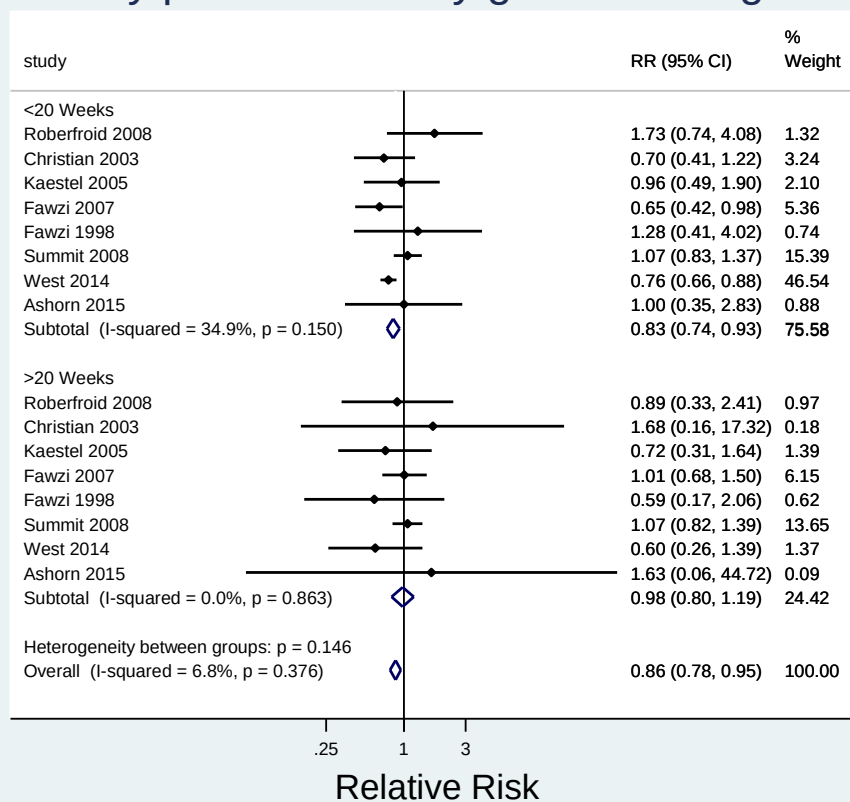
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by trimester at enrollment



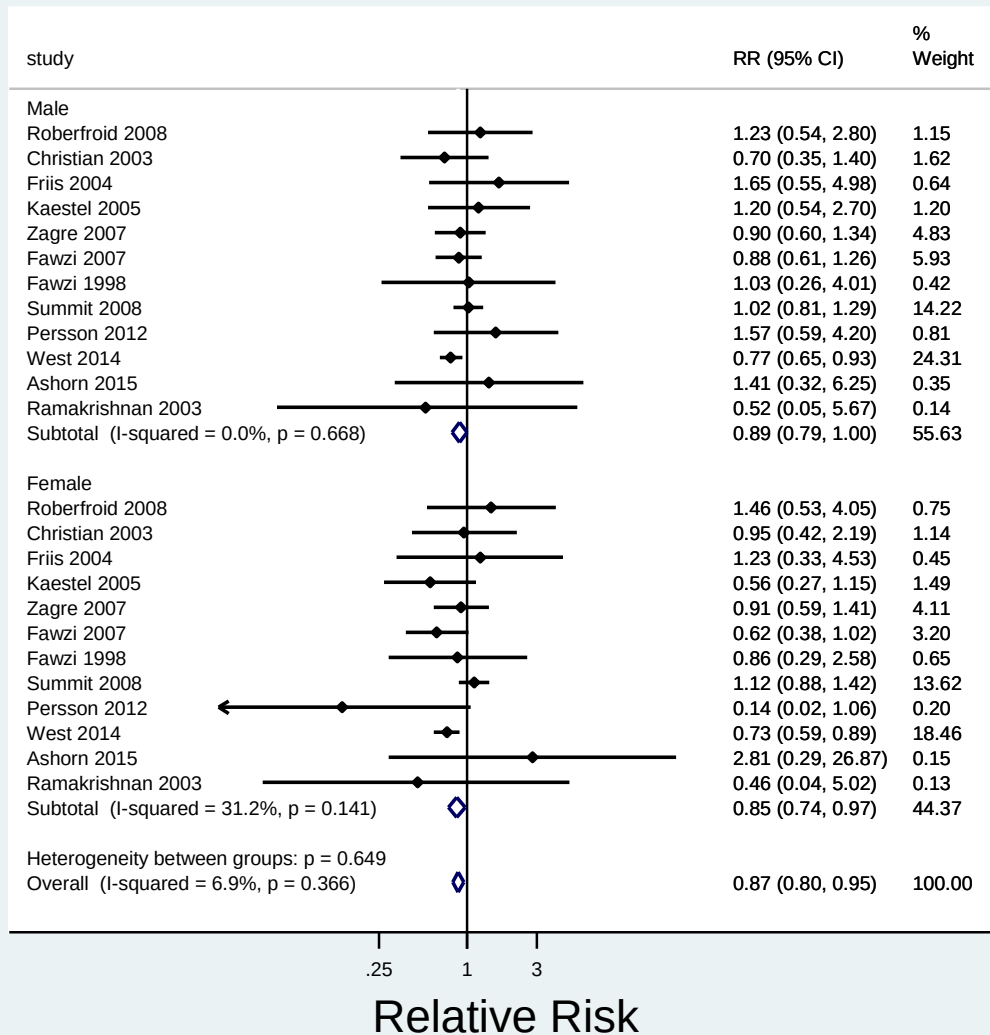
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by gestational age at enrollment



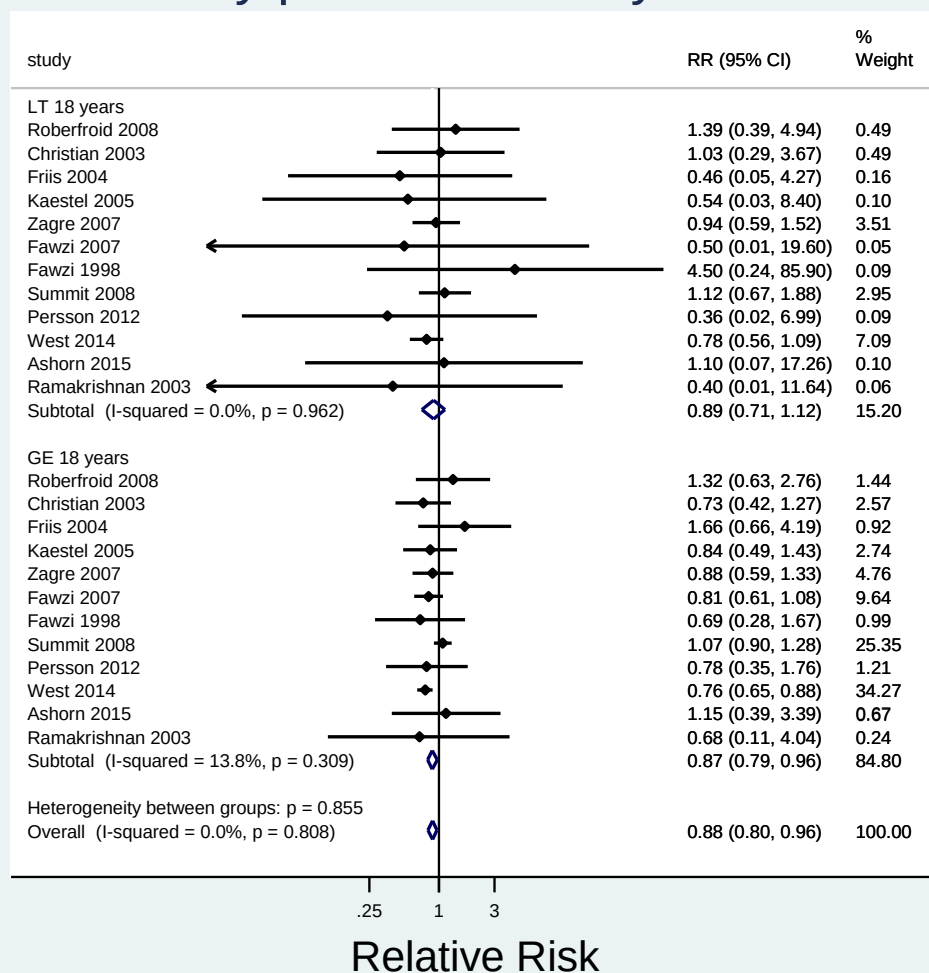
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by sex



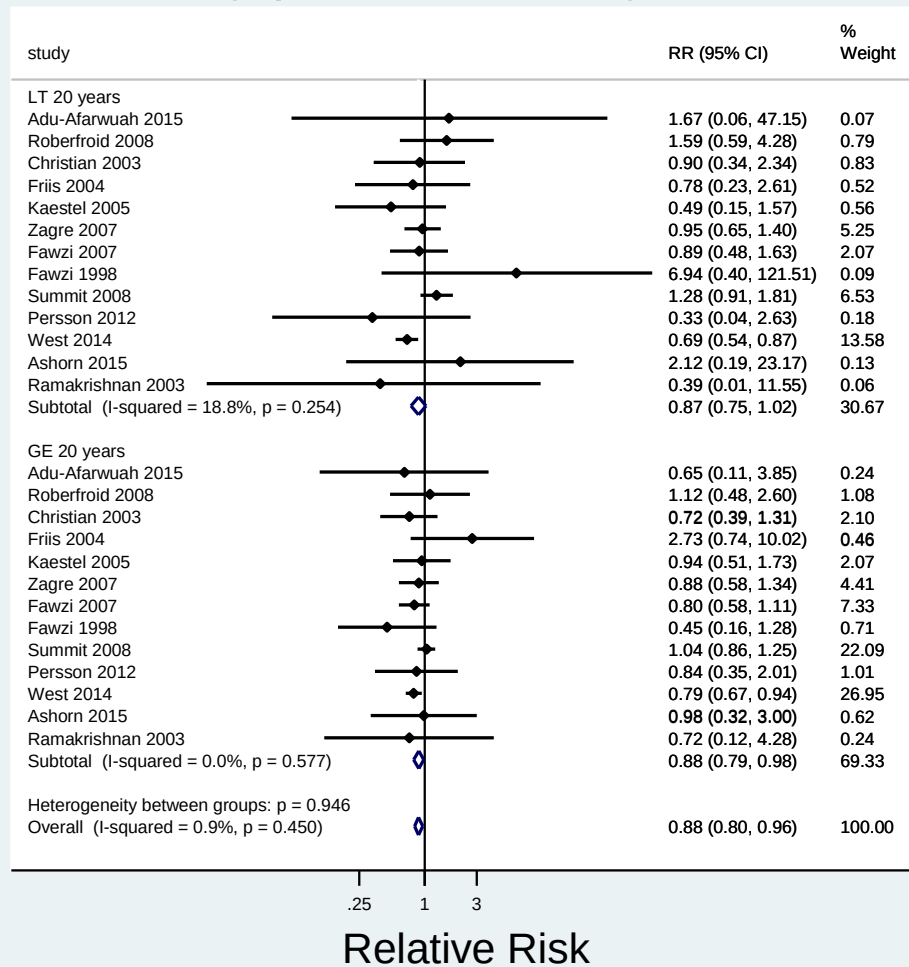
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by maternal age <18



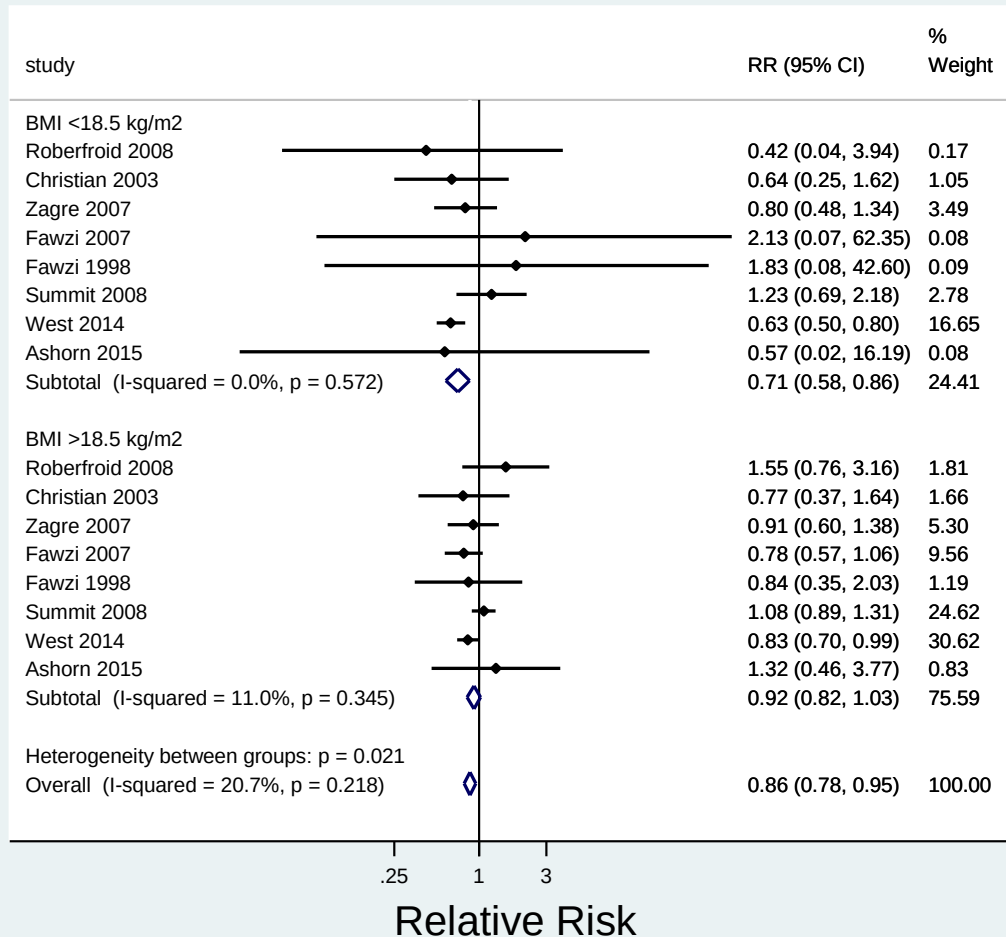
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by maternal age <20



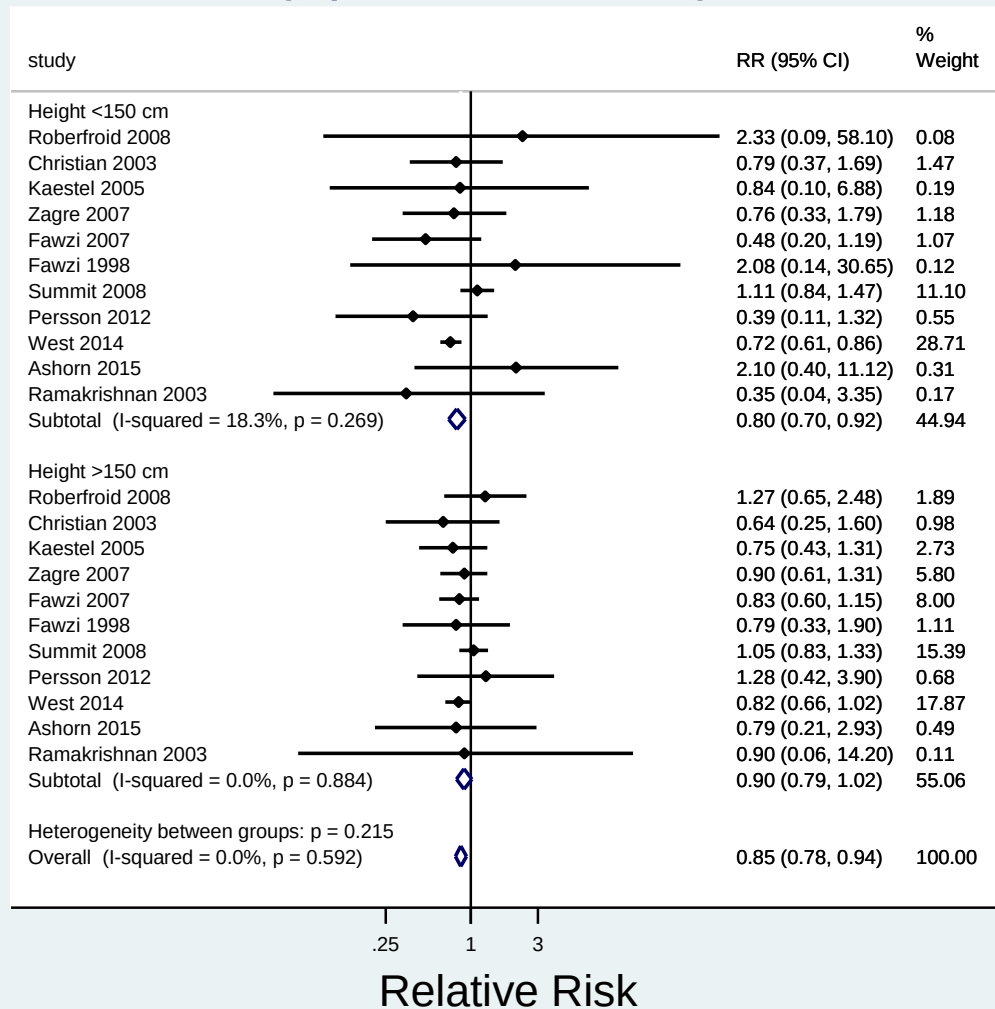
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by maternal BMI



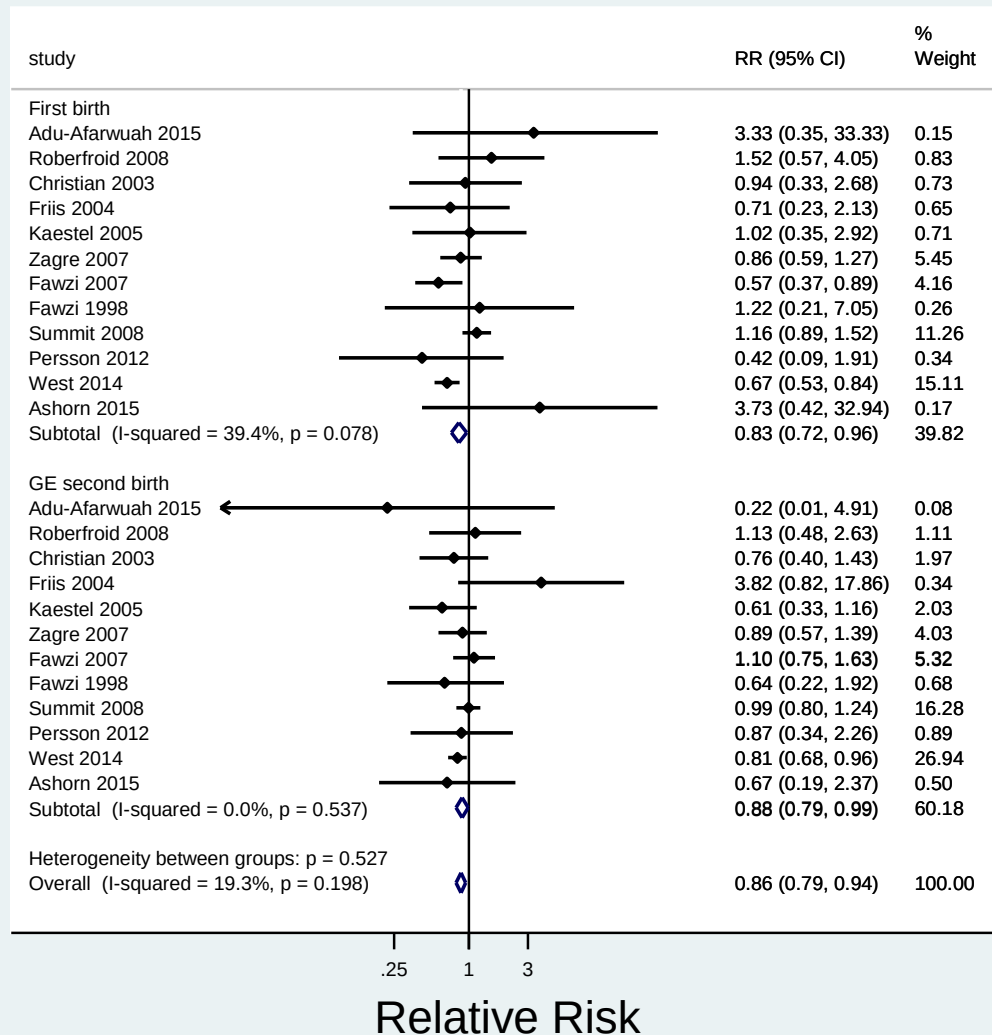
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by maternal height



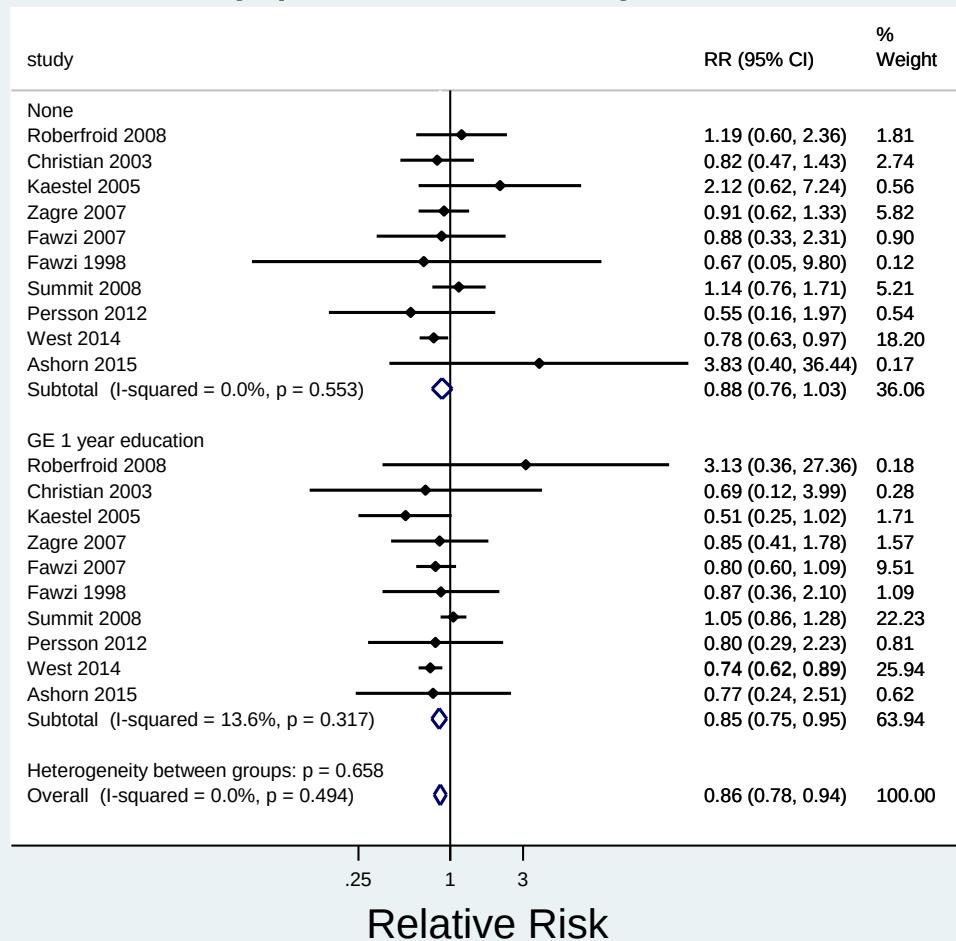
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by parity



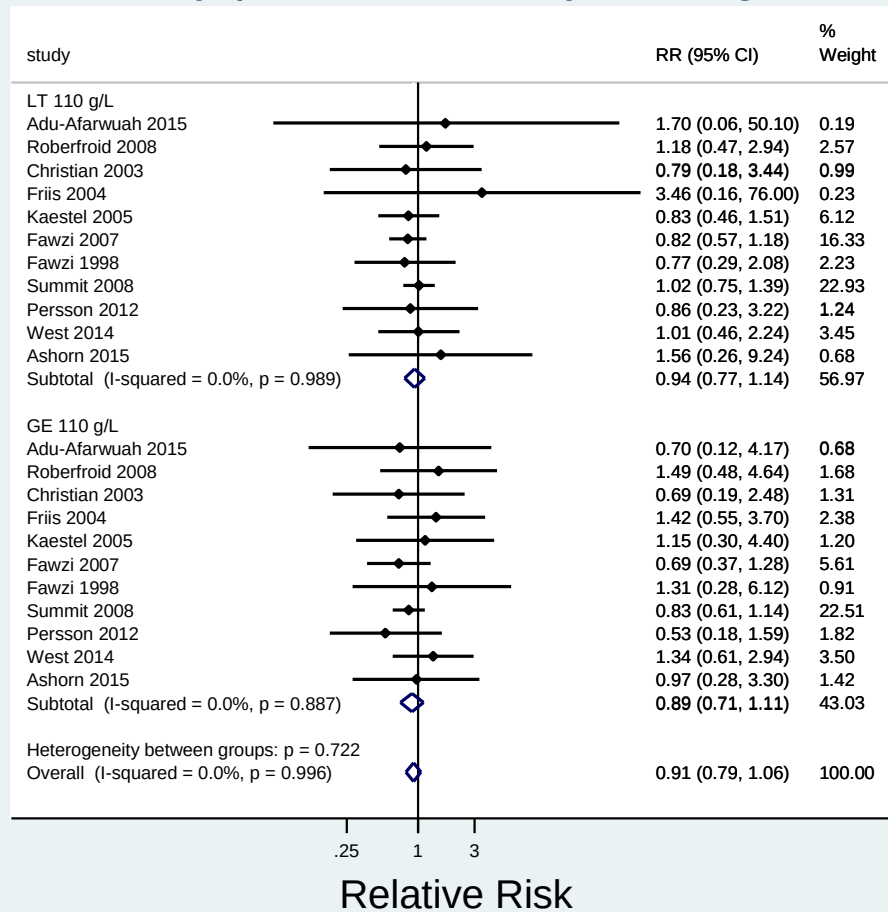
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by maternal education



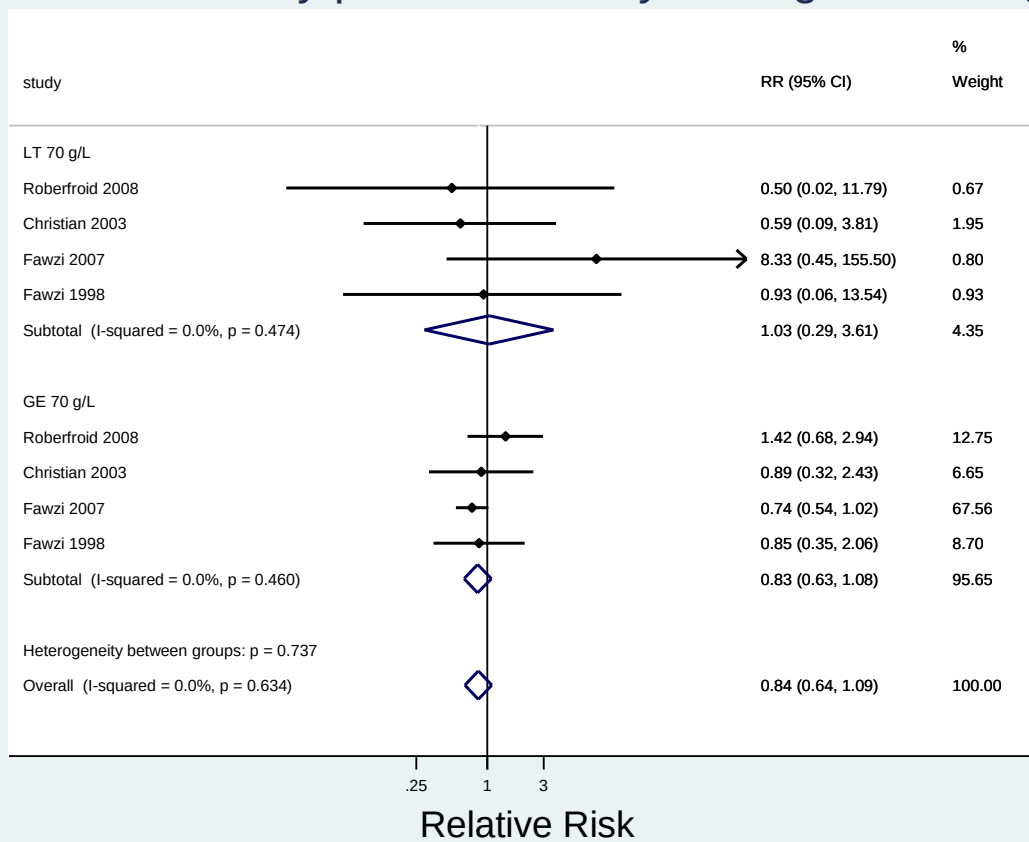
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by hemoglobin <110 g/L



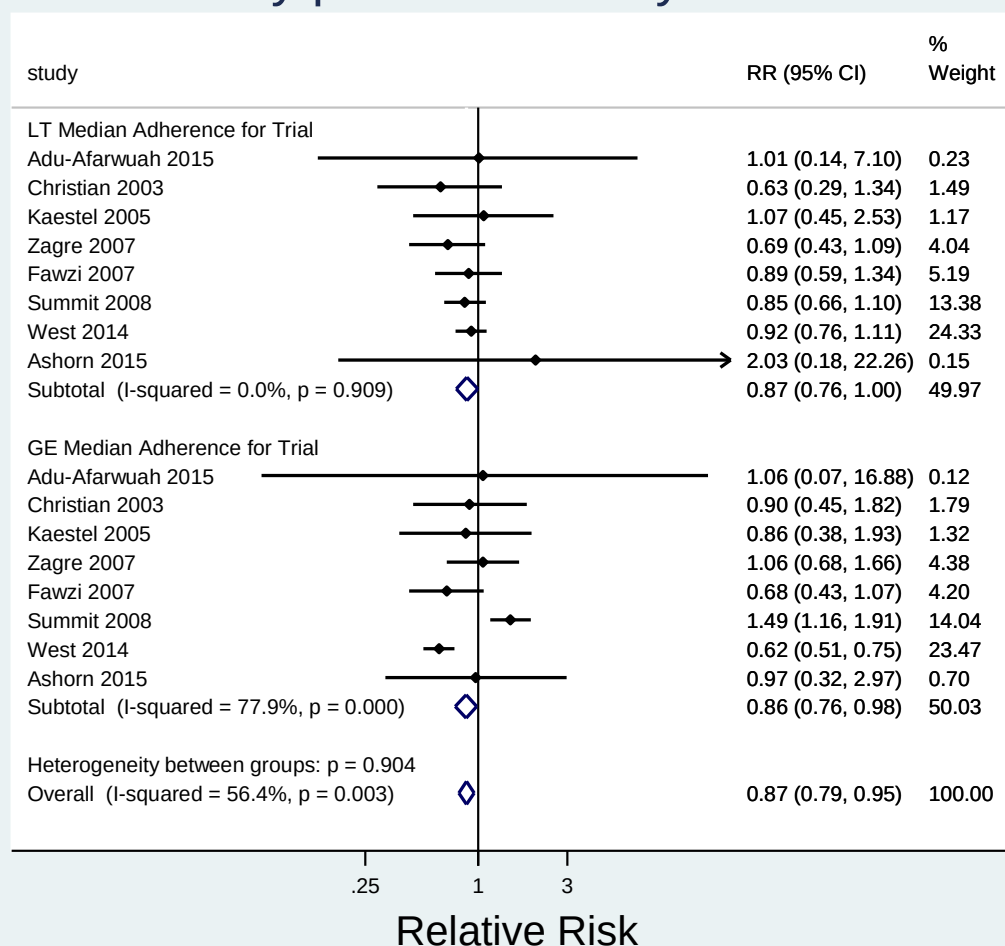
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by hemoglobin <70 g/L



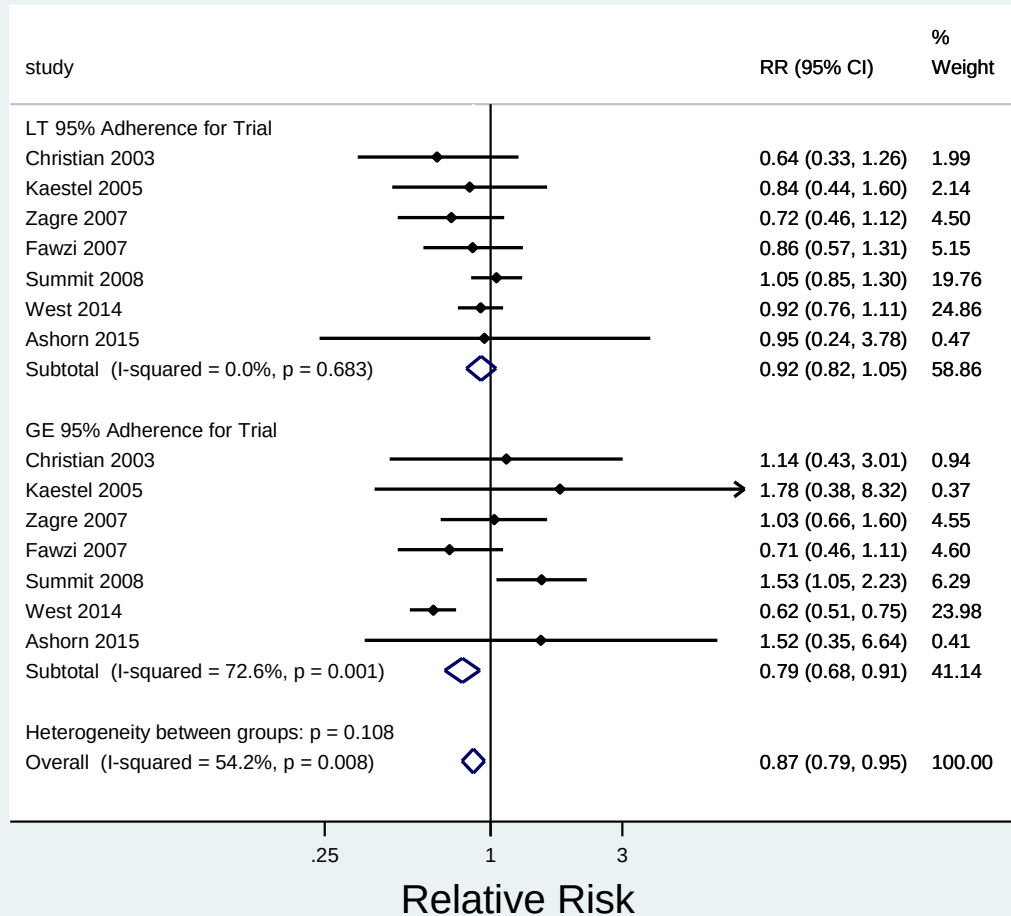
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by median adherence



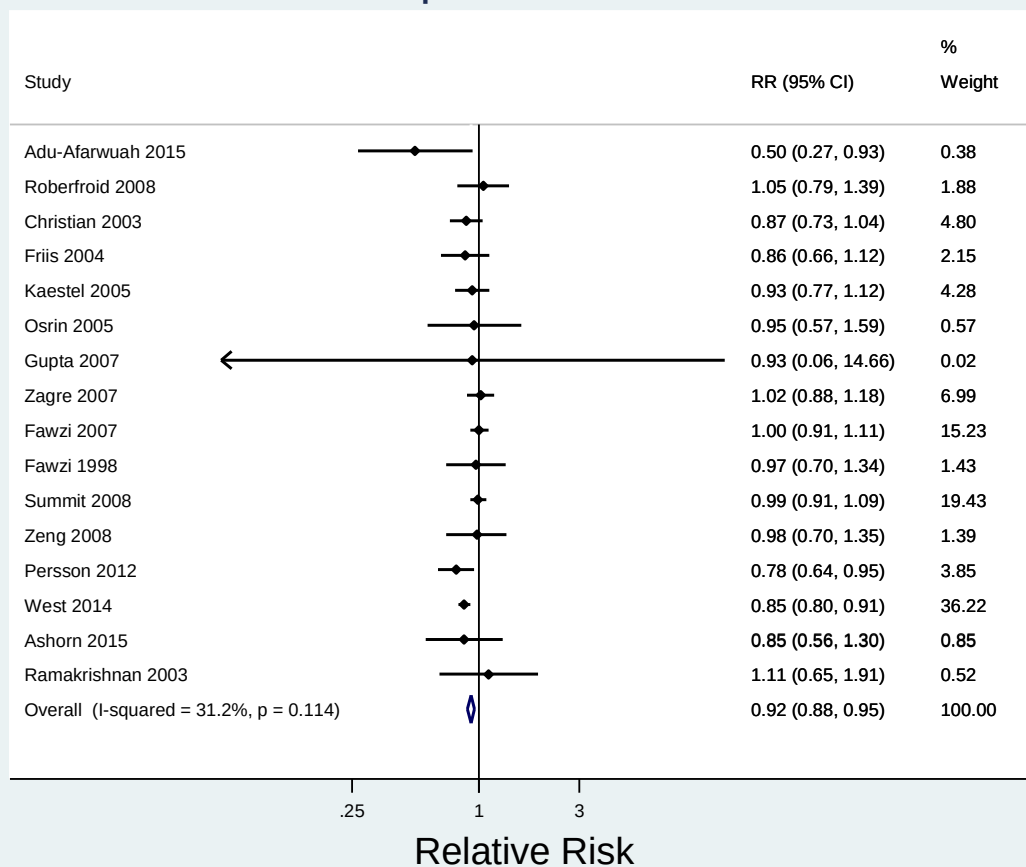
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk very preterm birth by 95% adherence



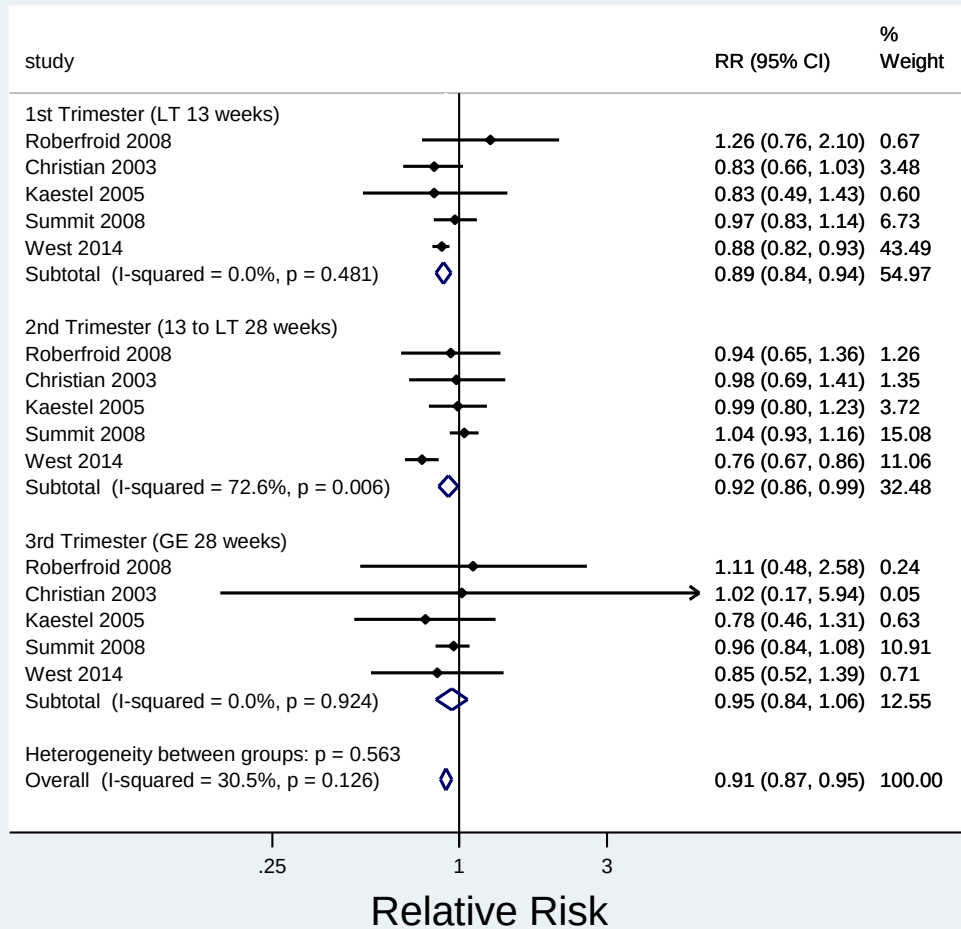
Outcome = Very Preterm Birth (<34 Weeks)

Relative risk of preterm birth - OVERALL



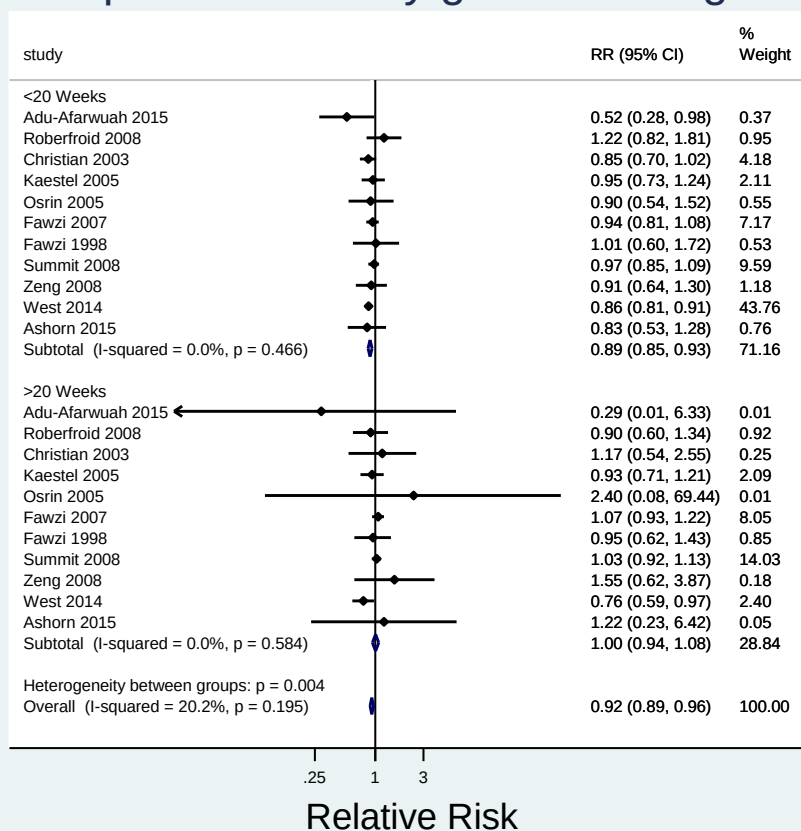
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by trimester at enrollment



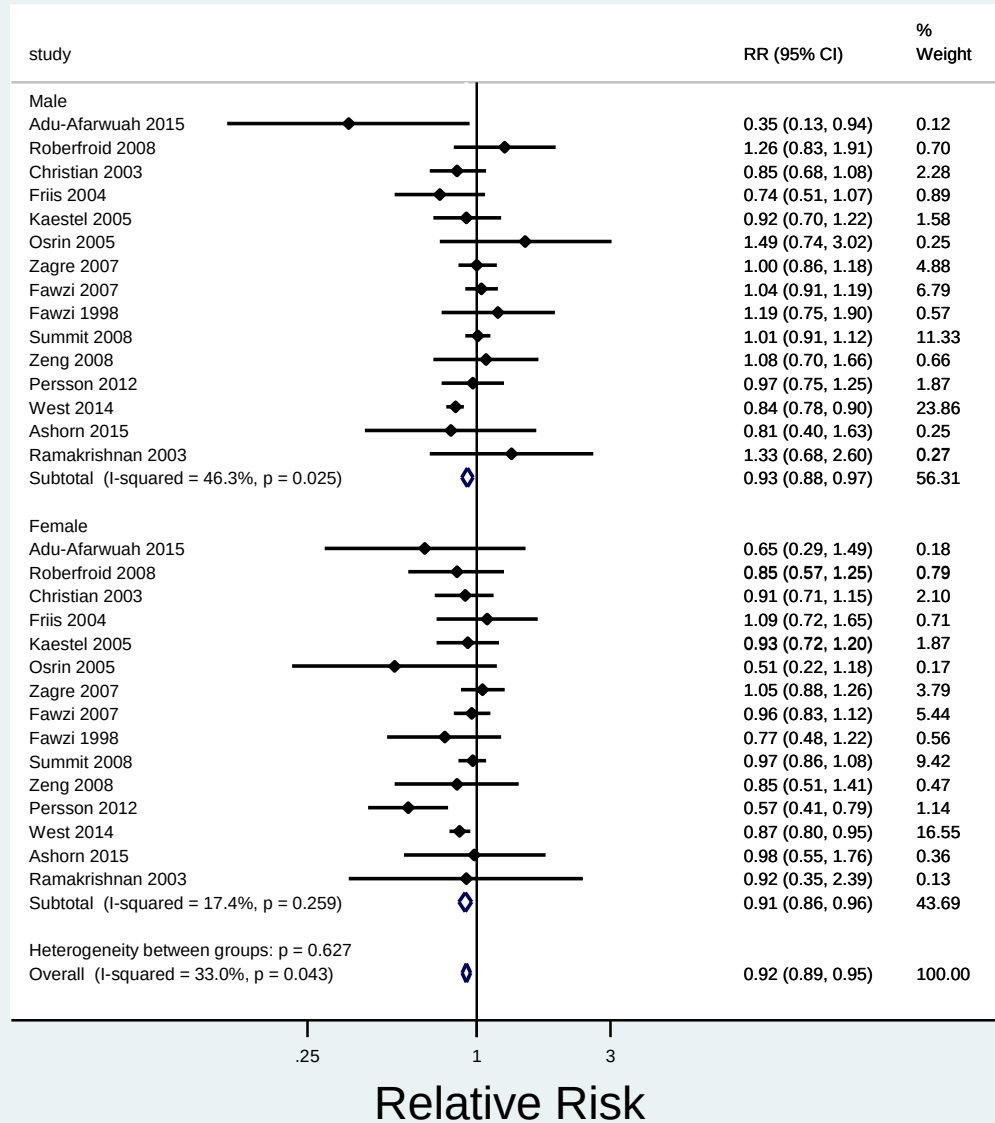
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by gestational age at enrollment



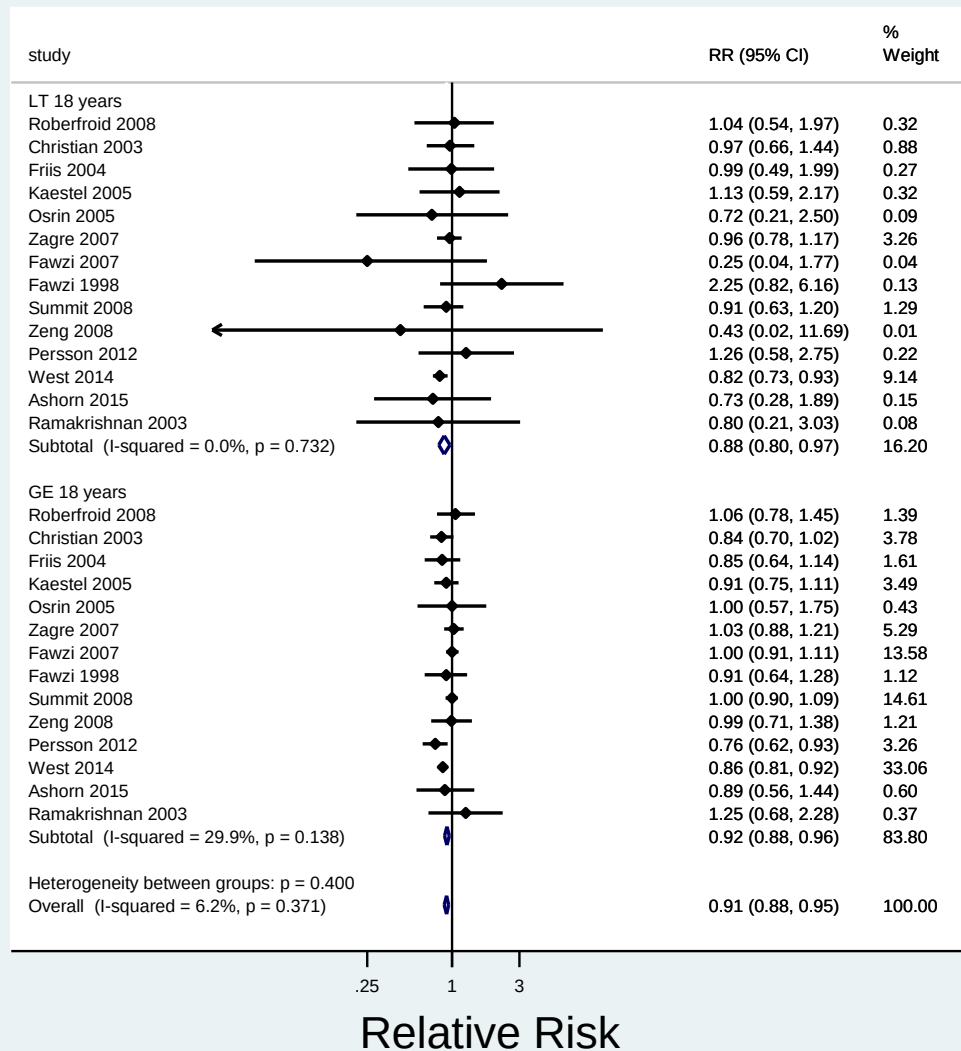
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by sex



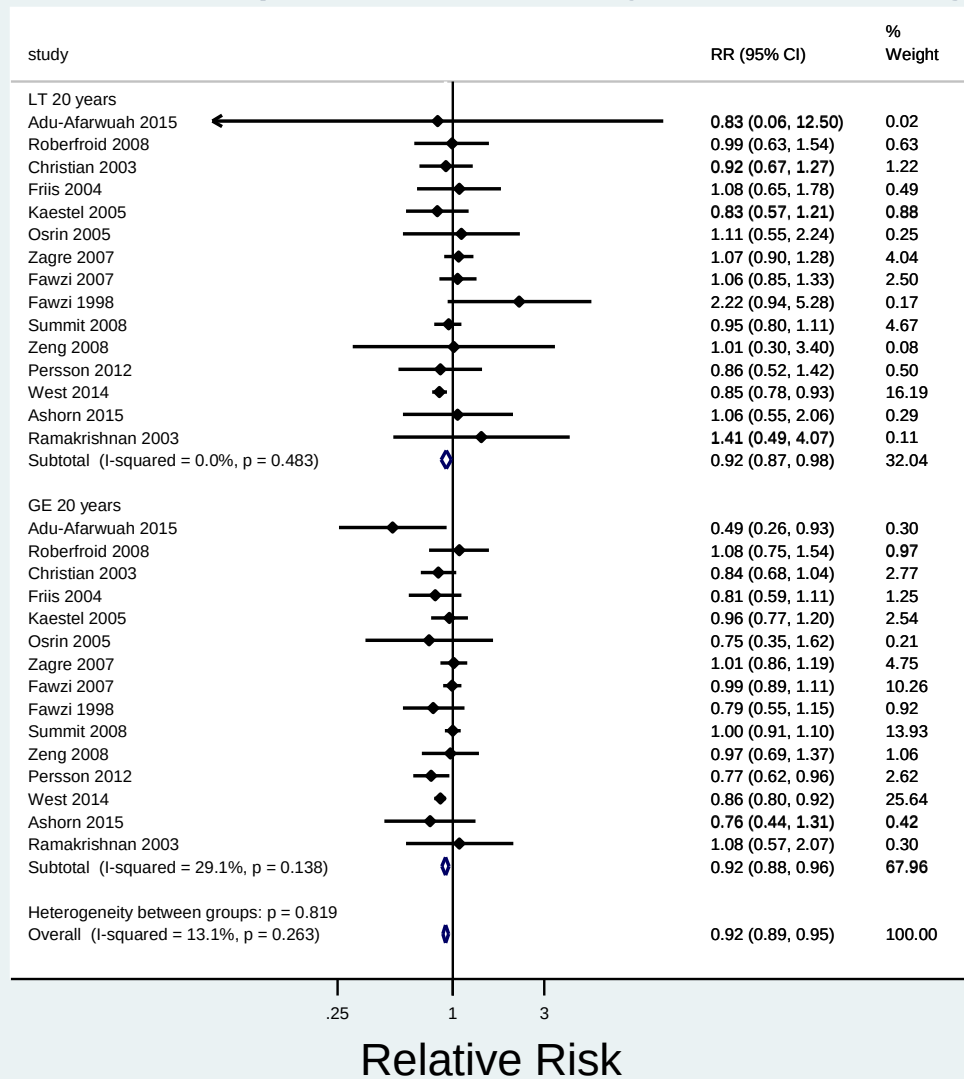
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by maternal age <18



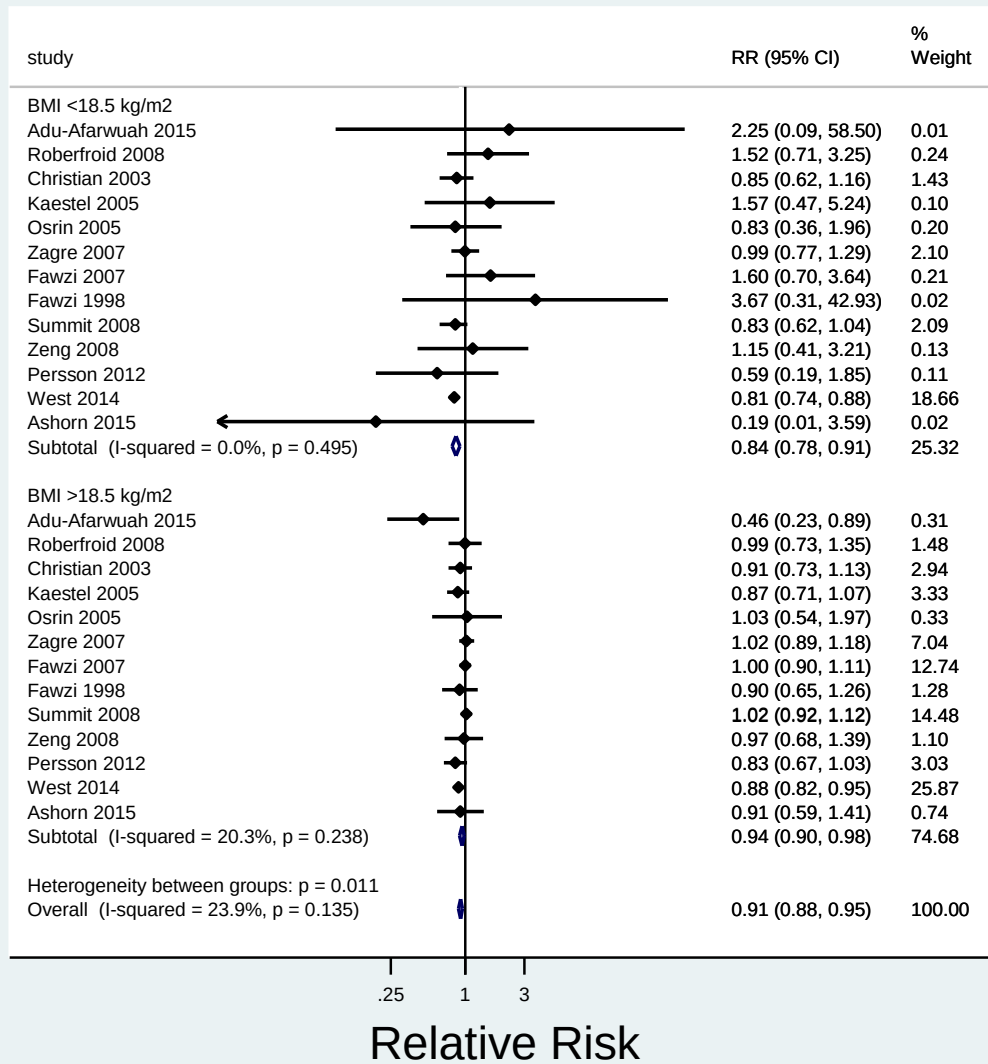
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by maternal age <20



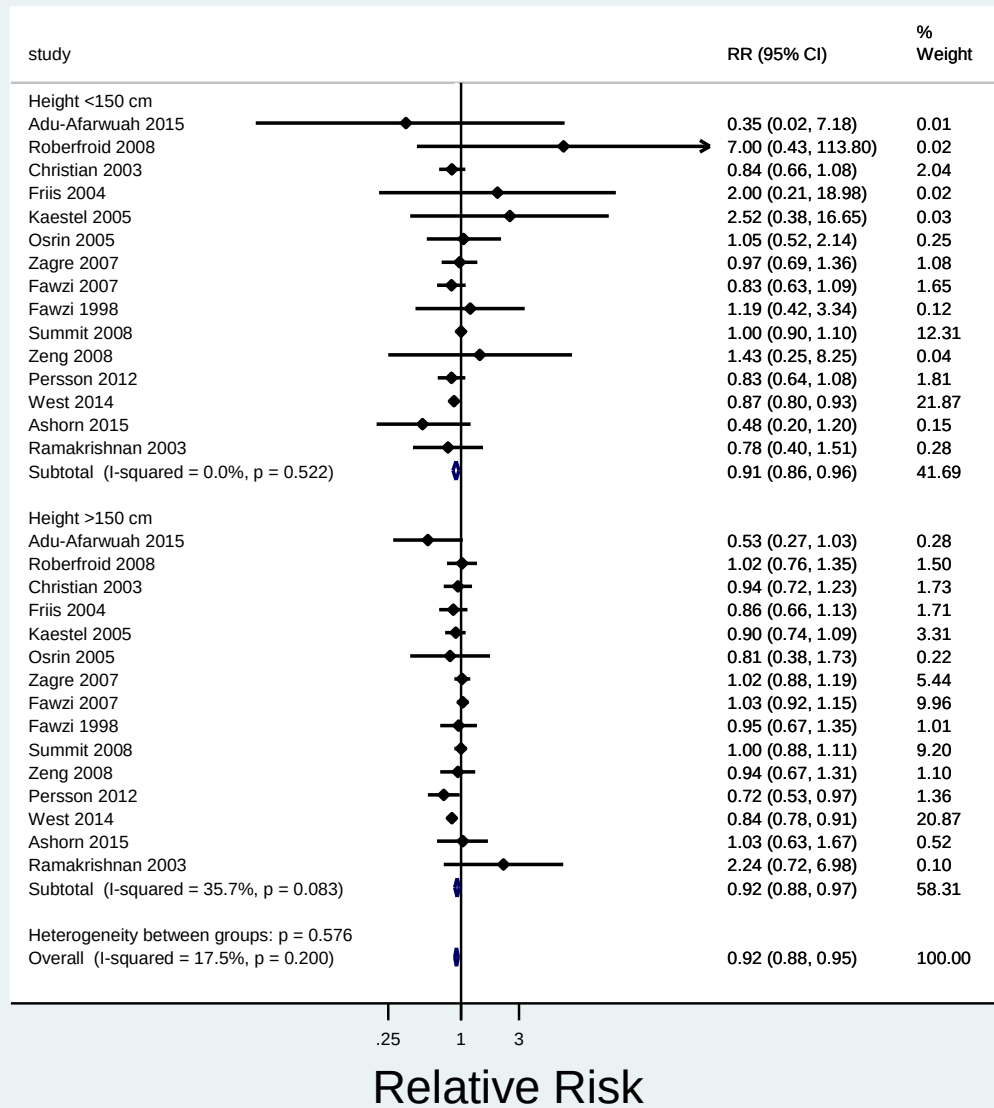
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by maternal BMI



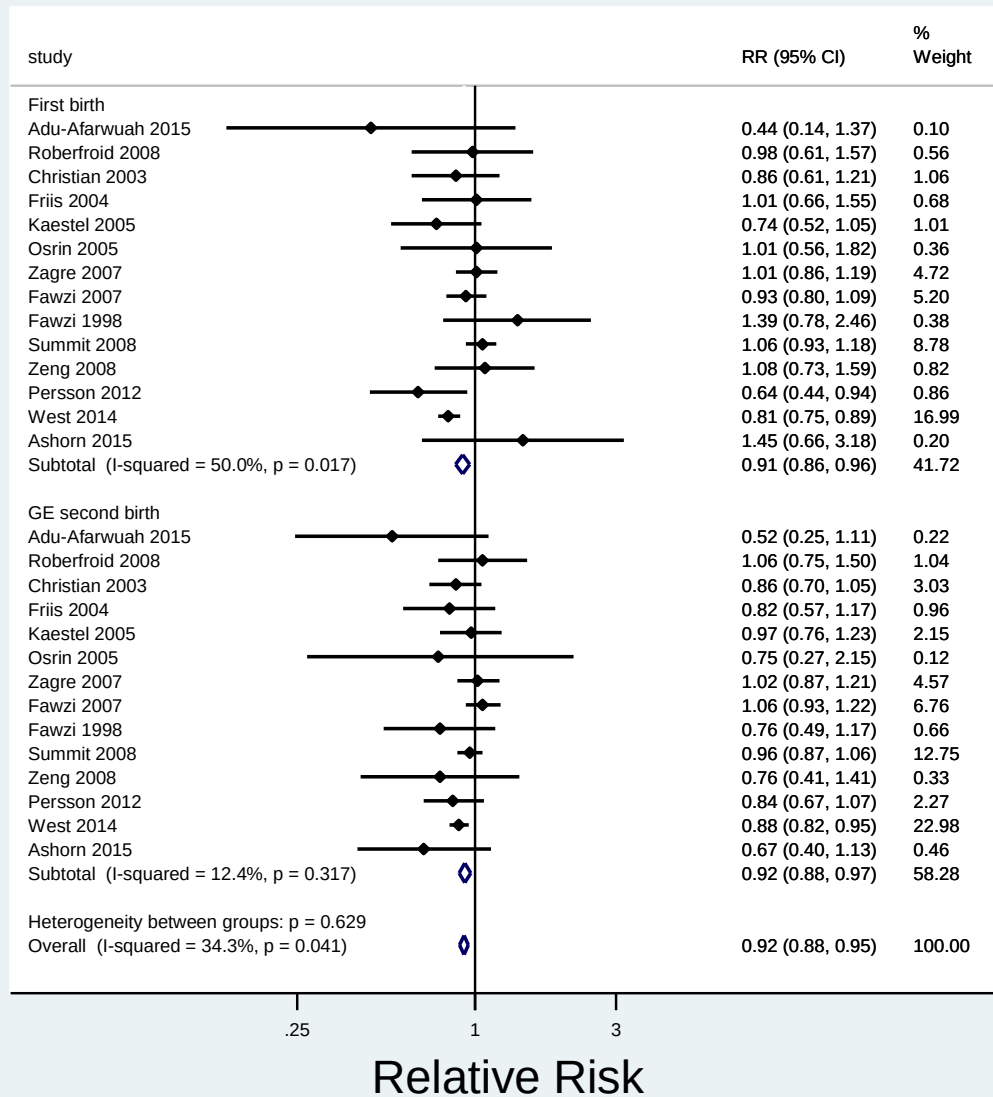
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by maternal height



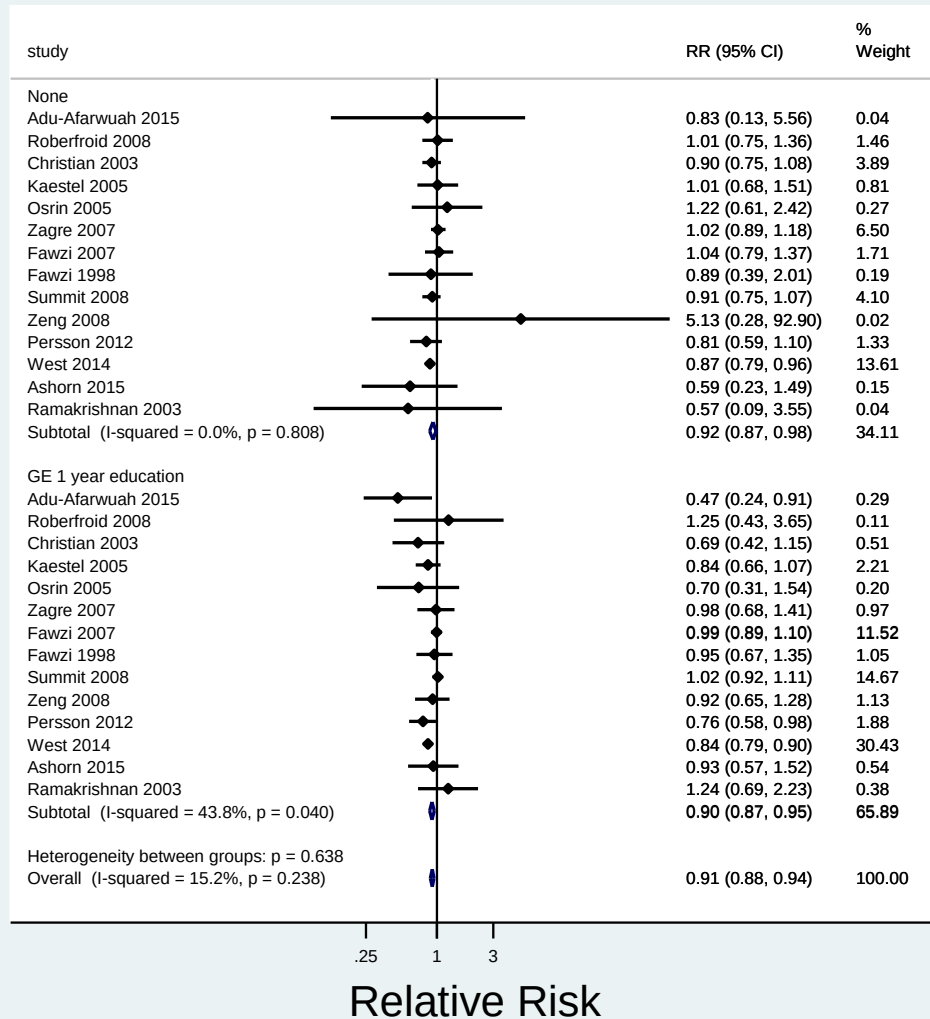
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by parity



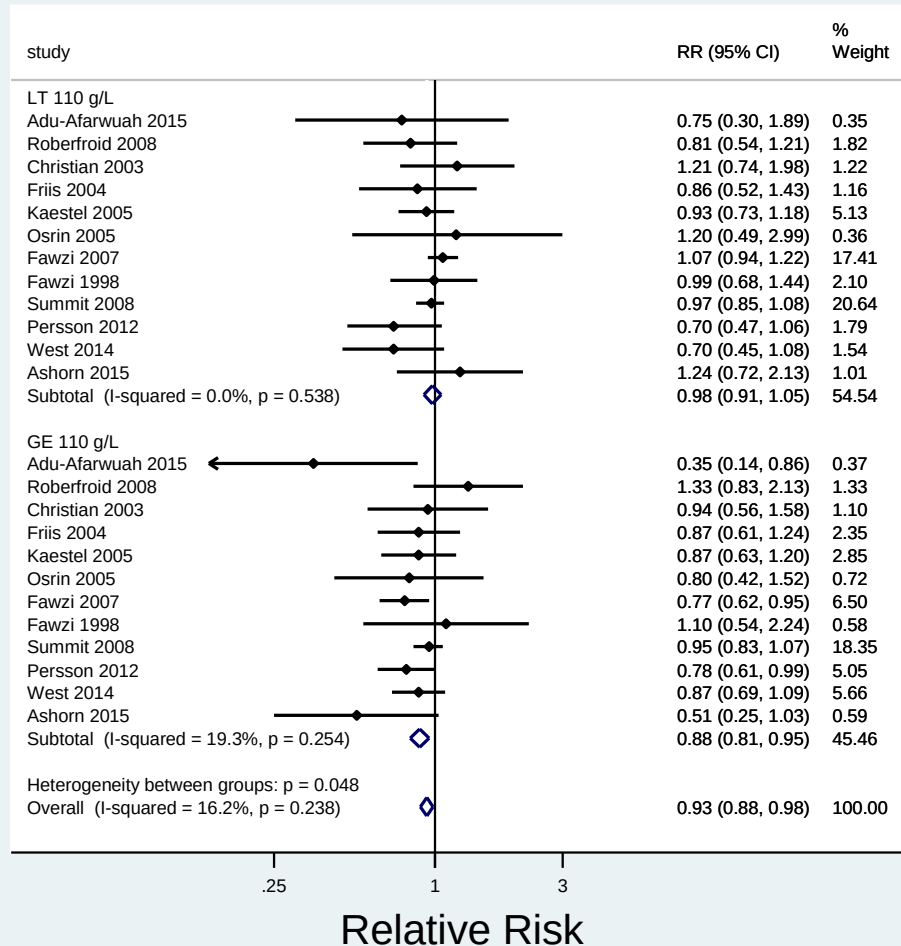
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by maternal education



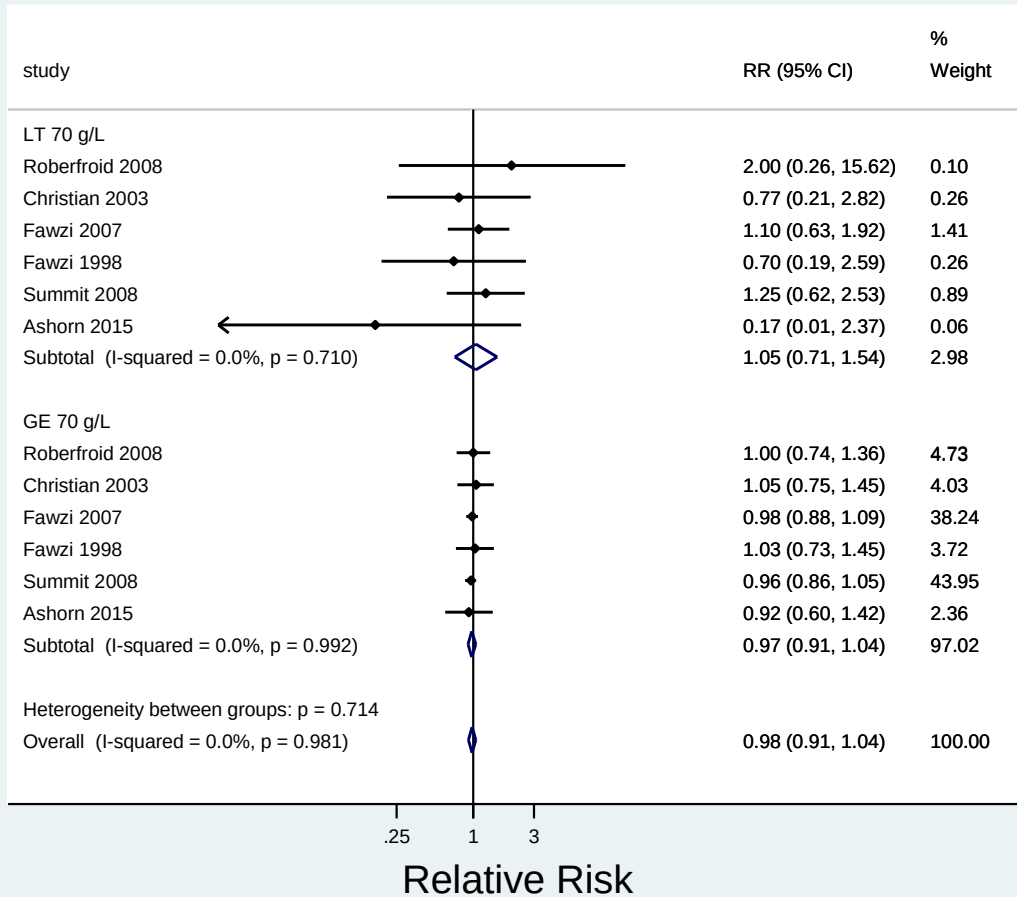
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by hemoglobin <110 g/L



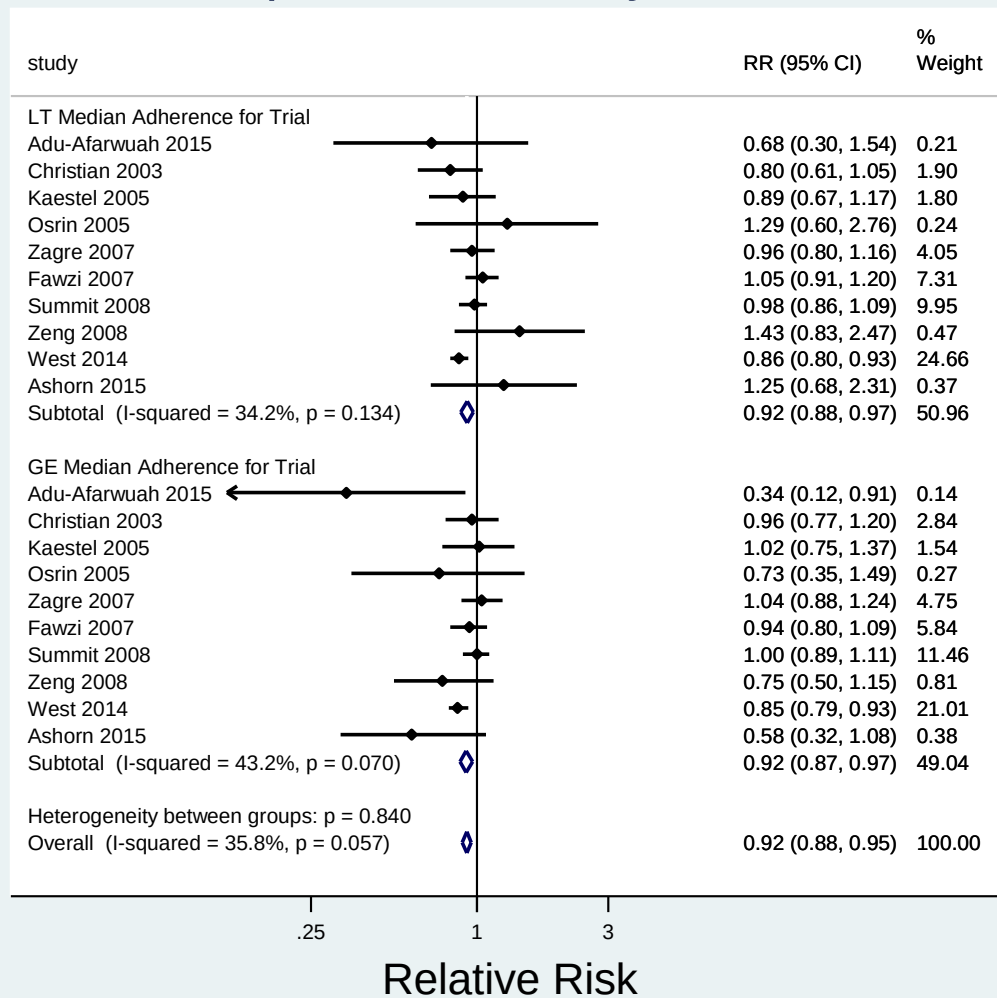
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by hemoglobin <70 g/L



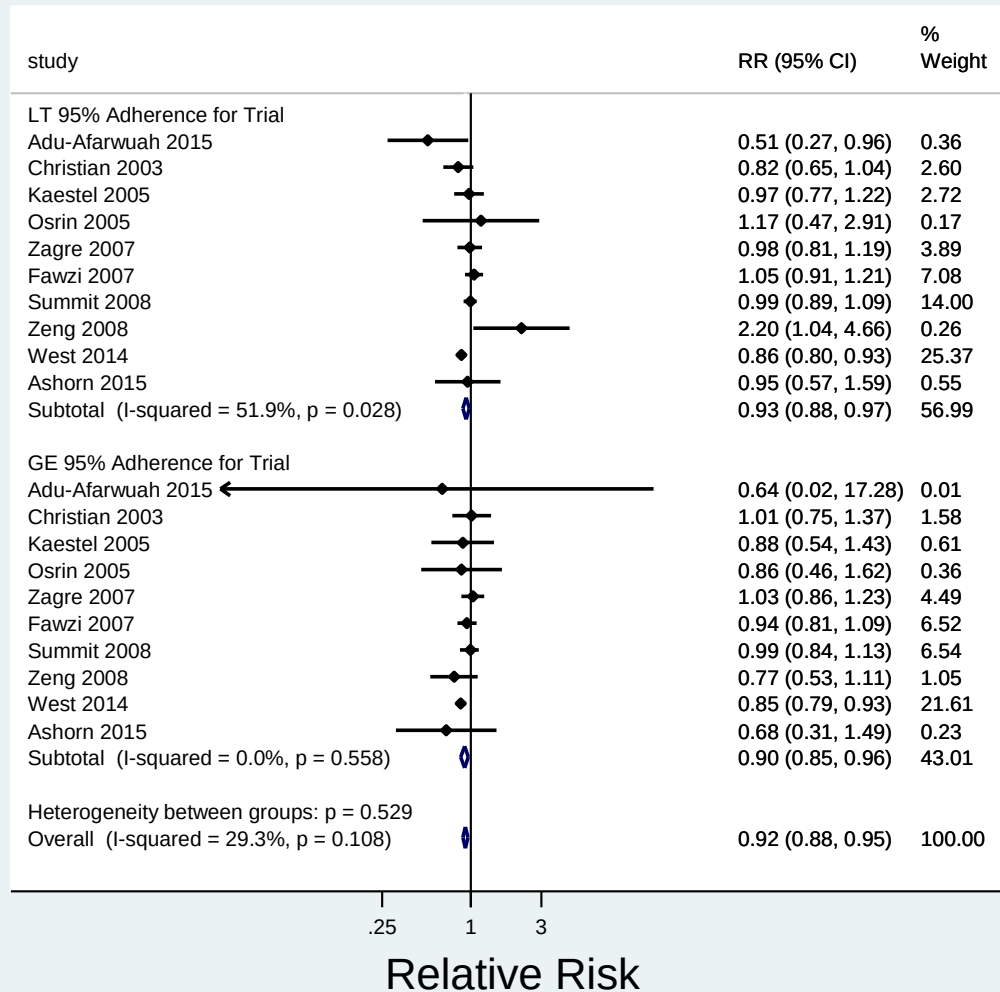
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by median adherence



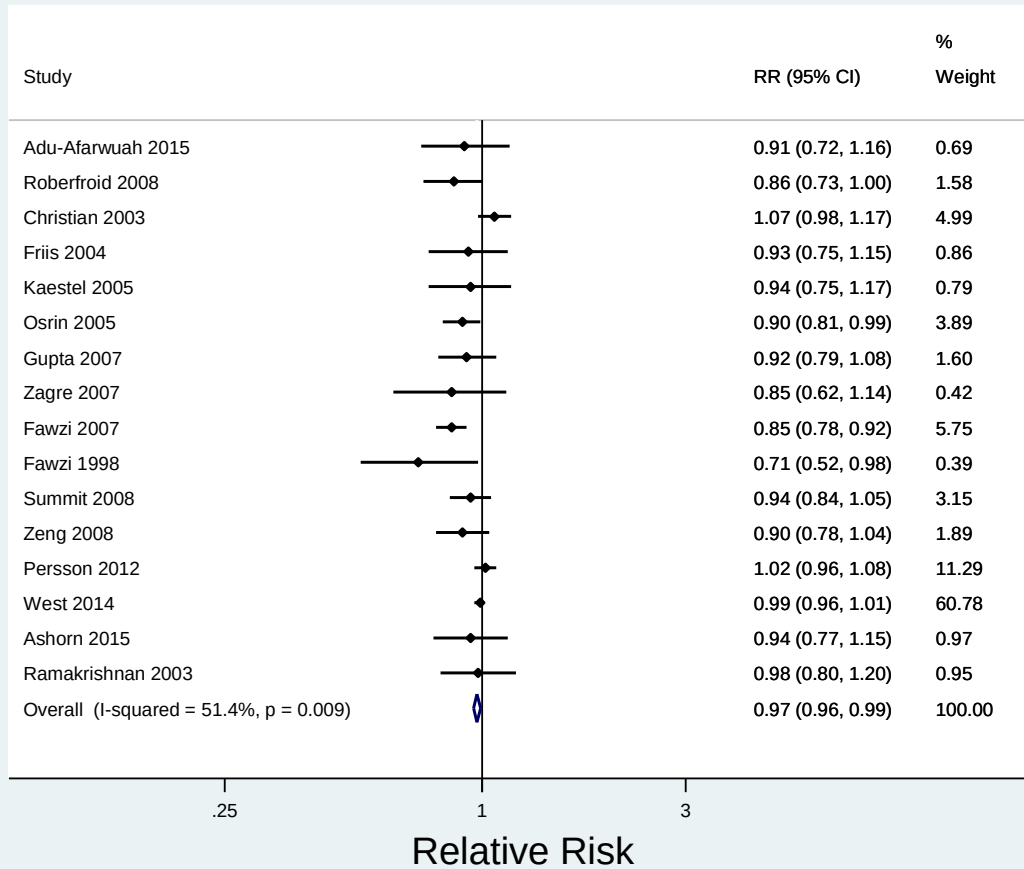
Outcome = Preterm Birth (<37 Weeks)

Relative risk of preterm birth by 95% adherence



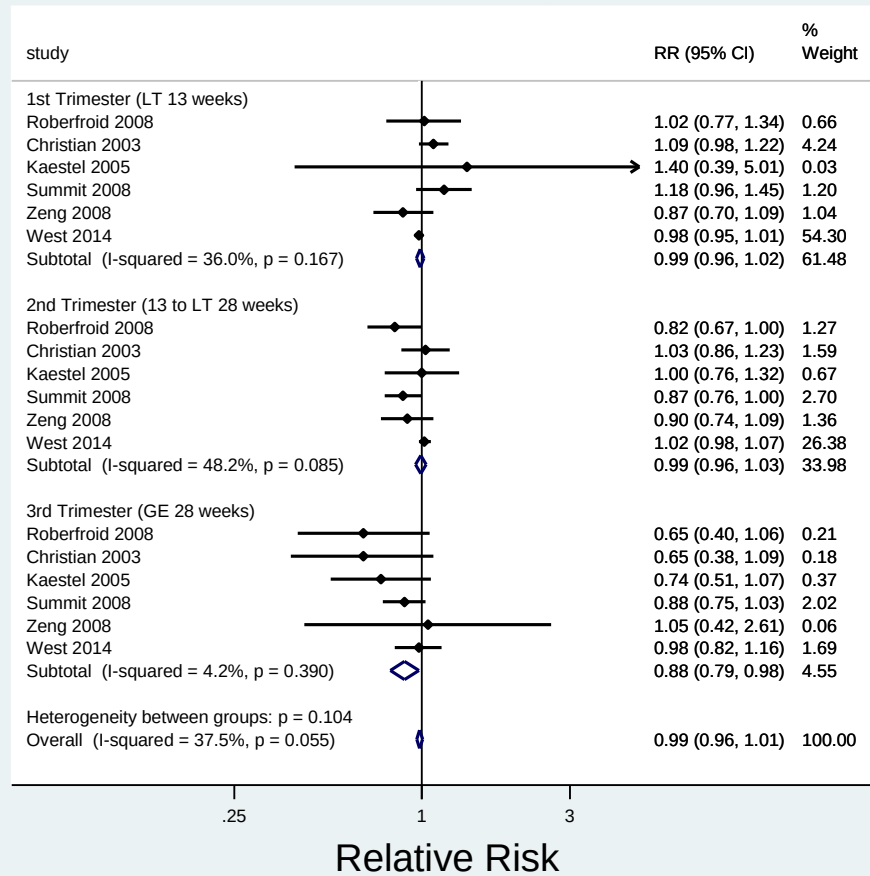
Outcome = Preterm Birth (<37 Weeks)

Relative risk of SGA Oken 10th - OVERALL



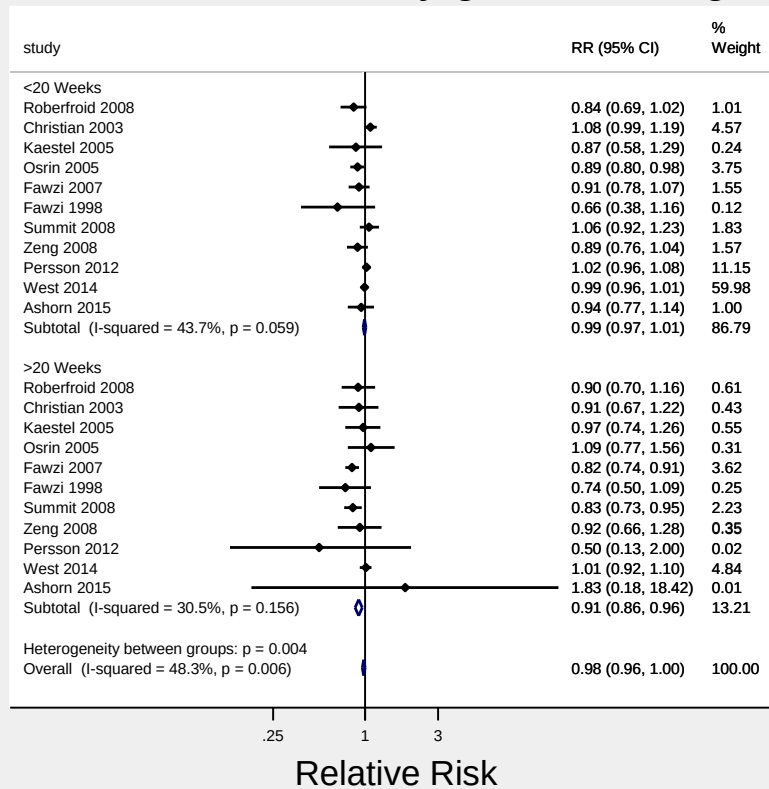
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by trimester at enrollment



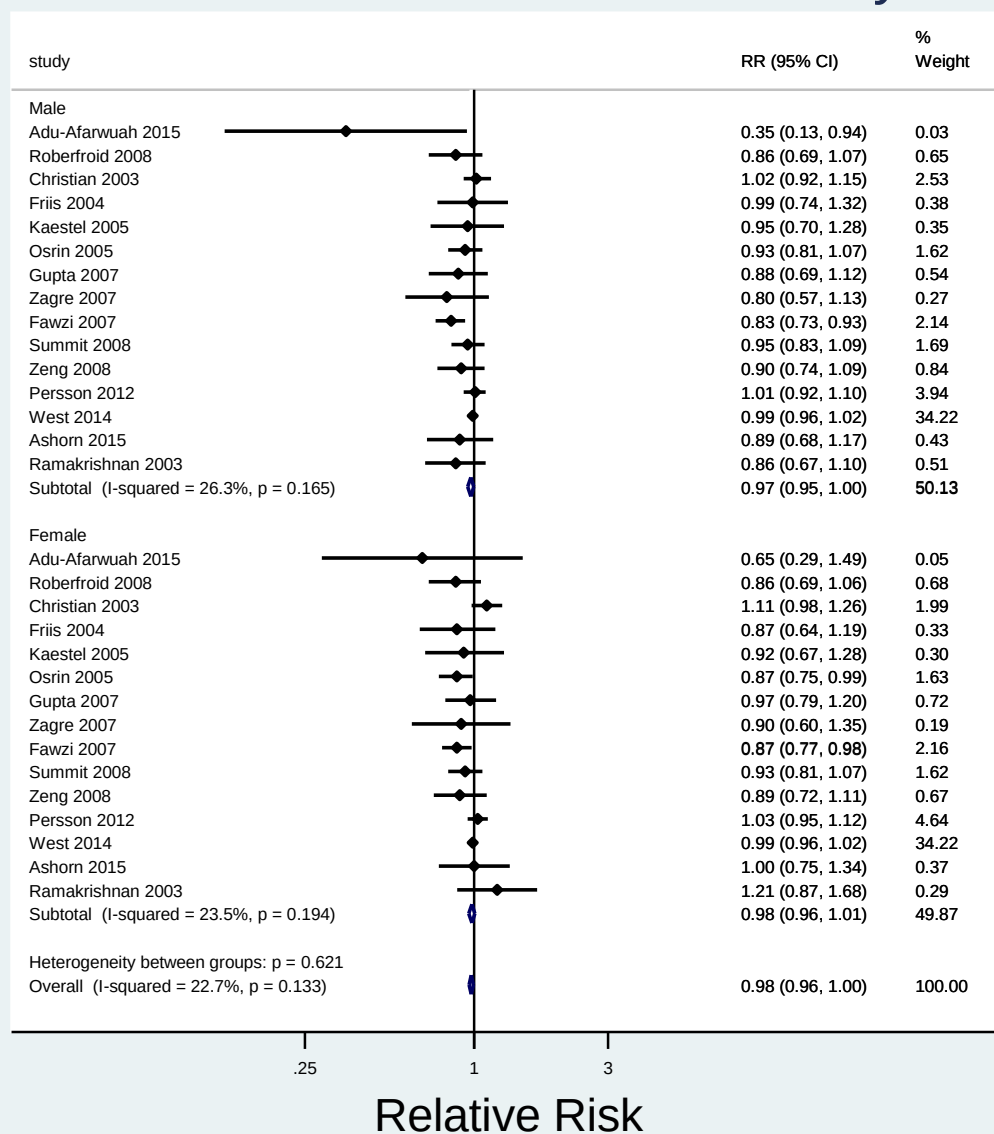
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by gestational age at enrollment



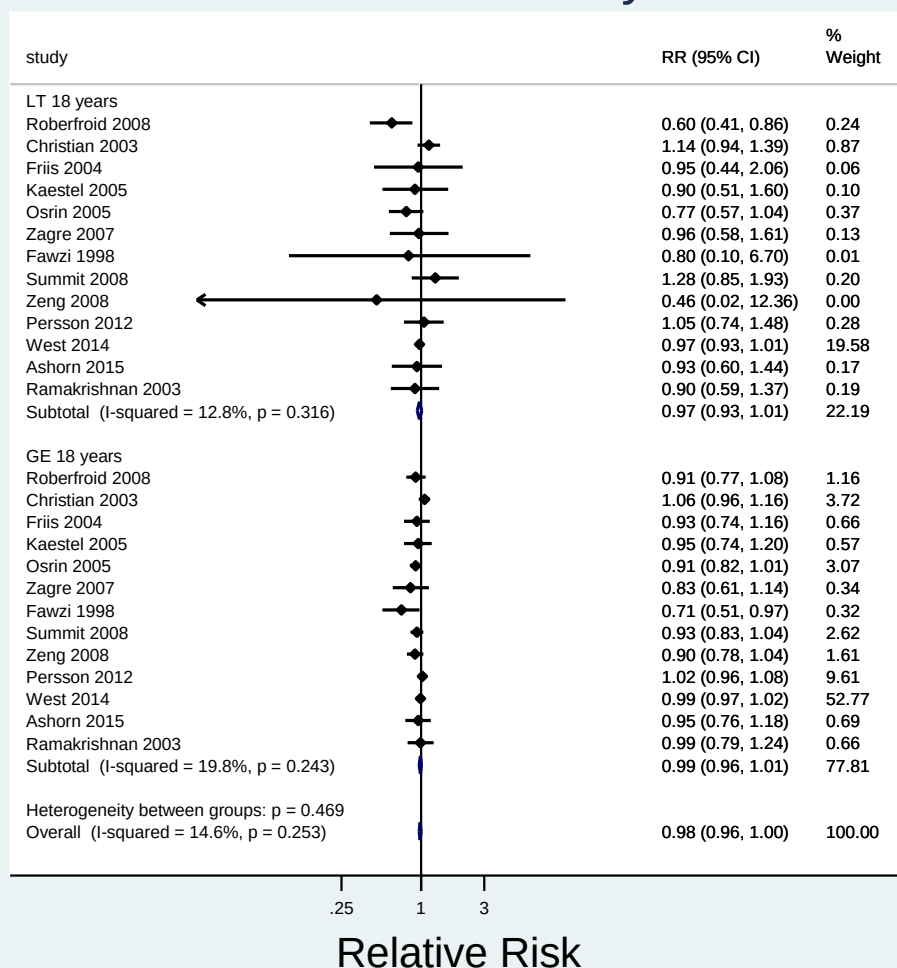
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by sex



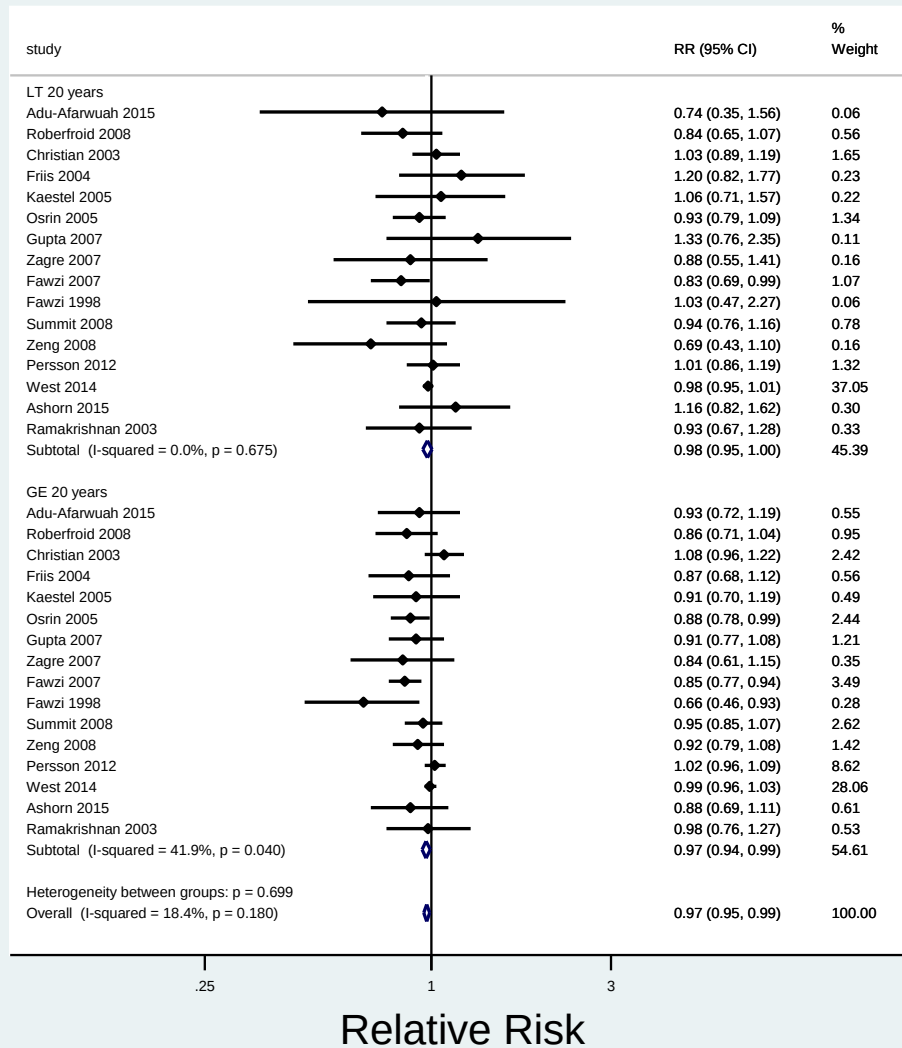
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by maternal age <18



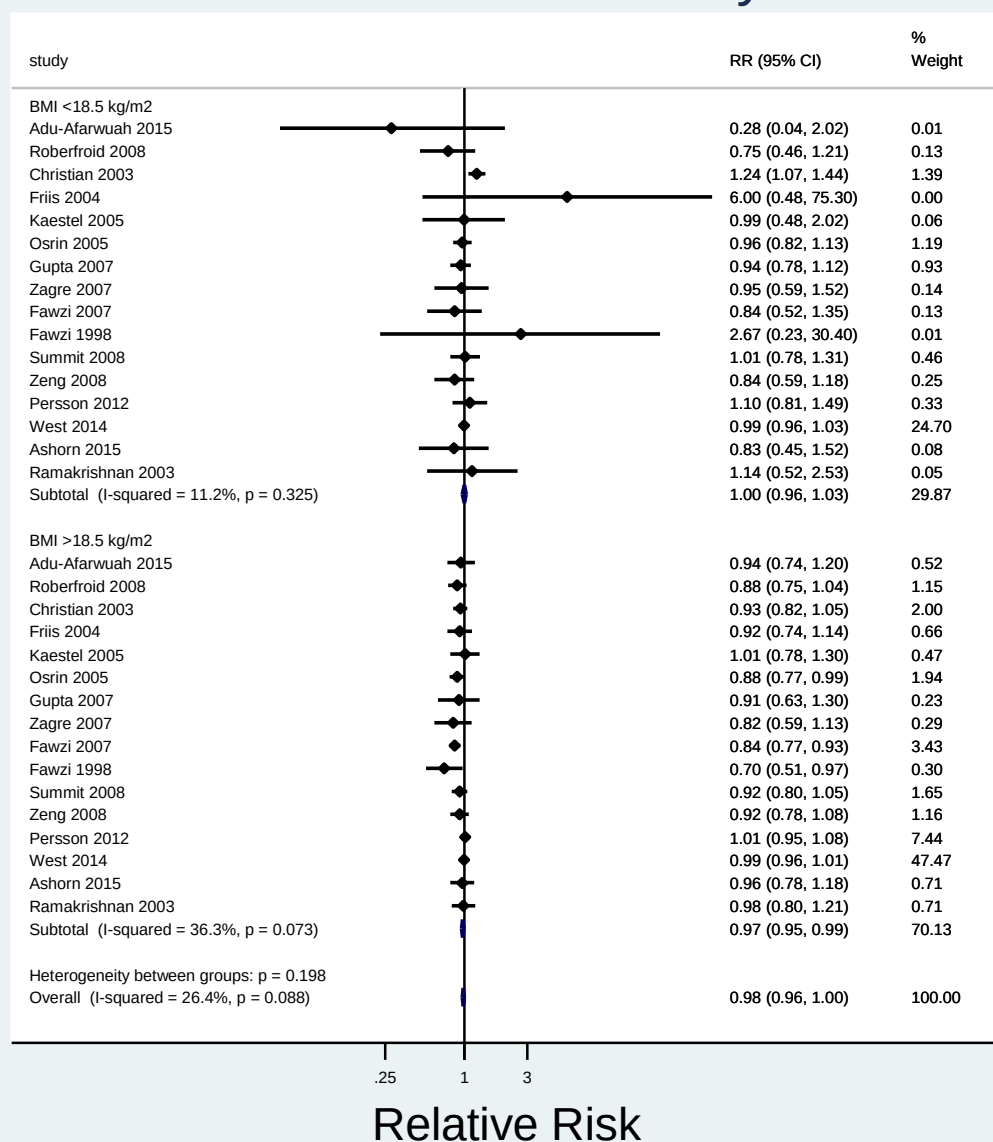
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by maternal age <20



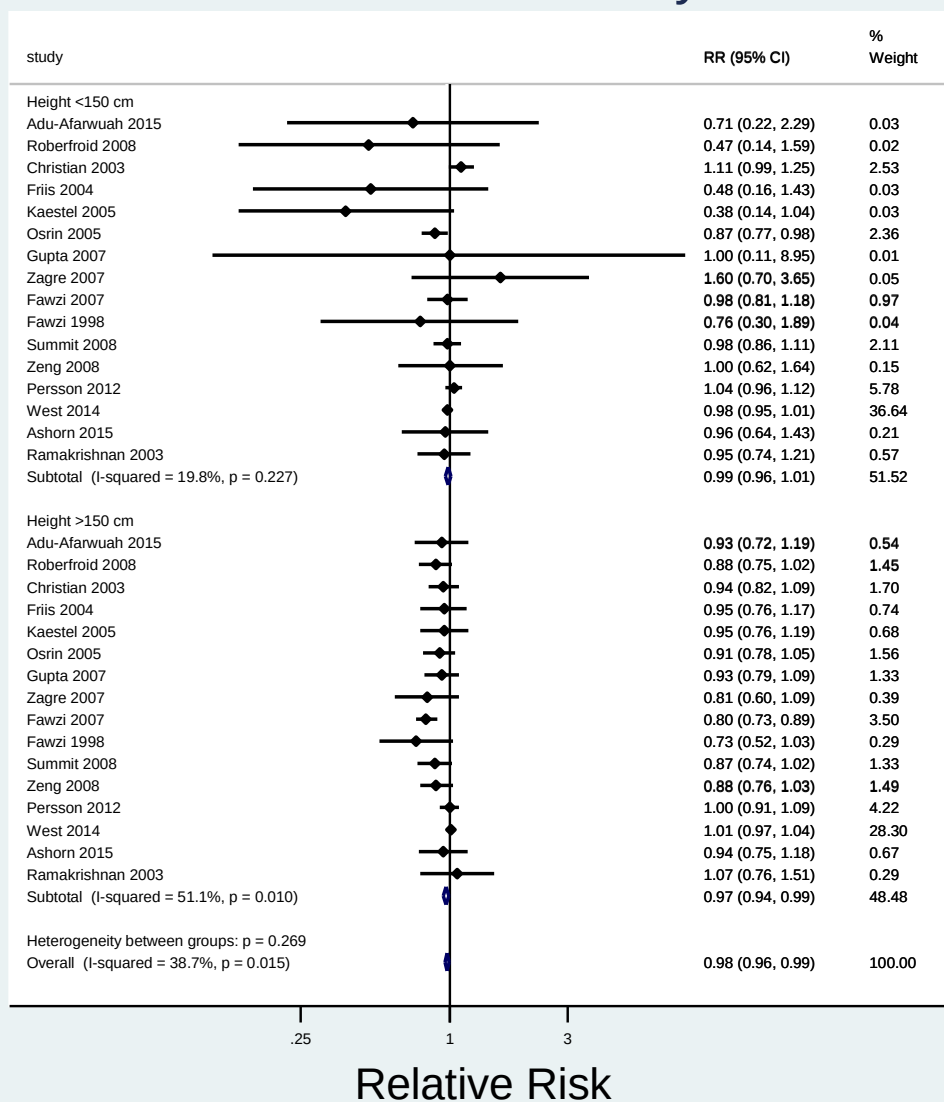
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by maternal BMI



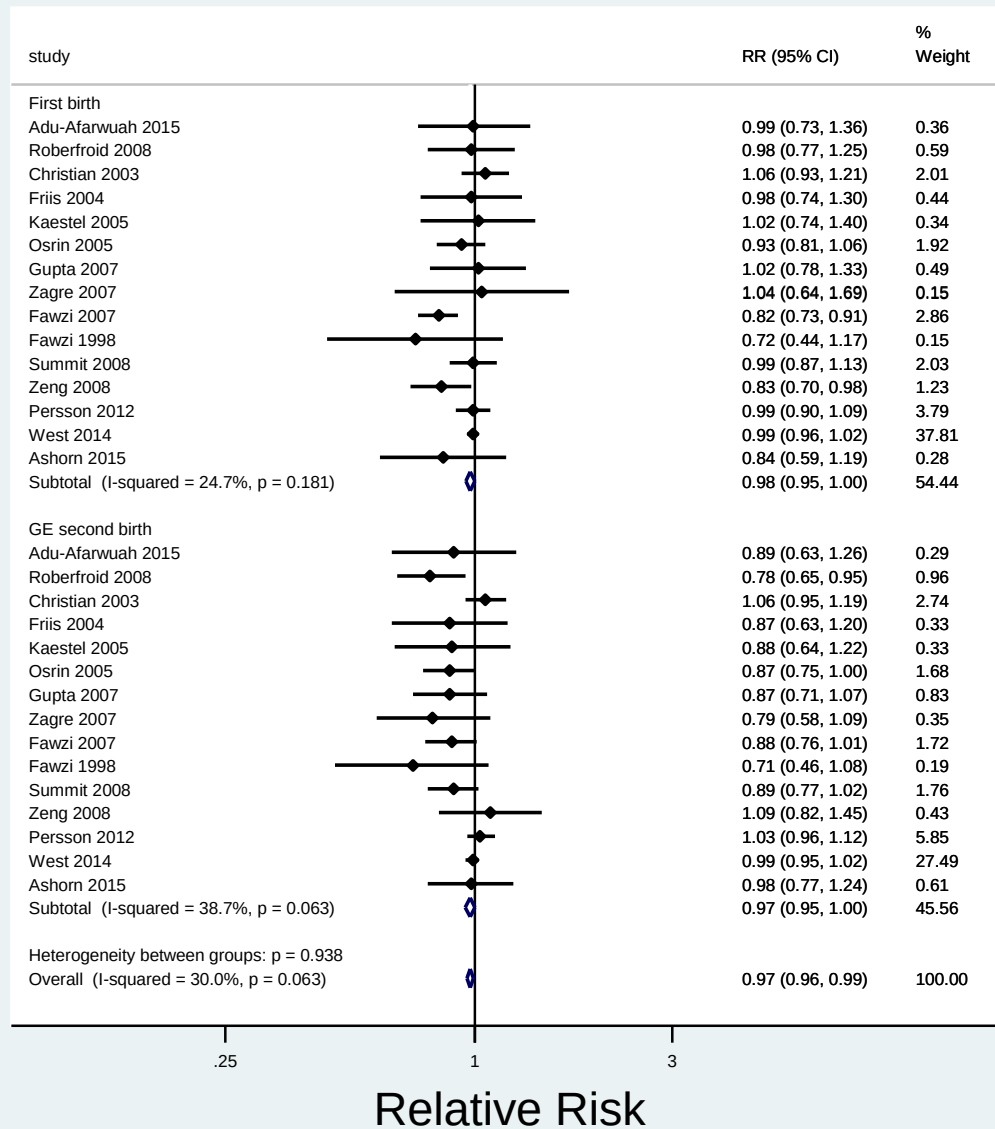
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by maternal height



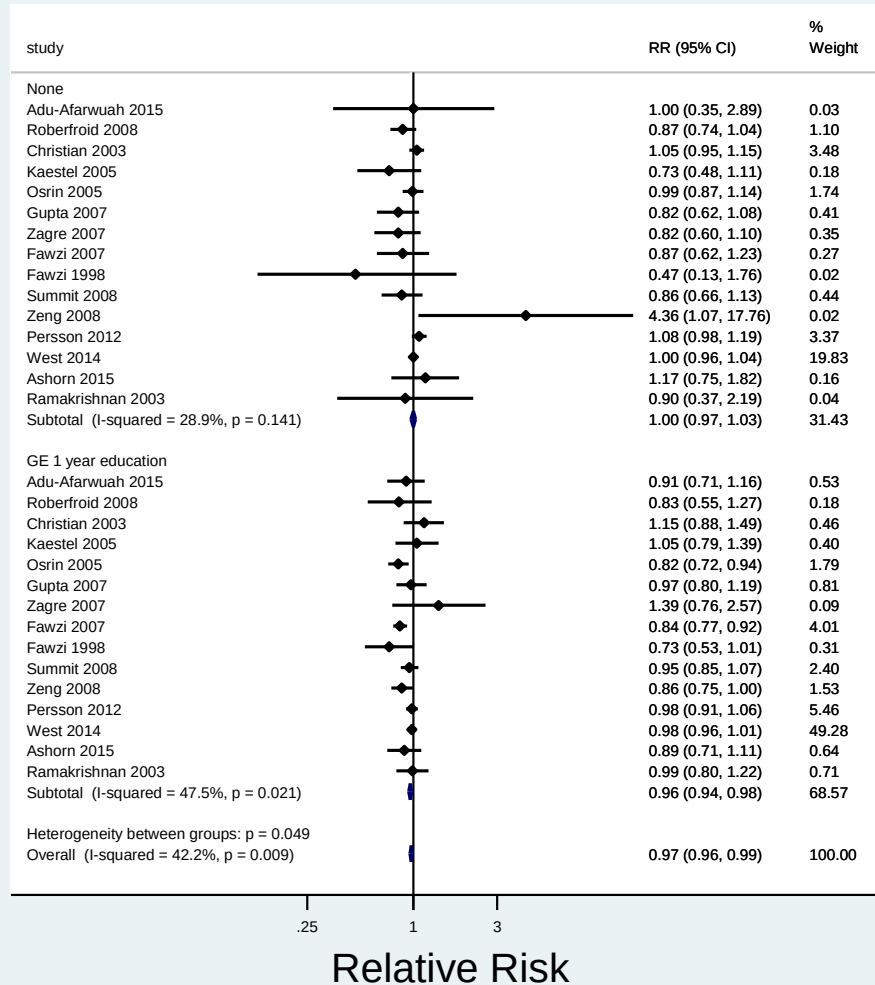
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by parity



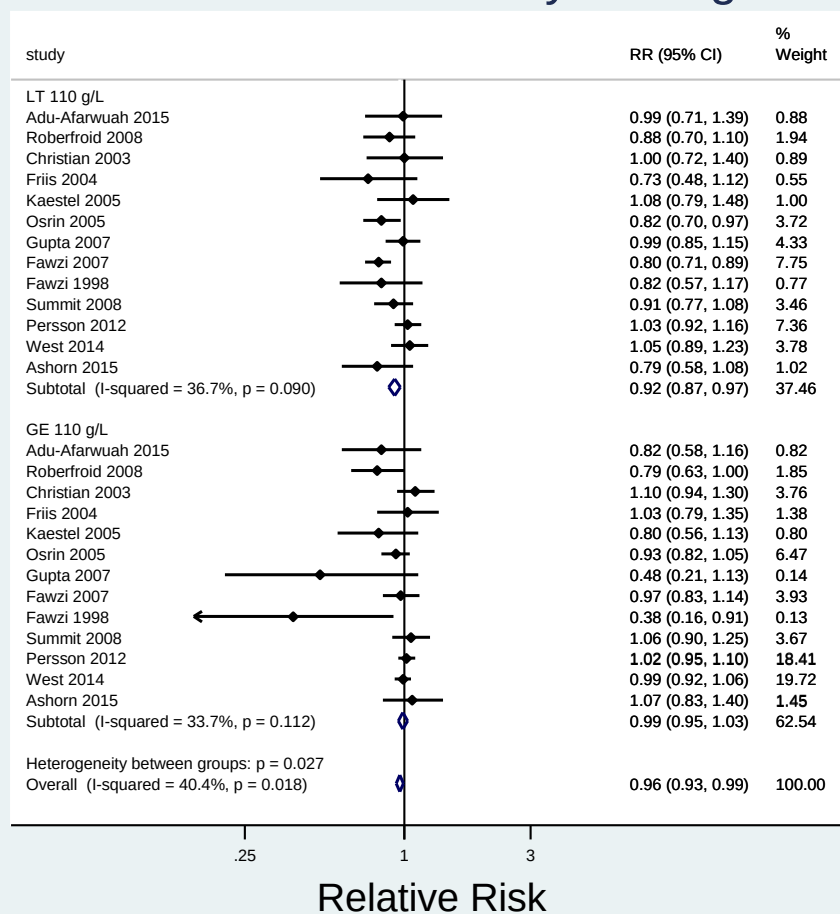
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by maternal education



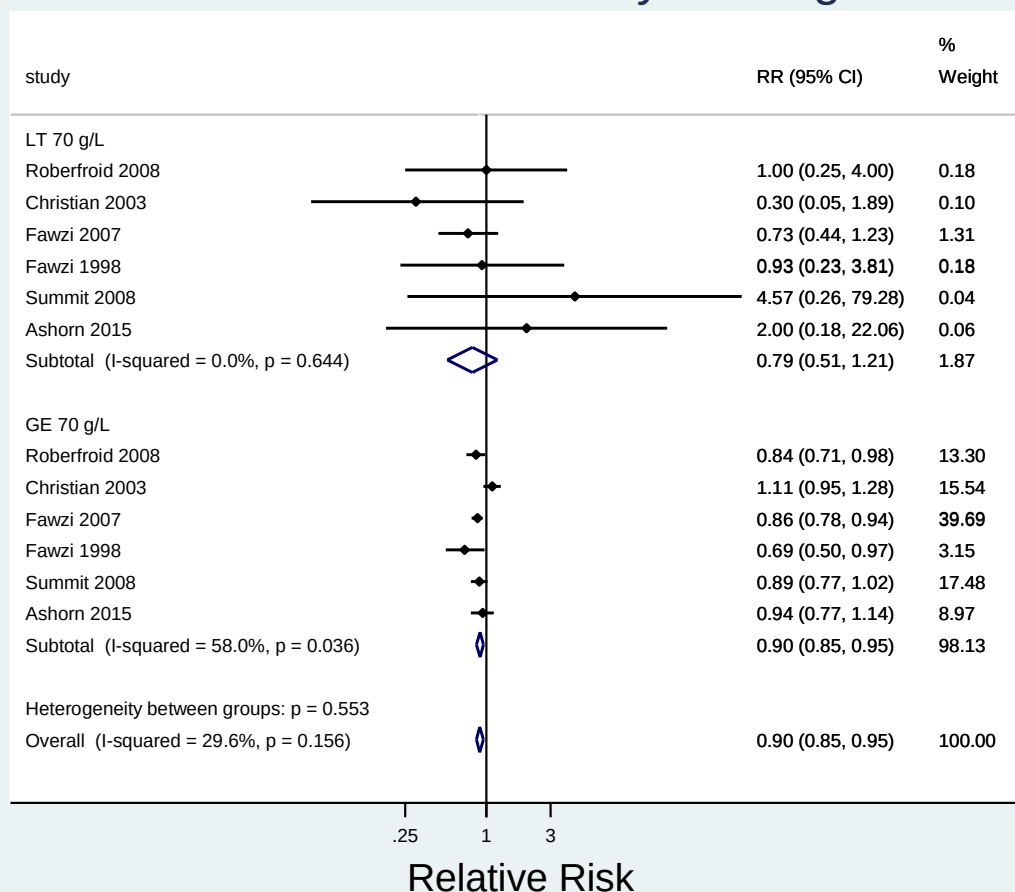
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by hemoglobin <110 g/L



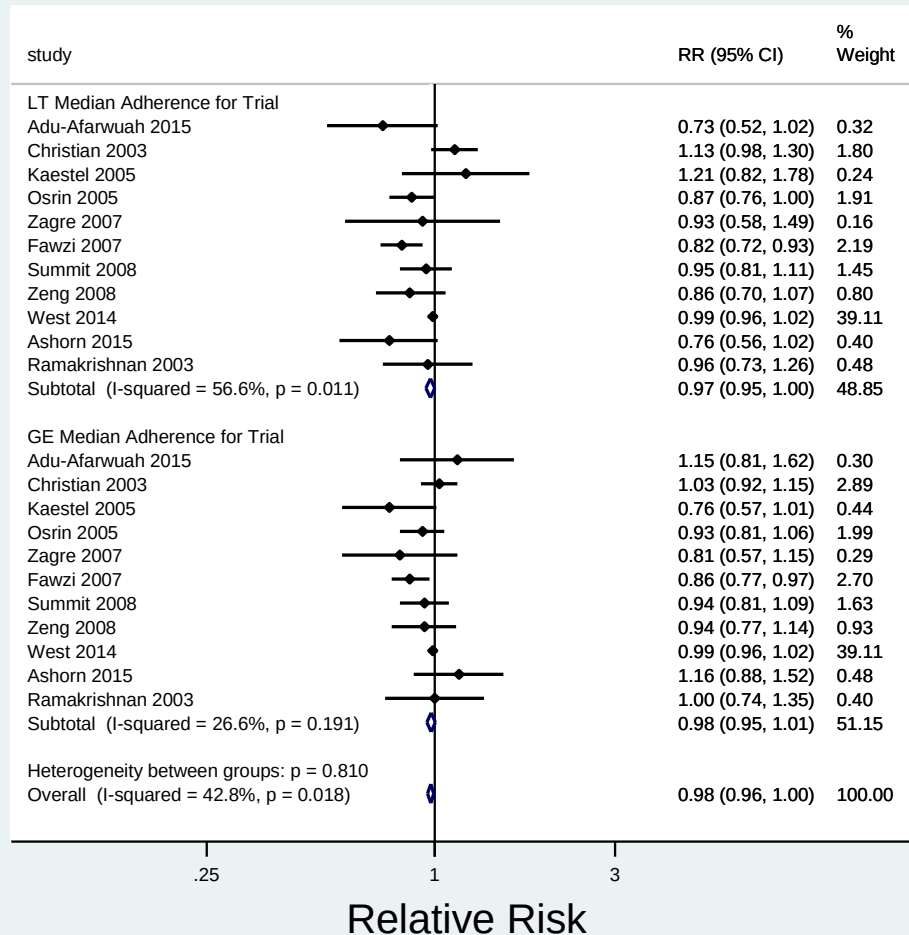
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by hemoglobin <70 g/L



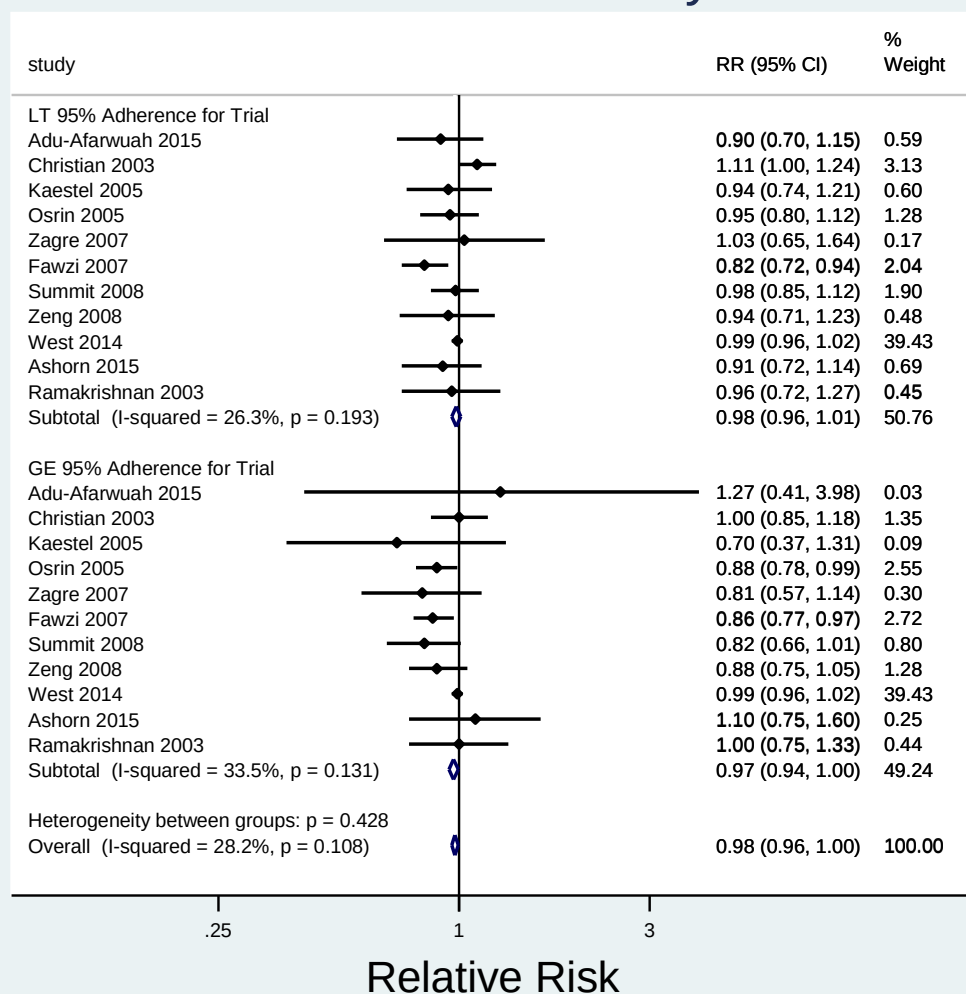
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by median adherence



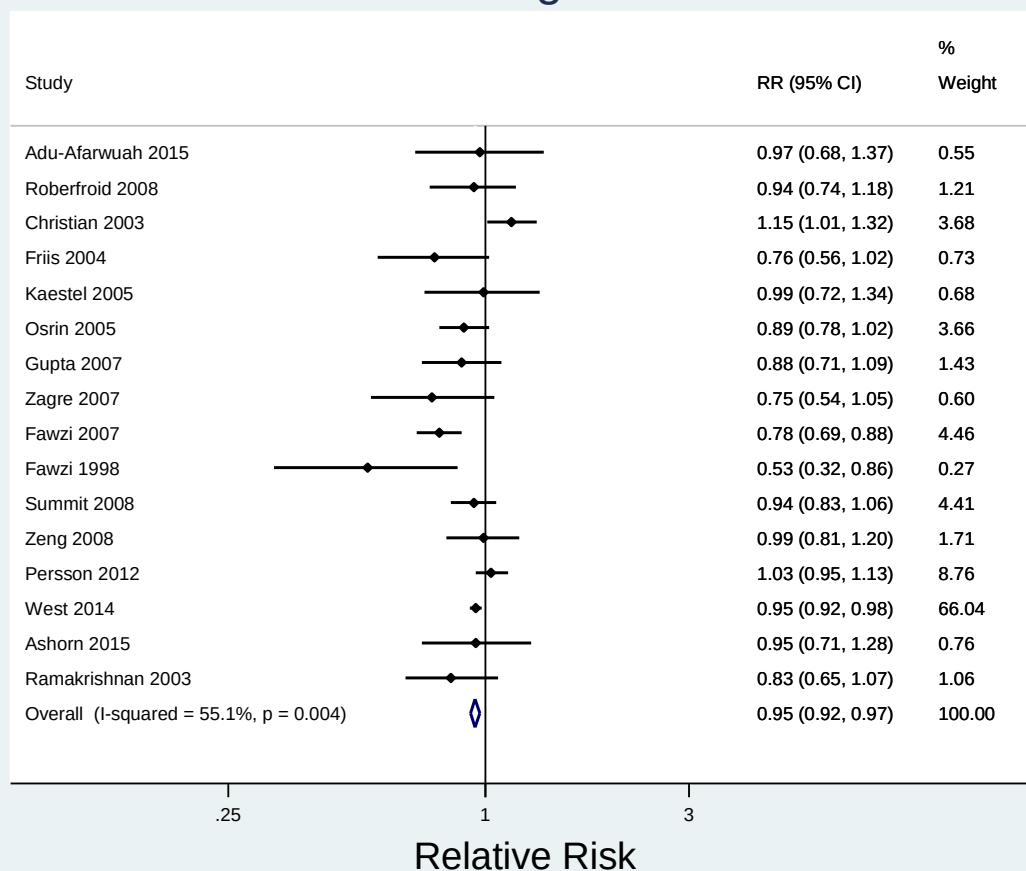
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Oken 10th by 95% adherence



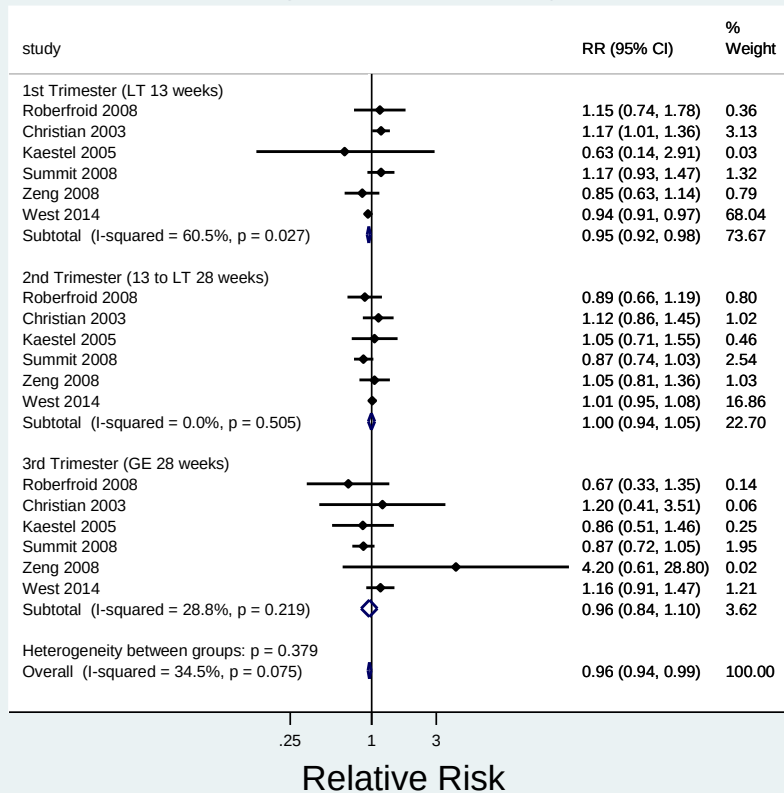
Outcome = SGA <10th Percentile (Oken)

Relative risk of SGA Intergrowth 10th - OVERALL



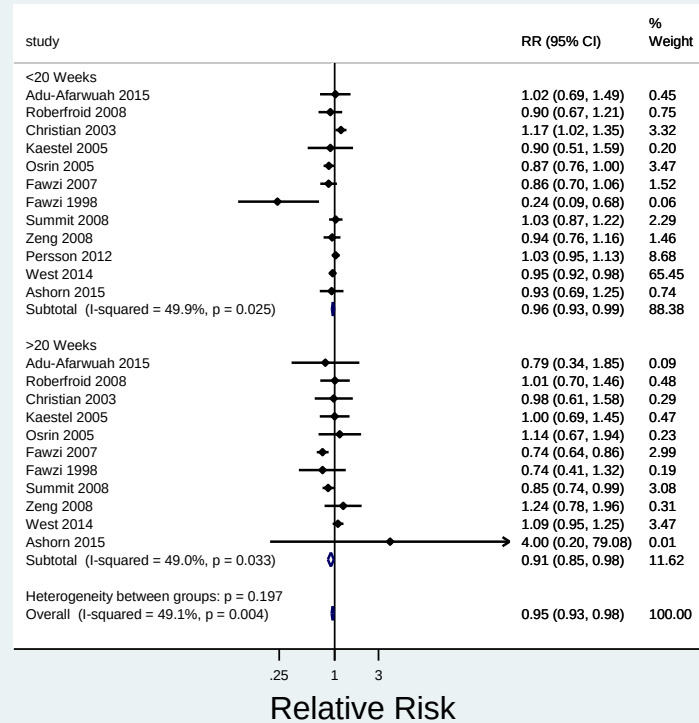
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by trimester at enrollment



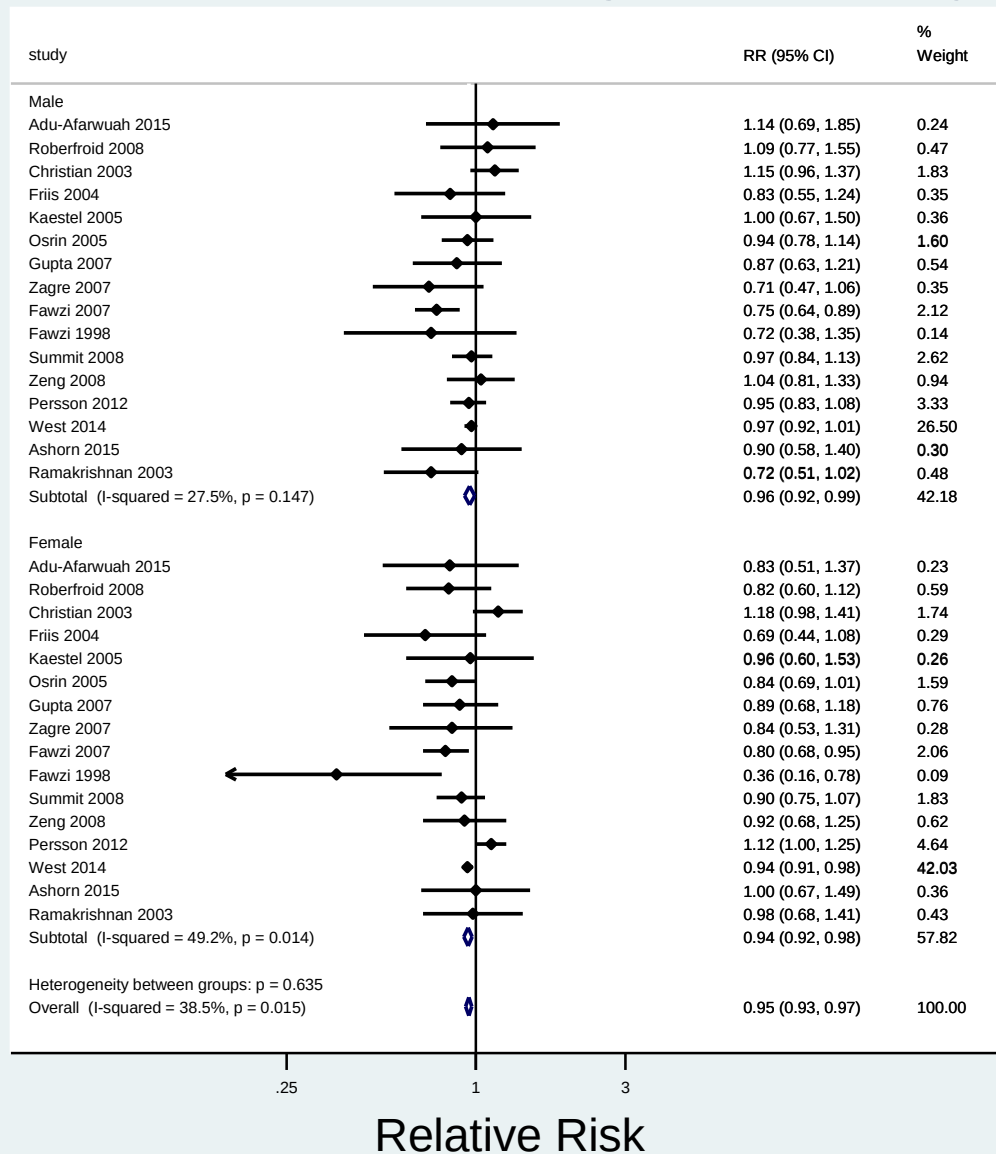
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by gestational age at enrollment



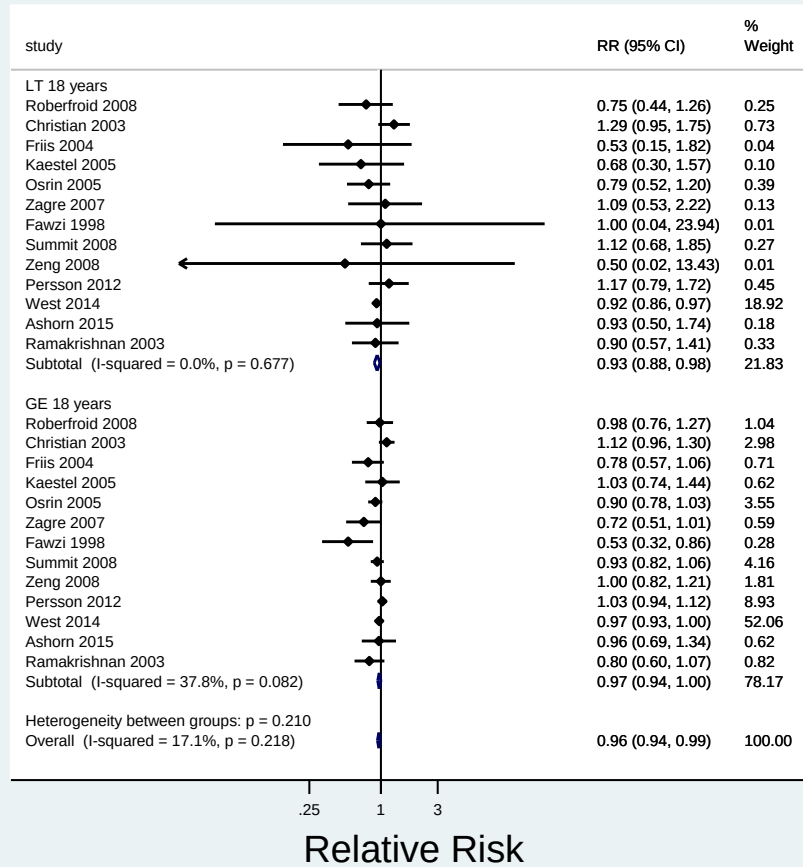
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by sex



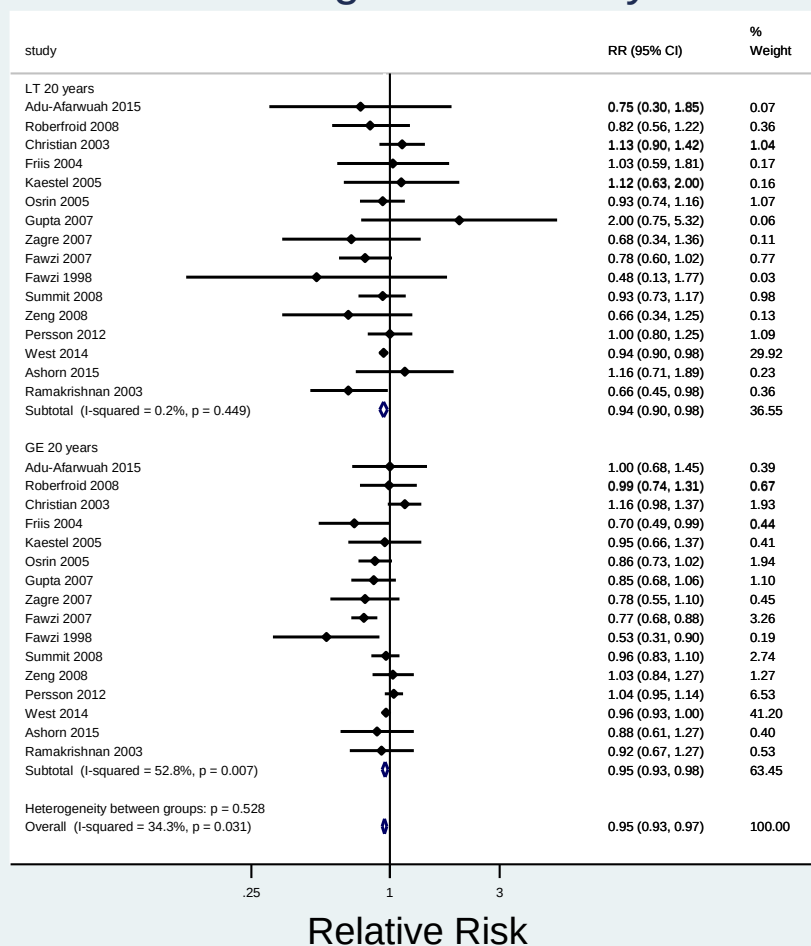
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by maternal age <18



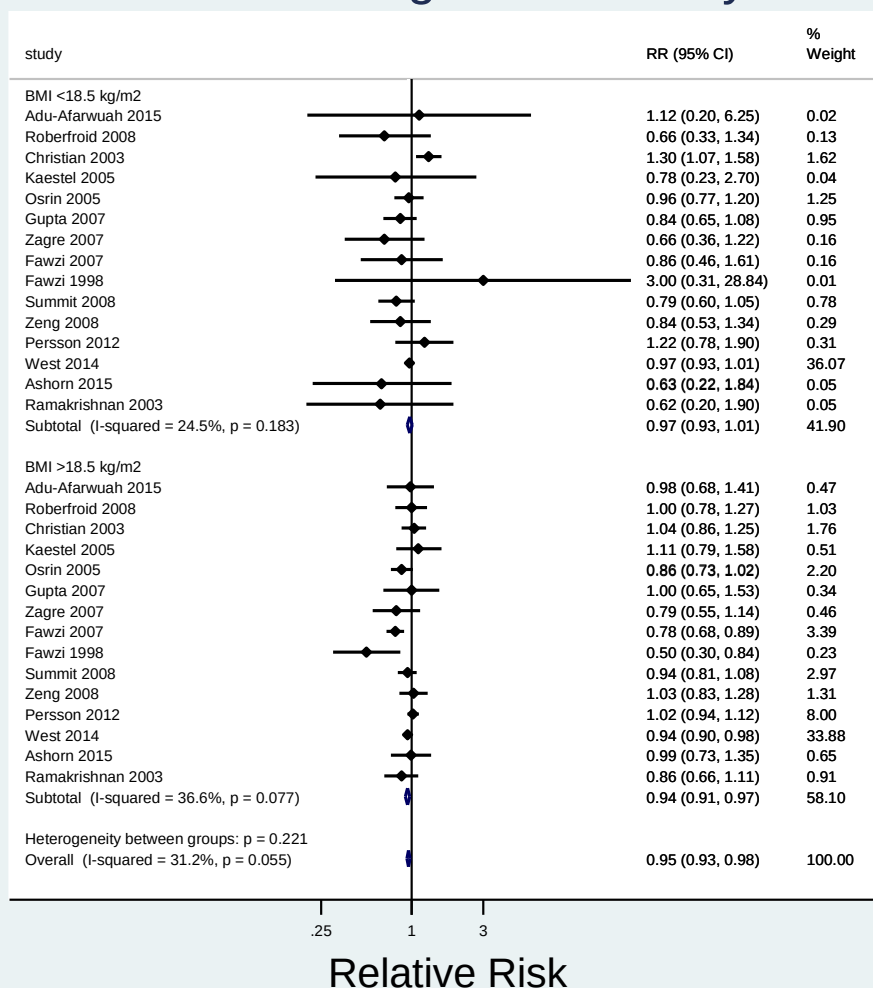
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by maternal age <20



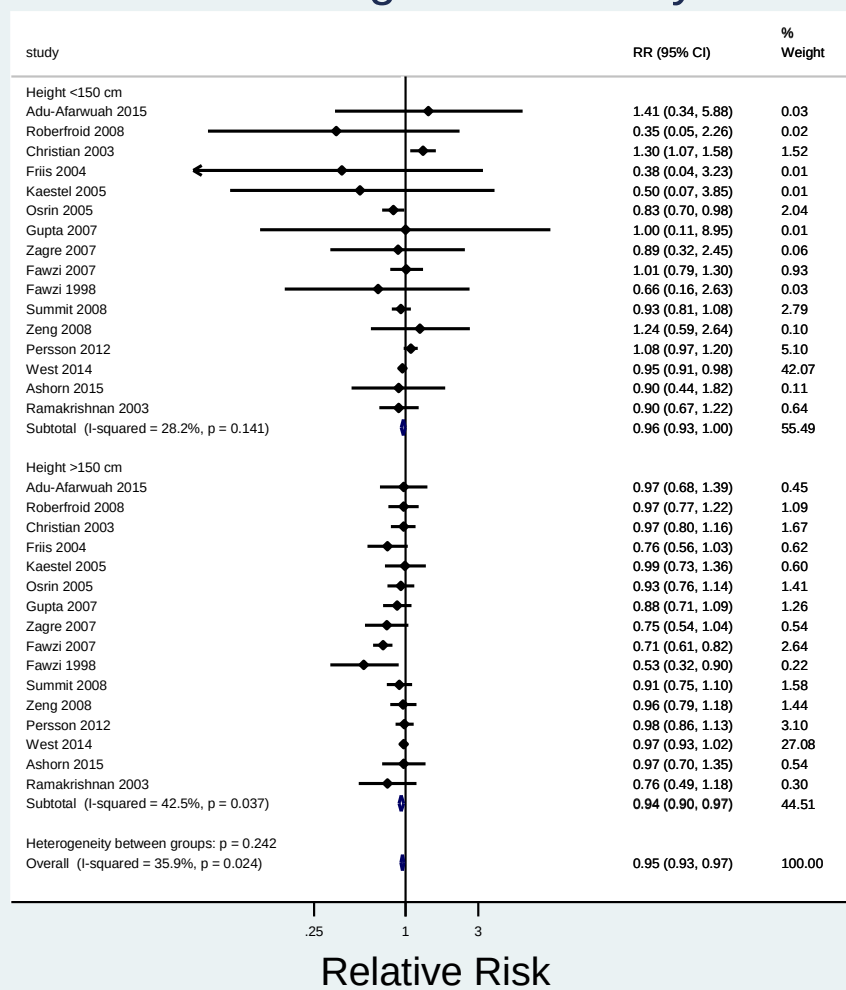
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by maternal BMI



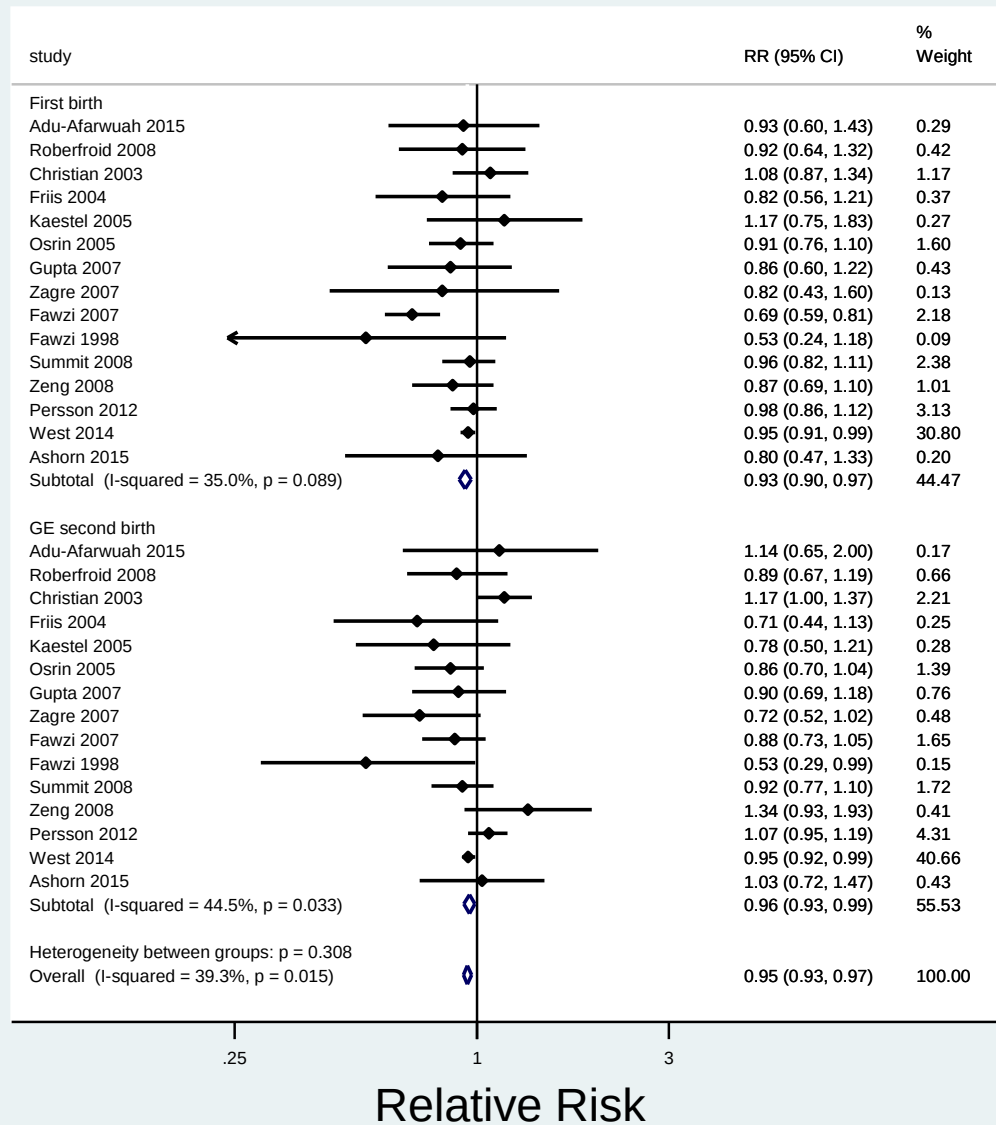
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by maternal height



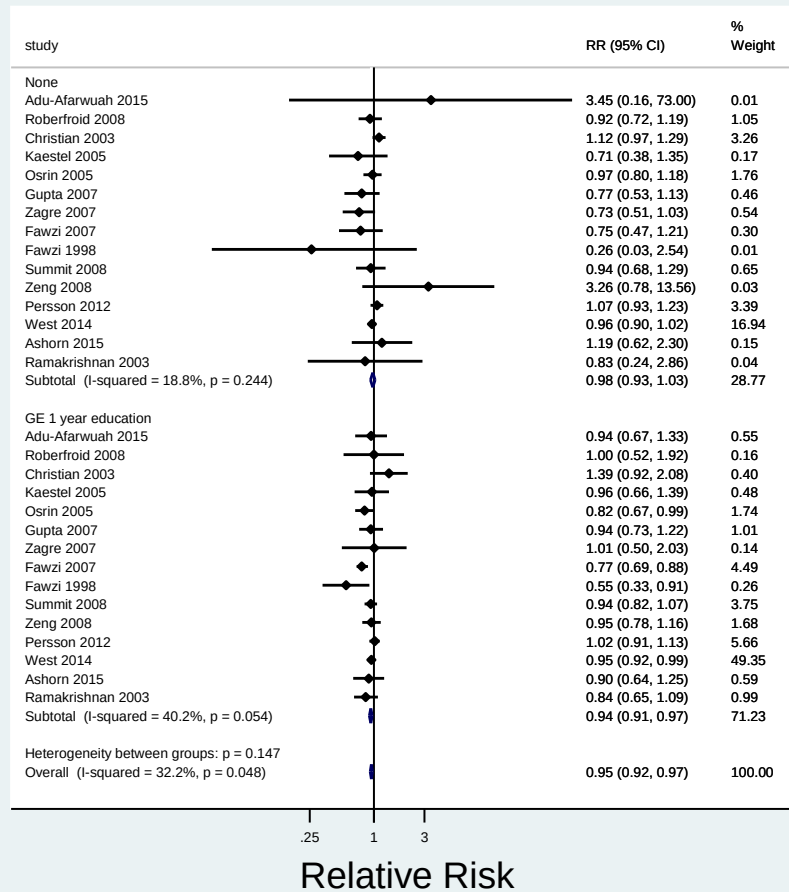
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by parity



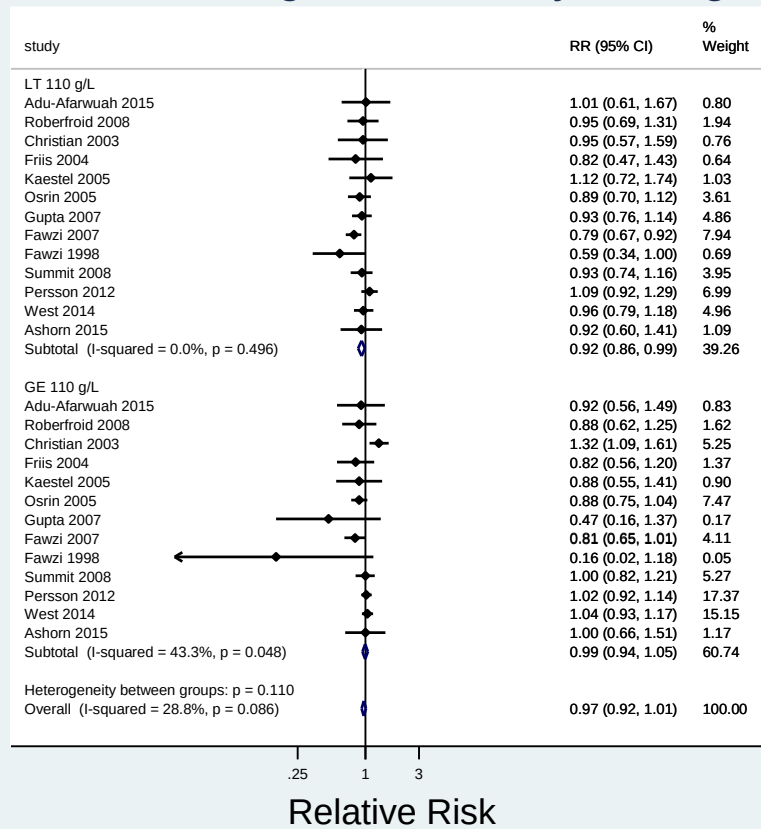
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by maternal education



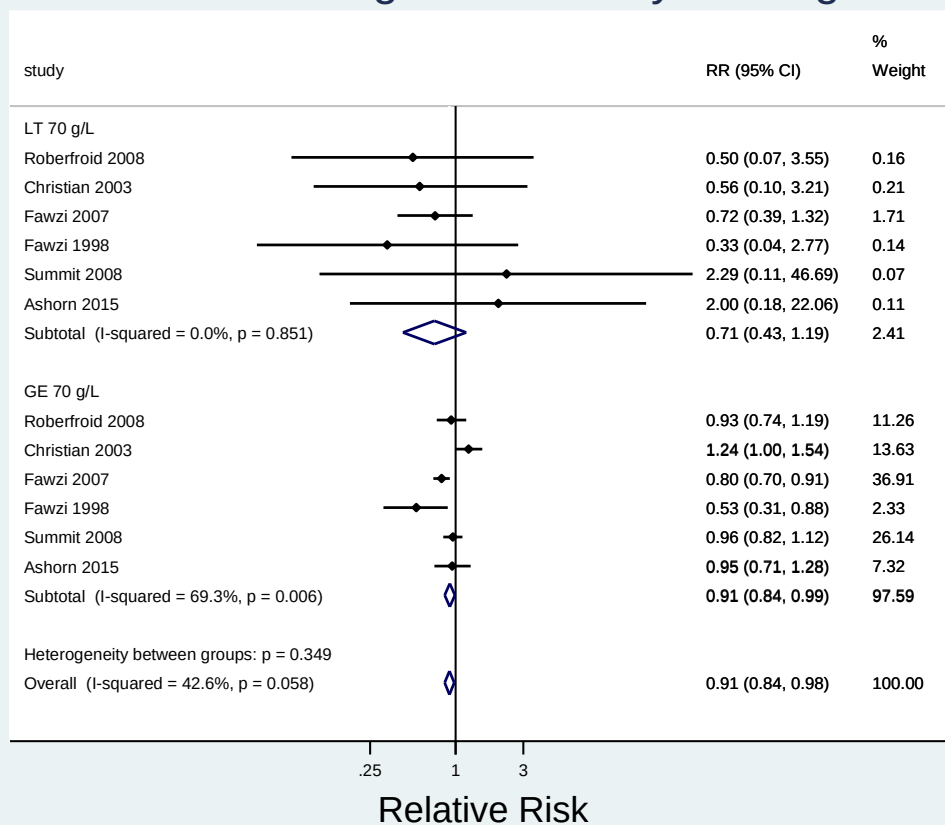
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by hemoglobin <110 g/L



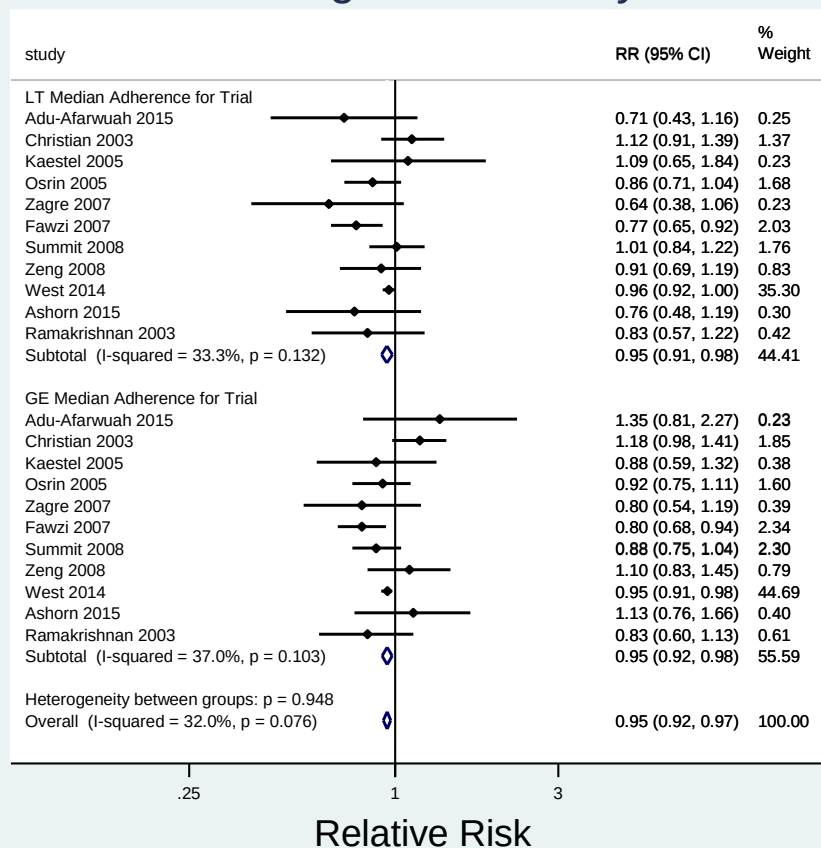
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by hemoglobin <70 g/L



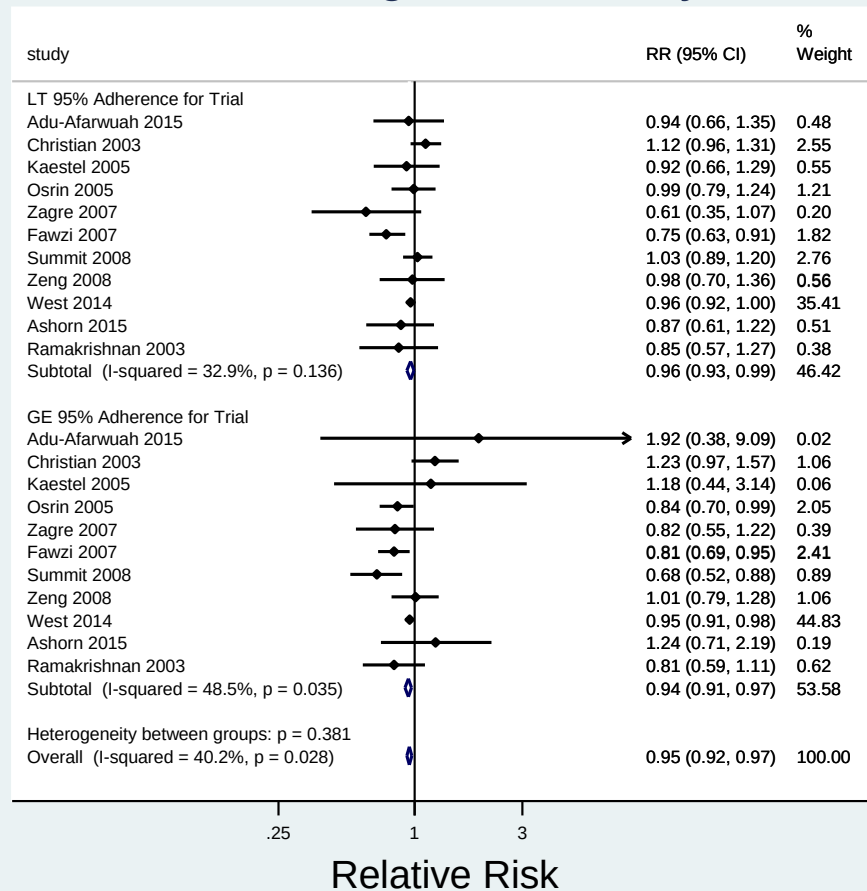
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by median adherence



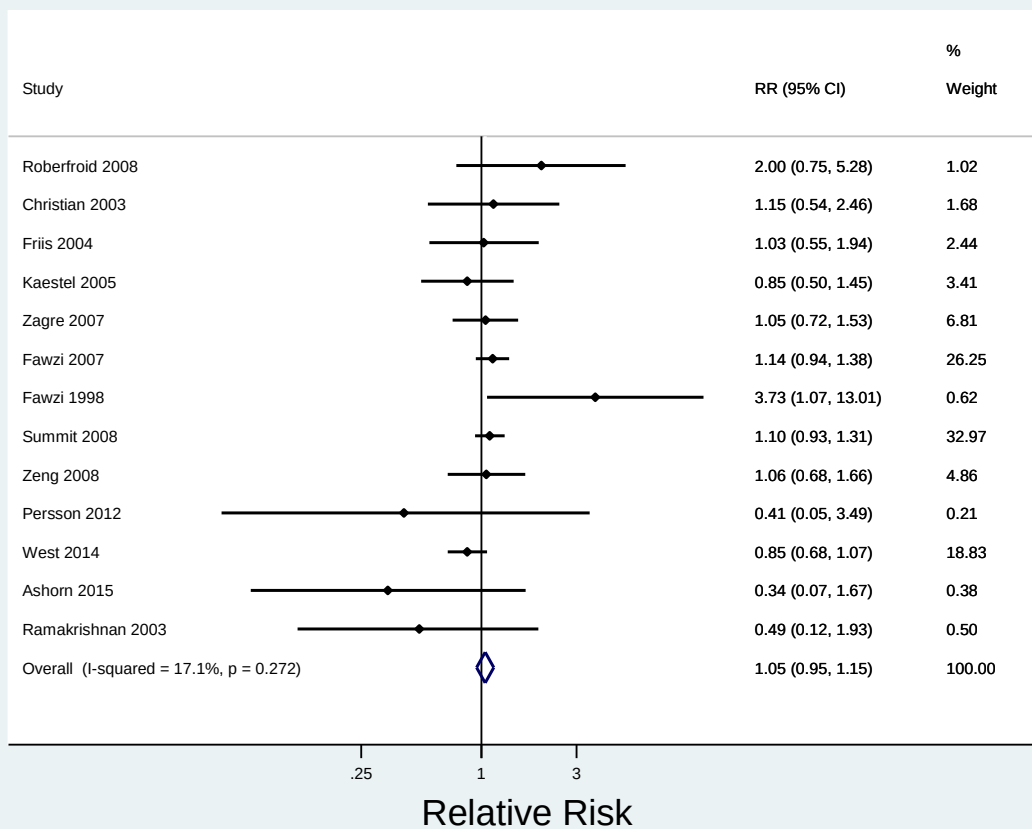
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of SGA Intergrowth 10th by 95% adherence



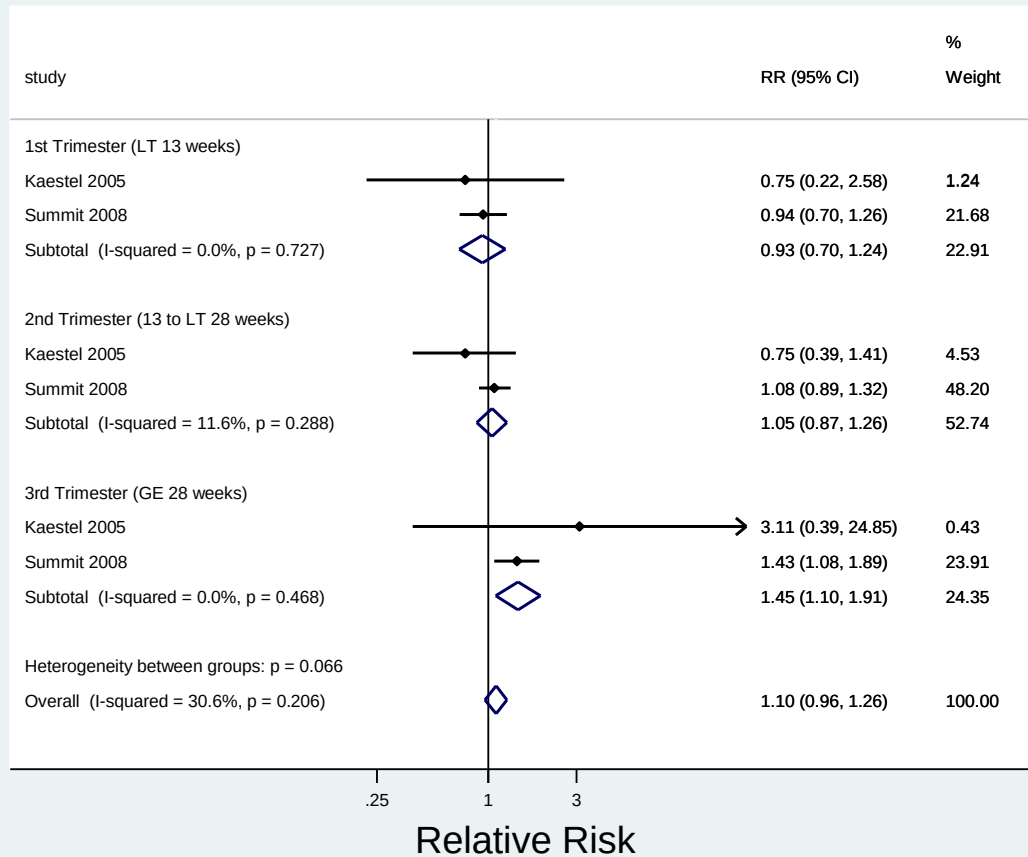
Outcome = SGA <10th Percentile (Intergrowth)

Relative risk of LGA Oken 90th - OVERALL



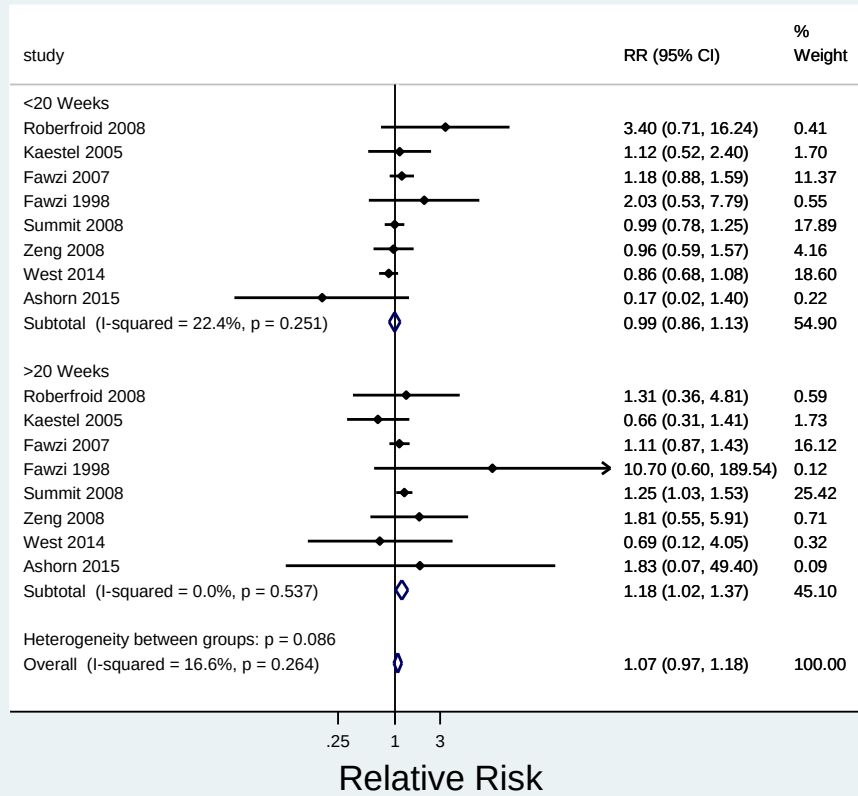
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by trimester at enrollment



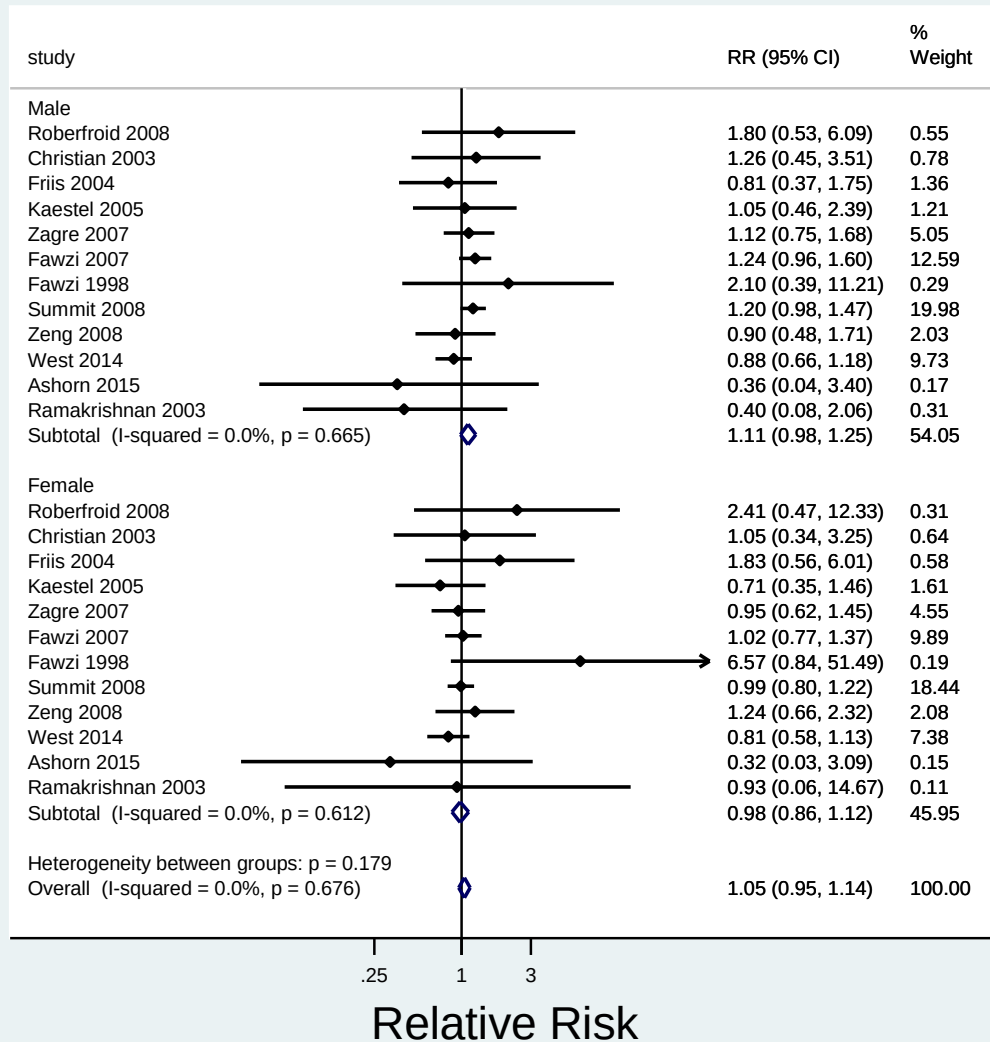
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by gestational age at enrollment



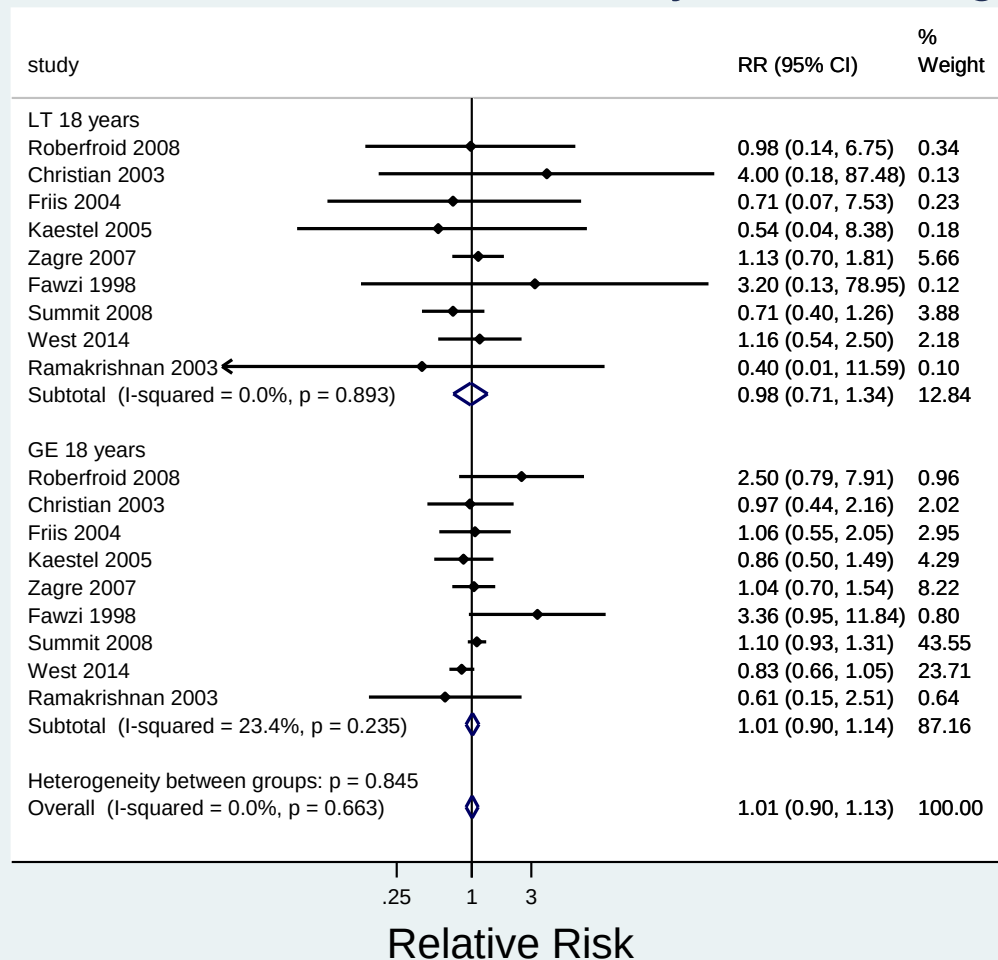
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by sex



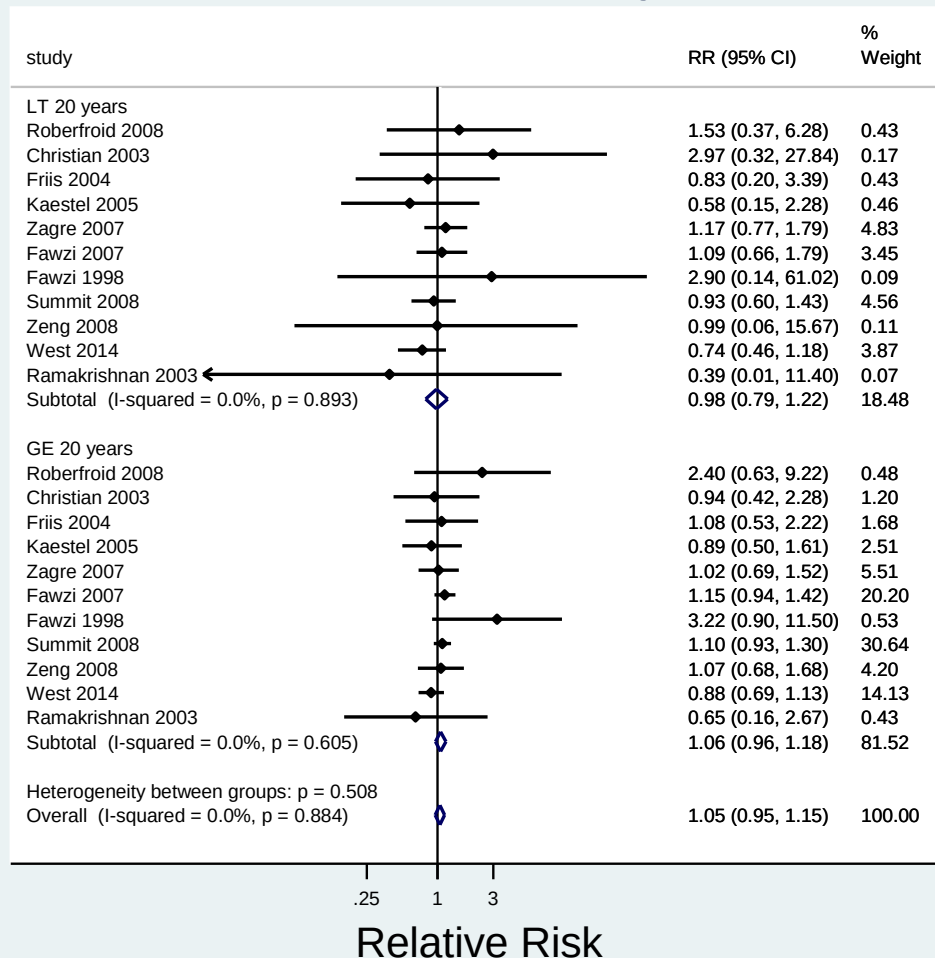
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by maternal age <18



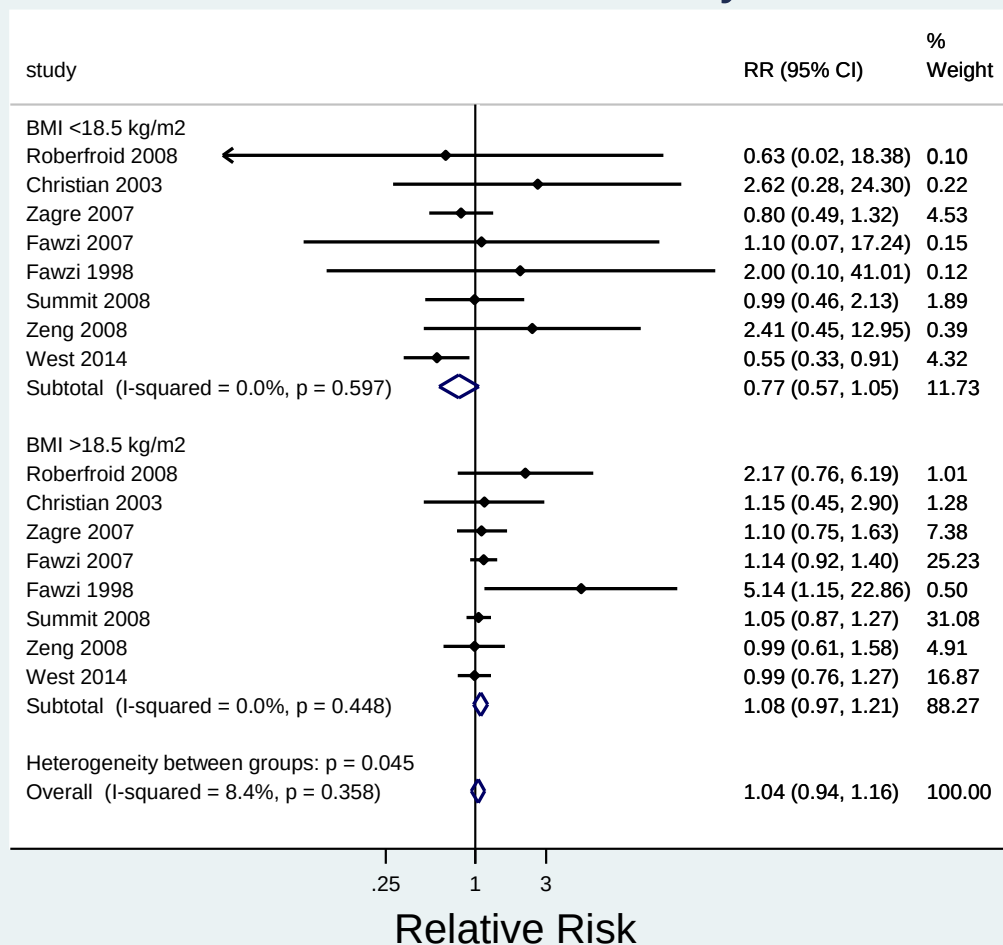
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by maternal age <20



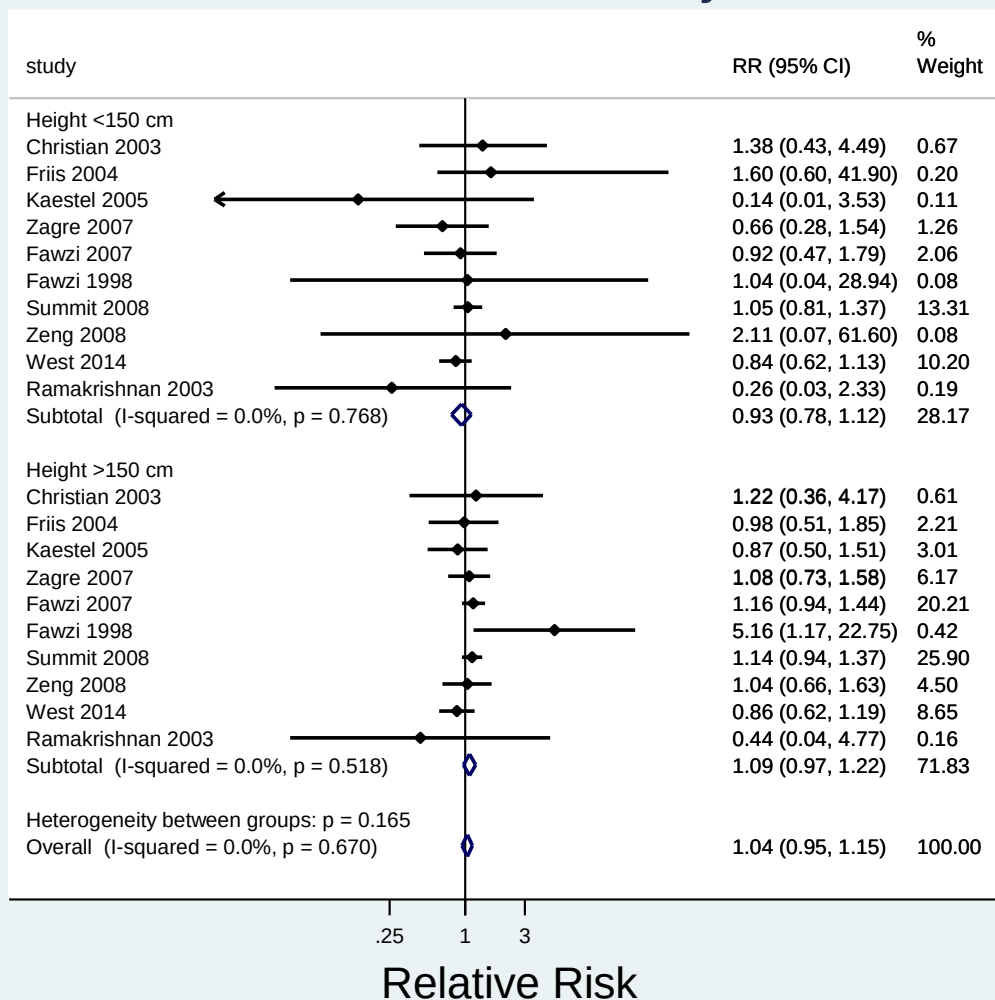
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by maternal BMI



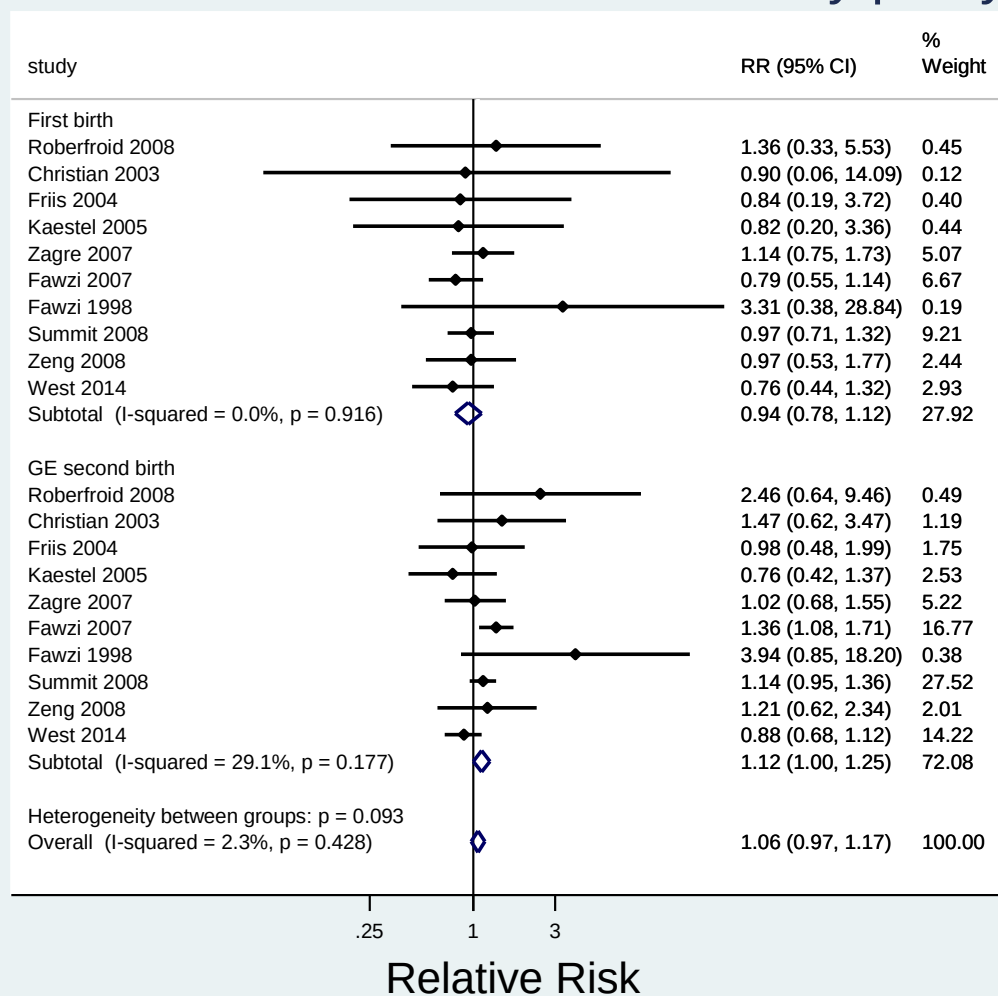
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by maternal height



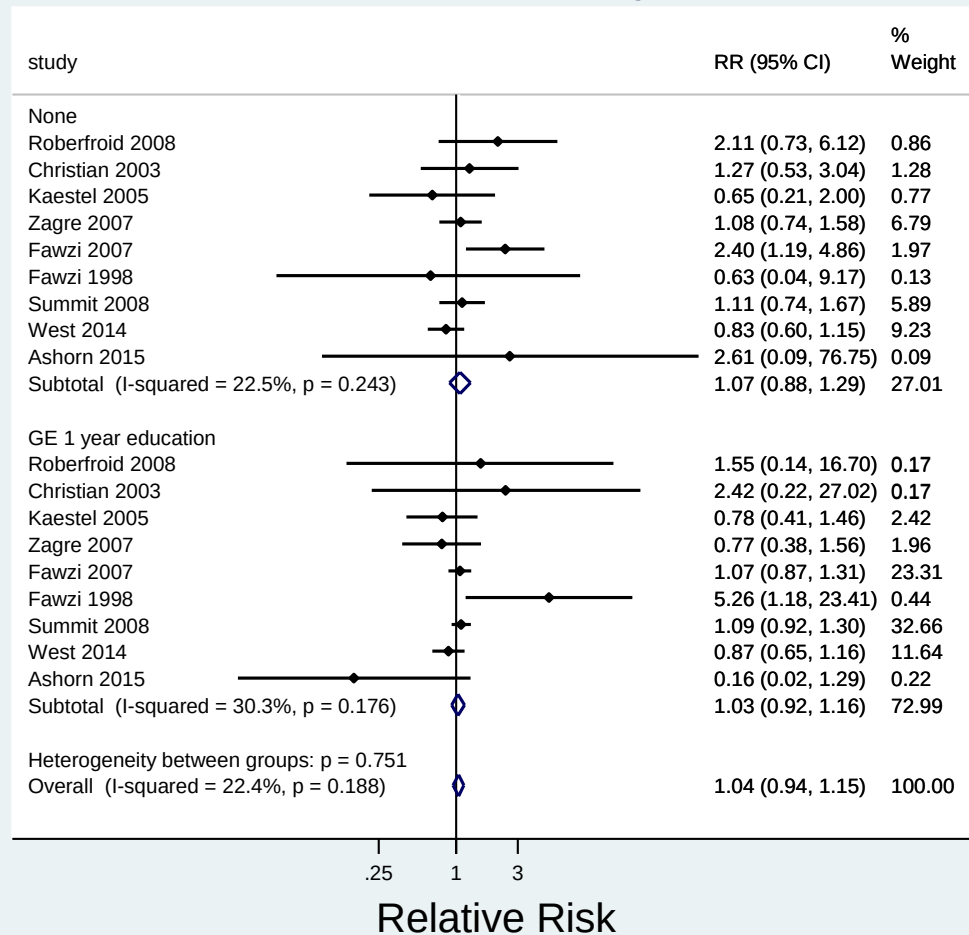
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by parity



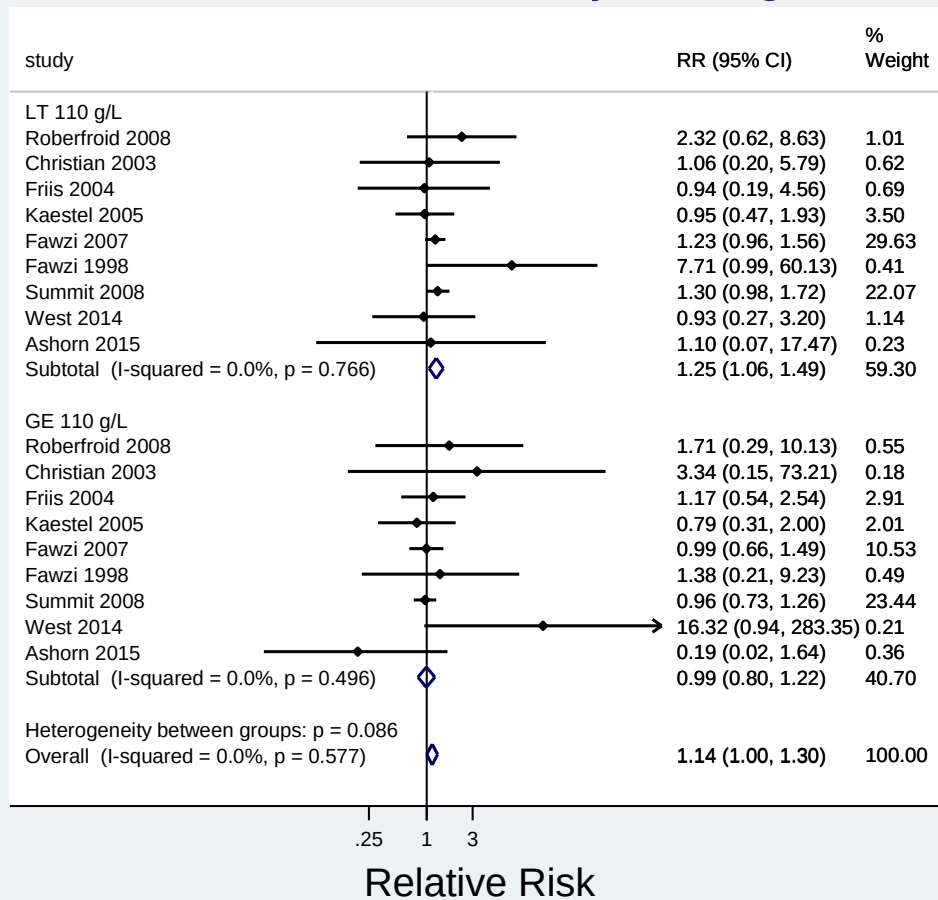
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by maternal education



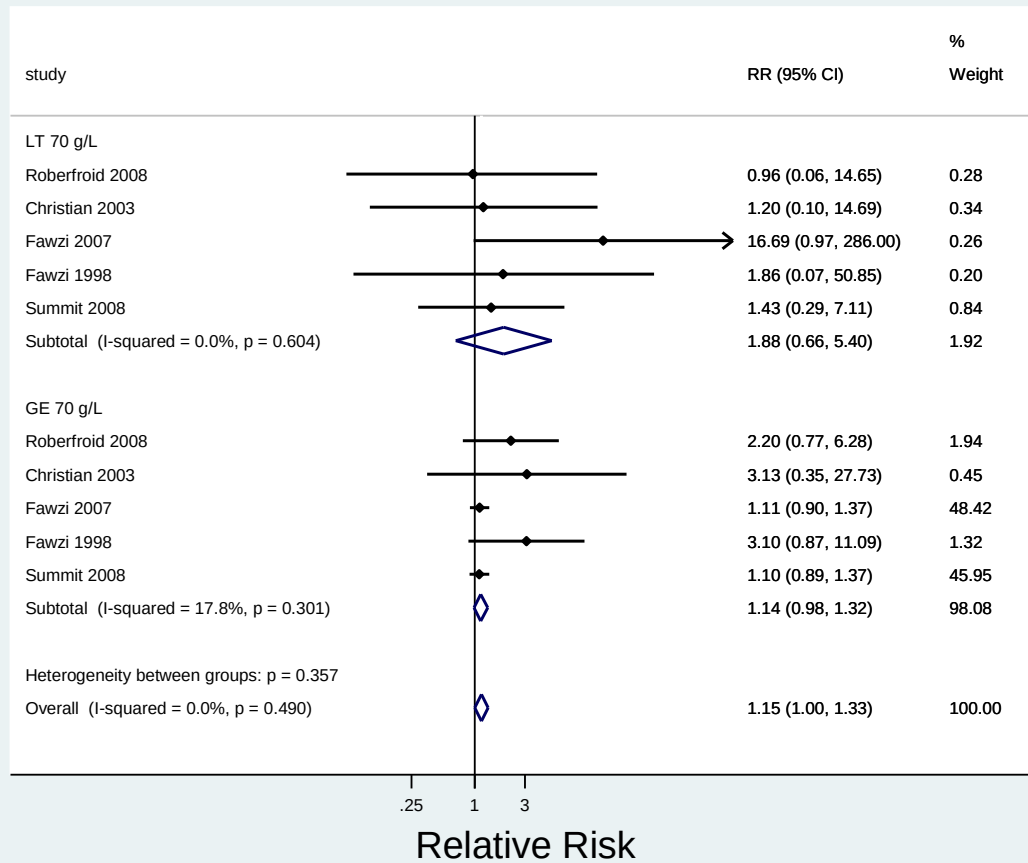
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by hemoglobin <110 g/L



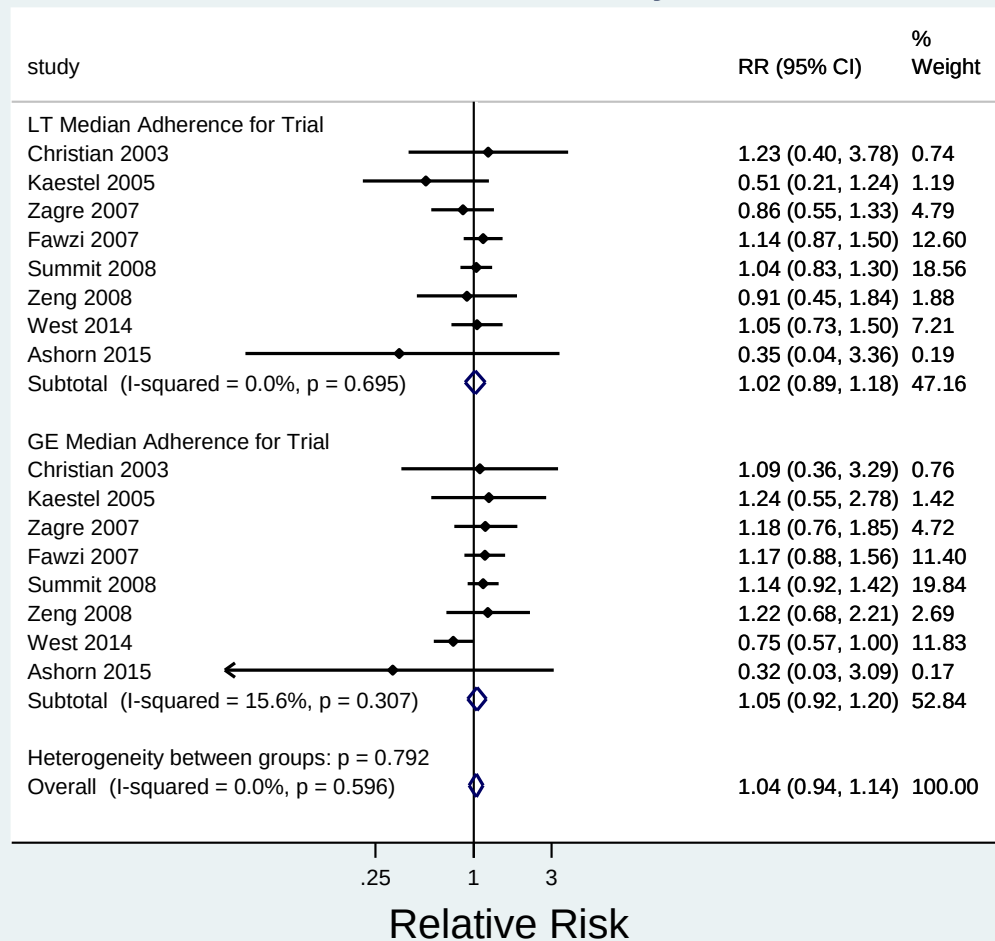
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by hemoglobin <70 g/L



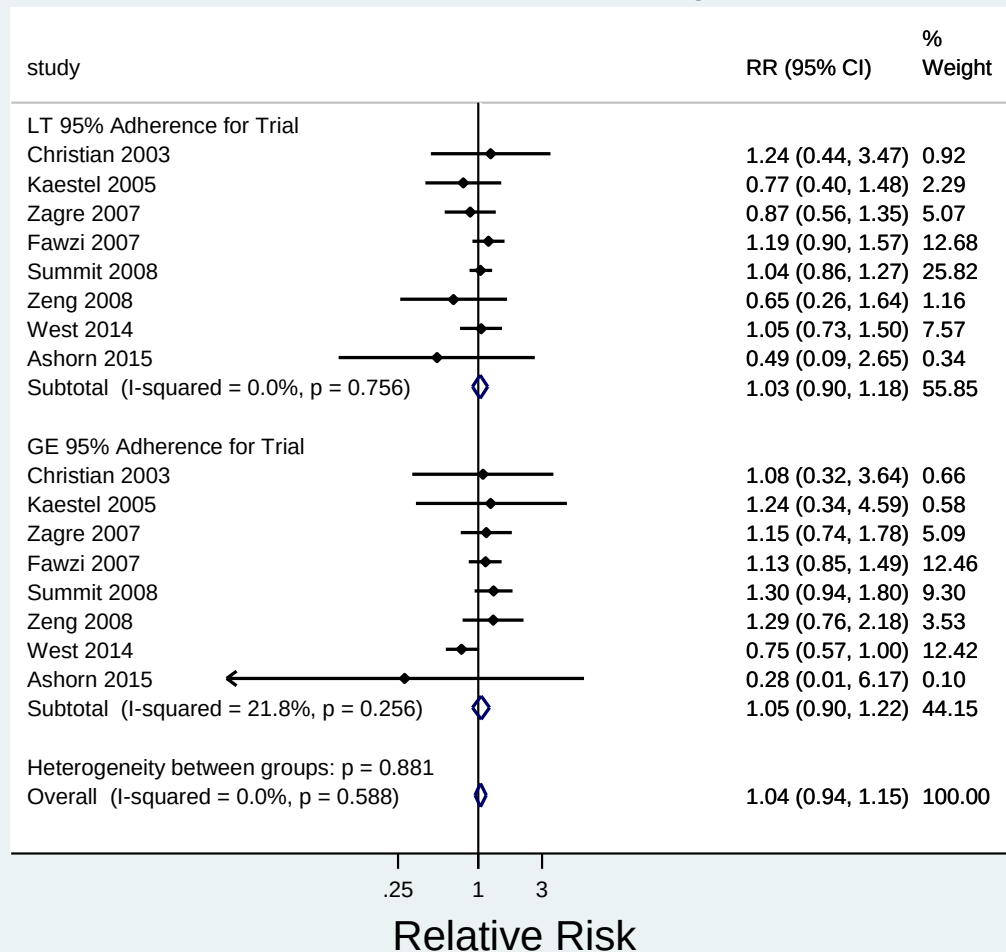
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by median adherence



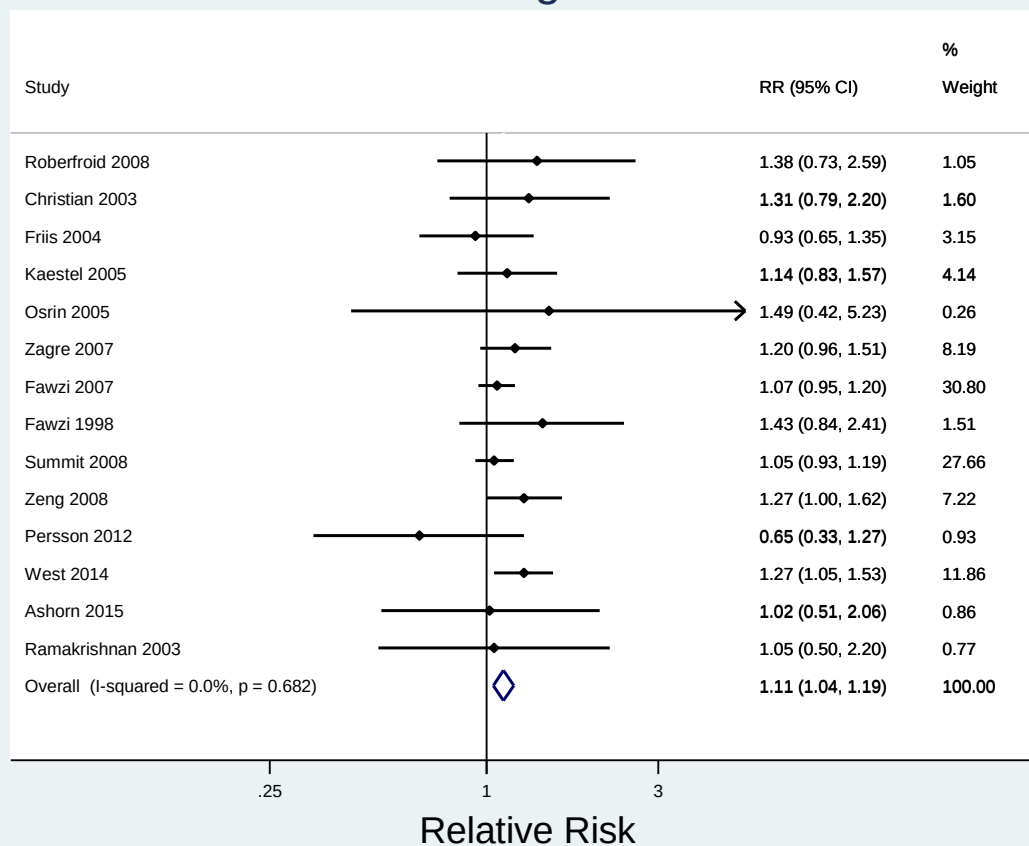
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Oken 90th by 95% adherence



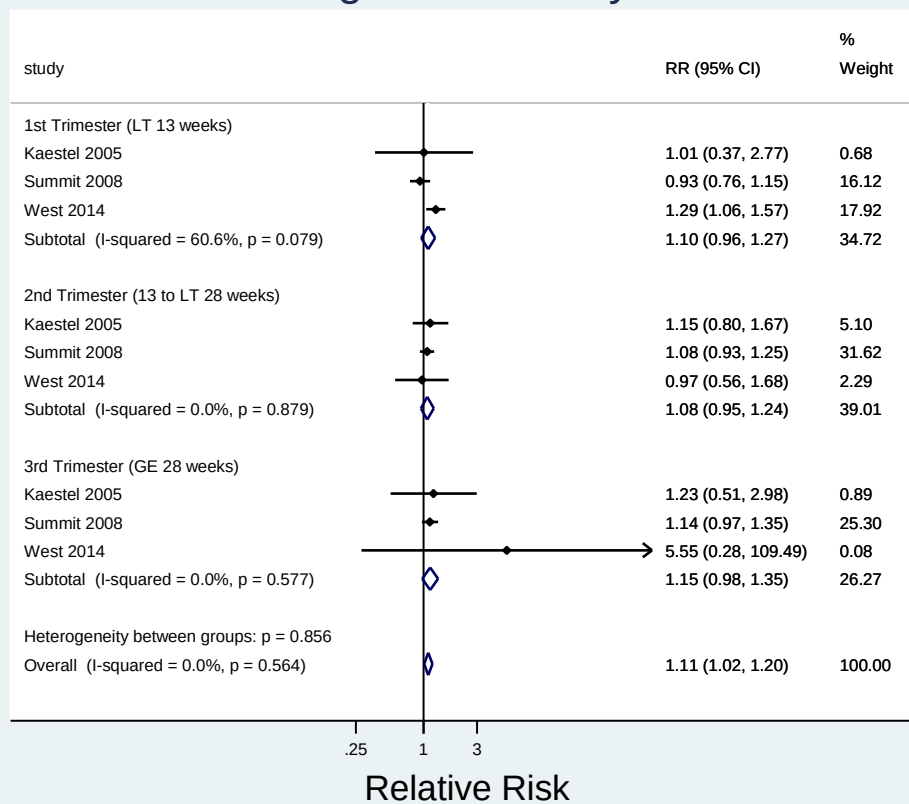
Outcome = LGA >90th Percentile (Oken)

Relative risk of LGA Intergrowth 90th - OVERALL



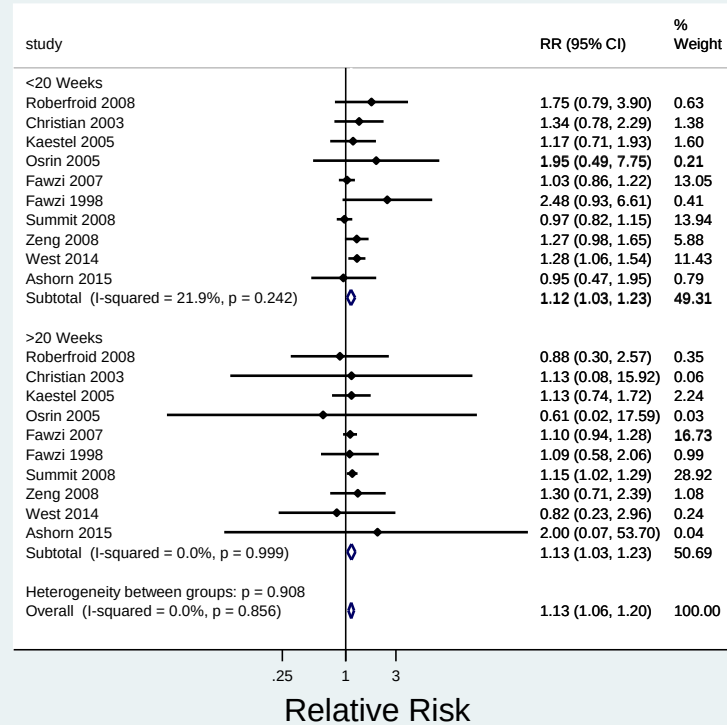
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by trimester at enrollment



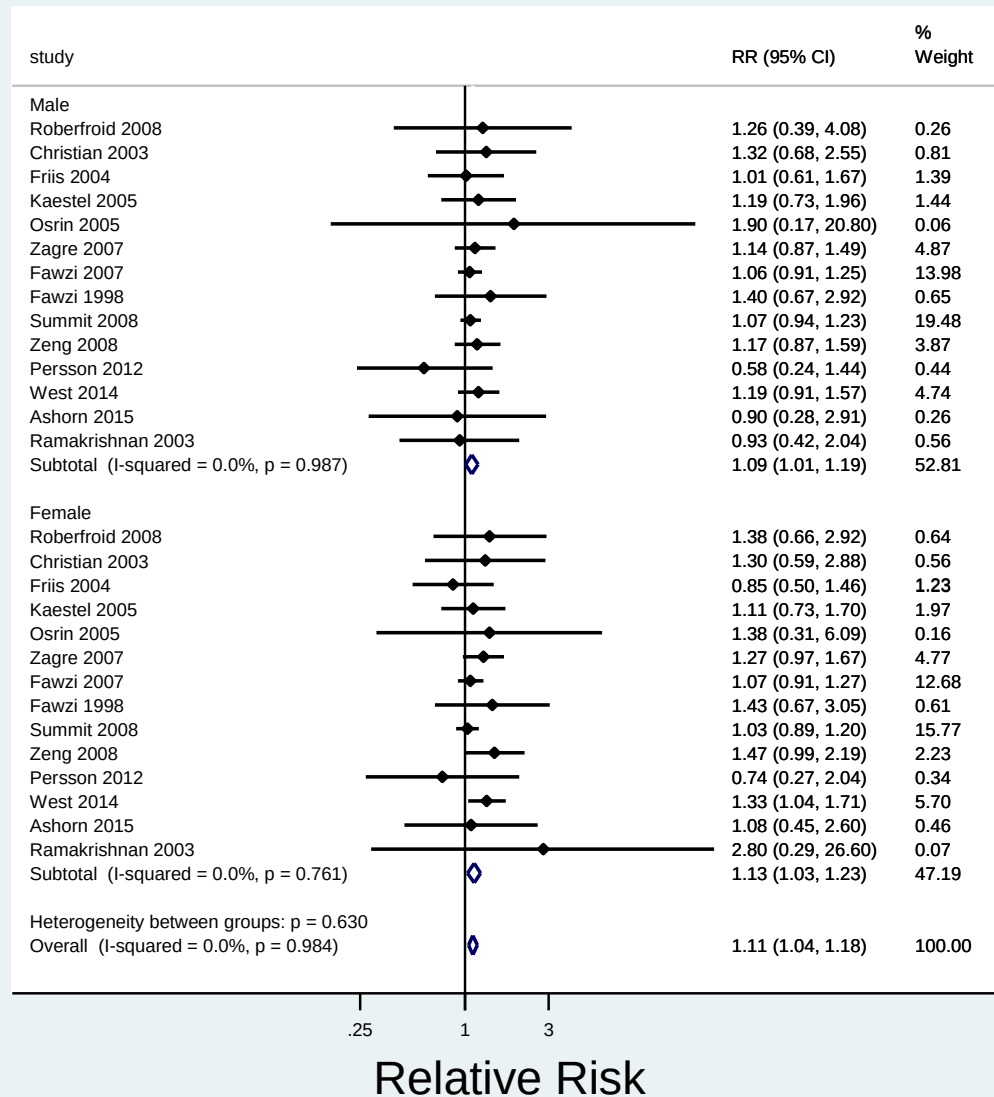
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by gestational age at enrollment



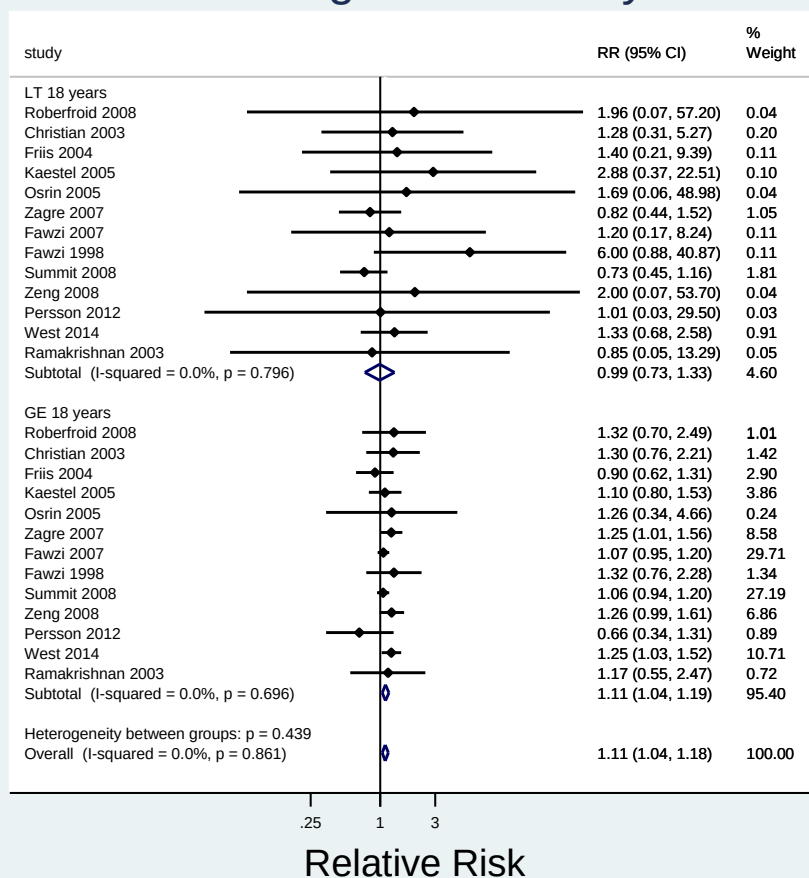
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by sex



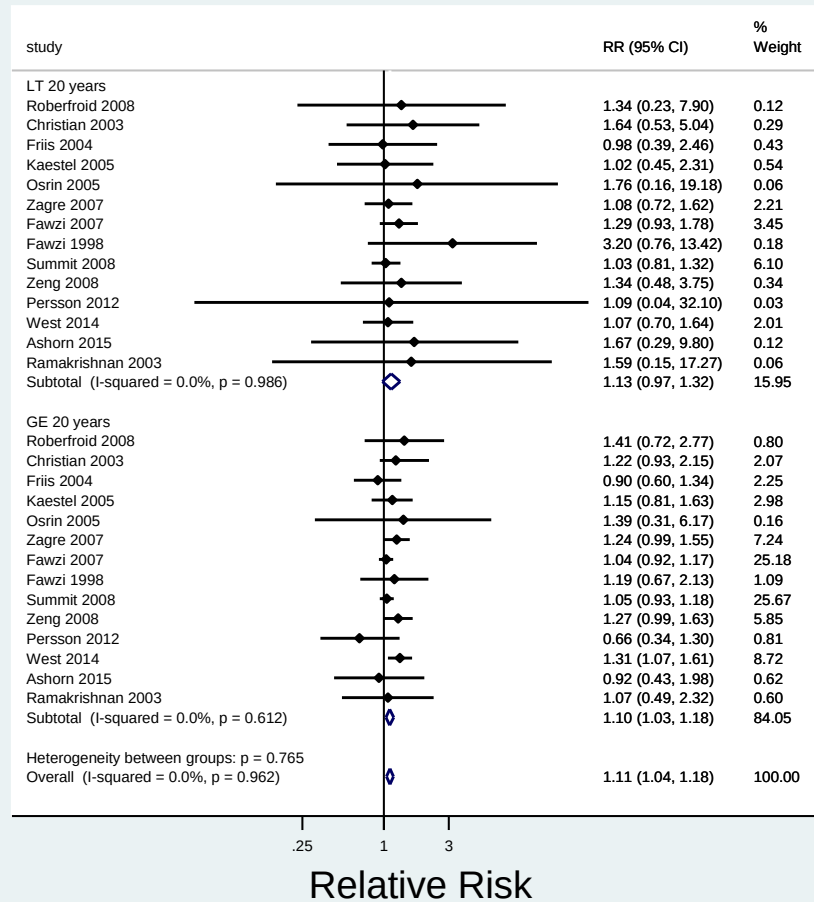
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by maternal age <18



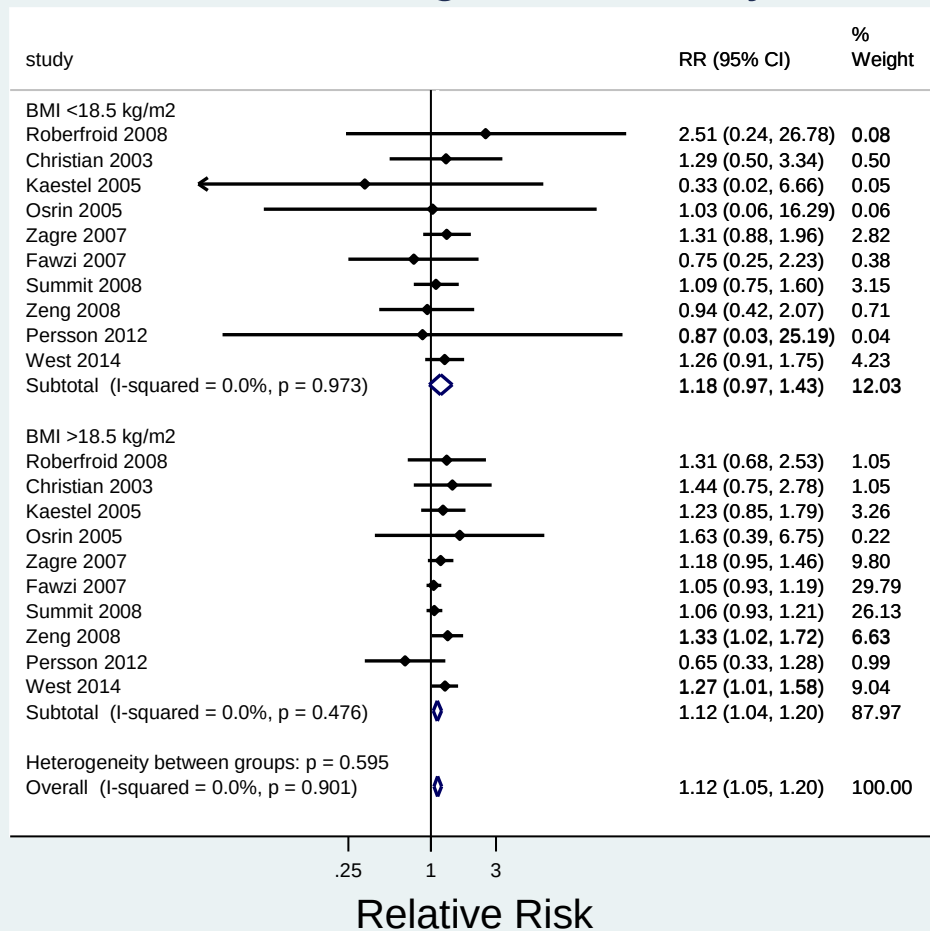
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by maternal age <20



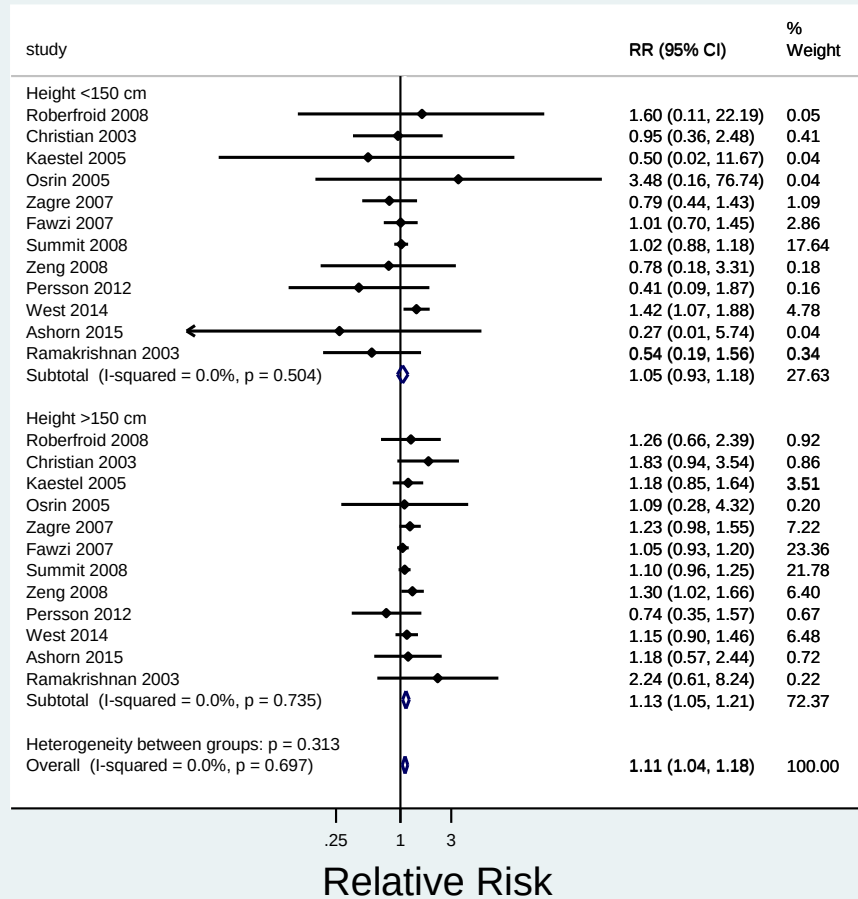
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by maternal BMI



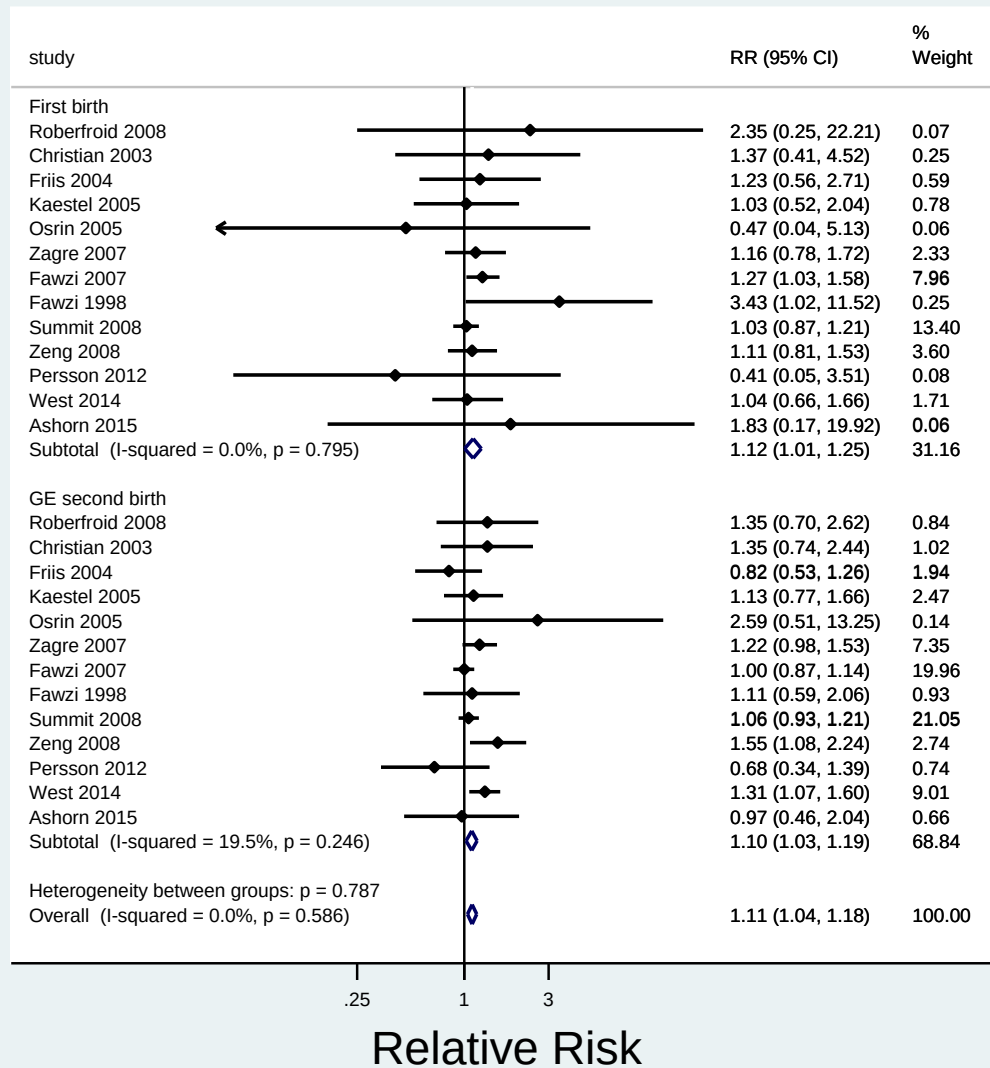
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by maternal height



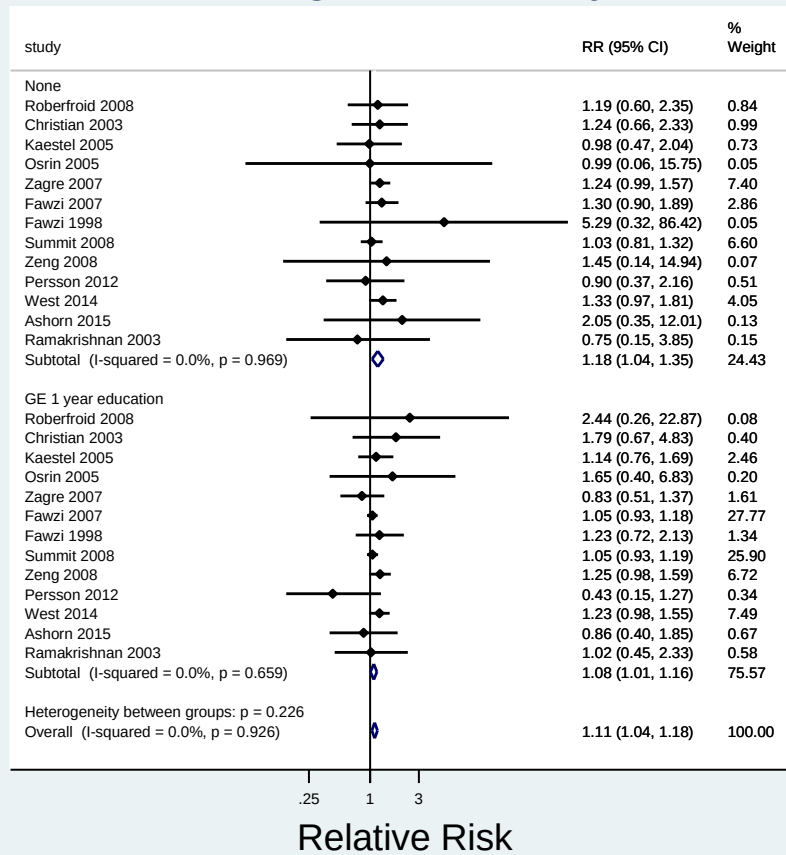
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by parity



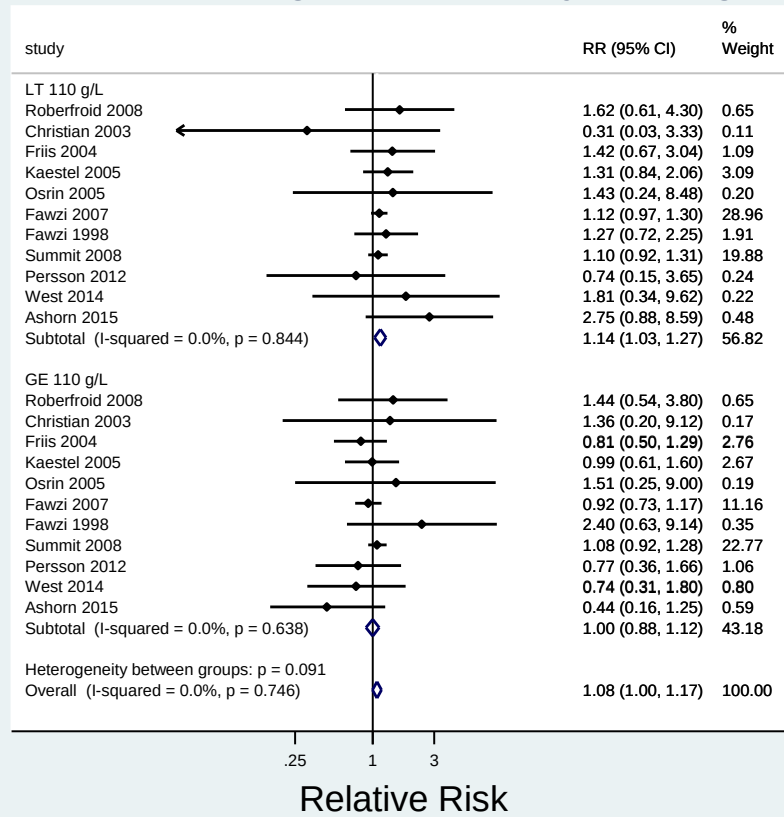
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by maternal education



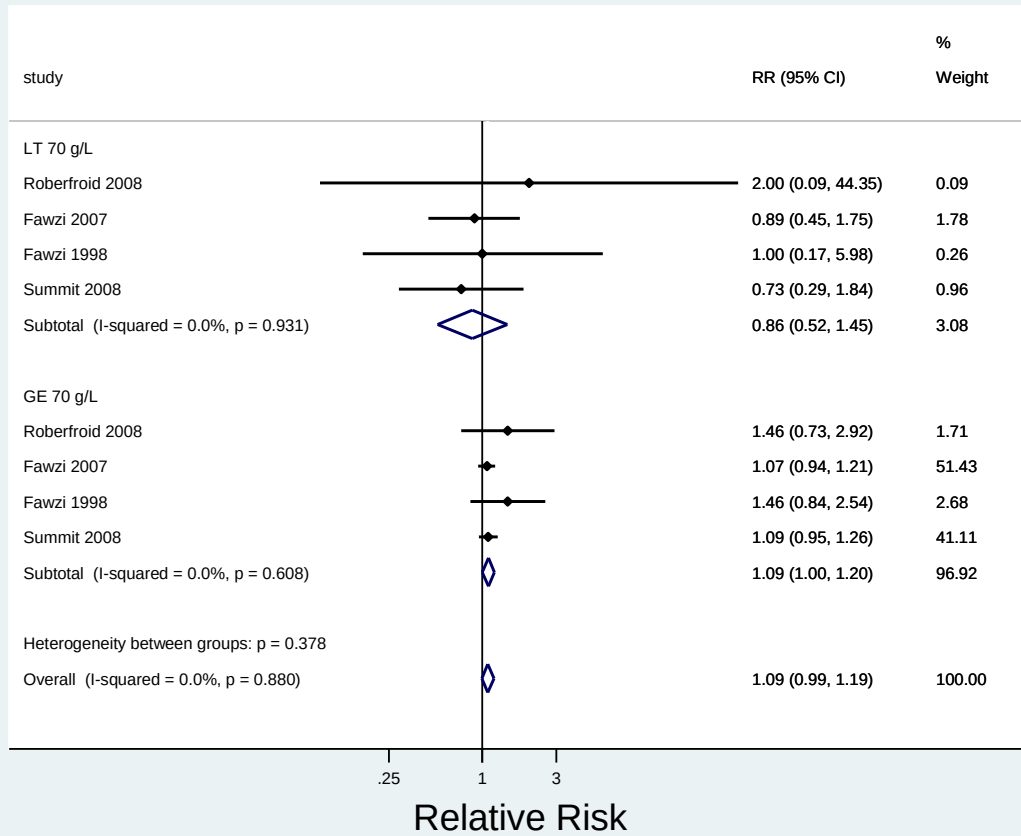
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by hemoglobin <110 g/L



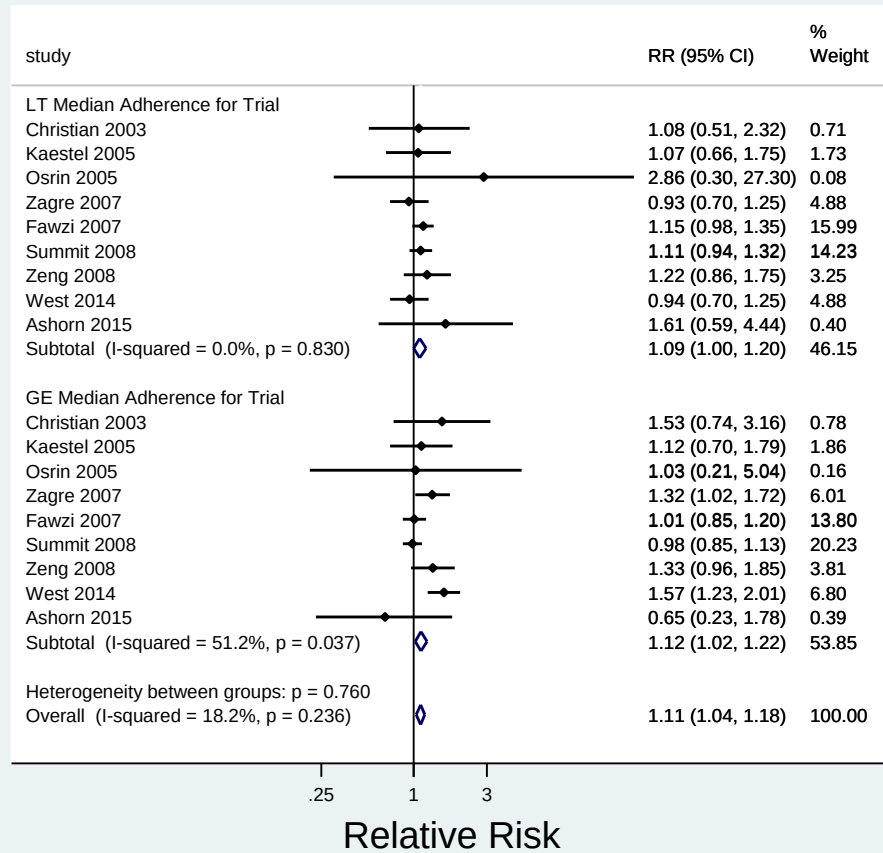
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by hemoglobin <70 g/L



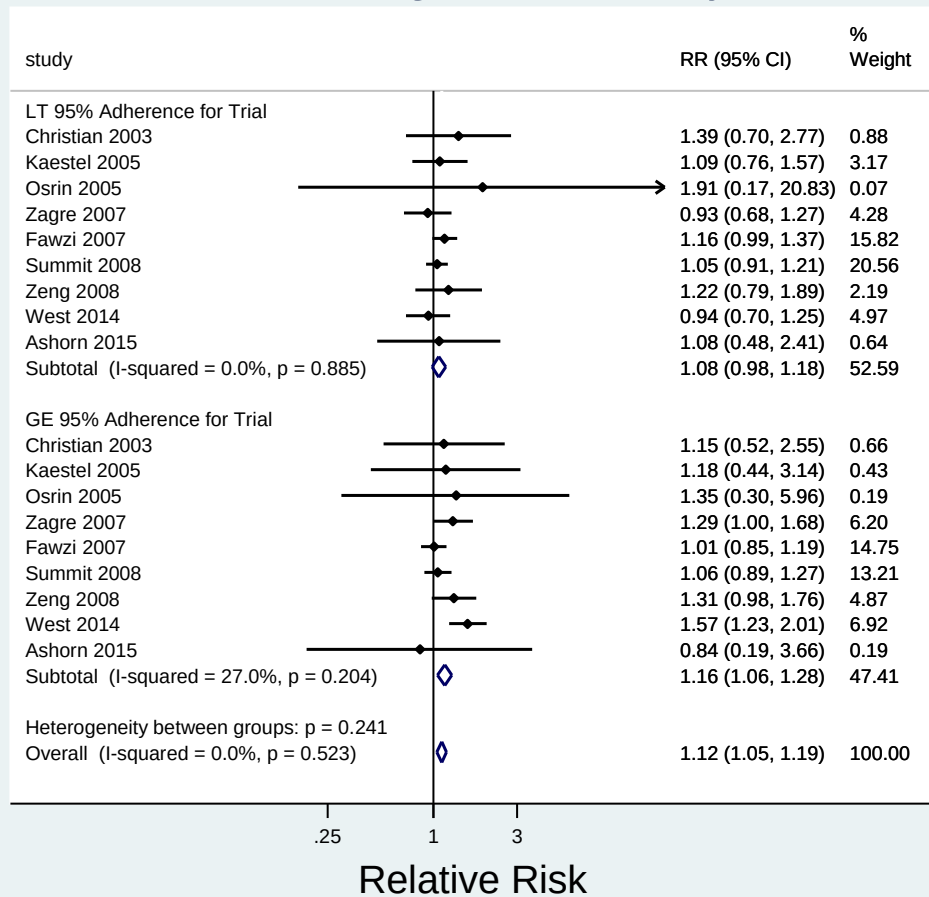
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by median adherence



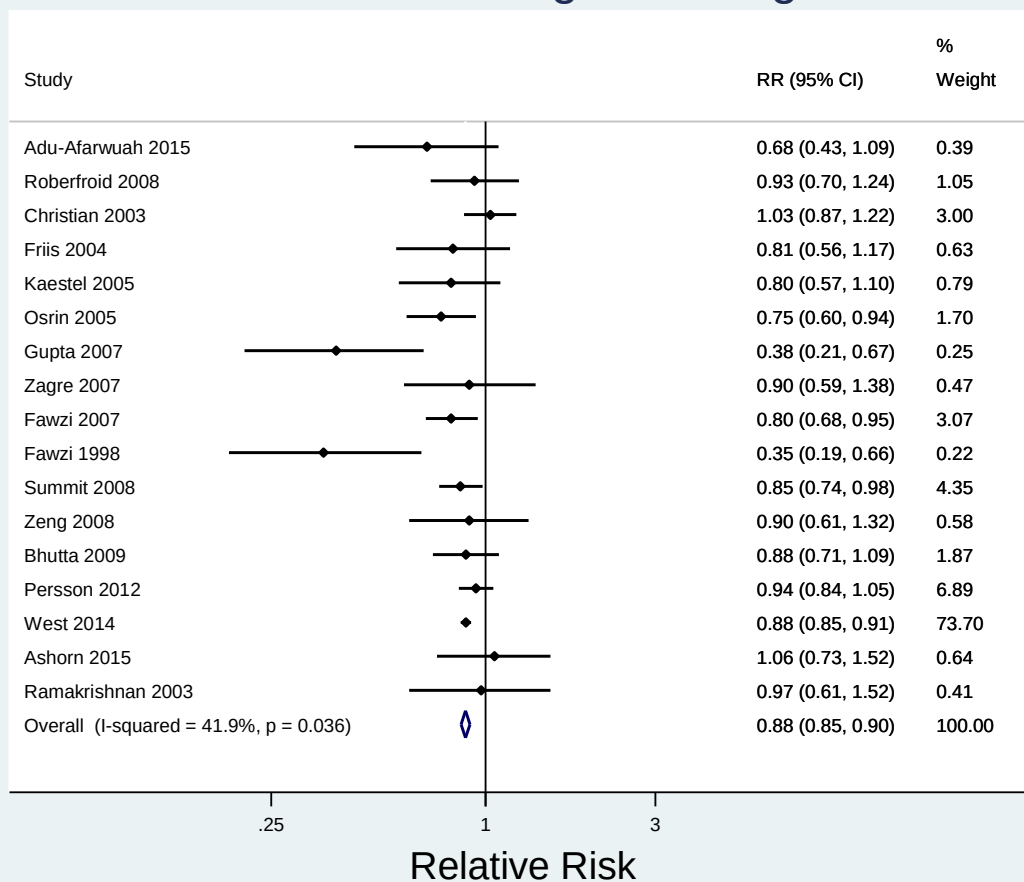
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of LGA Intergrowth 90th by 95% adherence



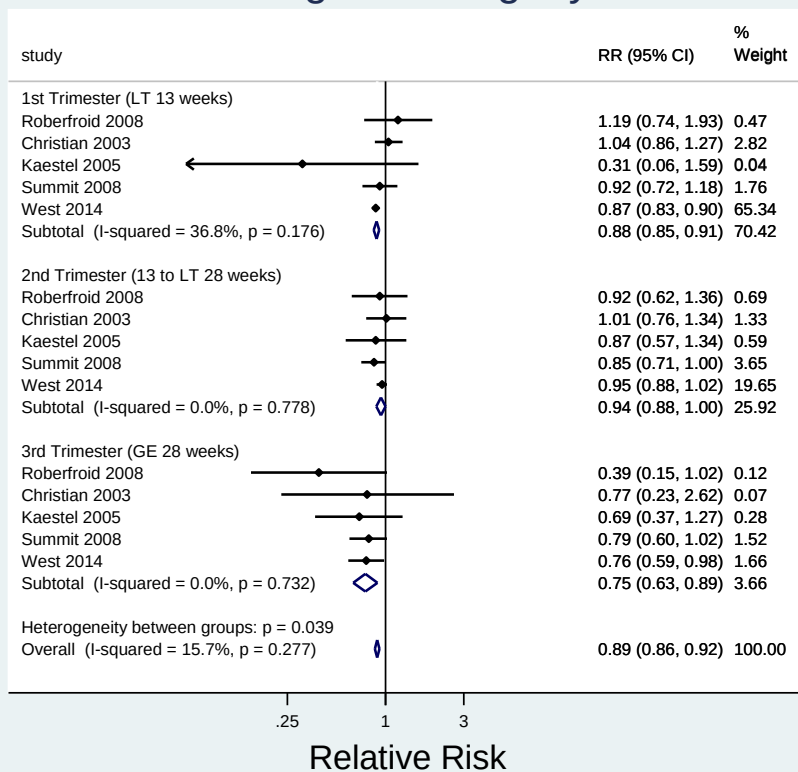
Outcome = LGA >90th Percentile (Intergrowth)

Relative risk of low birthweight <2500g - OVERALL



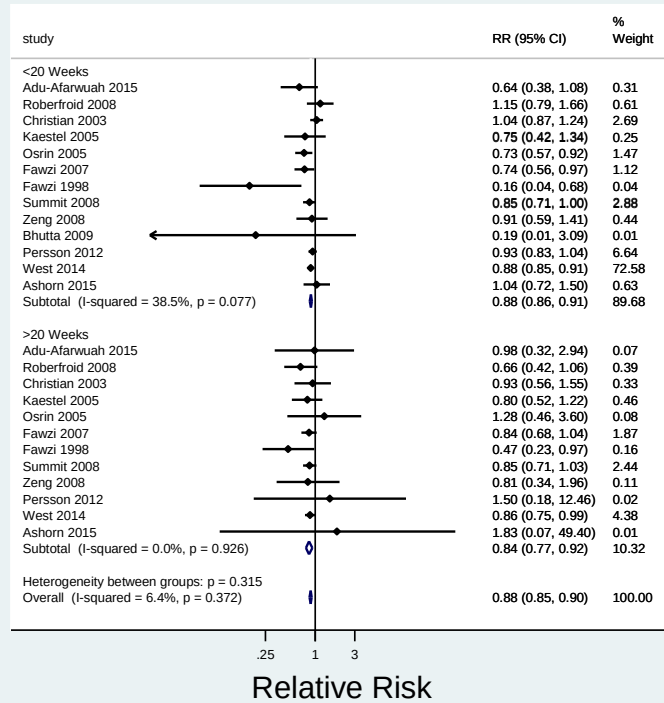
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by trimester at enrollment



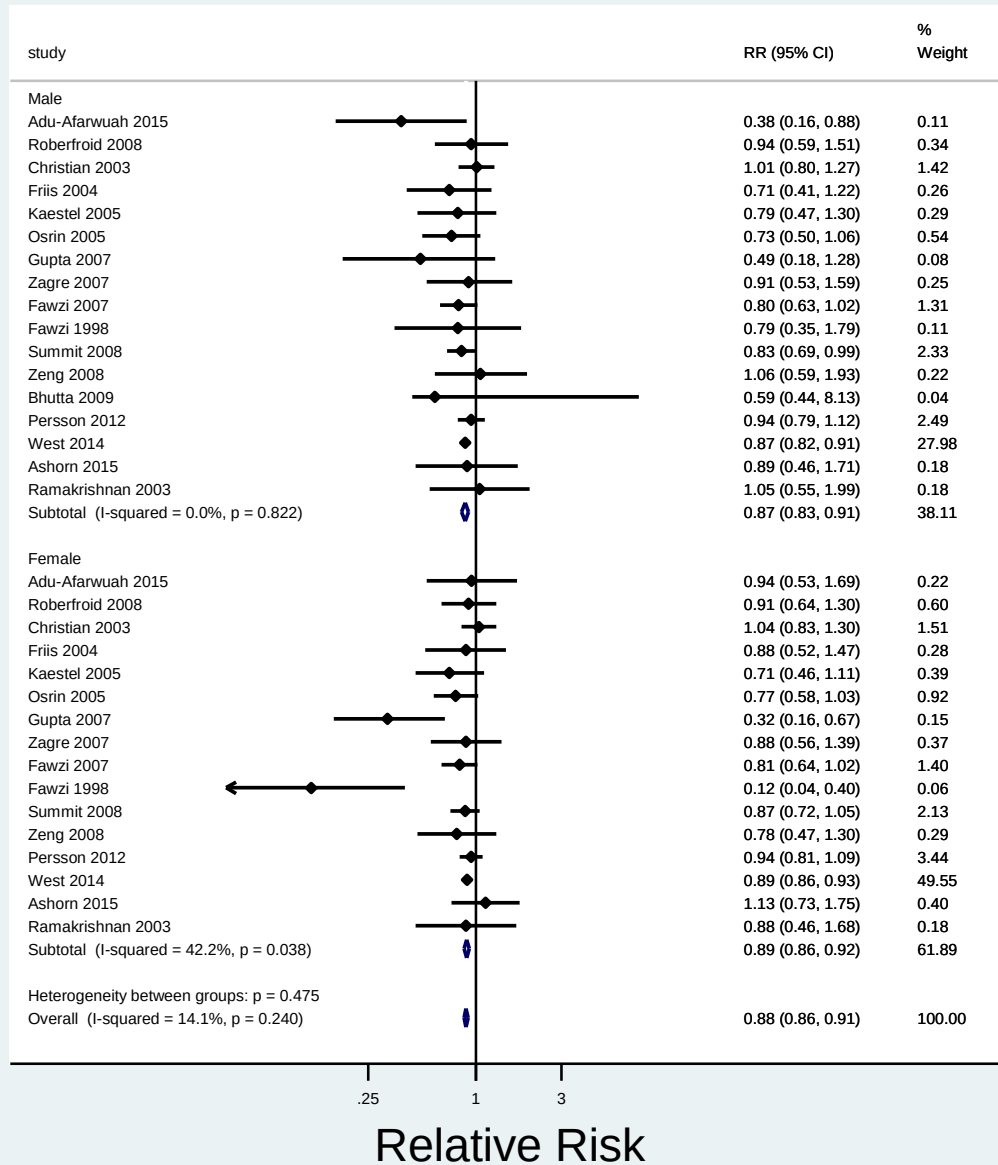
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by gestational age at enrollment



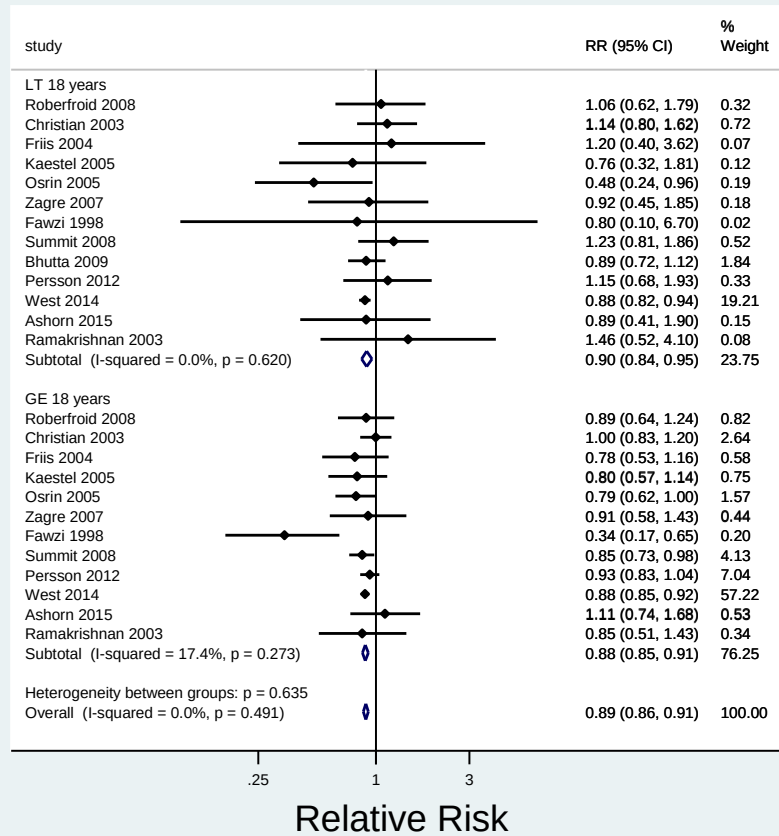
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by sex



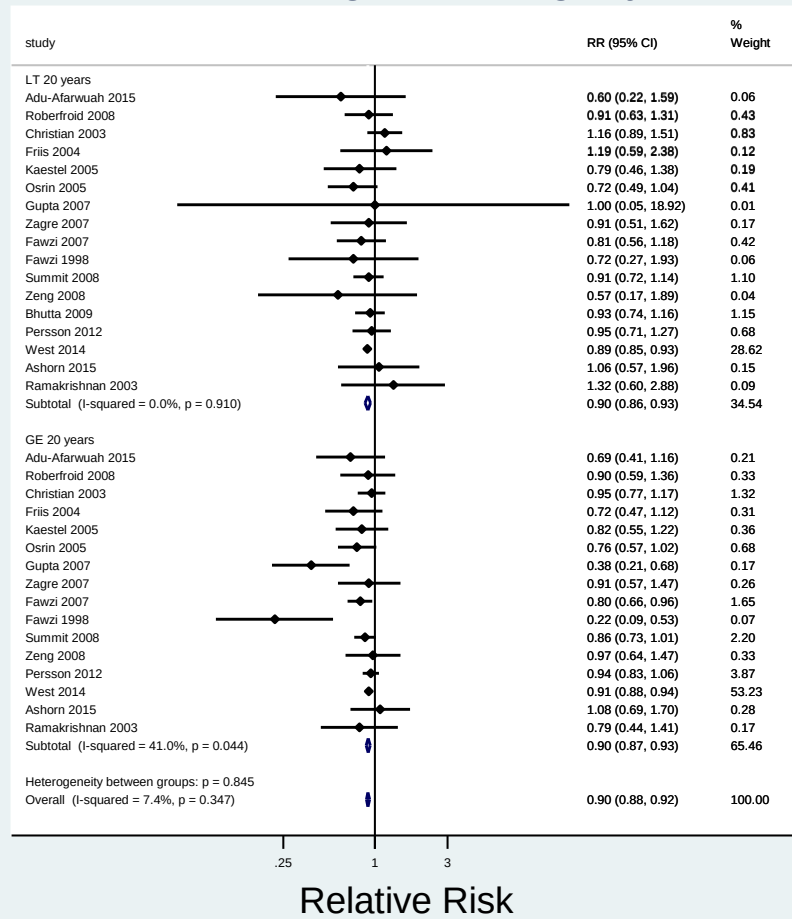
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by maternal age <18



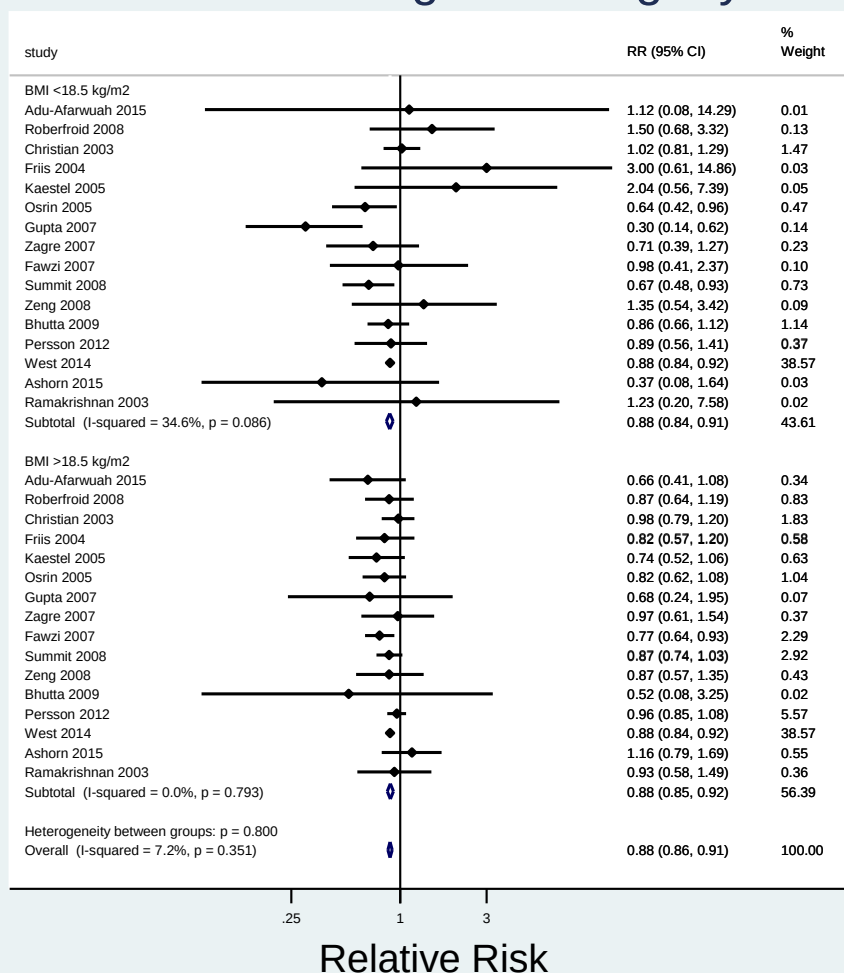
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by maternal age <20



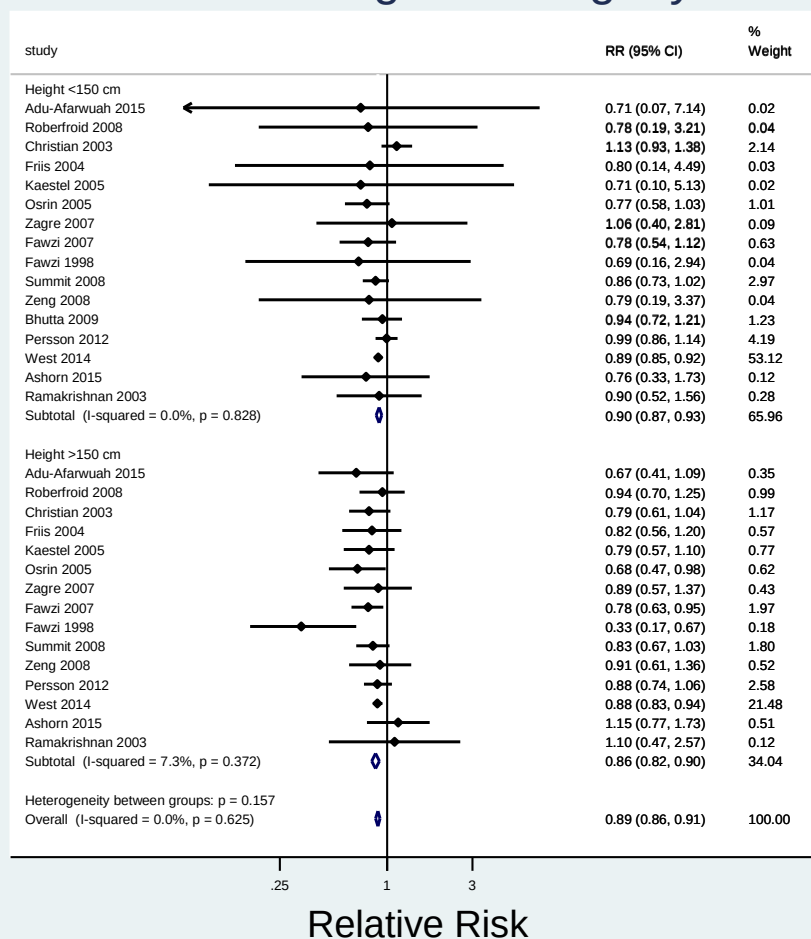
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by maternal BMI



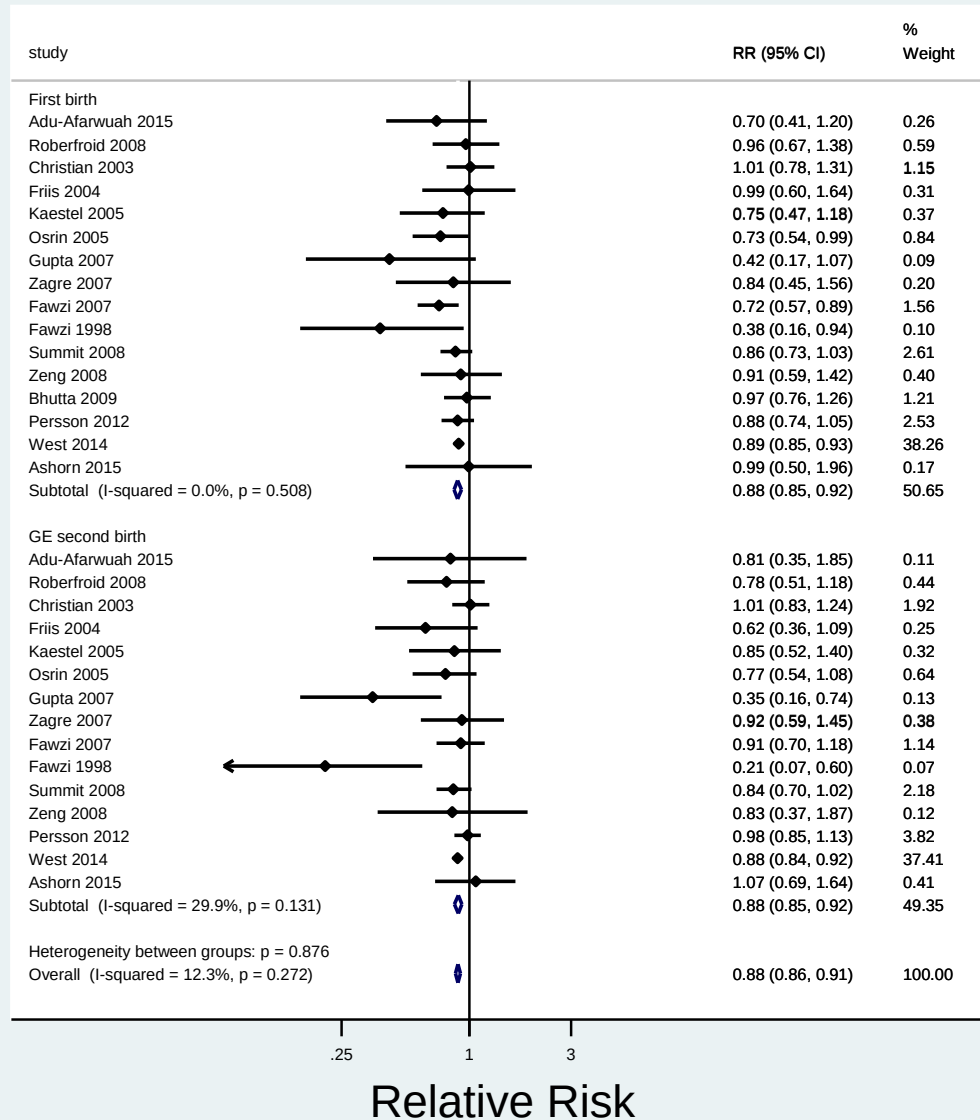
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by maternal height



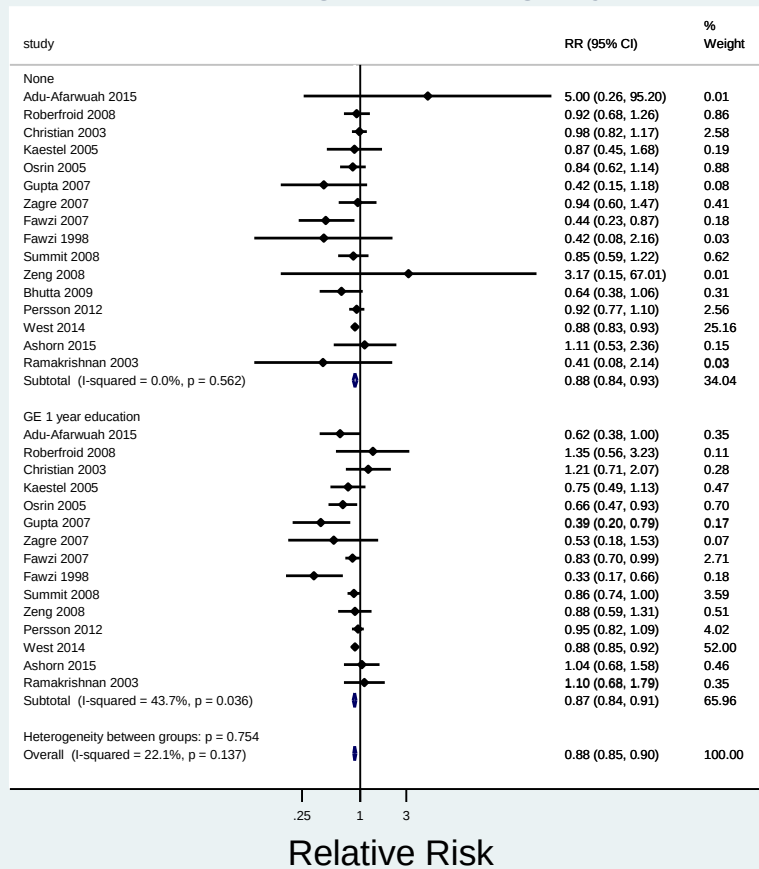
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by parity



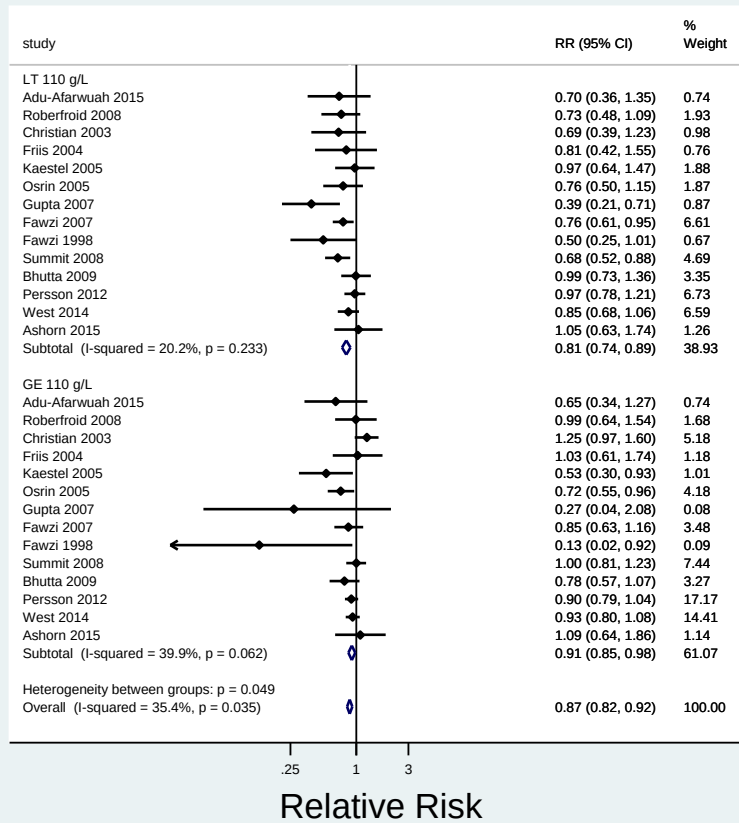
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by maternal education



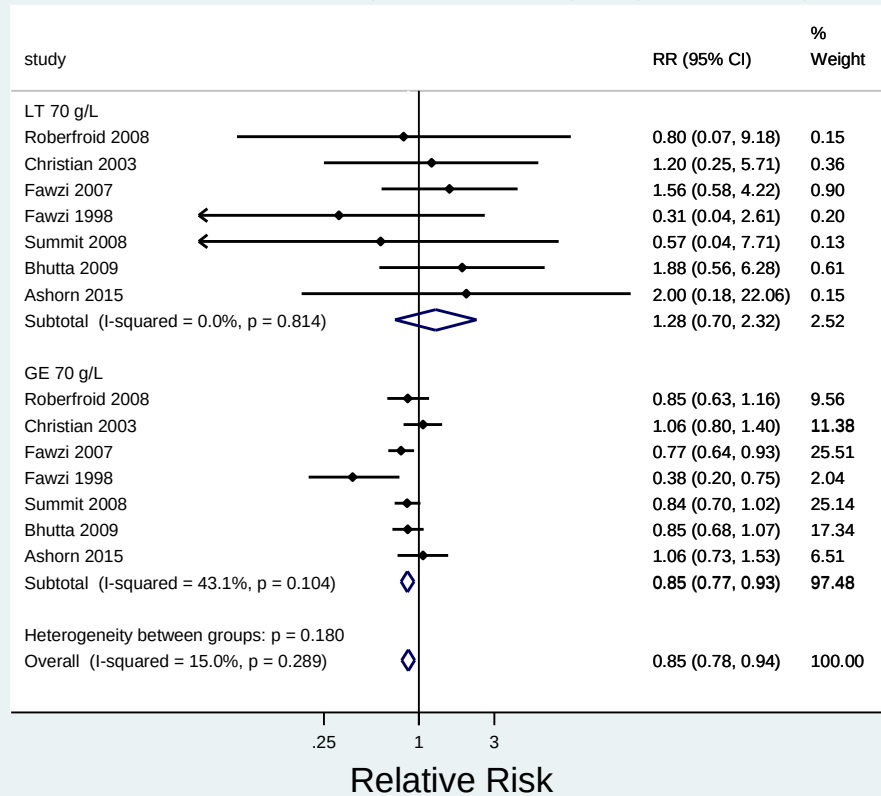
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by hemoglobin <110 g/L



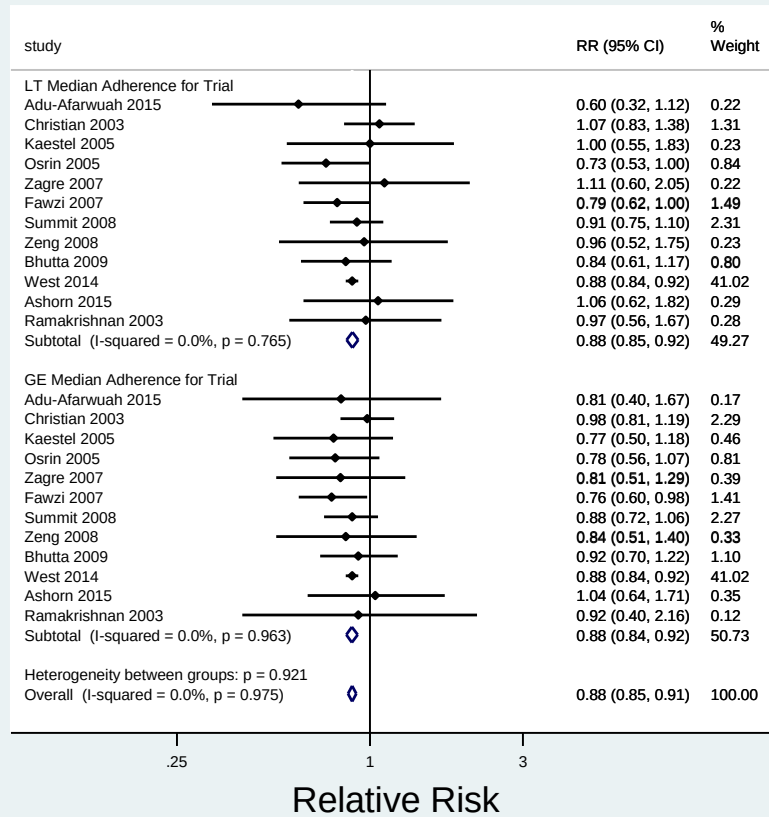
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by hemoglobin <70 g/L



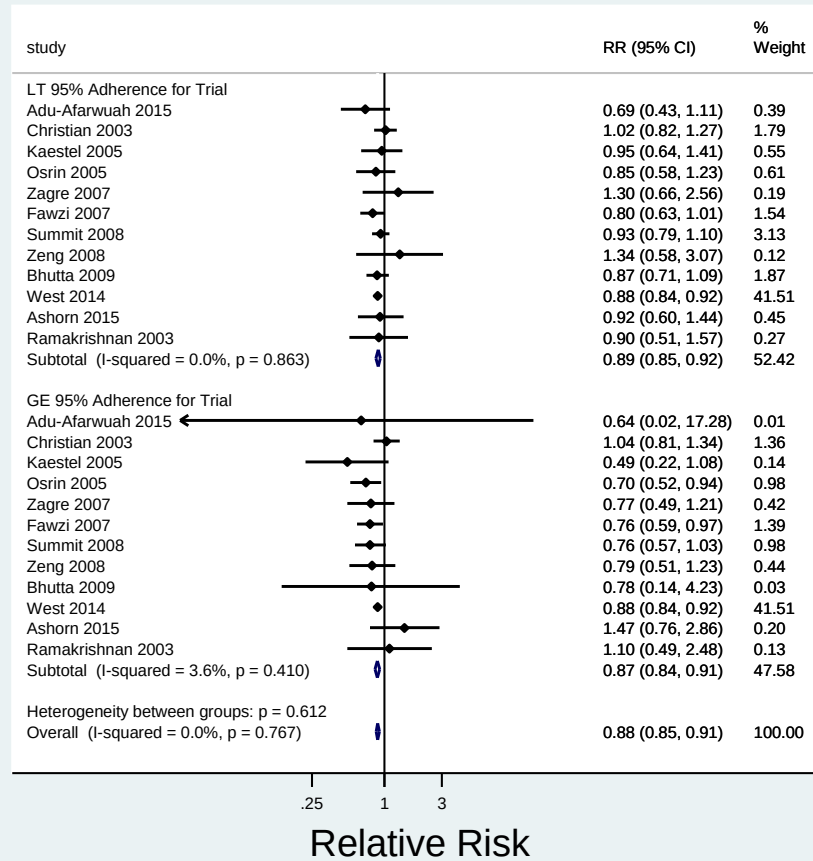
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by median adherence



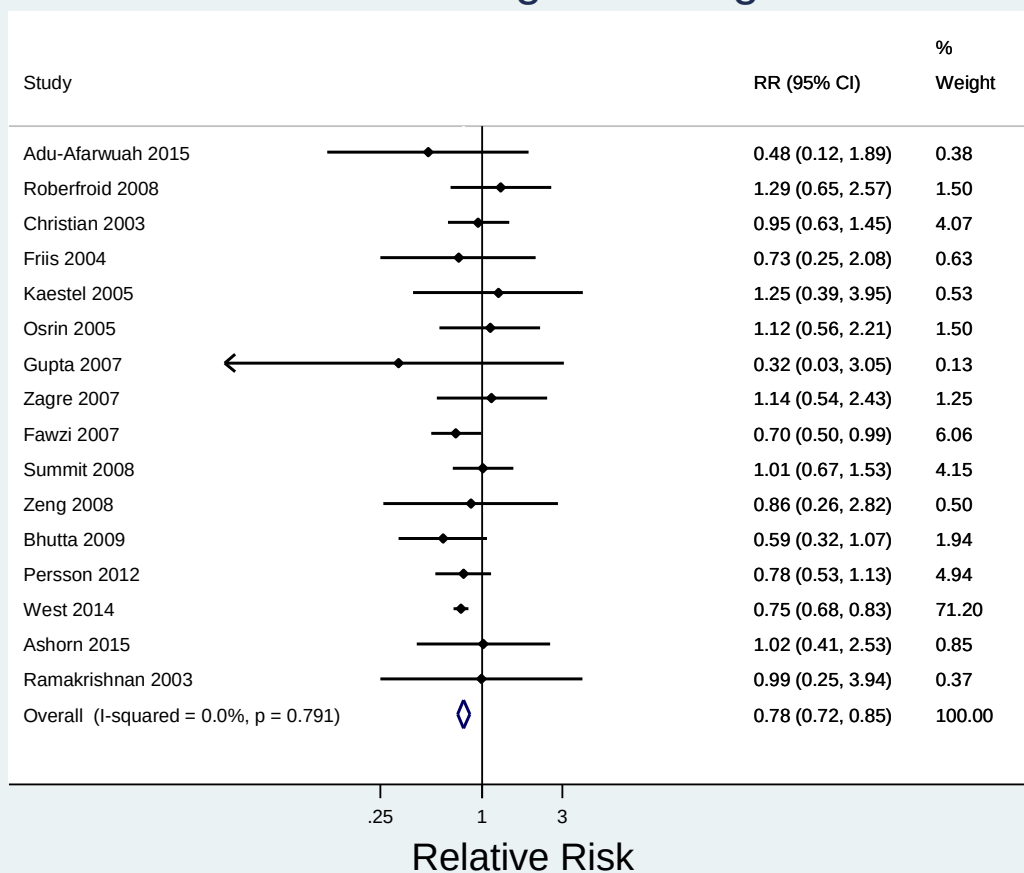
Outcome = Low Birthweight (<2500 g)

Relative risk of low birthweight <2500g by 95% adherence



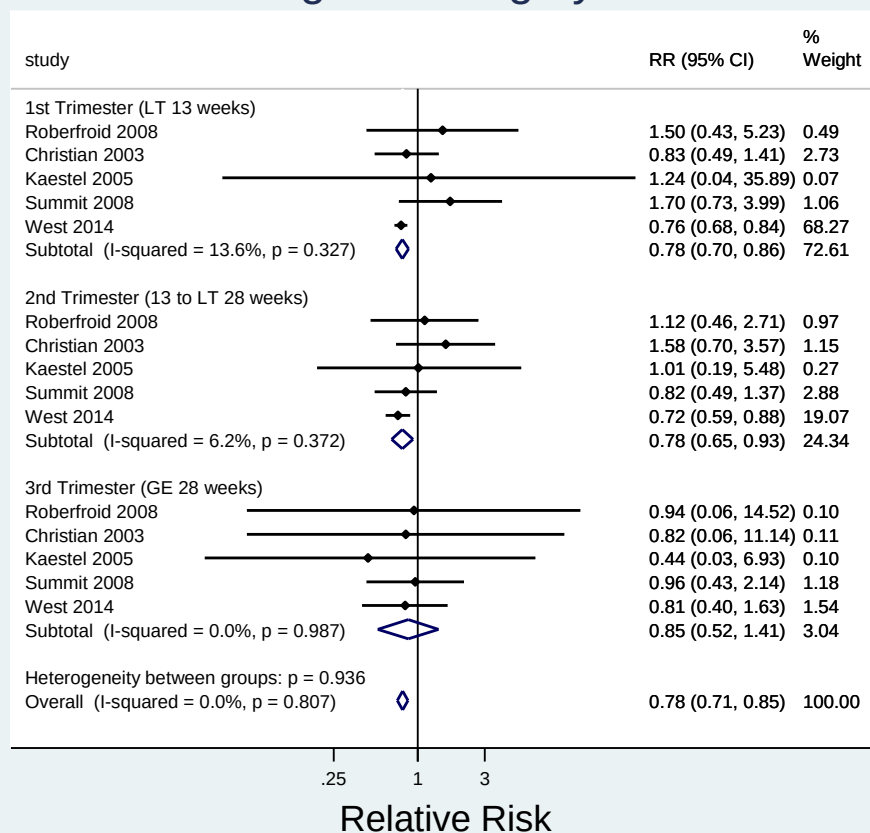
Outcome = Low Birthweight (<2500 g)

Relative risk of birthweight <2000g - OVERALL



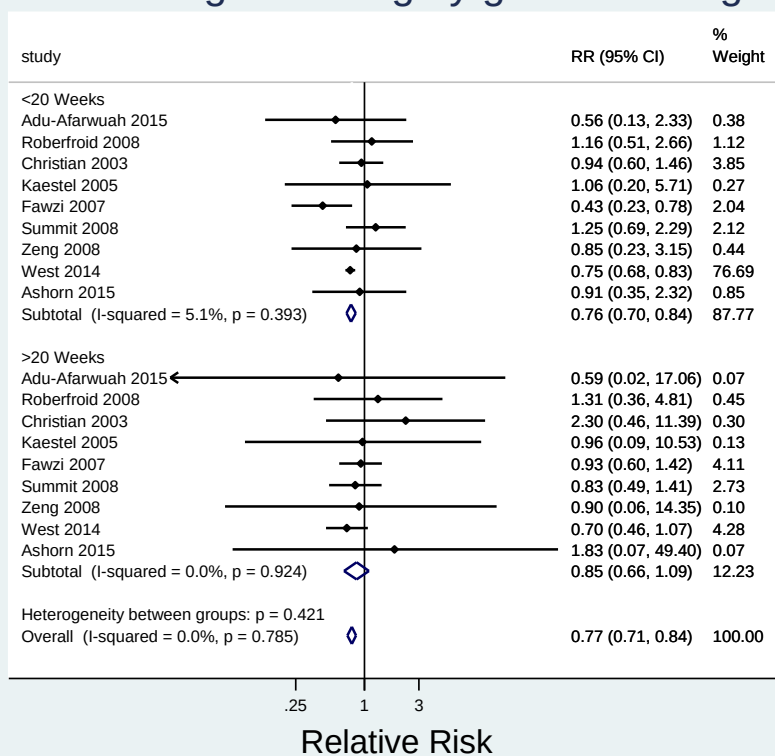
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by trimester at enrollment



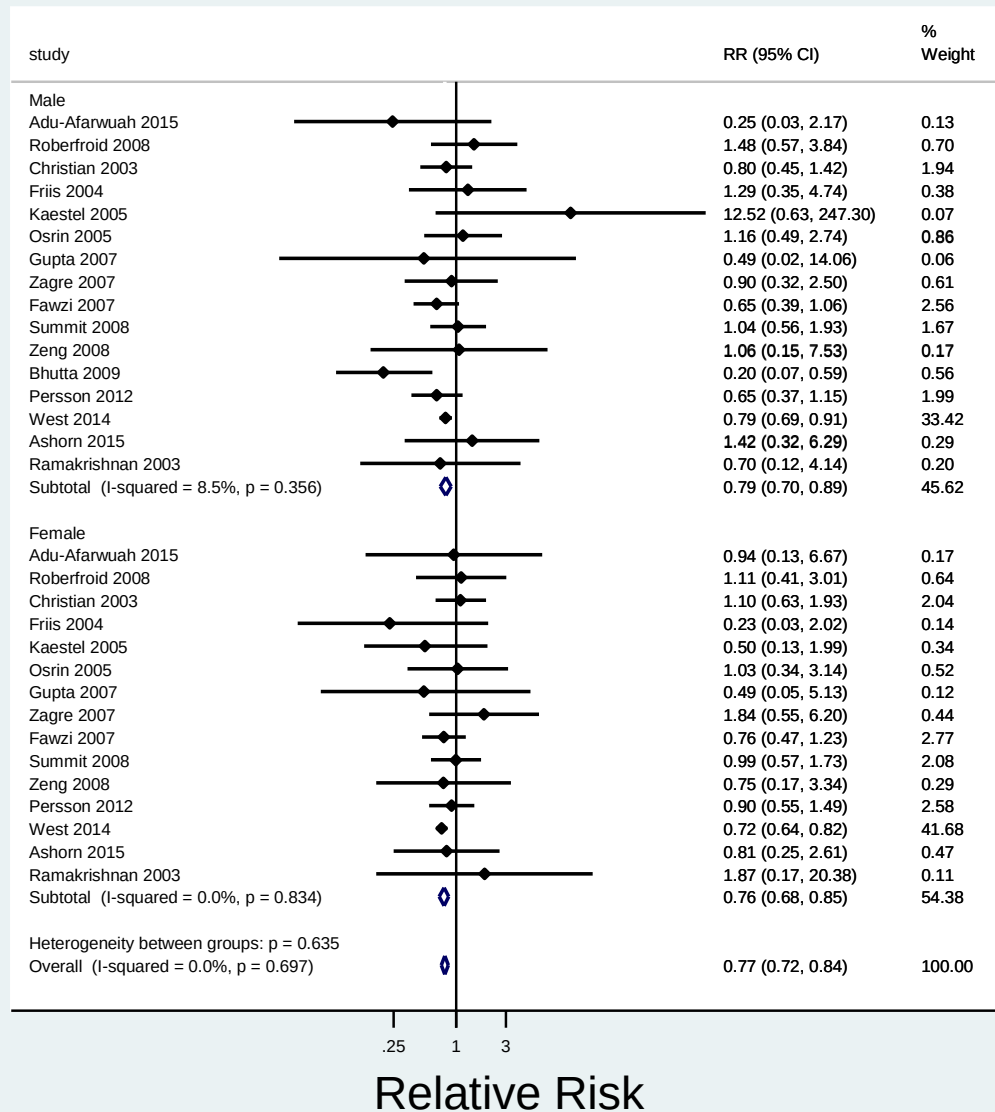
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by gestational age at enrollment



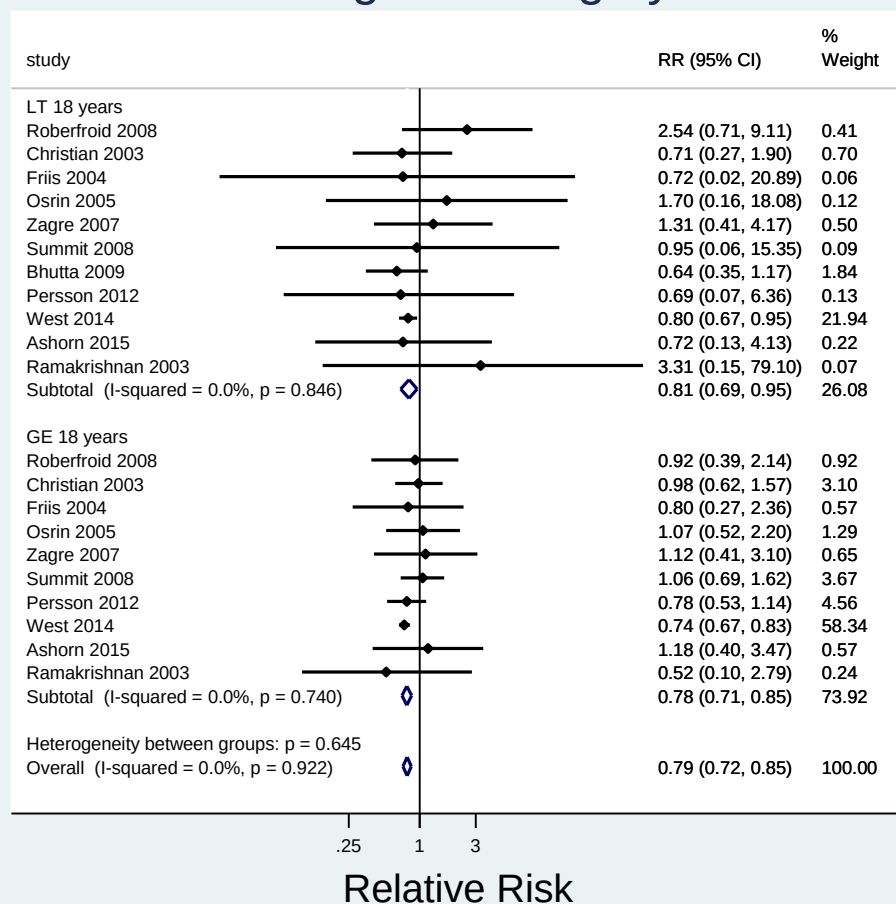
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by sex



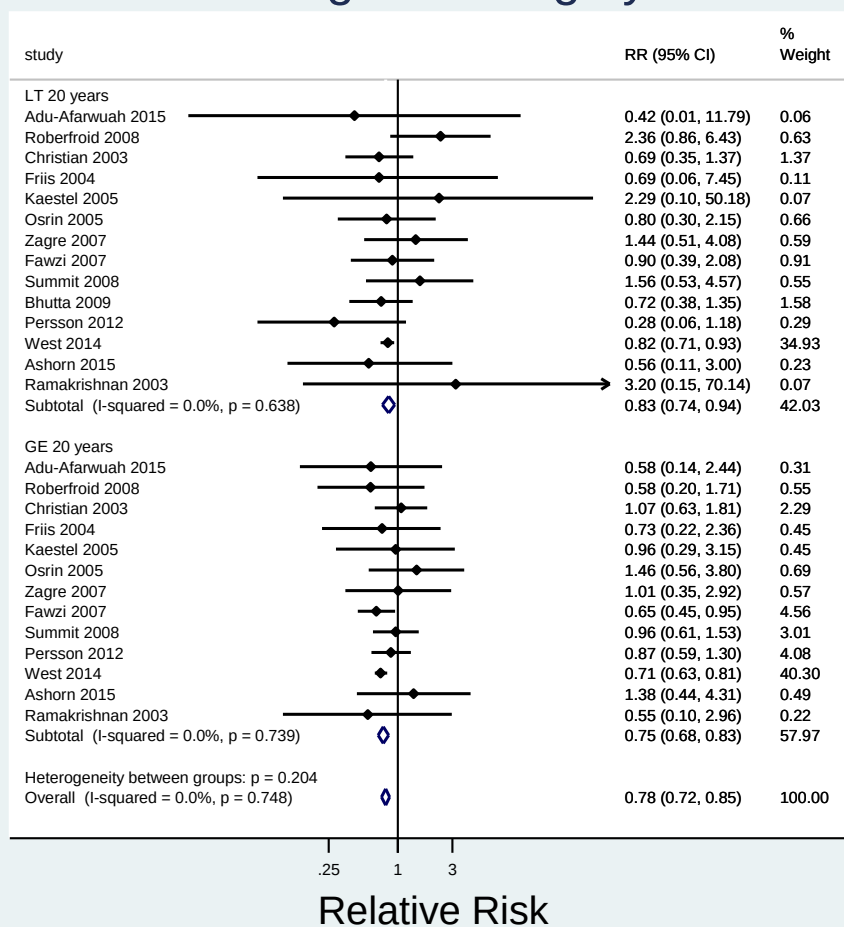
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by maternal age <18



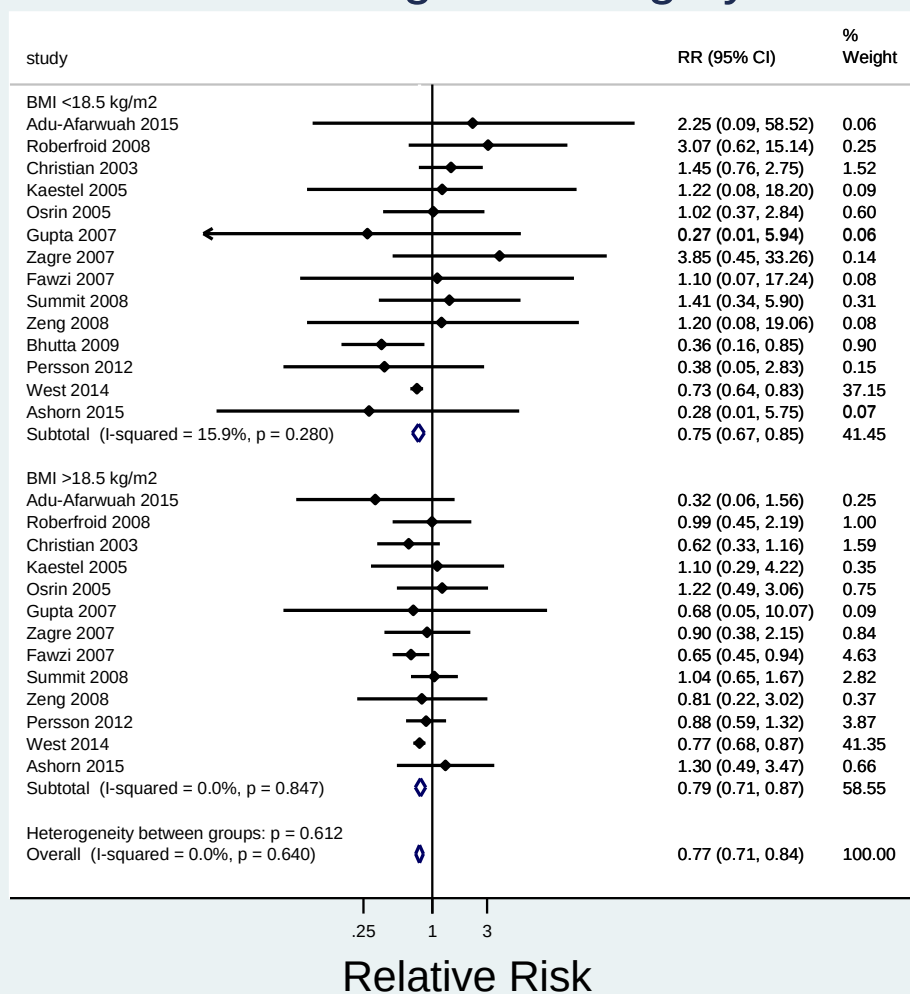
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by maternal age <20



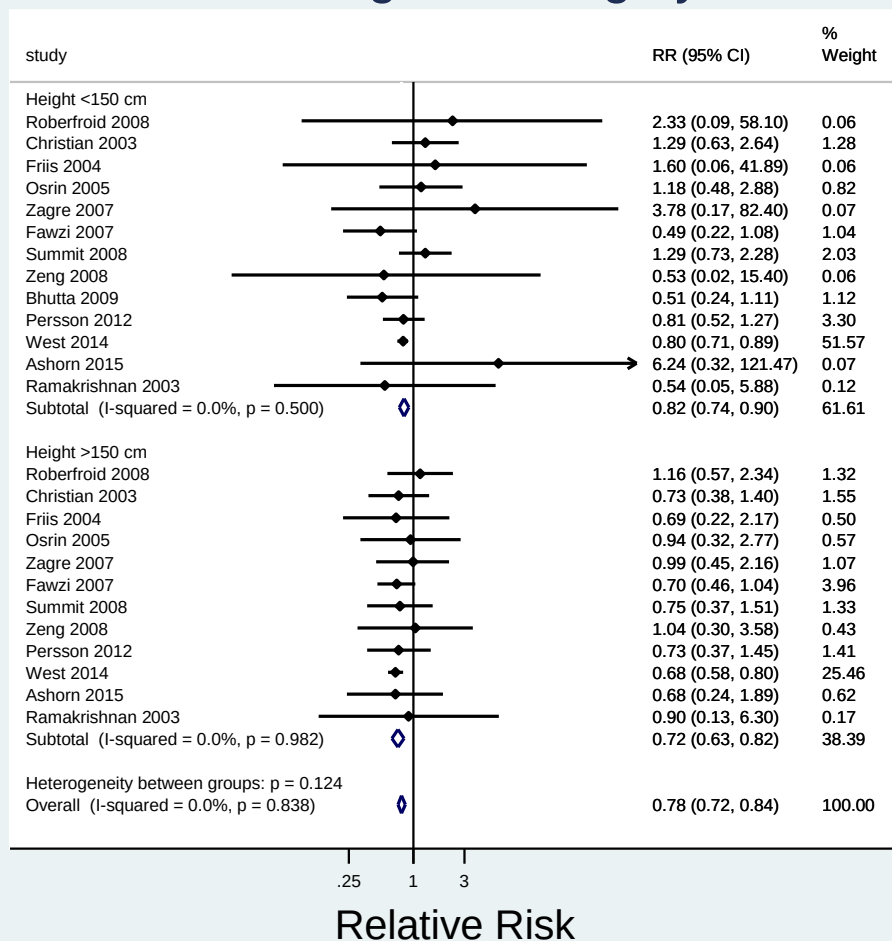
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by maternal BMI



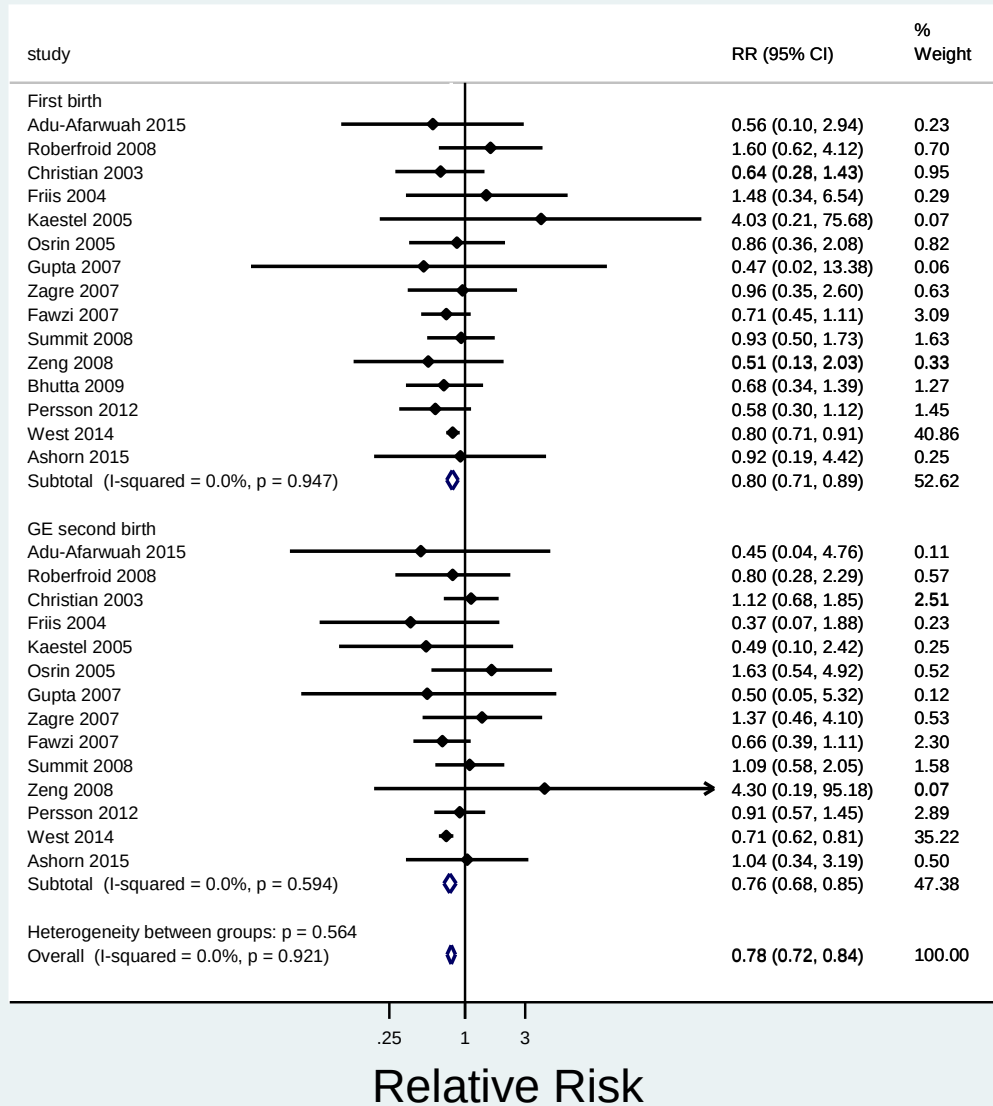
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by maternal height



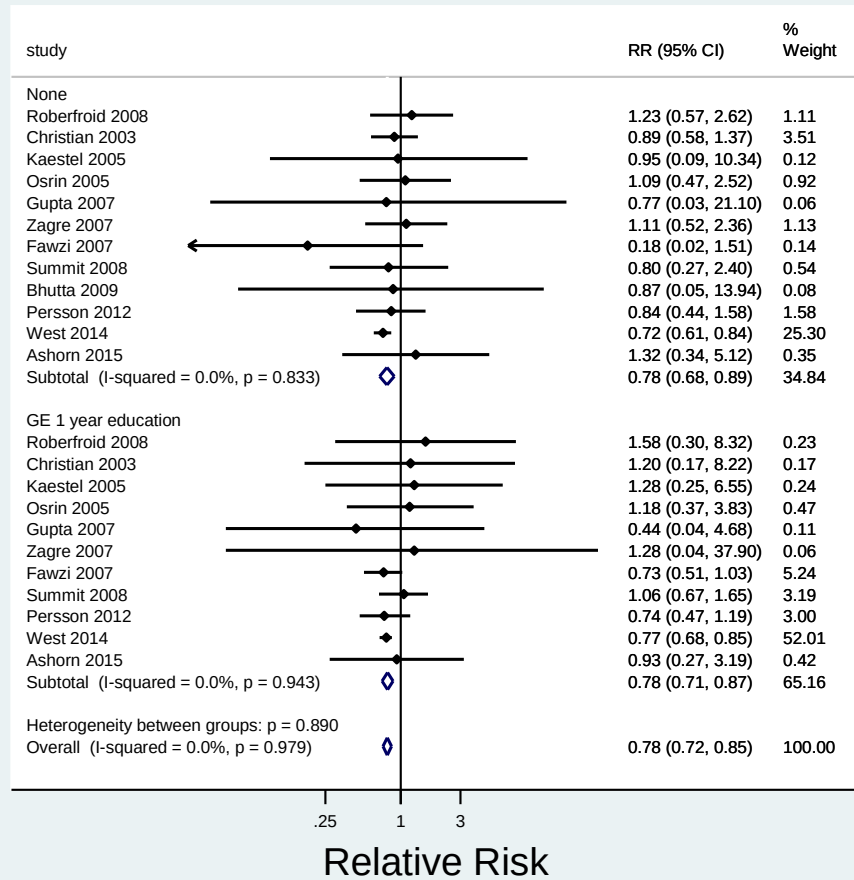
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by parity



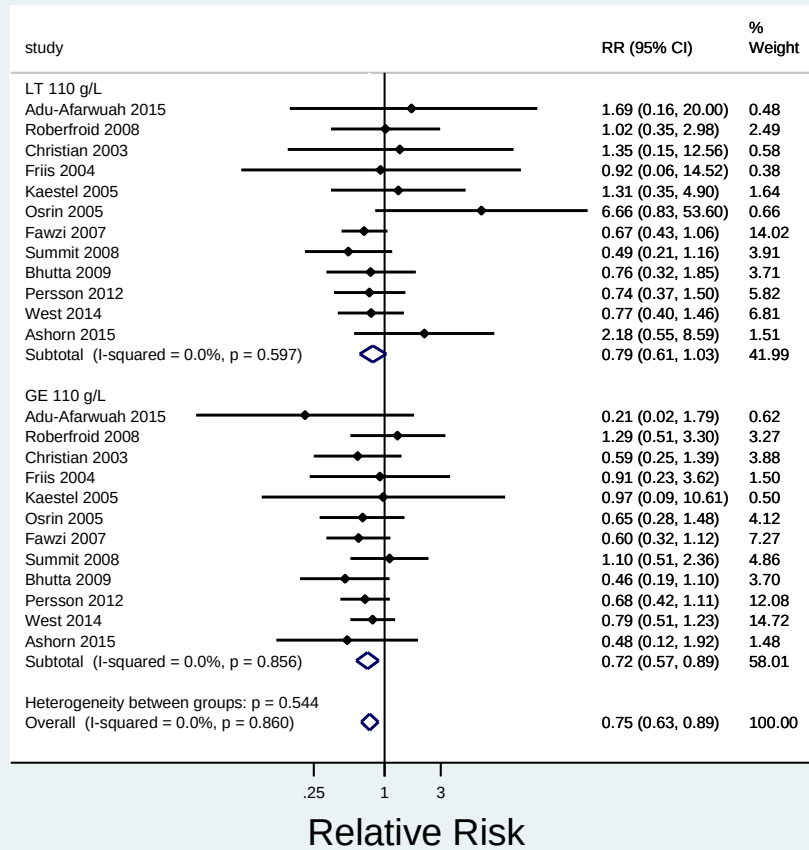
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by maternal education



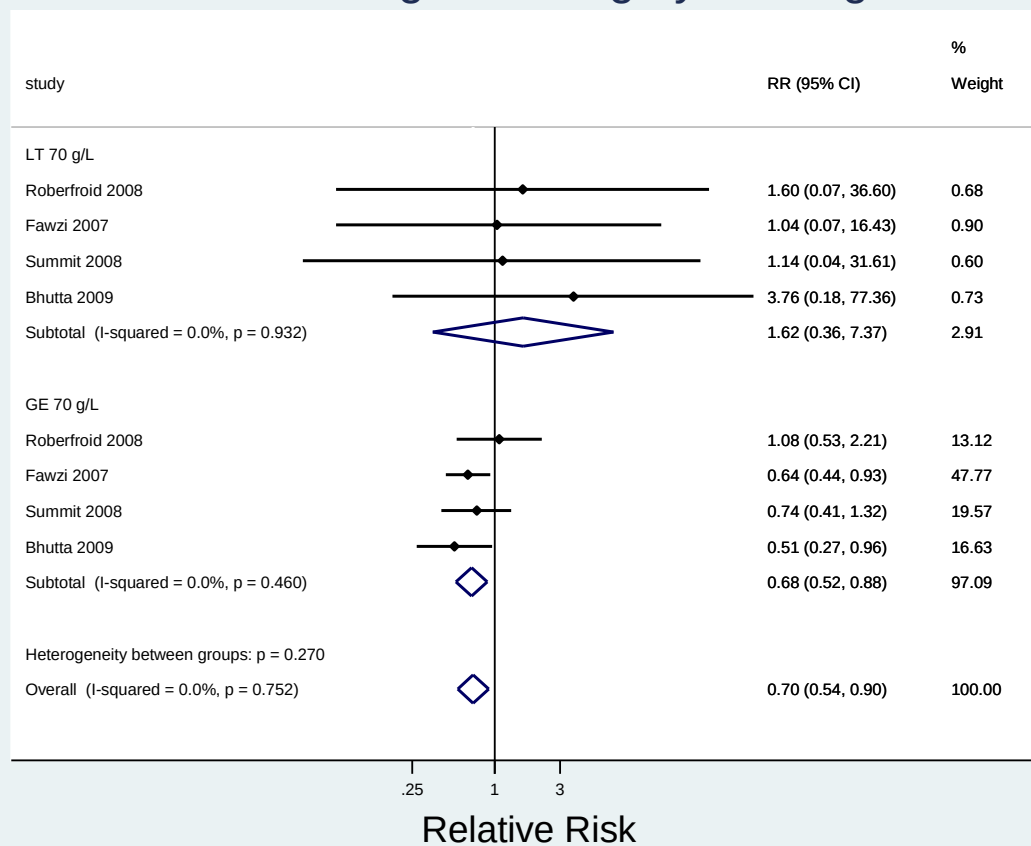
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by hemoglobin <110 g/L



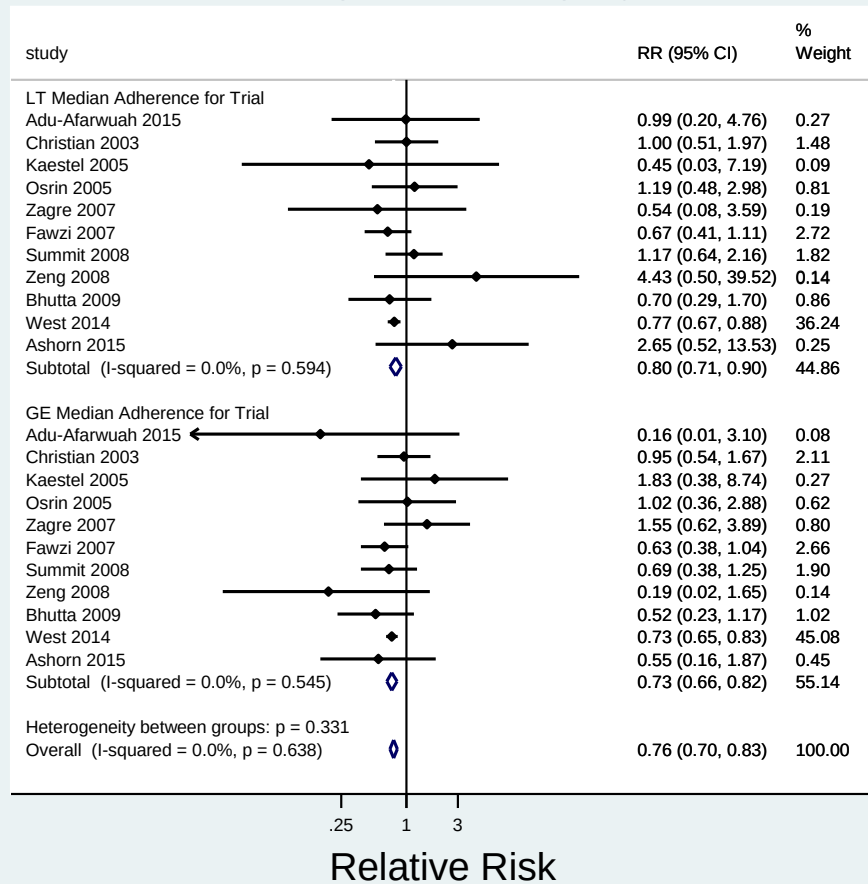
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by hemoglobin <70 g/L



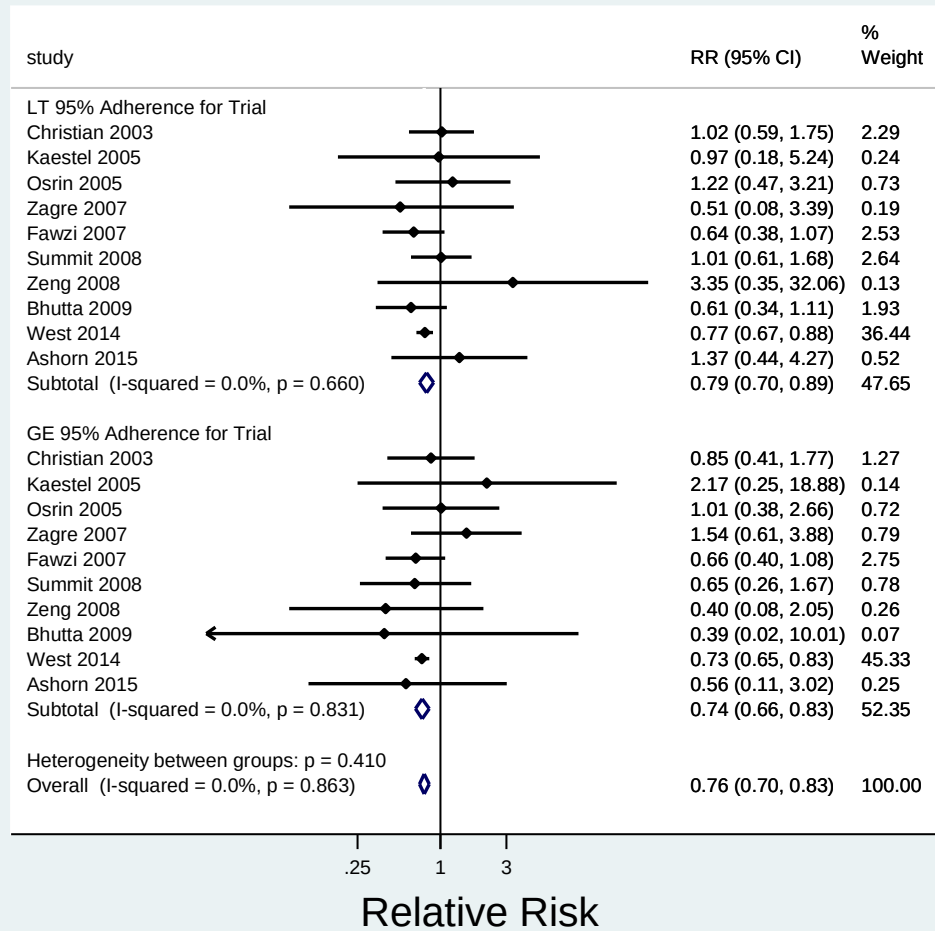
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by median adherence



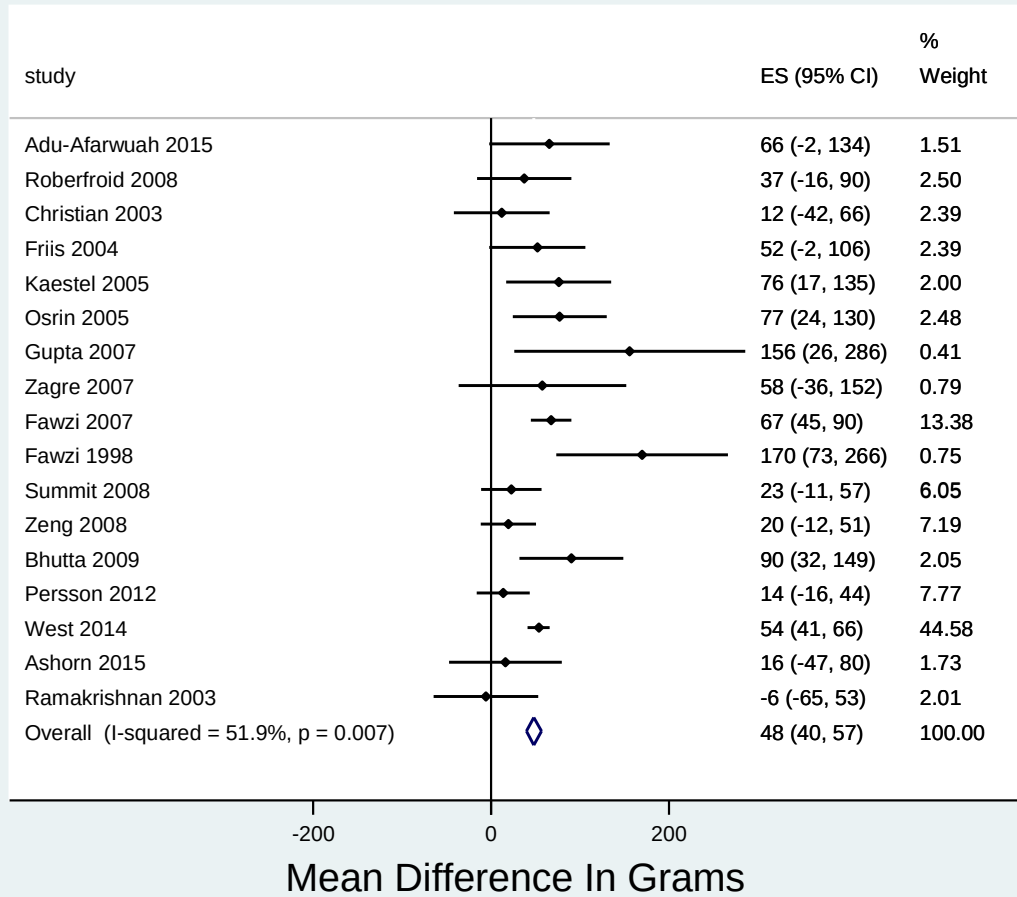
Outcome = Very Low Birthweight (<2000 g)

Relative risk of birthweight <2000g by 95% adherence



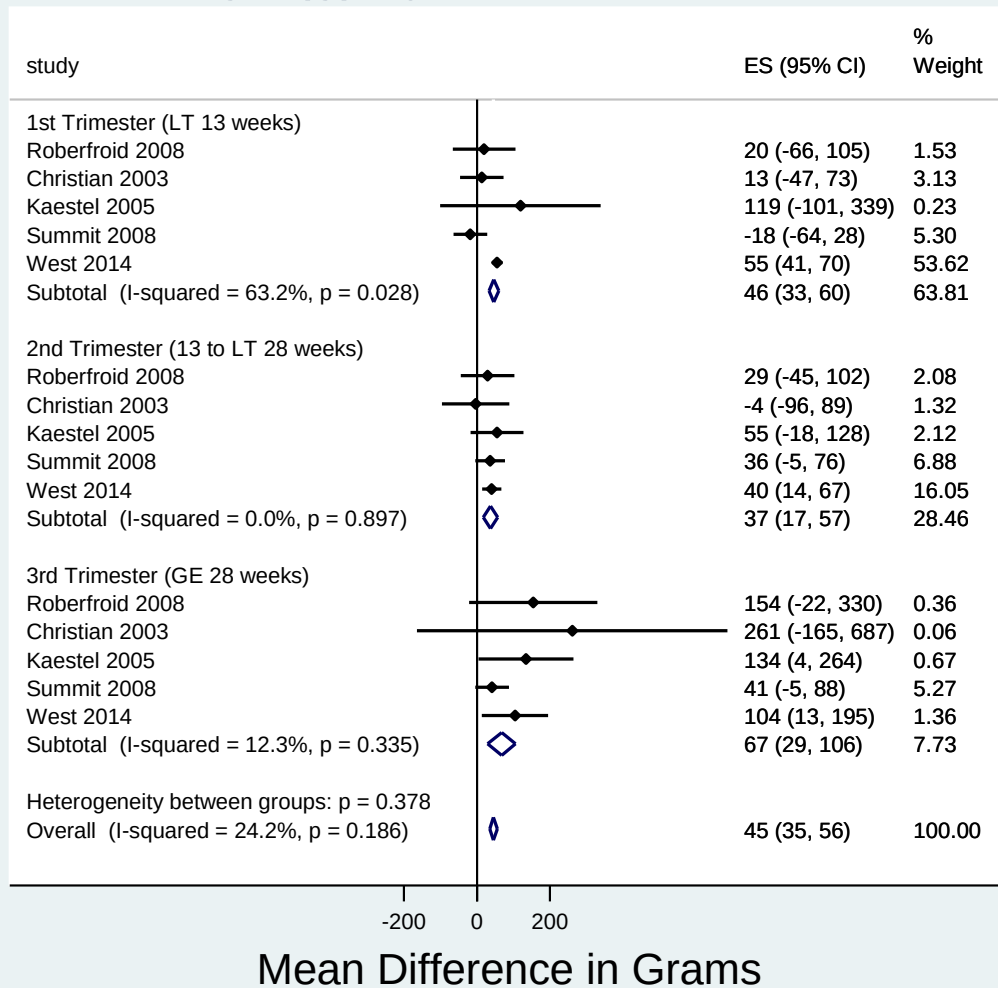
Outcome = Very Low Birthweight (<2000 g)

Birthweight(g) - OVERALL



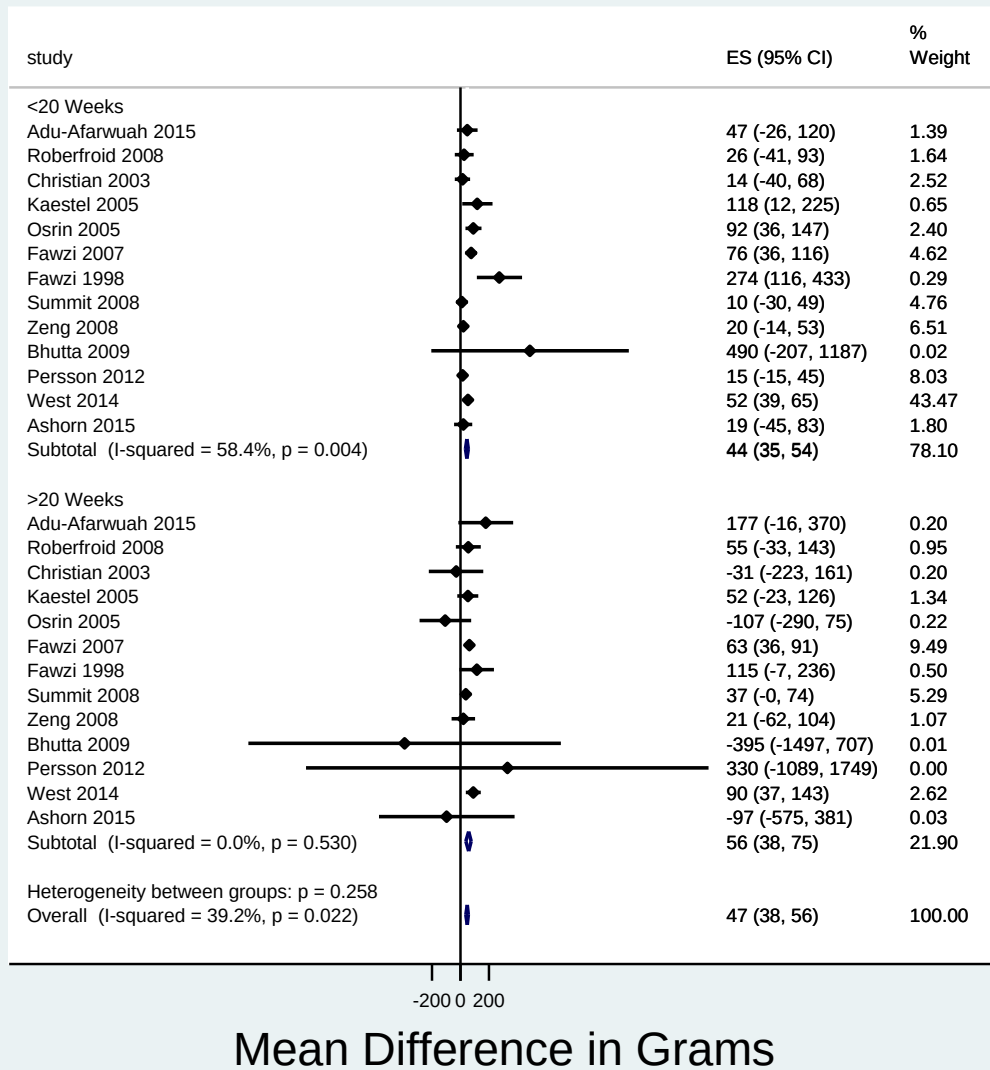
Outcome = Continuous Birthweight (g)

Birthweight(g) by trimester at enrollment



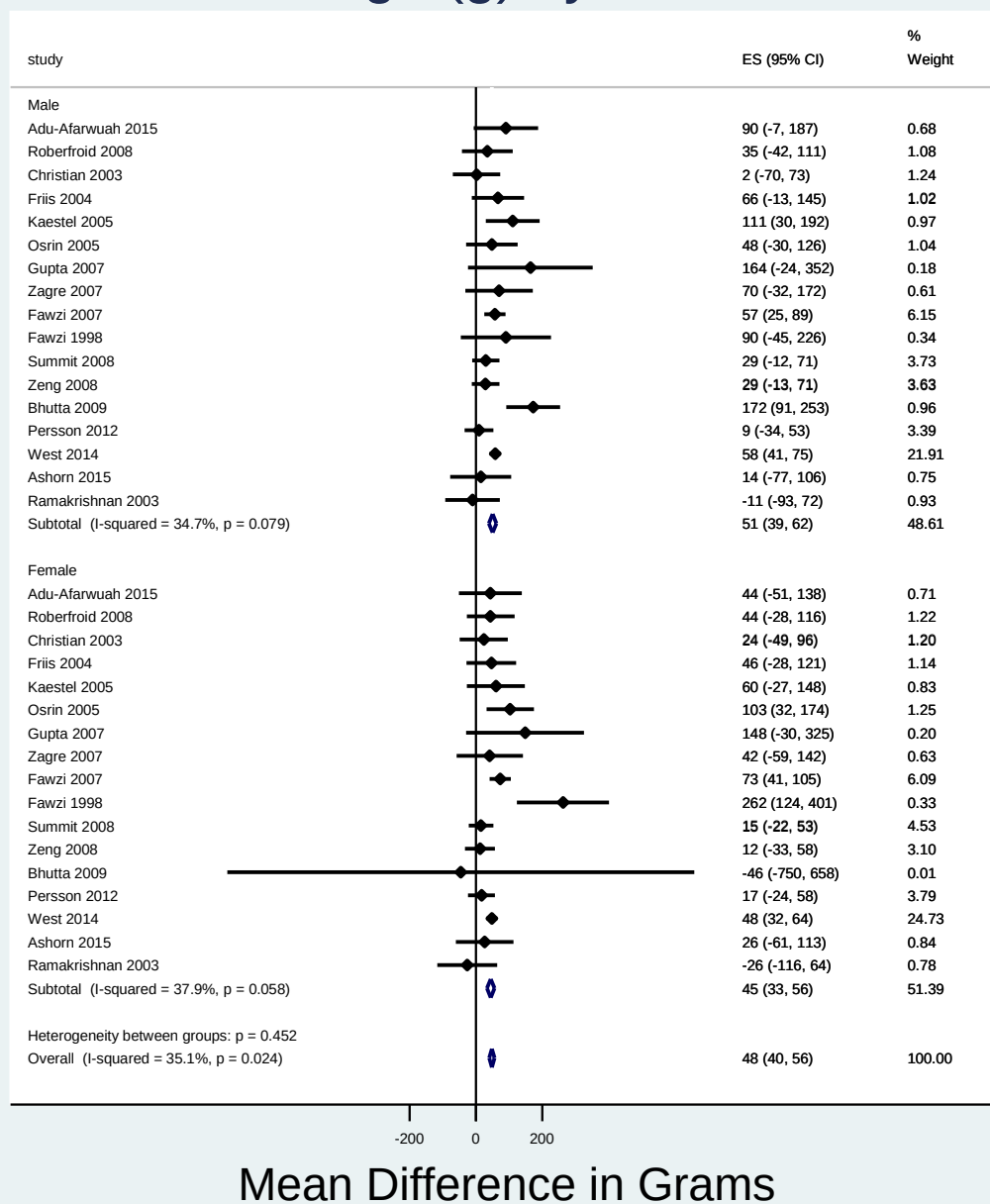
Outcome = Continuous Birthweight (g)

Birthweight(g) by gestational age at enrollment



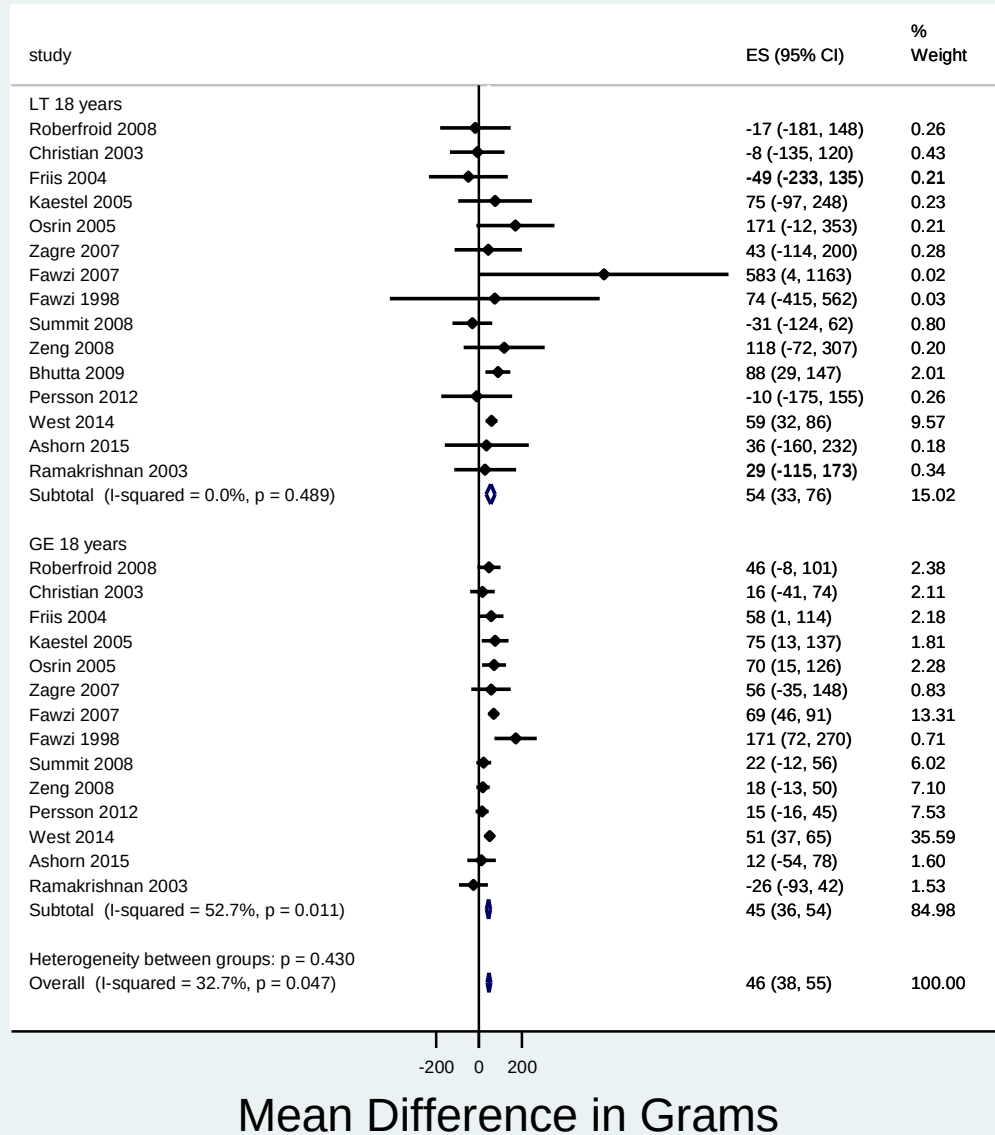
Outcome = Continuous Birthweight (g)

Birthweight(g) by infant sex



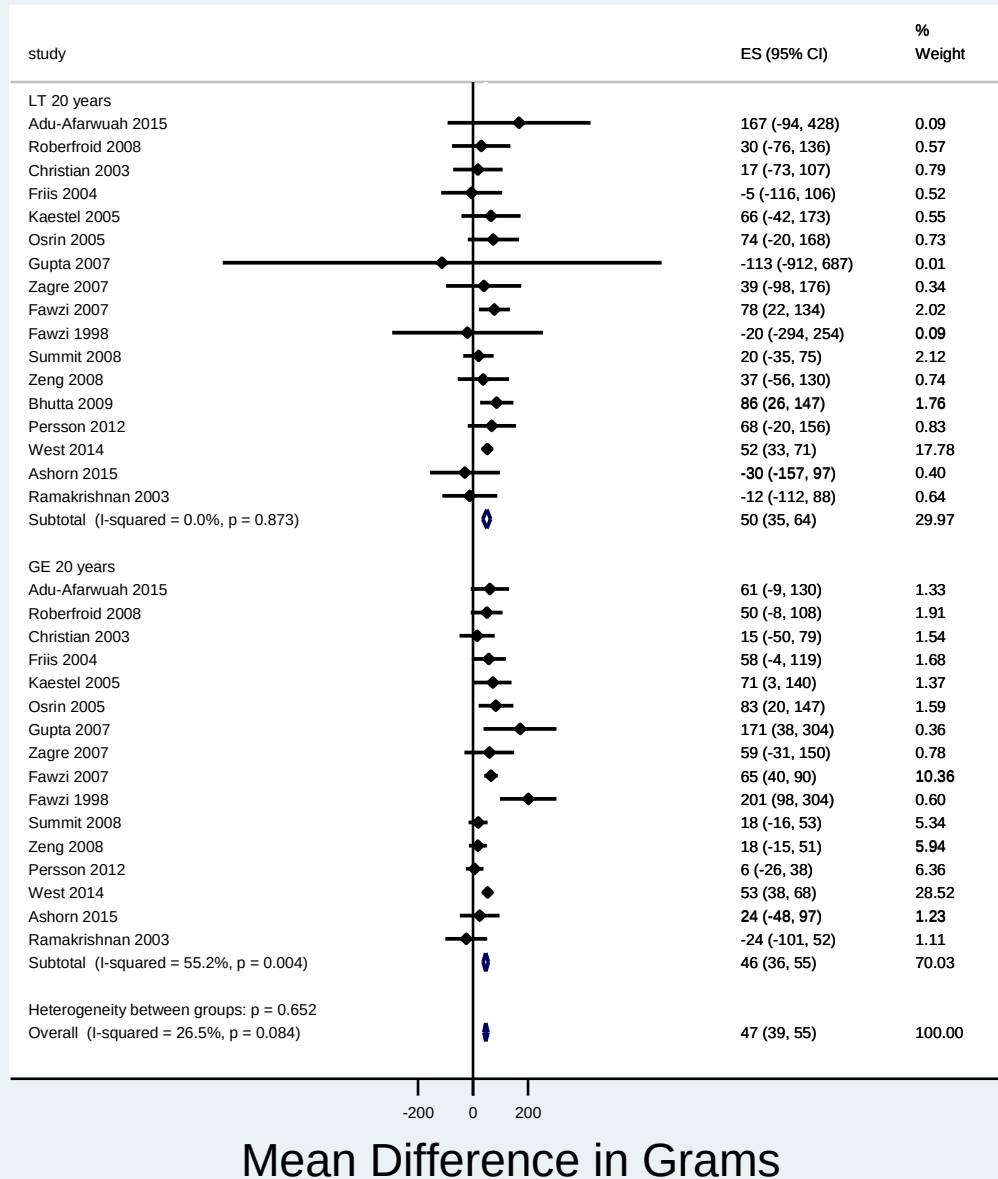
Outcome = Continuous Birthweight (g)

Birthweight(g) by maternal age <18 yrs



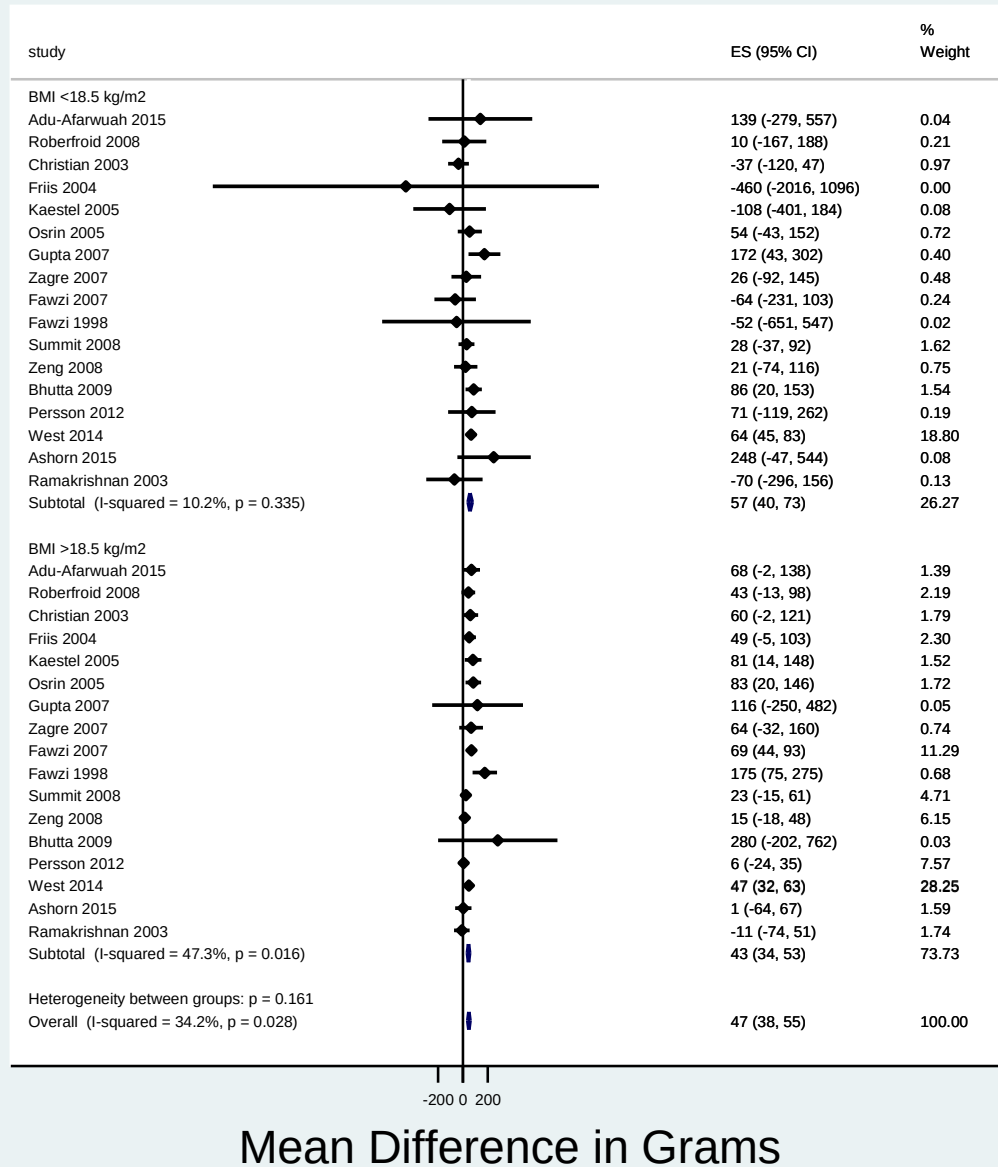
Outcome = Continuous Birthweight (g)

Birthweight(g) by maternal age <20 yrs



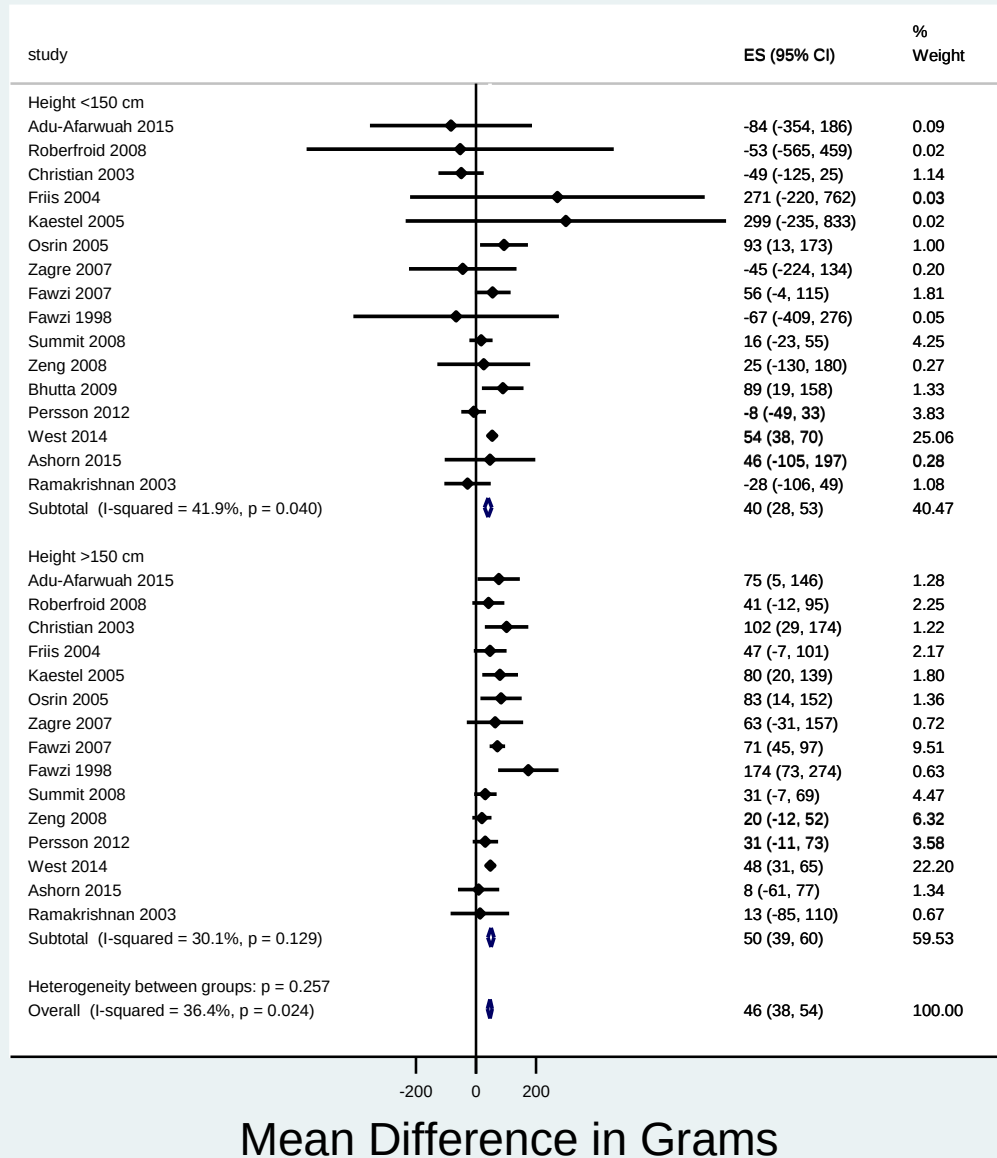
Outcome = Continuous Birthweight (g)

Birthweight(g) by maternal BMI



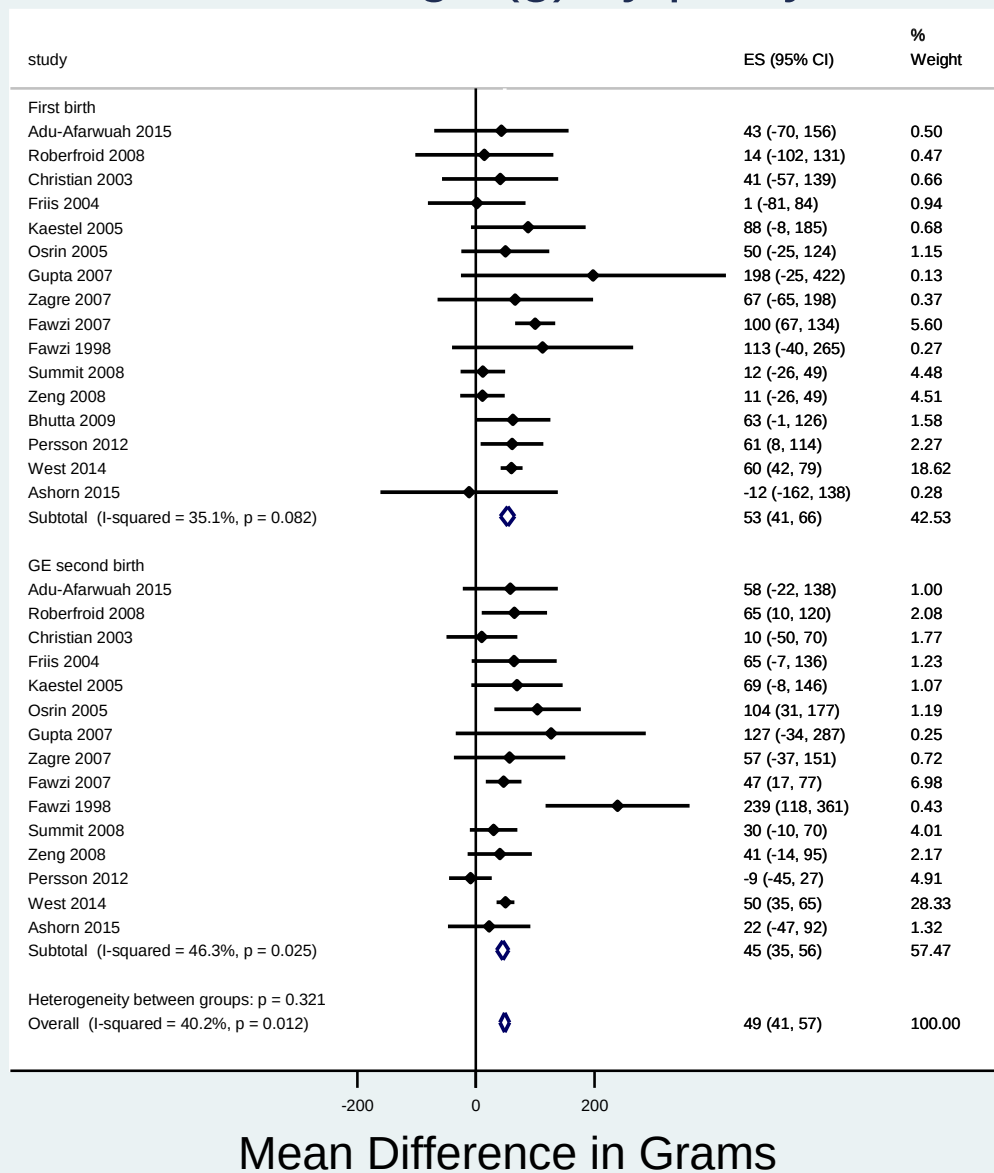
Outcome = Continuous Birthweight (g)

Birthweight(g) by mat height



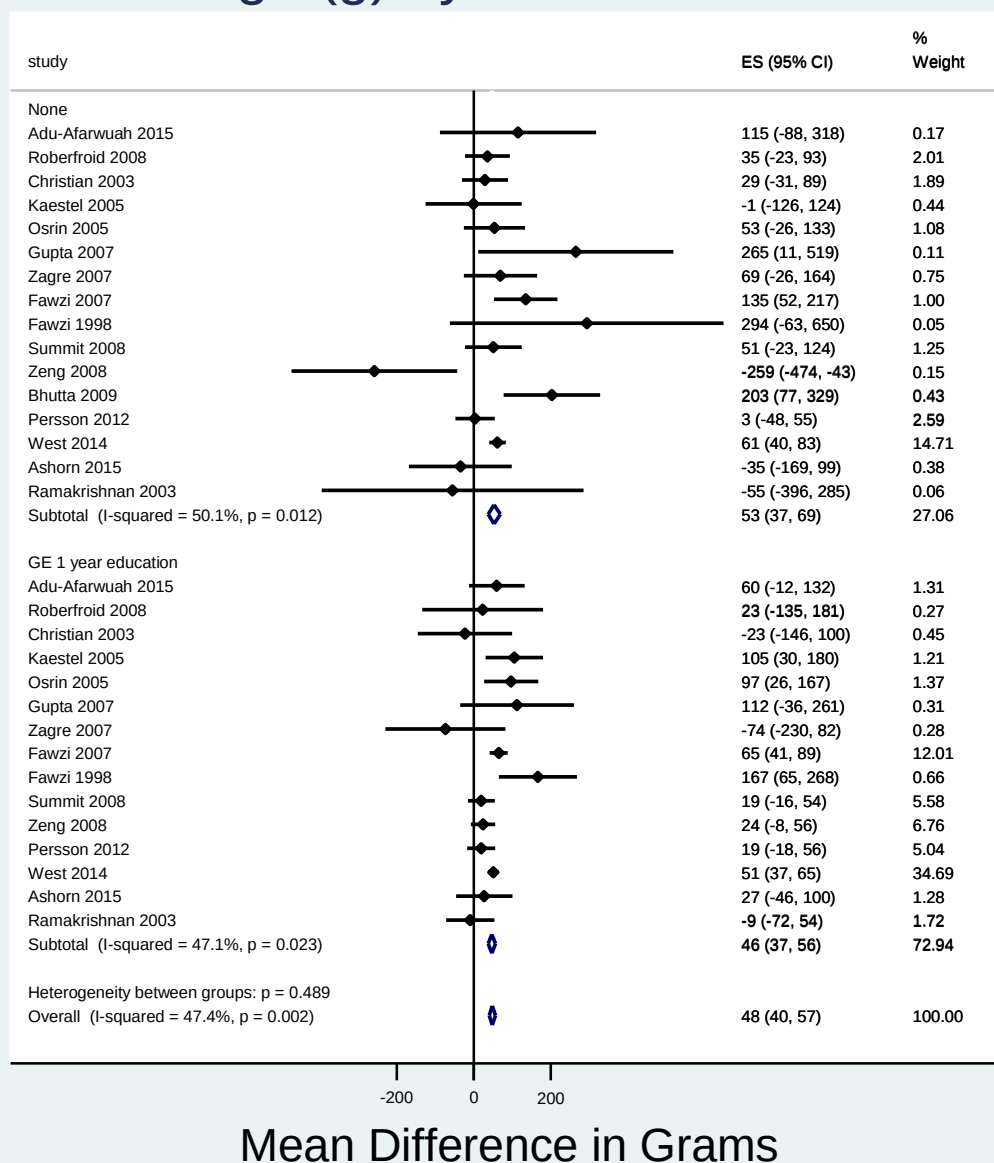
Outcome = Continuous Birthweight (g)

Birthweight(g) by parity



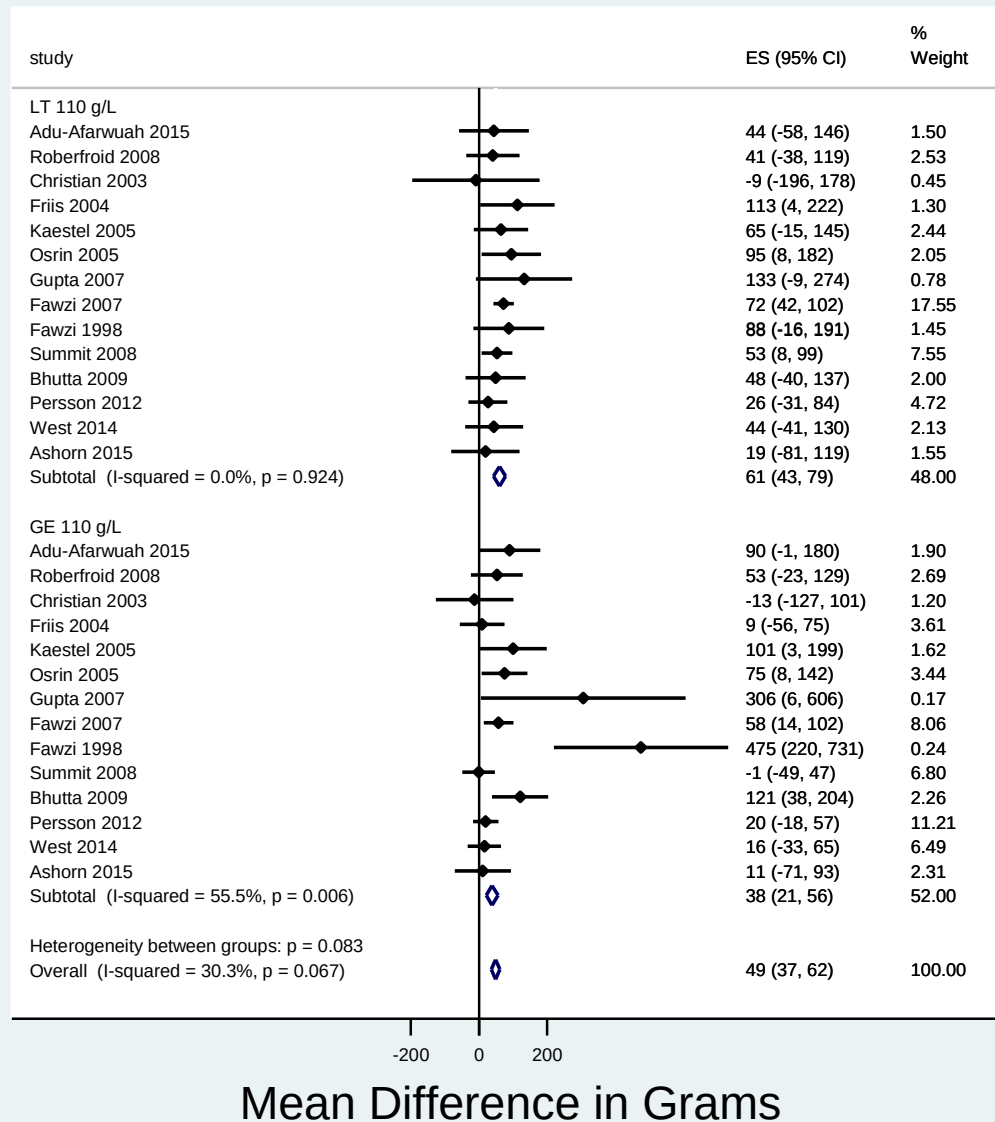
Outcome = Continuous Birthweight (g)

Birthweight(g) by maternal education



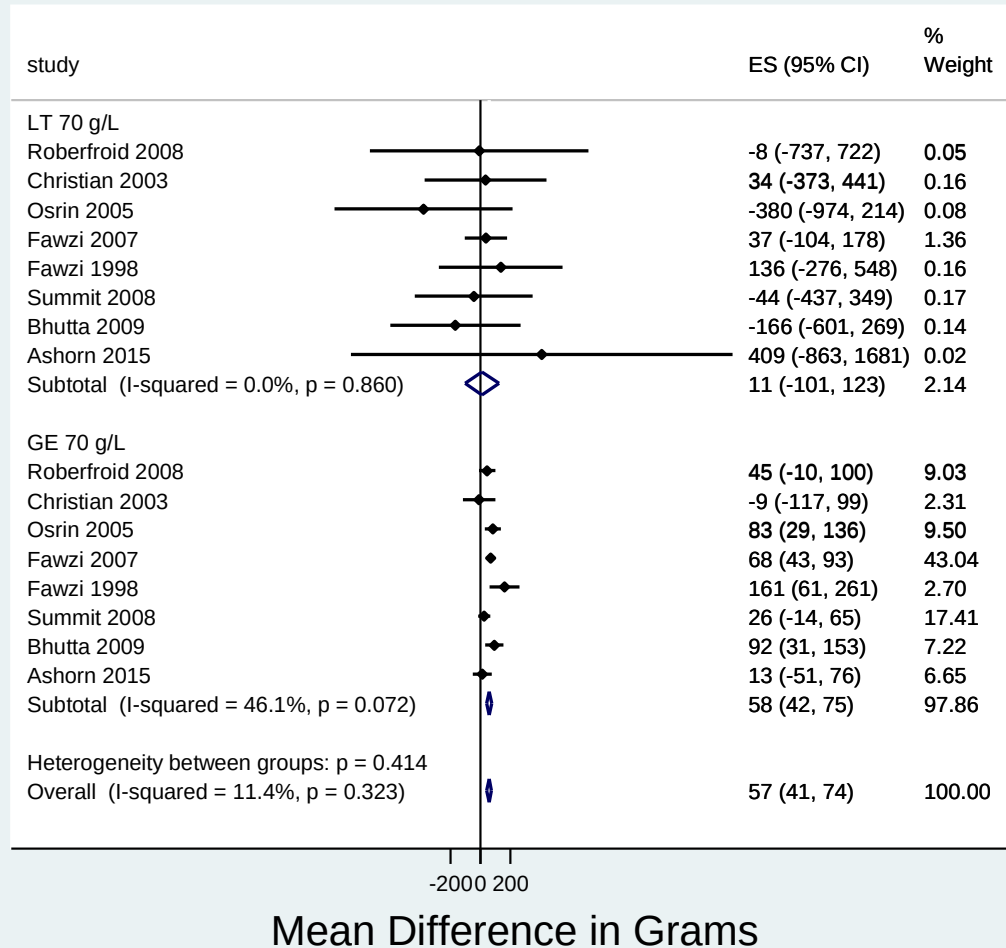
Outcome = Continuous Birthweight (g)

Birthweight(g) by hemoglobin <110 g/L



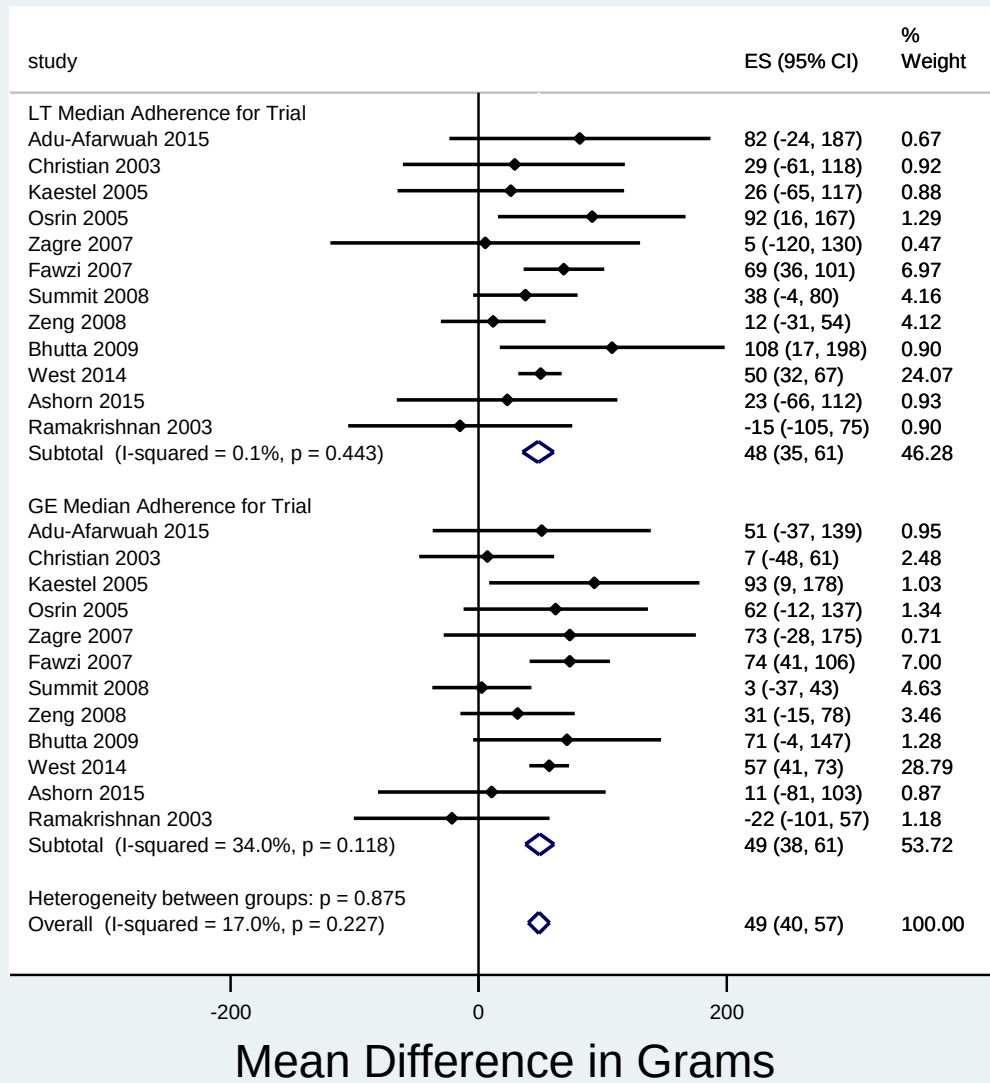
Outcome = Continuous Birthweight (g)

Birthweight(g) by hemoglobin <70 g/L



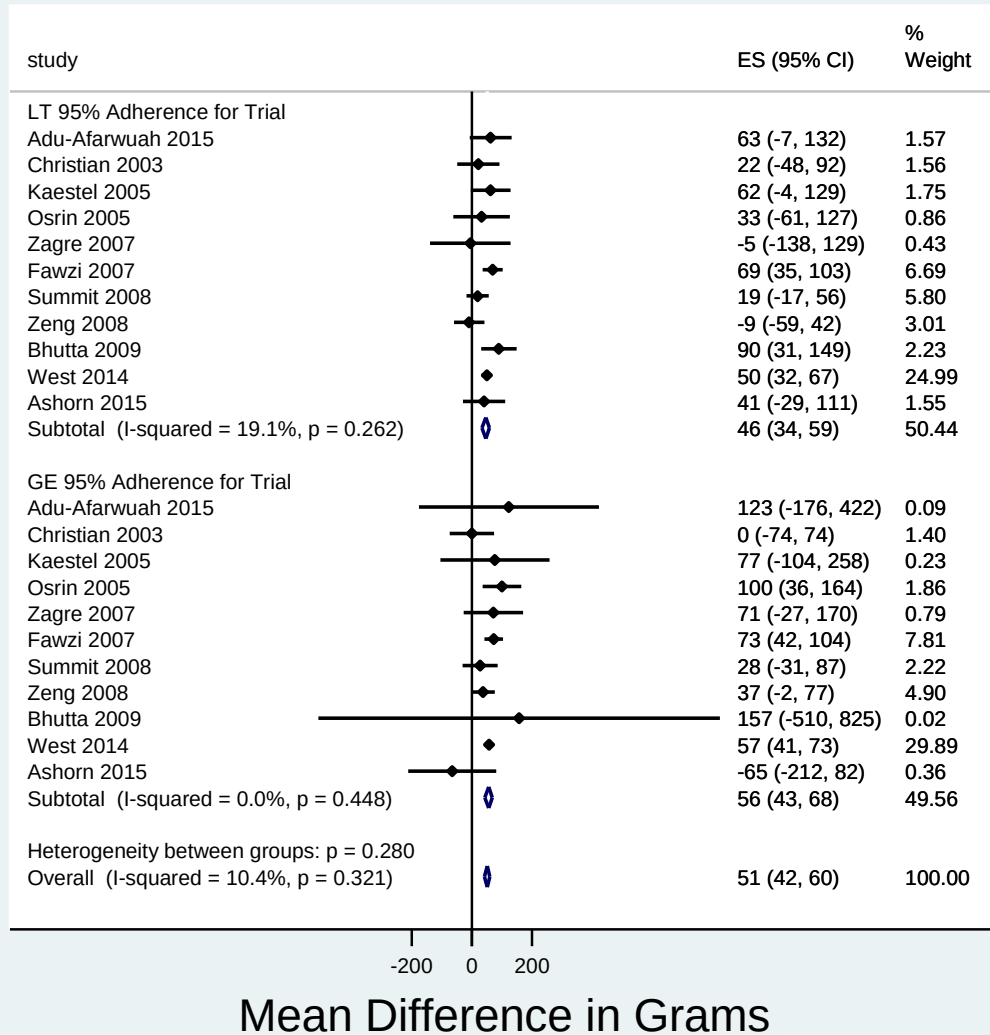
Outcome = Continuous Birthweight (g)

Birthweight(g) by median adherence



Outcome = Continuous Birthweight (g)

Birthweight(g) by 95% adherence



Outcome = Continuous Birthweight (g)

Supplemental Table 5. Random effect models for MMS on stillbirth, neonatal mortality, mortality to six months, and infant mortality stratified by effect modifiers

	Stillbirth		Neonatal Mortality		Mortality to Six		Infant Mortality (≤ 365	
	N ¹	Relative risk (95% CI)	N ¹	Relative risk (95% CI)	N ¹	Relative risk (95% CI)	N ¹	Relative risk (95% CI)
Overall-Random Effects	16	0.97 (0.85-1.11)	12	0.99 (0.89-1.09)	9	0.93 (0.86-1.00)	8	0.97 (0.88-1.06)
Infant Sex								
Male		0.92 (0.82-1.03)		1.06 (0.95-1.17)		0.98 (0.89-1.08)		1.05 (0.93-1.18)
Female	16	0.91 (0.80-1.03)	12	0.85 (0.75-0.96)	9	0.85 (0.75-0.95)	8	0.87 (0.77-0.99)
Gestational Age at Enrollment								
<20 Weeks		1.00 (0.90-1.12)		1.02 (0.83-1.25)		0.96 (0.87-1.05)		0.99 (0.89-1.11)
≥ 20 Weeks	10	0.82 (0.70-0.95)	10	0.94 (0.81-1.10)	7	0.83 (0.67-1.03)	7	0.93 (0.59-1.46)
Maternal adherence to regimen								
< 95% Adherence		0.95 (0.82-1.09)		1.07 (0.91-1.26)		0.97 (0.85-1.11)		1.06 (0.94-1.20)
$\geq 95\%$ Adherence	11	0.92 (0.82-1.04)	9	0.88 (0.77-1.01)	6	0.85 (0.74-0.97)	5	0.85 (0.74-0.98)
Maternal Age								
< 20 years		1.03 (0.76-1.40)		0.95 (0.83-1.10)		0.94 (0.84-1.09)		0.98 (0.86-1.13)
≥ 20 years	16	0.93 (0.82-1.06)	9	1.08 (0.91-1.29)	8	0.95 (0.81-1.12)	8	1.02 (0.83-1.24)
Parity								
First birth		1.01 (0.90-1.14)		0.93 (0.83-1.04)		0.94 (0.84-1.05)		0.97 (0.85-1.10)
Second + birth	15	0.91 (0.79-1.03)	12	1.06 (0.88-1.28)	9	0.92 (0.79-1.07)	8	0.99 (0.82-1.20)
Maternal Underweight at enrollment								
BMI <18.5		0.90 (0.78-1.04)		1.01 (0.86-1.20)		0.96 (0.83-1.12)		0.97 (0.84-1.13)
BMI ≥ 18.5	12	0.97 (0.84-1.13)	11	1.00 (0.87-1.16)	8	0.93 (0.83-1.05)	7	0.98 (0.88-1.09)
Maternal stature								
Height <150 cm		0.98 (0.83-1.15)		0.98 (0.81-1.19)		0.92 (0.83-1.02)		0.98 (0.87-1.11)
Height ≥ 150 cm	14	0.91 (0.77-1.08)	10	0.96 (0.86-1.08)	7	0.91 (0.81-1.02)	6	0.93 (0.81-1.06)
Maternal hemoglobin at enrollment								
Anemic <110 g/L		0.83 (0.66-1.05)		0.94 (0.74-1.18)		0.85 (0.61-1.18)		1.00 (0.73-1.38)
Non-anemic ≥ 110 g/L	13	0.94 (0.79-1.14)	10	0.94 (0.79-1.11)	8	0.93 (0.78-1.11)	7	1.01 (0.79-1.30)
Maternal education								
None		0.97 (0.83-1.14)		1.13 (0.97-1.31)		0.99 (0.86-1.13)		1.02 (0.88-1.18)
≥ 1 year formal education	14	0.92 (0.80-1.05)	12	0.92 (0.83-1.01)	8	0.88 (0.76-1.01)	7	0.95 (0.69-1.30)
Skilled birth attendant								
Yes		0.87 (0.79-0.98)		1.00 (0.91-1.11)		0.98 (0.83-1.17)		1.06 (0.95-1.20)
No	10	1.03 (0.87-1.22)	10	0.96 (0.80-1.15)	7	0.82 (0.74-0.92)	6	0.93 (0.71-1.24)

¹ N Number of studies included in subgroup analysis

Supplemental Table 6. Random effect models for MMS on low birthweight (<2500 g), preterm birth (<37 weeks), small-for-gestational-age (SGA) (<10th percentile Oken and Intergrowth) - stratified by potential effect modifiers

	Low Birthweight		Preterm (<37 weeks)		SGA (Oken)		LGA (Oken)	
	N ¹	Relative risk (95% CI)	N ¹	Relative risk (95% CI)	N ¹	Relative risk (95% CI)	N ¹	Relative risk (95% CI)
Overall-Random Effects	17	0.86 (0.81-0.92)	16	0.93 (0.97-0.98)	16	0.94 (0.90-0.98)	13	1.04 (0.92-1.18)
Infant Sex								
Male				0.96 (0.88-1.04)		0.95 (0.91-0.99)	12	1.11 (0.98-1.25)
Female	17	0.87 (0.83-0.91) 0.87 (0.79-0.95)	15	0.91 (0.85-0.97)	15	0.97 (0.93-1.01)		0.98 (0.86-1.12)
Gestational Age at Enrollment								
<20 Weeks		0.88 (0.81-0.95)	11	0.89 (0.85-0.93)	12	0.97 (0.93-1.02)	8	1.01 (0.84-1.20)
≥20 Weeks	13	0.84 (0.77-0.92)		1.00 (0.94-1.08)		0.90 (0.83-0.97)		1.18 (1.02-1.37)
Maternal adherence to regimen								
< 95% Adherence		0.89 (0.85-0.92)	10	0.95 (0.86-1.04)	11	0.97 (0.92-1.02)	8	1.03 (0.90-1.18)
≥ 95% Adherence	12	0.86 (0.81-0.92)		0.90 (0.85-0.96)		0.93 (0.97-0.98)		1.06 (0.88-1.28)
Maternal Age								
< 20 years		0.90 (0.86-0.93)	15	0.92 (0.87-0.98)	16	0.98 (0.95-1.00)	11	0.98 (0.79-1.22)
≥ 20 years	15	0.86 (0.79-0.93)		0.92 (0.86-0.98)		0.94 (0.90-0.98)		1.06 (0.96-1.18)
Parity								
First birth		0.88 (0.85-0.92)	14	0.93 (0.84-1.03)	15	0.96 (0.91-1.00)	10	0.94 (0.78-1.12)
Second + birth	16	0.88 (0.81-0.96)		0.93 (0.88-0.98)		0.94 (0.89-0.99)		1.11 (0.95-1.30)
Maternal Underweight at enrollment								
BMI <18.5		0.85 (0.74-0.97)	13	0.84 (0.78-0.91)	16	1.00 (0.94-1.06)	8	0.77 (0.57-1.05)
BMI ≥18.5	16	0.88 (0.85-0.92)		0.94 (0.89-0.99)		0.94 (0.90-0.97)		1.08 (0.97-1.21)
Maternal stature								
Height <150 cm		0.90 (0.87-0.93)	15	0.91 (0.86-0.96)	16	0.99 (0.94-1.03)	10	0.93 (0.78-1.12)
Height ≥150 cm	16	0.85 (0.80-0.91)		0.93 (0.87-1.00)		0.92 (0.87-0.97)		1.09 (0.97-1.22)
Maternal hemoglobin at enrollment								
Anemic <110 g/L		0.80 (0.72-0.90)	12	0.98 (0.91-1.05)	13	0.92 (0.86-0.99)	9	1.25 (1.06-1.49)
Non-anemic ≥110 g/L	14	0.90 (0.80-1.00)		0.86 (0.79-0.96)		0.98 (0.92-1.04)		0.99 (0.80-1.22)
Maternal education								
None		0.88 (0.84-0.93)	14	0.92 (0.87-0.98)	15	0.98 (0.93-1.04)	9	1.12 (0.87-1.43)
≥1 year formal education	16	0.84 (0.77-0.93)		0.90 (0.83-0.98)		0.93 (0.88-0.96)		1.00 (0.84-1.20)

¹ N Number of studies included in subgroup analysis

Supplemental Table 7. Effect of MMS on stillbirth, mortality, and birth outcomes stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Trials using same dose iron in MMS and IFA control arms ²		Trials using lower dose iron in MMS arm as compared to IFA control arm ³		p-value for heterogeneity by iron dose
	N ¹	Relative risk (95% CI)	N ¹	Relative risk (95% CI)	
Stillbirth	7	0.90 (0.83-0.97)	8	1.08 (0.90-1.30)	0.07
Early neonatal mortality	6	0.95 (0.97-1.04)	7	1.24 (0.96-1.60)	0.06
Neonatal mortality	4	0.96 (0.88-1.04)	7	1.16 (0.92-1.45)	0.12
Six month mortality	3	0.92 (0.84-1.00)	5	1.14 (0.85-1.52)	0.17
Infant mortality	2	0.96 (0.87-1.05)	5	1.15 (0.89-1.49)	0.19
Low birthweight (<2500g)	8	0.88 (0.85-0.90)	8	0.85 (0.77-0.95)	0.62
Very low birthweight	8	0.77 (0.70-0.84)	8	0.94 (0.70-1.26)	0.18
Very preterm birth	7	0.85 (0.77-0.95)	6	0.97 (0.74-1.25)	0.39
Preterm birth	8	0.92 (0.89-0.96)	7	0.97 (0.88-1.06)	0.30
SGA Oken	8	0.98 (0.96-1.00)	7	0.90 (0.85-0.96)	0.01
SGA Intergrowth	8	0.94 (0.91-0.96)	7	0.92 (0.85-1.00)	0.64
LGA Oken	7	1.05 (0.95-1.17)	5	1.02 (0.80-1.30)	0.82
LGA Intergrowth	7	1.10 (1.02-1.18)	6	1.21 (1.06-1.40)	0.20

¹ Number of studies included in subgroup analysis

² Christian 2003, Friis 2004, Gupta 2007, Fawzi 2007, Fawzi 1998, Ramakrishnan 2003, SUMMIT 2008, West 2014

³ Ashorn 2015, Adu-Afarwuah 2015, Bhutta 2009, Kaestel 2005, Osrin 2005, Roberfroid 2008, Zagre 2007, Zeng 2008

Supplemental Table 8. Effect of MMS on stillbirth stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Same dose iron			Lower dose iron in MMS		
	Stillbirth			Stillbirth		
	N ¹	Relative risk (95% CI)	p value heterogeneity	N ¹	Relative risk (95% CI)	p value heterogeneity
Overall-Fixed Effects	7	0.90 (0.83-0.97)	-	8	1.08 (0.90-1.30)	-
Overall-Random Effects	7	0.91 (0.79-1.04)	-	8	1.04 (0.79-1.38)	-
Infant Sex						
Male	7	0.90 (0.79-1.02)	0.99	8	1.07 (0.76-1.51)	0.75
Female		0.90 (0.78-1.03)			0.99 (0.68-1.43)	
Gestational Age at Enrollment						
<20 Weeks	5	0.94 (0.86-1.04)	0.10	5	1.19 (0.94-1.52)	0.43
≥20 Weeks		0.80 (0.69-0.98)			0.95 (0.56-1.59)	
Maternal adherence to regimen						
< 95% Adherence	4	0.89 (0.80-0.99)	0.86	7	1.06 (0.83-1.34)	0.90
≥ 95% Adherence		0.91 (0.80-1.03)			1.03 (0.75-1.27)	
Maternal Age						
< 20 years	7	0.95 (0.81-1.12)	0.14	8	1.44 (0.84-2.49)	0.22
≥ 20 years		0.88 (0.80-0.96)			1.01 (0.82-1.23)	
Parity						
First birth	6	0.97 (0.85-1.10)	0.14	8	1.38 (1.00-1.90)	0.24
Second + birth		0.86 (0.78-0.95)			0.96 (0.76-1.22)	
Maternal Underweight at enrollment						
BMI <18.5	5	0.86 (0.74-1.00)	0.41	7	1.33 (0.85-2.08)	0.35
BMI ≥18.5		0.93 (0.84-1.03)			1.05 (0.86-1.29)	
Maternal stature						
Height <150 cm	6	0.97 (0.86-1.09)	0.14	7	0.94 (0.62-1.45)	0.64
Height ≥150 cm		0.85 (0.76-0.96)			1.06 (0.85-1.31)	
Maternal hemoglobin at enrollment						
Anemic <110 g/L	6	0.72 (0.59-0.89)	0.10	6	0.92 (0.64-1.31)	0.76
Non-anemic ≥110 g/L		0.93 (0.75-1.16)			0.99 (0.69-1.42)	
Maternal education						
None	5	0.91 (0.77-1.06)	0.85	8	1.10 (0.85-1.44)	0.83
≥1 year formal education		0.89 (0.81-0.98)			1.15 (0.87-1.53)	
Skilled birth attendant						
Yes	4	0.85 (0.76-0.96)	0.12	5	1.29 (0.90-1.86)	0.36
No		0.98 (0.85-1.15)			1.02 (0.72-1.45)	

¹ N Number of studies included in subgroup analysis

Supplemental Table 9. Effect of MMS on early neonatal mortality stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Same dose iron			Lower dose iron in MMS		
	Early neonatal mortality			Early neonatal mortality		
	N ¹	Relative risk (95% CI)	p value heterogeneity	N ¹	Relative risk (95% CI)	p value heterogeneity
Overall-Fixed Effects	6	0.95 (0.87-1.04)	-	7	1.24 (0.96-1.60)	-
Overall-Random Effects	6	0.95 (0.87-1.04)	-	7	1.29 (0.92-1.81)	-
Infant Sex						
Male						
Female	6	1.03 (0.91-1.17) 0.83 (0.72-0.96)	0.03	6	1.12 (0.77-1.63) 1.24 (0.78-1.96)	0.74
Gestational Age at Enrollment						
<20 Weeks						
≥20 Weeks	4	0.92 (0.83-1.02) 1.05 (0.88-1.27)	0.21	6	1.52 (1.07-2.17) 0.80 (0.51-1.26)	0.03
Maternal adherence to regimen						
< 95% Adherence						
≥ 95% Adherence	5	1.02 (0.90-1.16) 0.85 (0.73-0.99)	0.07	5	1.44 (1.03-2.01) 1.17 (0.68-2.01)	0.52
Maternal Age						
< 20 years						
≥ 20 years	6	0.99 (0.85-1.16) 0.93 (0.83-1.05)	0.53	6	1.09 (0.69-1.72) 1.35 (0.98-1.85)	0.45
Parity						
First birth						
Second + birth	5	0.93 (0.82-1.06) 0.98 (0.85-1.12)	0.59	7	0.99 (0.65-1.52) 1.49 (1.05-2.12)	0.15
Maternal Underweight at enrollment						
BMI <18.5						
BMI ≥18.5	5	0.92 (0.77-1.11) 0.95 (0.85-1.06)	0.32	6	1.51 (0.67-3.41) 1.21 (0.91-1.61)	0.61
Maternal stature						
Height <150 cm						
Height ≥150 cm	4	0.96 (0.84-1.09) 0.92 (0.80-1.06)	0.72	6	1.49 (0.74-2.99) 1.11 (0.81-1.53)	0.46
Maternal hemoglobin at enrollment						
Anemic <110 g/L						
Non-anemic ≥110 g/L	4	0.85 (0.68-1.06) 0.92 (0.72-1.18)	0.64	6	1.29 (0.87-1.91) 1.15 (0.76-1.72)	0.68
Maternal education						
None						
≥1 year formal education	5	1.08 (0.89-1.31) 0.85 (0.70-1.02)	0.14	6	1.34 (0.90-2.00) 1.14 (0.75-1.74)	0.58
Skilled birth attendant						
Yes						
No	4	1.01 (0.90-1.14) 0.86 (0.74-1.00)	0.09	4	1.59 (0.95-2.67) 0.93 (0.63-1.36)	0.10

¹ N Number of studies included in subgroup analysis

Supplemental Table 10. Effect of MMS on neonatal mortality stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Same dose iron			Lower dose iron in MMS		
	Neonatal mortality			Neonatal mortality		
	N ¹	Relative risk (95% CI)	p value heterogeneity	N ¹	Relative risk (95% CI)	p value heterogeneity
Overall-Fixed Effects	4	0.96 (0.88-1.04)	-	7	1.16 (0.92-1.45)	-
Overall-Random Effects	4	0.96 (0.85-1.08)	-	7	1.16 (0.92-1.45)	-
Infant Sex						
Male	4	1.05 (0.94-1.18)	0.01	7	1.16 (0.84-1.62)	0.50
Female		0.84 (0.73-0.95)			0.97 (0.66-1.44)	
Gestational Age at Enrollment						
<20 Weeks	4	0.96 (0.87-1.06)	0.98	6	1.29 (0.95-1.76)	0.11
≥20 Weeks		0.96 (0.81-1.13)			0.84 (0.54-1.30)	
Maternal adherence to regimen						
< 95% Adherence	4	1.01 (0.90-1.14)	0.11	5	1.34 (1.00-1.79)	0.38
≥ 95% Adherence		0.87 (0.76-1.00)			1.04 (0.66-1.66)	
Maternal Age						
< 20 years	3	0.96 (0.83-1.11)	0.77	6	0.88 (0.47-1.63)	0.32
≥ 20 years		0.98 (0.88-1.10)			1.23 (0.96-1.57)	
Parity						
First birth	4	0.92 (0.82-1.04)	0.46	7	0.95 (0.65-1.38)	0.13
Second + birth		0.98 (0.87-1.11)			1.38 (1.02-1.87)	
Maternal Underweight at enrollment						
BMI <18.5	4	1.00 (0.84-1.19)	0.45	6	1.14 (0.63-2.06)	0.96
BMI ≥18.5		0.93 (0.83-1.03)			1.16 (0.90-1.51)	
Maternal stature						
Height <150 cm	4	0.95 (0.85-1.07)	0.92	5	1.15 (0.65-2.02)	0.94
Height ≥150 cm		0.94 (0.83-1.07)			1.12 (0.86-1.46)	
Maternal hemoglobin at enrollment						
Anemic <110 g/L	4	0.78 (0.64-0.96)	0.43	6	1.20 (0.85-1.71)	0.70
Non-anemic ≥110 g/L		0.88 (0.71-1.10)			1.09 (0.77-1.55)	
Maternal education						
None	4	1.09 (0.91-1.30)	0.09	7	1.25 (0.89-1.76)	0.49
≥1 year formal education		0.91 (0.83-1.01)			1.06 (0.76-1.47)	
Skilled birth attendant						
Yes	4	1.02 (0.91-1.14)	0.09	5	1.48 (0.95-2.29)	0.08
No		0.87 (0.76-1.00)			0.90 (0.65-1.25)	

¹ N Number of studies included in subgroup analysis

Supplemental Table 11. Effect of MMS on six month mortality stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Same dose iron			Lower dose iron in MMS		
	Six month mortality			Six month mortality		
	N ¹	Relative risk (95% CI)	p value heterogeneity	N ¹	Relative risk (95% CI)	p value heterogeneity
Overall-Fixed Effects	3	0.92 (0.84-1.00)	-	5	1.14 (0.85-1.52)	-
Overall-Random Effects	3	0.92 (0.79-1.08)	-	5	1.14 (0.85-1.52)	-
Infant Sex						
Male	3	0.97 (0.88-1.08)	0.07	5	1.23 (0.81-1.86)	0.61
Female		0.84 (0.74-0.95)			1.05 (0.67-1.64)	
Gestational Age at Enrollment						
<20 Weeks	3	0.94 (0.86-1.04)	0.13	4	1.18 (0.81-1.71)	0.43
≥20 Weeks		0.82 (0.69-0.96)			0.83 (0.38-1.80)	
Maternal adherence to regimen						
< 95% Adherence	3	0.98 (0.88-1.09)	0.08	3	0.94 (0.55-1.62)	0.52
≥ 95% Adherence		0.82 (0.72-0.95)			1.20 (0.74-1.92)	
Maternal Age						
< 20 years	3	0.94 (0.82-1.07)	0.82	4	1.23 (0.72-2.10)	0.76
≥ 20 years		0.92 (0.83-1.02)			1.11 (0.77-1.59)	
Parity						
First birth	3	0.92 (0.82-1.04)	0.86	5	1.17 (0.73-1.85)	0.82
Second + birth		0.91 (0.81-1.02)			1.09 (0.74-1.60)	
Maternal Underweight at enrollment						
BMI <18.5	3	0.95 (0.82-1.11)	0.49	4	1.13 (0.52-2.45)	0.96
BMI ≥18.5		0.89 (0.81-0.99)			1.15 (0.83-1.60)	
Maternal stature						
Height <150 cm	3	0.92 (0.82-1.02)	0.91	3	0.94 (0.46-1.92)	0.66
Height ≥150 cm		0.91 (0.81-1.02)			1.13 (0.75-1.71)	
Maternal hemoglobin at enrollment						
Anemic <110 g/L	3	0.67 (0.55-0.82)	0.04	4	1.25 (0.74-2.10)	0.58
Non-anemic ≥110 g/L		0.91 (0.74-1.13)			1.03 (0.65-1.62)	
Maternal education						
None	3	0.97 (0.84-1.12)	0.38	4	1.09 (0.68-1.74)	0.79
≥1 year formal education		0.90 (0.81-0.99)			0.99 (0.59-1.66)	
Skilled birth attendant						
Yes	3	1.01 (0.90-1.13)	0.01	3	1.50 (0.72-3.11)	0.36
No		0.81 (0.72-0.91)			1.02 (0.69-1.50)	

¹ N Number of studies included in subgroup analysis

Supplemental Table 12. Effect of MMS on infant mortality stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Same dose iron			Lower dose iron in MMS		
	Infant mortality			Infant mortality		
	N ¹	Relative risk (95% CI)	p value heterogeneity	N ¹	Relative risk (95% CI)	p value heterogeneity
Overall-Fixed Effects	2	0.96 (0.87-1.05)	-	5	1.15 (0.89-1.49)	-
Overall-Random Effects	2	1.04 (0.77-1.40)		5	1.15 (0.89-1.49)	-
Infant Sex						
Male	2	1.04 (0.91-1.19)	0.05	5	1.24 (0.85-1.79)	0.56
Female		0.86 (0.75-0.99)			1.06 (0.72-1.55)	
Gestational Age at Enrollment						
<20 Weeks	2	0.96 (0.87-1.05)	0.79	5	1.20 (0.90-1.61)	0.30
≥20 Weeks		0.90 (0.61-1.35)			0.85 (0.47-1.53)	
Maternal adherence to regimen						
< 95% Adherence	2	1.06 (0.93-1.21)	0.05	3	1.08 (0.66-1.77)	0.96
≥ 95% Adherence		0.83 (0.72-0.96)			1.06 (0.67-1.69)	
Maternal Age						
< 20 years	2	0.95 (0.82-1.10)	0.80	5	1.34 (0.84-2.15)	0.46
≥ 20 years		0.97 (0.86-1.11)			1.09 (0.81-1.47)	
Parity						
First birth	2	0.95 (0.83-1.09)	0.98	5	1.24 (0.80-1.92)	0.65
Second + birth		0.96 (0.84-1.09)			1.09 (0.79-1.51)	
Maternal Underweight at enrollment						
BMI <18.5	2	0.96 (0.82-1.12)	0.92	4	1.08 (0.58-2.01)	0.91
BMI ≥18.5		0.95 (0.84-1.07)			1.16 (0.86-1.56)	
Maternal stature						
Height <150 cm	2	0.98 (0.86-1.11)	0.67	3	1.00 (0.49-2.03)	0.83
Height ≥150 cm		0.94 (0.81-1.09)			1.09 (0.74-1.60)	
Maternal hemoglobin at enrollment						
Anemic <110 g/L	2	0.99 (0.56-1.75)	0.82	4	1.19 (0.78-1.82)	0.69
Non-anemic ≥110 g/L		1.08 (0.68-1.70)			1.05 (0.70-1.59)	
Maternal education						
None	2	1.02 (0.87-1.21)	0.31	4	1.00 (0.68-1.47)	0.84
≥1 year formal education		0.92 (0.82-1.03)			1.07 (0.67-1.71)	
Skilled birth attendant						
Yes	2	1.09 (0.96-1.25)	0.38	3	1.66 (0.86-3.20)	0.17
No		0.80 (0.68-0.93)			1.00 (0.72-1.39)	

¹ N Number of studies included in subgroup analysis

Supplemental Table 13. Effect of MMS on low birthweight (<2500g) stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Same dose iron			Lower dose iron in MMS		
	Low birthweight (<2500g)			Low birthweight (<2500g)		
	N ¹	Relative risk (95% CI)	p value heterogeneity	N ¹	Relative risk (95% CI)	p value heterogeneity
Overall-Fixed Effects	8	0.88 (0.85-0.90)	-	8	0.85 (0.77-0.95)	-
Overall-Random Effects	8	0.83 (0.73-0.94)	-	8	0.85 (0.77-0.95)	-
Infant Sex						
Male	8	0.87 (0.83-0.91)	0.49	7	0.81 (0.67-0.99)	0.70
Female		0.89 (0.85-0.92)			0.85 (0.73-0.99)	
Gestational Age at Enrollment						
<20 Weeks	5	0.88 (0.85-0.91)	0.44	7	0.85 (0.73-0.98)	0.66
≥20 Weeks		0.85 (0.77-0.93)			0.79 (0.60-1.04)	
Maternal adherence to regimen						
< 95% Adherence	5	0.89 (0.85-0.92)	0.79	7	0.89 (0.77-1.03)	0.79
≥ 95% Adherence		0.88 (0.84-0.92)			0.76 (0.63-0.93)	
Maternal Age						
< 20 years	8	0.90 (0.86-0.94)	0.91	7	0.87 (0.75-1.01)	0.88
≥ 20 years		0.90 (0.87-0.93)			0.86 (0.73-1.00)	
Parity						
First birth	7	0.88 (0.85-0.92)	0.88	8	0.86 (0.75-0.99)	0.95
Second + birth		0.88 (0.84-0.92)			0.86 (0.72-1.02)	
Maternal Underweight at enrollment						
BMI <18.5	7	0.88 (0.84-0.92)	0.95	8	0.84 (0.69-1.02)	0.87
BMI ≥18.5		0.88 (0.84-0.91)			0.86 (0.75-0.98)	
Maternal stature						
Height <150 cm	7	0.89 (0.86-0.93)	0.23	8	0.86 (0.71-1.02)	0.99
Height ≥150 cm		0.86 (0.81-0.91)			0.86 (0.84-0.99)	
Maternal hemoglobin at enrollment						
Anemic <110 g/L	7	0.74 (0.65-0.83)	<0.001	6	0.88 (0.74-1.04)	0.30
Non-anemic ≥110 g/L		0.98 (0.88-1.08)			0.77 (0.66-0.91)	
Maternal education						
None	7	0.88 (0.84-0.93)	0.01	8	0.88 (0.74-1.04)	0.37
≥1 year formal education		0.87 (0.84-0.91)			0.78 (0.66-0.93)	

* N Number of studies included in subgroup analysis

Supplemental Table 14. Effect of MMS on preterm birth stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Same dose iron			Lower dose iron in MMS		
	Preterm birth			Preterm birth		
	N ¹	Relative risk (95% CI)	p value heterogeneity	N ¹	Relative risk (95% CI)	p value heterogeneity
Overall-Fixed Effects	8	0.92 (0.88-0.96)	-	7	0.97 (0.88-1.06)	-
Overall-Random Effects	8	0.93 (0.86-1.00)	-	7	0.97 (0.88-1.06)	-
Infant Sex						
Male	7	0.91 (0.87-0.96)	0.94	7	1.00 (0.89-1.13)	0.60
Female		0.91 (0.86-0.97)			0.95 (0.84-1.08)	
Gestational Age at Enrollment						
<20 Weeks	5	0.88 (0.84-0.93)	0.003	6	0.92 (0.79-1.08)	0.83
≥20 Weeks		1.01 (0.94-1.09)			0.95 (0.77-1.18)	
Maternal adherence to regimen						
< 95% Adherence	4	0.92 (0.87-0.97)	0.56	6	0.97 (0.85-1.12)	0.77
≥ 95% Adherence		0.90 (0.84-0.95)			0.95 (0.82-1.10)	
Maternal Age						
< 20 years	7	0.90 (0.84-0.97)	0.75	7	1.02 (0.89-1.18)	0.49
≥ 20 years		0.91 (0.78-0.96)			0.96 (0.86-1.07)	
Parity						
First birth	6	0.90 (0.85-0.96)	0.58	7	0.97 (0.86-1.11)	0.87
Second + birth		0.92 (0.87-0.97)			0.96 (0.85-1.08)	
Maternal Underweight at enrollment						
BMI <18.5	5	0.82 (0.76-0.89)	0.004	7	1.03 (0.83-1.29)	0.37
BMI ≥18.5		0.94 (0.90-0.99)			0.95 (0.86-1.05)	
Maternal stature						
Height <150 cm	7	0.91 (0.86-0.96)	0.70	7	0.96 (0.72-1.26)	0.97
Height ≥150 cm		0.92 (0.87-0.97)			0.96 (0.87-1.06)	
Maternal hemoglobin at enrollment						
Anemic <110 g/L	6	1.00 (0.92-1.09)	0.08	5	0.94 (0.78-1.13)	0.51
Non-anemic ≥110 g/L		0.90 (0.82-0.98)			0.85 (0.68-1.06)	
Maternal education						
None	6	0.89 (0.83-0.96)	0.61	7	1.02 (0.90-1.14)	0.10
≥1 year formal education		0.91 (0.87-0.96)			0.86 (0.74-1.01)	

¹ N Number of studies included in subgroup analysis

Supplemental Table 15. Effect of MMS on SGA based on Oken standard stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Same dose iron			Lower dose iron in MMS		
	SGA (Oken)			SGA (Oken)		
	N ¹	Relative risk (95% CI)	p value heterogeneity	N ¹	Relative risk (95% CI)	p value heterogeneity
Overall-Fixed Effects	8	0.98 (0.96-1.00)	-	7	0.90 (0.85-0.96)	-
Overall-Random Effects	8	0.95 (0.89-1.10)	-	7	0.90 (0.85-0.96)	-
Infant Sex						
Male	7	0.98 (0.95-1.01)	0.67	7	0.89 (0.82-0.98)	0.87
Female		0.99 (0.96-1.01)			0.89 (0.81-0.97)	
Gestational Age at Enrollment						
<20 Weeks	5	0.99 (0.97-1.02)	0.002	6	0.88 (0.82-0.95)	0.34
≥20 Weeks		0.90 (0.85-0.95)			0.96 (0.83-1.11)	
Maternal adherence to regimen						
< 95% Adherence	5	0.99 (0.96-1.02)	0.61	6	0.94 (0.85-1.03)	0.41
≥ 95% Adherence		0.98 (0.95-1.01)			0.88 (0.81-0.97)	
Maternal Age						
< 20 years	8	0.98 (0.95-1.01)	0.75	7	0.92 (0.82-1.03)	0.62
≥ 20 years		0.97 (0.94-1.00)			0.89 (0.83-0.96)	
Parity						
First birth	7	0.98 (0.95-1.01)	0.91	7	0.92 (0.84-1.00)	0.47
Second + birth		0.98 (0.95-1.01)			0.88 (0.80-0.96)	
Maternal Underweight at enrollment						
BMI <18.5	8	1.00 (0.97-1.03)	0.19	7	0.92 (0.81-1.04)	0.90
BMI ≥18.5		0.97 (0.95-1.00)			0.91 (0.85-0.97)	
Maternal stature						
Height <150 cm	8	0.99 (0.96-1.01)	0.55	7	0.88 (0.79-0.98)	0.71
Height ≥150 cm		0.97 (0.94-1.00)			0.90 (0.84-0.97)	
Maternal hemoglobin at enrollment						
Anemic <110 g/L	7	0.90 (0.84-0.96)	0.02	5	0.87 (0.78-0.97)	0.64
Non-anemic ≥110 g/L		1.00 (0.95-1.06)			0.90 (0.82-0.99)	
Maternal education						
None	7	1.00 (0.96-1.04)	0.16	7	0.93 (0.85-1.03)	0.29
≥1 year formal education		0.97 (0.95-0.99)			0.87 (0.81-0.94)	

* N Number of studies included in subgroup analysis

Supplemental Table 16. Effect of MMS on LGA based on Oken standard stratified by iron dose provided in the MMS arm as compared to the IFA comparison arm

	Same dose iron			Lower dose iron in MMS		
	LGA (Oken)			LGA (Oken)		
	N ¹	Relative risk (95% CI)	p value heterogeneity	N ¹	Relative risk (95% CI)	p value heterogeneity
Overall-Fixed Effects	7	1.05 (0.95-1.17)	-	5	1.02 (0.80-1.30)	-
Overall-Random Effects	7	1.10 (1.02-1.18)	-	5	1.02 (0.79-1.31)	-
Infant Sex						
Male	7	1.12 (0.97-1.28)	0.20	5	1.07 (0.79-1.44)	0.68
Female		0.98 (0.84-1.14)			0.97 (0.72-1.32)	
Gestational Age at Enrollment						
<20 Weeks	4	0.99 (0.85-1.14)	0.07	4	1.02 (0.69-1.51)	0.89
≥20 Weeks		1.20 (1.03-1.40)			0.97 (0.55-1.71)	
Maternal adherence to regimen						
< 95% Adherence	4	1.08 (0.94-1.25)	0.55	4	0.79 (0.57-1.11)	0.09
≥ 95% Adherence		1.05 (0.94-1.17)			1.19 (0.86-1.64)	
Maternal Age						
< 20 years	7	0.92 (0.71-1.19)	0.31	4	1.12 (0.77-1.65)	0.75
≥ 20 years		1.07 (0.95-1.19)			1.04 (0.80-1.35)	
Parity						
First birth	6	0.88 (0.71-1.09)	0.04	4	1.08 (0.78-1.49)	0.81
Second + birth		1.14 (1.01-1.28)			1.02 (0.76-1.37)	
Maternal Underweight at enrollment						
BMI <18.5	5	0.71 (0.47-1.07)	0.06	3	0.87 (0.54-1.39)	0.38
BMI ≥18.5		1.08 (0.95-1.22)			1.11 (0.83-1.49)	
Maternal stature						
Height <150 cm	7	0.95 (0.79-1.15)	0.21	3	0.63 (0.28-1.39)	0.26
Height ≥150 cm		1.10 (0.97-1.25)			1.02 (0.78-1.32)	
Maternal hemoglobin at enrollment						
Anemic <110 g/L	6	1.26 (1.05-1.51)	0.12	3	1.16 (0.63-2.12)	0.41
Non-anemic ≥110 g/L		1.01 (0.81-1.25)			0.77 (0.35-1.66)	
Maternal education						
None	5	1.05 (0.83-1.32)	0.96	4	1.11 (0.79-1.56)	0.15
≥1 year formal education		1.05 (0.94-1.19)			0.74 (0.47-1.16)	

¹ N Number of studies included in subgroup analysis

Table 17A. The effect of MMS on neonatal mortality (1 month) by sex, omitting single studies, one at a time

Study Omitted	Female ES (95% CI)	Male ES (95% CI)
Summit 2008	0.88 (0.77,1.01)	1.06 (0.94,1.20)
West 2014	0.86 (0.72,1.03)	1.03 (0.90,1.18)
All Studies	0.85 (0.75,0.96)	1.06 (0.95,1.17)

Table 17B. The effect of MMS on neonatal mortality (1 month) by maternal education, omitting single studies, one at a time

Study Omitted	No Education ES (95% CI)	≥ 1 Year Education ES (95% CI)
Summit 2008	1.12 (0.93,1.34)	0.94 (0.84,1.06)
West 2014	1.29 (1.00,1.66)	0.88 (0.76,1.01)
All Studies	1.13 (0.97,1.31)	0.92 (0.83,1.01)

Table 17C. The effect of MMS on 6-month mortality by anemia (<110g/L), omitting single studies, one at a time

Study Omitted	Hgb < 110 g/L ES (95% CI)	Hgb ≥ 110 g/L ES (95% CI)
Summit 2008	1.03 (0.72,1.48)	1.02 (0.78,1.33)
West 2014	0.68 (0.57,0.83)	0.93 (0.77,1.12)
All Studies	0.71 (0.60,0.86)	0.93 (0.78,1.11)

Table 17D. The effect of MMS on 6-month mortality by skilled birth attendant (SBA), omitting single studies, one at a time

Study Omitted	SBA ES (95% CI)	No SBA ES (95% CI)
Summit 2008	1.09 (0.96,1.23)	0.81 (0.70,0.93)
West 2014	0.88 (0.75,1.04)	0.88 (0.75,1.04)
All Studies	1.00 (0.90,1.11)	0.82 (0.74,0.92)

Table 17E. The effect of MMS on infant mortality by sex, omitting single studies, one at a time

Study Omitted	Female ES (95% CI)	Male ES (95% CI)
Summit 2008	n/a	n/a
West 2014	0.97 (0.73,1.30)	1.15 (0.90,1.49)
All Studies	0.87 (0.77,0.99)	1.05 (0.93,1.18)

Table 17F. The effect of MMS on infant mortality by skilled birth attendant (SBA), omitting single studies, one at a time

	SBA	No SBA
Study Omitted	ES (95% CI)	ES (95% CI)
Summit 2008	n/a	n/a
West 2014	1.01 (0.78,1.30)	1.07 (0.77,1.49)
All Studies	1.06 (0.95,1.20)	0.82 (0.71,0.95)

Table 17G. The effect of MMS on low birth weight (<2500g) by anemia (<110 g/L), omitting single studies, one at a time

	Hgb < 110 g/L	Hgb ≥ 110 g/L
Study Omitted	ES (95% CI)	ES (95% CI)
Summit 2008	0.86 (0.76,0.92)	0.90 (0.86,0.97)
West 2014	0.81 (0.73,0.89)	0.90 (0.86,0.98)
All Studies	0.81 (0.74,0.89)	0.91 (0.85,0.98)

Table 17H. The effect of MMS on preterm birth by maternal BMI omitting single studies, one at a time

	BMI <18.5 kg/m ²	BMI ≥18.5 kg/m ²
Study Omitted	ES (95% CI)	ES (95% CI)
Summit 2008	0.84 (0.78,0.91)	0.92 (0.88,0.96)
West 2014	0.94 (0.81,1.09)	0.97 (0.92,1.03)
All Studies	0.84 (0.78,0.91)	0.94 (0.90,0.98)

Table 17I. The effect of MMS on preterm birth by gestational age at enrollment, omitting single studies, one at a time

	<20 Weeks	≥20 Weeks
Study Omitted	ES (95% CI)	ES (95% CI)
Summit 2008	0.88 (0.83,0.92)	0.98 (0.88,1.09)
West 2014	0.93 (0.86,1.01)	1.03 (0.95,1.12)
All Studies	0.89 (0.85,0.93)	1.00 (0.94,1.08)

Table 17J. The effect of MMS on SGA (Oken) by gestational age at enrollment, omitting single studies, one at a time

	<20 wks	≥20 Weeks
Study Omitted	ES (95% CI)	ES (95% CI)
Summit 2008	0.99 (0.96,1.01)	0.92 (0.87,0.98)
West 2014	0.99 (0.95,1.03)	0.85 (0.80,0.91)
All Studies	0.99 (0.97,1.01)	0.91 (0.86,0.96)

Table 17K. The effect of MMS on LGA (Oken) by maternal BMI, omitting single studies, one at a time

	BMI <18.5 kg/m ²	BMI ≥18.5 kg/m ²
Study Omitted	ES (95% CI)	ES (95% CI)
Summit 2008	0.74 (.53,1.04)	1.10 (0.96,1.27)
West 2014	0.94 (0.64,1.39)	1.11 (0.98,1.25)
All Studies	0.77 (0.57,1.05)	1.08 (0.97,1.21)

Table 17L. The effect of MMS on SGA (Oken) by maternal anemia status, omitting single studies, one at a time

	Hgb < 110 g/L	Hgb ≥ 110 g/L
Study Omitted	ES (95% CI)	ES (95% CI)
Summit 2008	0.92 (0.87,0.97)	0.99 (0.95,1.03)
West 2014	0.91 (0.86,0.96)	0.99 (0.94,1.04)
All Studies	0.92 (0.87,0.97)	0.99 (0.95,1.03)

